FAST MOVING TECHNOLOGY

STÄUBLI

CS8C Controller

Instruction manual



D.280.812.04-F - 24/07/2019
CS8C - Master
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A "readme.pdf" document may be delivered on the robot DVD. It contains the documentation addenda and errata.

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1 - INTRODUCTION

1.1 - FOREWORD

M0000242.1

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Electrical drawings provided in this manual are for information only. The reference for robot electrical drawings is provided in a separate manual. The photos are used to make the document easier to understand, they cannot be construed as being of a contractual nature.

Please report errors and omissions to the Stäubli Robotics Customer Support contact for your country, that can be found at:

- http://www.staubli.com/en/contacts/division/robotics/

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1.1.1 - OBJECTIVE OF THIS MANUAL

M0000548.1

The objective of this manual is to provide information concerning the installation, use and maintenance of the Stäubli CS8C controller. It provides help for people working on the equipment, for reference purposes only. Stäubli recommends attending training courses for easier understanding of this manual, better use and robot maintenance.

1.1.2 - SPECIAL MESSAGES CONCERNING SAFETY, DANGERS, CAUTIONS AND INFORMATION

M0000549.1

In this document, there are several icons for important warnings.

These warnings are as follows (by decreasing order of importance):



DANGER
Instructions drawing the reader's attention to the risks of accidents that could lead to serious bodily harm if the steps shown are not complied with. In general, this type of indication describes the potential danger, its possible effects and the necessary steps to reduce the danger. It is essential to comply with the instructions to ensure personal safety.



SAFETY
Instructions drawing the reader's attention that they are responsible if the steps shown are not complied with. It is essential to comply with the instructions to maintain the robot safety level.



Instructions directing the reader's attention to the risks of material damage or failure if the steps shown are not complied with. It is essential to comply with these instructions to ensure equipment reliability and performance levels.



Supplies further information, or underlines a point or an important procedure. This information must be memorized to make it easier to apply and ensure correct sequencing of the operations described.



ESD
This symbol is a reminder of the possible consequences of touching electrostatically sensitive components (see chapter 3.4).

1.2 - DEFINITIONS AND GLOSSARY

M0000594.1

In this manual, a specific vocabulary is used according to our products. This chapter allows an easier general understanding.

People in contact with the product are designated by different names:

Person: General term identifying all individuals likely to come close to the Stäubli machine.

Staff: Identifies people specifically employed and trained to install, operate, and service the Stäubli machine.

User: Refers to people or the company responsible for operating the Stäubli machine.

Operator: Refers to the person who starts or stops the robot, or controls its operation.

Integrator: Refers to people or the company responsible for the design and manufacture of the robot cell.

Item	Definition
ABZ	Input board for incremental ABZ encoders
ARPS	Auxiliary Robot Power Supply (logical voltages)
BIO	Basic Inputs Outputs
BRB	Brake Release Board
BRK	Brake
COD	Arm Encoder
CPT	Computer drawer
DIG	Digital part of the amplifier

Item	Definition
DOOR	Door contact
DSI	Dual Sensor Interface board in Arm
EV	Solenoid valve
FAN	Fan
Fieldbus	Industrial bus to connect Inputs/Outputs devices
IC	Interconnect Cable (between controller and arm)
LSW	Limit Switch
MCP	Manual Control Pendant
MCPES	Manual Control Pendant Emergency Stop
MOT	Motor
PSM	Power Supply Module (power voltage)
PWR	Power part of the amplifier
REGEN	Regeneration resistance
RPS	Robot Power Supply
RSI	Robot Safety Interface
S1	Main switch
SRC	Stäubli Robotics Controls: The firmware running on the Stäubli controllers
SRS	Stäubli Robotics Suite: The PC software to configure, develop, tune and maintain a robotics application
STARC	Stäubli Advanced Robot Control, a patented architecture
Th	Thermal Sensor
UESA	User Emergency Stop
UESB	User Safety Stop
USEREN	User validation
VAL 3	The Stäubli robotics programing language for Stäubli controllers
uniVAL	A SRC control mode allowing to drive the robot from an external computer through an industrial fieldbus
WMS	Working Modes Selection control panel
WMSES	Working Mode Selection Emergency Stop

2 - DESCRIPTION OF THE CONTROLLER

2.1 - IDENTIFICATION

M0000243.1

Robots are manufactured by

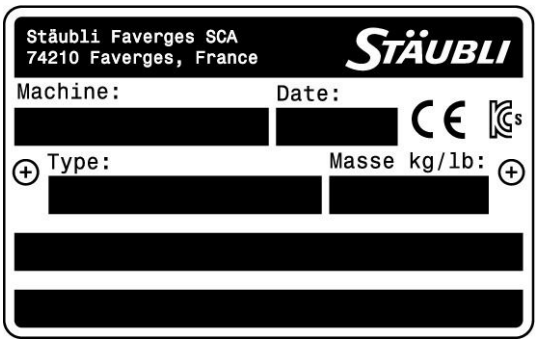
Stäubli Faverges SCA

Place Robert Stäubli

74210 Faverges France

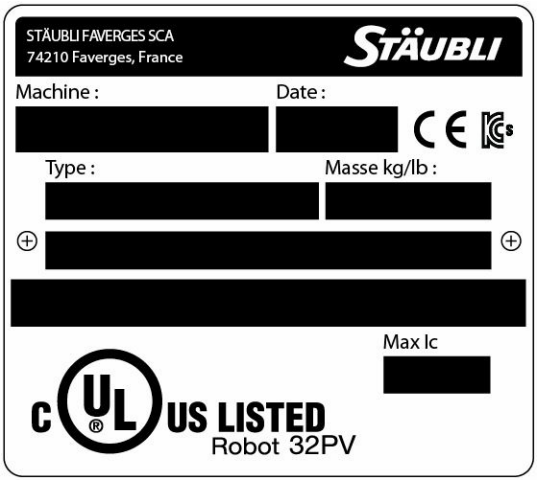
They are identified by a plate attached to the controller and to the arm.

Example:



I0000124

Figure 2.1 : Standard version



I0000014

Figure 2.2 : UL version

The CE mark placed on the identification plate refers to the "Electromagnetic Compatibility" directive.

SAFETY
The manufacturer's identification plate must remain present and legible throughout the robot service life.

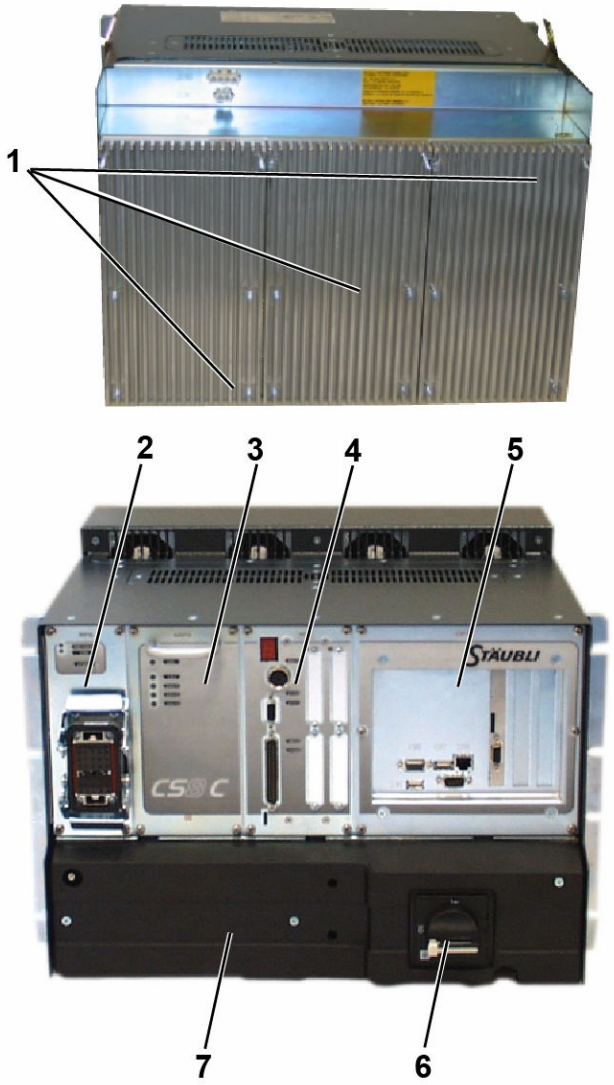
Whenever requesting information, ordering spare parts, or requesting intervention, please state the type and serial number of the machine concerned, as shown on the identification plate.

You can find the telephone number of your nearest contact on a label placed on the controller. You can also find our list of contacts on the following site: http://www.staubli.com/contacts.

2.2 - LOCATION AND DESCRIPTION OF THE MAIN COMPONENTS

2.2.1 - THE CONTROLLER

M0000201.1



I0000402

Figure 2.3

The CS8C controller is made up of a processor **(5)**, the intelligent part of the installation.

The processor controls the robot via digital power amplifiers **(1)** dedicated to each axis of the arm.

The electrical power is converted by the PSM **(7)** power section, the RPS **(2)** power supply, and the ARPS **(3)** power supply which supplies to each of the above elements the voltage required for correct operation from the mains voltage delivered by the electrical network.

The functions required for electrical safety are grouped together on the RSI board.

To disconnect the system from the power supply, set the master switch **(6)**, located on the front panel of the controller, to 0. Before doing so, you must stop the arm motion and switch off arm power supply.

2.2.2 - THE MCP

M0000241.1

General description

The MCP (Manual Control Pendant) can be used to enable arm power supply and control its movements.

Location and handling of the MCP



I0000192

Figure 2.4

The MCP is provided with a holder that must be fixed outside of the cell. It has two functions:

- To ensure that the MCP is easily accessible for the operator.
- To detect the presence of the MCP outside the cell to allow arm power on after an emergency stop.

It is necessary to use the enabling device (#) located under the MCP to switch on arm power. The button location enables it to be used by right-handed or left-handed people alike.

2.2.3 - WMS FRONT PANEL

M0000384.1

Modes of operation are selected from the WMS front panel which has to be installed permanently outside of the cell. The removable 3-position keyswitch prevents from changing the mode when it is not allowed.

3 - SAFETY

3.1 - SAFETY REGULATIONS

M0000550.1

Intended use

A robot is not a ready-to-use product. It is a "partly completed machinery" designed to be integrated in a "machine": the robot cell. This machine as a whole shall be compliant to the applicable safety regulations before it can be operated.

The Stäubli robot is designed for industrial applications, where the operator is safely separated from a moving robot with protective devices. Temporary proximity between the operator and the robot is possible for collaborative operation, or for tuning and maintenance tasks, with additional risks that shall be handled by the integrator.

European regulations

In the European Union, the applicable safety regulations for a machine are defined by the Machinery Directive 2006/42/EC. This directive specifies that a risk assessment shall be carried out in order to determine the health and safety requirements which apply to the machinery. The machinery must then be designed and constructed taking into account the results of the risk assessment. The directive defines essential health and safety requirements that are mandatory.

International standards have been developed to provide a framework for the correct application of the Machinery Directive. The Machinery Directive, and harmonized standards list, can be found at http://eur-lex.europa.eu. For robot integration, application of the specifications of the EN ISO 10218-2:2011 standard on the basis of the integrator risk assessment confers a presumption of conformity with the essential health and safety requirements of the Machinery Directive covered by the standard.

The Machinery Directive defines the procedures to evaluate the machinery conformity. For completed machinery, the CE mark accompanied by the EC conformity declaration confers presumption of compliance with the Directive. For partly completed machinery, the incorporation declaration defines the essential health and safety requirements and other applicable Directives that are satisfied.

The robot is delivered with its incorporation declaration. An electronic copy including several translations is available on the robot DVD. It can also be requested from the Stäubli Customer Support contact.

The robot incorporation declaration is based on the applicable requirements implementation of the following harmonized standards:

■ EN ISO 10218-1:2011	Robots and robotic devices - Safety requirements for industrial robots - Part 1
■ EN ISO 13849-1:2015	Safety of machinery - Safety-related parts of control systems - Part 1
■ EN 60204-1:2006 and EN 60204-1/A1:2009	Safety of machinery - Electrical equipment
■ EN 61000-6-4:2007 / A1:2011	Electromagnetic compatibility - Generic emissions standard
■ EN 61000-6-2:2005	Electromagnetic compatibility - Generic immunity standard

In the United States, there are no standardized Occupational Safety and Health Administration regulatory standards for robots. However, standard ANSI/RIA R15.06-2012 is referenced as the standard applicable to robot systems. That standard is harmonized to make it basically identical to standards ISO 10218-1:2011 and ISO 10218-2:2011, in the same way as standard CAN/CSA-Z434-03 (R2013) for Canada. These ISO standards for industrial robots have also been adopted in Korea.

Each country can draw up further safety standards, and implement specific local rules. For example, the UL standard for robots and robot equipment UL 1740 sets out further requirements covering the robot and electrical safety.

For the UL version, the robot system has been evaluated to meet the following standards:

- Standard UL 1740 Robots and Robotics Equipment
- Standard RIA15-06 American National Standard for Industrial Robots and Robot Systems. Safety Requirements.
- Standard CSA Z434-03 Industrial Robots and Robot Systems. General Safety Requirements.
- Standard NFPA 79 Electrical standard for industrial machinery
- Standard NFPA 70 National Electrical Code

For relative information about REACH 1907/2006 regulation (CE) consult the following internet address: https://www.staubli.com/en/robotics-reach/

3.2 - SAFETY DIRECTIVES CONCERNING TO THE WORK ENVIRONMENT

3.2.1 - ANALYSIS OF SAFETY AROUND THE MACHINE

M0000142.1

Safety must be taken into account for the machine from the design and development stage on.

Before planning the installation of the machine, it is necessary to study the following points:

- Plan the safety strategies that reduce risks to an acceptable level.
- Define the tasks required for the foreseeable applications and assess the access and/or approach requirements.
- Identify the sources of risks including the failures and the failure modes associated with each of the tasks. The risks can involve:
  - the cell itself
  - its association with other items of equipment
  - the interactions between people and the cell.
- Assess and estimate the risks stemming from cell operation:
  - programming risks
  - operating risks
  - risks during use
  - maintenance risks for the machine.
- Select the protective methods:
  - use of protective devices
  - installation of signalling means
  - compliance with safe working procedures.

These points are taken from the standards applicable to robots.

This list is not exhaustive. Above all, it is necessary to comply with the standards in force in your country.

3.2.2 - RULES CONCERNING THE ROBOT'S WORK AREA

M0000143.1

The controlled area or isolation area in which the robot moves must be determined using protective devices (protective elements).

Protective elements are devices protecting people from a dangerous area. See the standards currently in force concerning safety for industrial handling equipment.

DANGER

At the time of an emergency stop, the final position of the arm can never be determined precisely because of the kinetic energy involved. It is thus necessary to make sure that no persons or obstructions are present in the robot's work area when the arm is powered up.

3.3 - STAFF REQUIREMENTS

3.3.1 - ABILITY

M0000743.1

Stäubli robots work with computer controlled mechanisms, capable of moving at high speed and exerting considerable force. Like all robots and most industrial equipment, they must be operated with great care by the user of the machine. All staff using Stäubli robots must be familiar with the warnings and recommendations given in this manual. The safety rules, warning, notes and recommendations provided in this safety chapter are only one part of the over all safety instructions. Other information is provided throughout the manual.

It is prohibited to modify a robot without a written agreement from Stäubli.

3.3.1.1 - Mechanical and electrical dangers

M0000545.1



This sign, applied on different parts of the robot, indicates that there is a potential electrical danger and that only qualified service personnel may install or service the robot system.



DANGER

- Only qualified service personnel may install or service the robot system.
- Ensure compliance with all local and national safety and electrical codes for the installation and operation of the robot system.
- Disconnect all the electrical and pneumatic power supplies before carrying out any work on the controller or the arm.
- Lockout and tagout equipment before servicing.

To turn off power, set the CS8C main switch to the "0" position.

To prevent inadvertent during the service operation, the main switch must be locked in 0 position using a padlock whose key is to be kept by the person carrying out the service operation. The locked status must be shown by a sign. For example, put a "Do not operate" sign in place.

Before powering up the system, make sure that all the electrical protection systems have been fitted and that there is no risk of electric shocks.



When the main switch is set to 0, voltage remains present between the input terminals (mains supply filter) and the main switch input.



DANGER

Each time the arm is powered on, keep one hand close to the "Emergency stop" button in order to be able to press it as quickly as possible in the event of a problem.

- For UL robots: When arm is powered-on, a light on the arm is on to indicate there is a potential danger. This light is also on when manual brake release is performed (on axis 1 on RX and TX robots, on axis 3 on Scara robots).
- Do not connect or disconnect components while the unit is under power. The connection between the controller and the robot arm can only be made if the controller has been switched off.
- Remove part or tool loaded on robot during maintenance operations.

- If unusual sounds or vibrations are noted on the robot arm, especially following a shock or some other incident, it is necessary to inspect the tool and gripper fastenings carefully and make diagnoses at low speed.

- If a crash of the arm occurs, all safety components involved in the safety have to be checked carefully to verify they are still operating and not damaged: hard stop devices on the arm, electrical limit switches, calibration of the robot. Don't hesitate to call Stäubli service for any doubt.
- Each time a calibration, adjustment or recovery procedure is done, the calibration of the arm has to be controlled carefully to verify that the robot is able to move in its expected angular range and not more. This verification has to be done at slow speed.

DANGER
All people are prohibited from remaining in the isolation area in which the robot arm operates. Certain robot working modes such as the "brake release" mode can lead to unforeseeable arm movements.

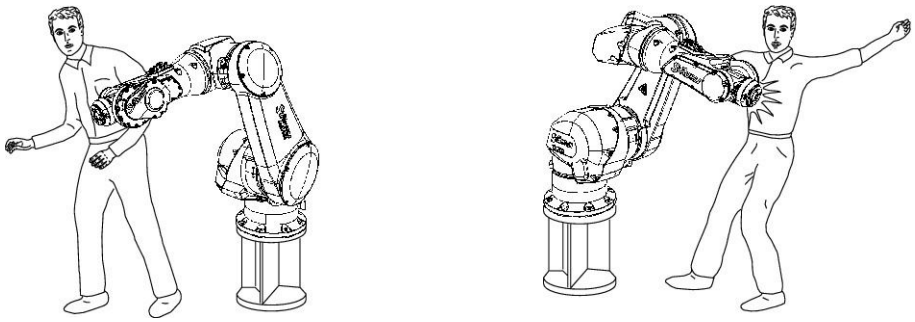


Figure 3.1

Following maintenance work, whether it involves mechanical, electrical, pneumatic or software operations, it is advisable to make sure that the robot functions correctly, first at low speed while the person stays outside the cell, and then under the normal conditions of use. In particular, make sure that all the protective and safety systems are correctly in place, and that calibration of the robot is correct.

3.3.1.2 - Machine safety devices

M0000546.1

The safety devices must form an integral part of the design and installation of the machine. Operator training and compliance with the operating procedures constitute a major element in setting up the safety devices and systems.

Stäubli robots feature various interfaces that help the user to develop safety systems and devices for the machine. These functions include the emergency stop circuits, the digital input/output lines, and the display system for error and warning messages (see the "Integration" chapter in the controller instruction manual). When the system is used without the MCP, these messages/warnings can only be consulted via the application program.

3.4 - SAFETY DIRECTIVES CONCERNING TO PROTECTION OF THE EQUIPMENT

M0000208.1

DANGER
To ensure reliability and precision in the robot's movements, the machine environment must comply with the levels of electrical disturbance set out in the safety standards.

3.4.1 - CONNECTIONS

M0000203.1

- Before connecting the controller to the power supply, make sure that its nominal voltage does indeed correspond to the network voltage.
- When connecting the controller, use a cable whose cross-section corresponds to the power rating shown on the manufacturer's plate.
- Before removing or inserting an electronic component, switch off arm power and then switch off controller power and comply with the procedure.
- Take care to avoid blocking the air inlets and outlets for the controller airflow path.
- Never use the emergency stop to power down the arm under normal conditions of use.

3.4.2 - INFORMATION ON ELECTROSTATIC DISCHARGES

M0000207.1

What is an electrostatic discharge?

Everyone has felt the effects of static electricity on their clothes or when they touch a metal object, without being aware of the damage that can be done to electronic components by static electricity.

Our desire to integrate the notions of quality and reliability in our products makes it necessary to prevent electrostatic discharges from causing damage to them. This means that all the staff and login users must be informed.

Storage of a charge

An electric charge is created simply by combining a conductor, a dielectric and the ground (lowest reference potential, usually the ground in the case of an electrostatic charge).

Example: People, printed circuits, integrated circuits, components, conducting mats when separated from the ground by a dielectric.

Electrostatic discharges or ESD

Most people have experienced ESD by receiving an electric shock when walking on a carpet and touching a doorknob or when getting out of a car.

In most cases, the following is true:

- To feel an ESD, a charge of at least 3 500 V is required.
- To hear one, a charge of at least 5 000 V is required.
- To see a spark, a charge of at least 10 000 V is required.

This shows that it is possible to develop charges of over 10 000 V before noticing an electrostatic discharge!

Risks created by an electrostatic discharge

A high ESD voltage (several thousand volts) creates danger for electronic components. A semi-conductor must be handled carefully to prevent destruction by ESD. It is estimated that ESD destroy only 10% of the components that they effect. The other 90% of components fall into the "deteriorated" category. A component may be damaged with simply 25% of the voltage required to destroy it.

These hidden faults can lead to problems that appear several days, weeks or even months after the incident. Components may also undergo a change in their operating characteristics. Initial tests are successfully passed but an intermittent error occurs under vibration or temperature constraints. The same components will pass the "on/off" test successfully, as carried out during repairs, but the problem will reappear again once on site.

Typical ESD voltages

Source	Low relative humidity 10 - 20%	Average relative humidity 40%	High relative humidity 65 - 90%
Walking on carpet	35 kV	15 kV	1.5 kV
Walking on vinyl	12 kV	5 kV	0.3 kV
Working at the workstation	6 kV	2.5 kV	0.1 kV
Plastified instructions	7 kV	2.6 kV	0.6 kV
Polyethylene bags	20 kV	2 kV	1.2 kV
Cellular polyurethane	18 kV	11 kV	1.5 kV

Charge sources

- Work surfaces
- Packaging
- Floors
- Handling
- Chairs
- Assembly
- Carriages
- Cleaning
- Clothes
- Repairing

Parts sensitive to static charges

- Electronic cards
- Power supplies
- Encoders
- etc

3.4.3 - PREVENTION OF DAMAGE DUE TO ELECTROSTATIC DISCHARGES M0000547.1

It is essential to guard against electrostatic discharges during an intervention concerning electronic components, subassemblies and complete systems.

Elimination of the danger due to ESD requires a combined team effort. By complying with the following instructions, the user can substantially reduce the potential damage caused by ESD and ensure long-term reliability for the robot:

- Inform the staff of the risks stemming from ESD.
- Know the critical zones sensitive to ESD.
- Know the rules and procedures to deal with ESD.
- Always carry components and boards in a tray to protect them from electrostatic charges.
- Always ground yourself before working on a workstation.
- Keep non-conducting equipment (static charge generators) away from components and boards.
- Use tools providing protection from ESD.

Stäubli workstation

To handle electronic cards, Stäubli workstations are given a grounded coating that dissipates static electricity.

Work zones

An antistatic bracelet is required to handle boards or electronic components.

Remove objects that generate static electrical charges from the work area.

Printed circuits, boards and electronic components must be kept in antistatic bags.

Antistatic wrist strap

The wrist straps are supplied as part of the standard equipment for the robot.



Use an antistatic wrist strap and mat connected to the frame of the controller or to the frame of the arm and the ground during all handling of electronic boards or electrical components.

3.5 - SAFETY FUNCTIONS OF THE ROBOT

3.5.1 - OVERVIEW OF THE SAFETY FUNCTIONS

M0000468.1

The controller implements the safety functions listed hereafter, according to EN ISO 13849-1:2008, and the corresponding IEC 62061: 2005 performance level with a proof test interval of 20 years:

Function	Level	PFHd	Chapter
EStop push button and other safety stop functions	Cat3, PLd / SIL2, HFT1	$\leq 1.0\ 10^{-7}$	3.5.3
Door	Cat3, PLd / SIL2, HFT1	$\leq 1.0\ 10^{-7}$	3.5.4
MCP enabling device	Cat3, PLd / SIL2, HFT1	$1.0\ 10^{-7}$	3.5.5
Limited speed in manual mode	Cat3, PLd / SIL2, HFT1	$2.7\ 10^{-7}$	3.5.6
Emergency stop output signal	Cat3, PLd / SIL2, HFT1	$5.2\ 10^{-7}$	3.5.7
Power status output signal	Cat3, PLd / SIL2, HFT1	$2.3\ 10^{-7}$	3.5.8
Axes limiting devices	Refer to arm instruction manual		

The performance level of the safety functions is valid with the delivered version of the Stäubli Robotics Controls software:

- s6.9 (or newer version) for Stäubli Robotics Controls s6.
- s7.3 (or newer version) for Stäubli Robotics Controls s7.

3.5.2 - RESIDUAL RISKS

M0000171.1

Whatever the safety functions implemented, there are some residual risks due to the type of machine:

Adequate safety performance level	A Category 3 PLd safety function is usually adequate for a temporary protection of the operator. A permanent protection of the operator may require a higher performance of the safety function. The required safety performance level can only be determined by an application-specific risk analysis.
Faultless conditions	The robot must only be operated in faultless conditions. Some safety functions require specific conditions of use to guaranty their integrity. It is recommended to keep records of all safety related actions on the robot and on the cell in a "safety book" or the like. Special care must be taken during maintenance and restarting phases.
Adequate conditions of use	Safe operation in the cell does not rely only on the safety functions of the robot. Refer to the instruction manuals of arm and controller for robot conditions of use. These recommendations do not prevent from a detailed risk analysis related to the specificities of the cell. Special care must be taken during installation and maintenance phases.
Risk signaling	ISO 10218-1, 5.7.3 recommends that "wherever possible, the manual mode of operation shall be performed with all persons outside the safeguarded space".
	The arm must be secured before any operation under its mechanical parts, in manual mode, maintenance, or when using the manual brake release function.
	Before operating near the robot it must be assured that the brakes are operational.
	Risks of impingement between the moving robot and the cell environment.
	Risks of contact with moving edges or hot parts of the robot, even at low speed.
	Risks of potentially dangerous situations (falling part when opening the gripper, command of other moving parts in the cell...) due to the activation of output signals.
	Risks of reduced attention to other hazards in the cell (steps, other moving parts, sharp edges...) when paying attention to the robot.
	Risk of a sudden operator reaction to an unexpected robot movement, even at low speed (incorrect command for an operation, major joint movement caused by a small movement of the tool centre, untoward program command).
	Risks of crushing during arm lifting or movement, or release of a joint brake.

3.5.3 - STOP FUNCTION

3.5.3.1 - Definition

M0000387.1

When one of the stop conditions is activated the robot is stopped safely followed by a power-off and brakes activation sequence (stop category 1 as defined by EN 60204-1). The safe stop is maintained until all stop conditions are released.

The stops inputs are described in figure 5.1 (Integration chapter):

- MCPES, WMSES, UESA, USEREN (in manual mode only), UESB.

3.5.3.2 - Limits and conditions of use

M0000388.1

- Refer to chapter 5.1 for correct integration.
- As the emergency stops are not the normal way to stop a robot, they may not be used very often. For E-Stop push buttons or other stop channels which are rarely used, it is requested to test them once a year with the robot operating at low speed and to keep records of these tests. Other limitations may be necessary according to the constraints of the application. The goal of this test is to ensure that both contacts of all stop signals are operating properly. If one of the contact is not operating, messages will be indicated on the MCP and a repair procedure will be necessary (see chapter 5.1).
- Manual brake release remains possible when the safe stop is maintained.
- It is strictly prohibited to bypass Stop contacts.
- E-Stop equipment must comply with EN ISO 13850.
- With LLI and uniVAL options: the deceleration and power-off control of the robot must be handled by the user motion control. If arm power is not shutdown within 0.5s, the Stäubli safety stop function is activated that forces the power-off and brakes activation sequence.
- With external axis option: With external control hardware (motor amplifier, brake control), the power-off and brake activation sequence of the external axis must be safely designed, based on the safe outputs of the controller.

3.5.3.3 - Performance

M0000389.1

The stop function has the following safety performance parameters:

STOP: EN ISO 13849-1:2008 Category 3 PLd

With $MTTF_d$ = 97.2 years, DC = 93.8%, CCF = 90 --> $PFH_d \leq 1.0\ 10^{-7}$

3.5.3.4 - Response time

M0000406.1

The response time is defined in arm manuals (time to stop, angle to stop) according to different speed and payload configurations.

With LLI and uniVAL options, the response time depends on the deceleration of the user motion control.

3.5.4 - DOOR FUNCTION

3.5.4.1 - Definition

M0000391.1

If the robot is not in manual mode, when the door contacts open the robot is stopped safely followed by a power-off and brakes activation sequence (stop category 1 as defined by EN 60204-1). The safe stop is maintained until all stop conditions are released.

The door inputs are described in figure 5.1 (Integration chapter): DOOR signals.

3.5.4.2 - Limits and conditions of use

M0000392.1

Refer to limits and conditions of use of the safety stop function.

3.5.4.3 - Performances

M0000393.1

The door function has the following safety performance parameters:

DOOR: EN ISO 13849-1:2008 Category 3 PLd

With $MTTF_d$ = 86.2 years, DC = 93.9%, CCF = 90 --> $PFH_d \leq 1.0\ 10^{-7}$

3.5.4.4 - Response time

M0000394.1

The response time is defined in arm manuals (time to stop, angle to stop) according to different speed and payload configurations.

3.5.5 - MCP ENABLING DEVICE FUNCTION

3.5.5.1 - Definition M0000396.1

If the robot is in manual mode, releasing or pressing completely the MCP enabling device creates a safe robot stop as defined with the safety stop function.

It is only possible to enable arm power if the MCP enabling device is pressed on its middle position.

3.5.5.2 - Limits and conditions of use M0000397.1

- This safety function is only applicable to a Stäubli MCP. Third party MCP or simulated MCP are out of the scope of this safety function.
- The panic position of the enabling device must be tested once a year with the robot operating at low speed and records of these tests must be kept.
- Refer to residual risks when operating near the robot (see chapter 3.5.2).
- With LLI, uniVAL and external axis options: Refer to limits and conditions of use of the safety stop function.

3.5.5.3 - Performances M0000398.1

The Stäubli MCP enabling device function has the following safety performance parameters:

Stäubli MCP enabling device: EN ISO 13849-1:2008 Category 3 PLd

With $MTTF_d$ = 48 years, DC = 93.7%, CCF = 70 --> PFH_d = 1.0 10^{-7}

3.5.5.4 - Response time M0000394.1

The response time is defined in arm manuals (time to stop, angle to stop) according to different speed and payload configurations.

3.5.6 - LIMITED SPEED IN MANUAL MODE FUNCTION

3.5.6.1 - Definition M0000401.1

If arm velocity exceeds safety limits the robot is stopped with a power-off and brakes activation sequence (stop category 0 as defined by EN 60204-1).

3.5.6.2 - Limits and conditions of use M0000475.1

- The Cartesian velocity of the flange and the TCP is limited to 250 mm/sec as required in chapters 5.6 and 5.7.3 of ISO 10218-1:2011.
- This safety function, coupled with a very fast reaction time, minimizes the risks from any sudden move of the robot in the eventuality of a system failure.
- Refer to residual risks when operating near the robot (see chapter 3.5.2).

3.5.6.3 - Performances M0000426.1

The limited speed function has the following safety performance parameters:

Limited speed in manual mode: EN ISO 13849-1:2008 Category 3 PLd

With $MTTF_d$ = 28.9 years, DC = 90.5%, CCF = 70 --> PFH_d = 2.7 10^{-7}

3.5.6.4 - Response time M0000394.1

The response time is defined in arm manuals (time to stop, angle to stop) according to different speed and payload configurations.

3.5.7 - EMERGENCY STOP OUTPUT SIGNALS

3.5.7.1 - Definition M0000403.1

When one of the MCPES, WMSES , UESA stop input is activated the ESOUT switches are kept open.

The stops inputs are described in figure 5.1 (Integration chapter).

3.5.7.2 - Limits and conditions of use

M0000404.1

- Refer to chapter 5.1 for correct integration.
- To meet the EN ISO 13849-1:2008 Category 3 PLd safety performance level required for emergency stop functions, the emergency stop output signals must be connected to a stop circuit performing a cross check monitoring of the ESOUT switches.
- A "long E-StopOut" configuration is available to include the USEREN (in manual mode) and DOOR (in automatic mode) signals in the emergency stop output signals. This configuration does not have the adequate safety performance level required for emergency stop signals. Therefore this configuration can only be used if the USEREN and DOOR signals are included for device protection only and not for safety risk reduction in the cell.

3.5.7.3 - Performances

M0000405.1

The emergency stop output signals have the following performance parameters:

$MTTF_d$ = 218 years, $PFH_d = 5.2\ 10^{-7}$

With these conditions: 2 cycles (2 E-Stop) per hour and 8760 hours per year.

--> $B_{10d} = MTTF_d \times 0.1 \times n_{op}$ = 380.000 cycles.

3.5.7.4 - Response time

M0000407.1

The response time is 4 ms maximum.

3.5.8 - POWER STATUS OUTPUT SIGNAL

3.5.8.1 - Definition

M0000437.1

When arm power is enabled, the USERPS switches are closed.

3.5.8.2 - Limits and conditions of use

M0000438.1

- Refer to chapter 5.1.2 for correct integration.
- To meet the EN ISO 13849-1: 2008 Category 3 PLd safety performance level, the power status output signals must be connected to a safety circuit performing a cross check monitoring of the USERPS switches.

3.5.8.3 - Performances

M0000439.1

The power output signals have the following performance parameters:

$MTTF_d$ = 490 years, $PFH_d = 2.3\ 10^{-7}$

With these conditions: 2 cycles (2 enable / disable power command) per hour and 8760 hours per year.

--> $B_{10d} = MTTF_d \times 0.1 \times n_{op}$ = 858.000 cycles.

3.5.8.4 - Response time

M0000440.1

The response time is 22 ms maximum.

4 - INSTALLATION

4.1 - ROBOTIZED CELL ENVIRONMENT

M0000210.1

Installation of the robot must be planned in accordance with the standard instructions

(see chapter 3.1).

The declaration of incorporation and conformity relates to the CE mark.

For UL robots, UL mark information can be obtained on UL web site : http://www.ul.com. In the proposed menu list, select "Online Certifications Directory" and fill in the "UL File Number" file number with e221459.

4.1.1 - POSITIVE SAFETY

M0000211.1

The robot implements safety functions with defined safety performance levels as documented in chapter 3.5. The robotized cell must have its own safety functions with adequate performance levels, that must be identified with a specific risk analysis. The safety functions of the cell can be based on the safety functions of the robot. The management of functional safety in the cell is regulated with international standards such as those listed in chapter 3.1.

4.1.2 - POWER SOURCES

M0000212.1

The electrical power supply of the robot and the machine must comply with the current standards (see chapter 3.1).

The characteristics of the power supply and the grounding terminals must comply with the manufacturers' specifications.

Each robotized cell must be equipped with means of separating each of its power sources.



SAFETY
Make sure that the voltage supplied corresponds to the voltage shown on the manufacturer's plate of the CS8C controller.

4.2 - ON-SITE PREPARATION

4.2.1 - ELECTRICAL NETWORK

M0000213.1

The system is linked to the mains power supply via a cable with 2 or 3 wires + ground connected to the controller. Various power supply voltages are possible: 200 V, 208 V, 230 V, 400 V, 440 V, 480 V (50/60 Hz) ± 10%. The choice of the power supply voltage and the type of network (single phase or three-phase) depend on the option selected and the type of arm used.

Power supply provided to the robot shall have an overvoltage category not exceeding level II as defined in IEC 60664-1:2007.



SAFETY

Make sure that the voltage supplied corresponds to the voltage shown on the manufacturer's plate of the CS8C controller.



When making the connections, the ground wire must be connected first.

Power rating to be installed:

TX40	1,5 kVA
TX60	1,7 kVA
TX90	2 kVA
RX160, TP80, TX250	3 kVA
TS40/60/80	1,7 kVA

If the controller is installed in an industrial chassis, to add 800 VA to the above values.

Minimum wire cross section: 14 AWG / 2 mm^2

Maximum wire cross section: 8 AWG / 6 mm^2 (flex wire)

Maximum wire cross section: 7 AWG / 10 mm^2 (solid wire)

Maximum tightening torque applicable on terminal block screw: 1 Nm; on M4 screw: 2.9 Nm

Interrupting Rating IR: 120 kA with Am fuses, 200 kA with UL fuses

IR = Interrupting current rating of protective device.



This value is true for short circuits occurring after the main fuses of the controller. If a short circuit occurs between the main input and the fuses, the short circuit current is only limited by the customer external protective device.

The main input line shall be protected from user side.

DANGER
Use a cable whose cross-section is suitable for the power rating shown on the manufacturer's plate and protect the line accordingly.

SAFETY
The main input line shall be protected from user side. Use a cable whose cross-section is suitable for the power rating shown on the manufacturer's plate and protect the line accordingly (Refer to Appendix 1: Protection of the power line for the CS8C controller).

The controller is fitted with a filter to limit the induced disturbances (disturbances caused by the controller). The filter can cause major leakage current peaks that have to be taken into account when selecting the elements protecting the power supply circuit (using a time delay ground fault circuit breaker). Leakage current can reach up to 250 mA for 3 ms. Nominal leakage current for 3-phase and single phase configurations is below 3 mA.

4.2.2 - PNEUMATIC NETWORK

M0000214.1

For the arm, and if solenoid valves are used, it is necessary to provide a supply of compressed air, lubricated or not, at a maximum pressure of 7 bar and filtered to 10 µm.

4.2.3 - WORK ENVIRONMENT

M0000215.1

- Working temperature: 5 to 40°C with controller cooling via a filtered fan.
- Storage temperature: -25 to 55° C.
- Humidity: 90% maximum without condensation.
- Maximum altitude: 2000 m or 800 mbar (absolute) minimum.
- Vibrations: Please consult us.
- Ingress protection: IP20.
- Clean environmental air: Class ISO7 (ISO 14644-1).
- Maximum allowable pollution degre: 2 (non conductive), according to IEC 60664-1; 2007.

If these conditions of cleanness and temperature in particular are not complied with, the controller must be integrated in a closed area in which the necessary conditions can be provided: A cooled industrial frame. If CS8C is enclosed in a IP54 cabinet, the door of this external chassis has to remain closed during operation.

The heat generated by the controller is 400 W.

Noise generated by the controller, measured at 1 m distance from the front face and at 1.6 m high, is below 70 dBA.

4.2.4 - CONTROLLER FOOT PRINT AND FITTINGS

M0000375.1

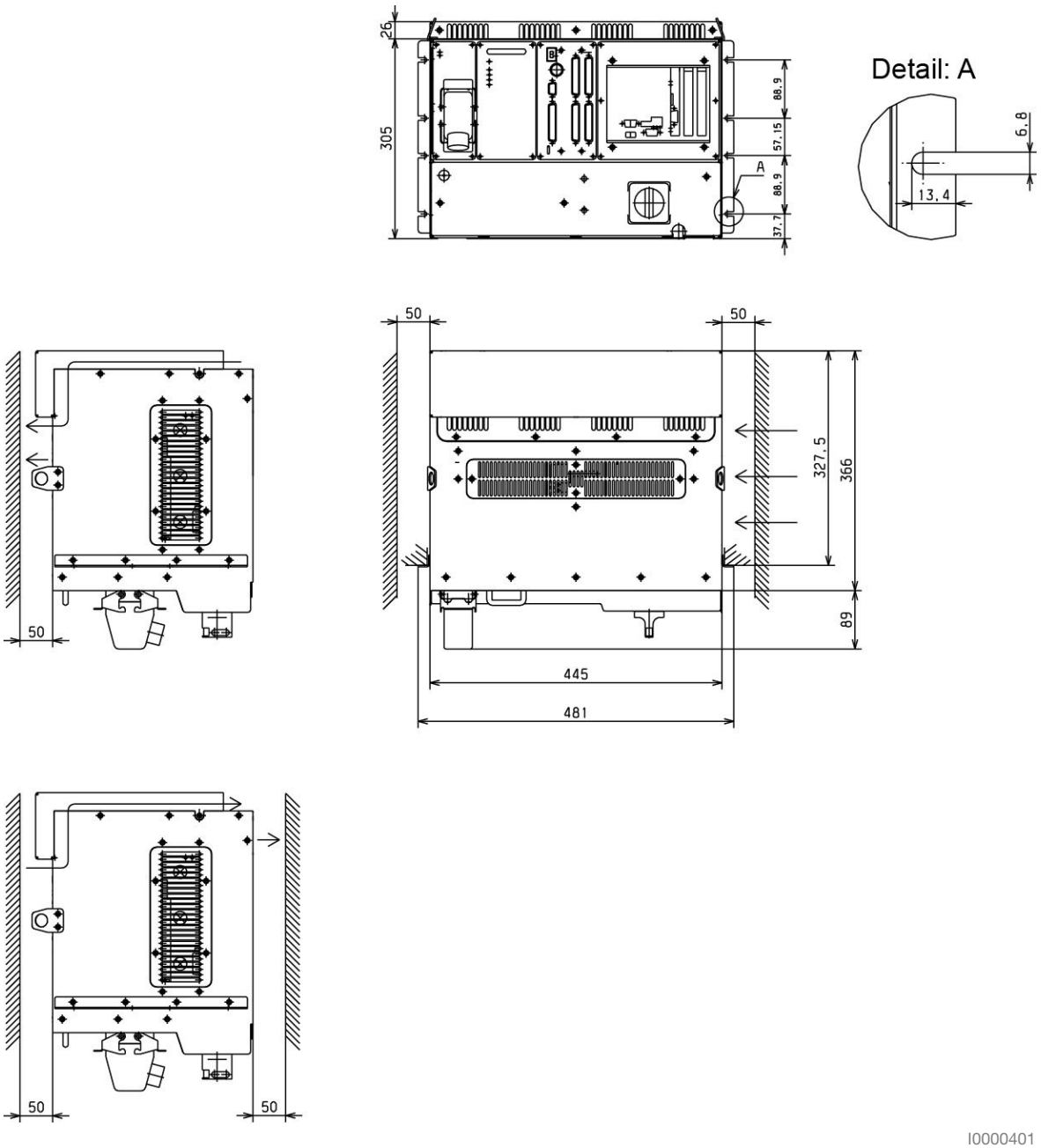
The CS8C controller can be installed directly in the cell if it is in compliance with its environmental constraints, or fitted in a 19" frame as shown in the layout hereafter, to meet the airflow constraints.

For easier maintenance, we recommend slides to hold the controller in place while it is not kept in a vertical position by its fastening points. The length of the Input/Output cables must also be taken into account.

Grounding the 19"-Rack and ensure a proper electrical connection to the brackets of the controller is a good way to create a common reference potential with all other equipments in the cell. This provides a better EMI protection and is not a replacement for the earth connection on the main input providing personal protection.

4.2.4.1 - CS8C for TX and TS arms

M0000260.1



I0000401

Figure 4.1

4.2.4.2 - CS8C for RX160, TP80, TX250 arms

M0000267.1

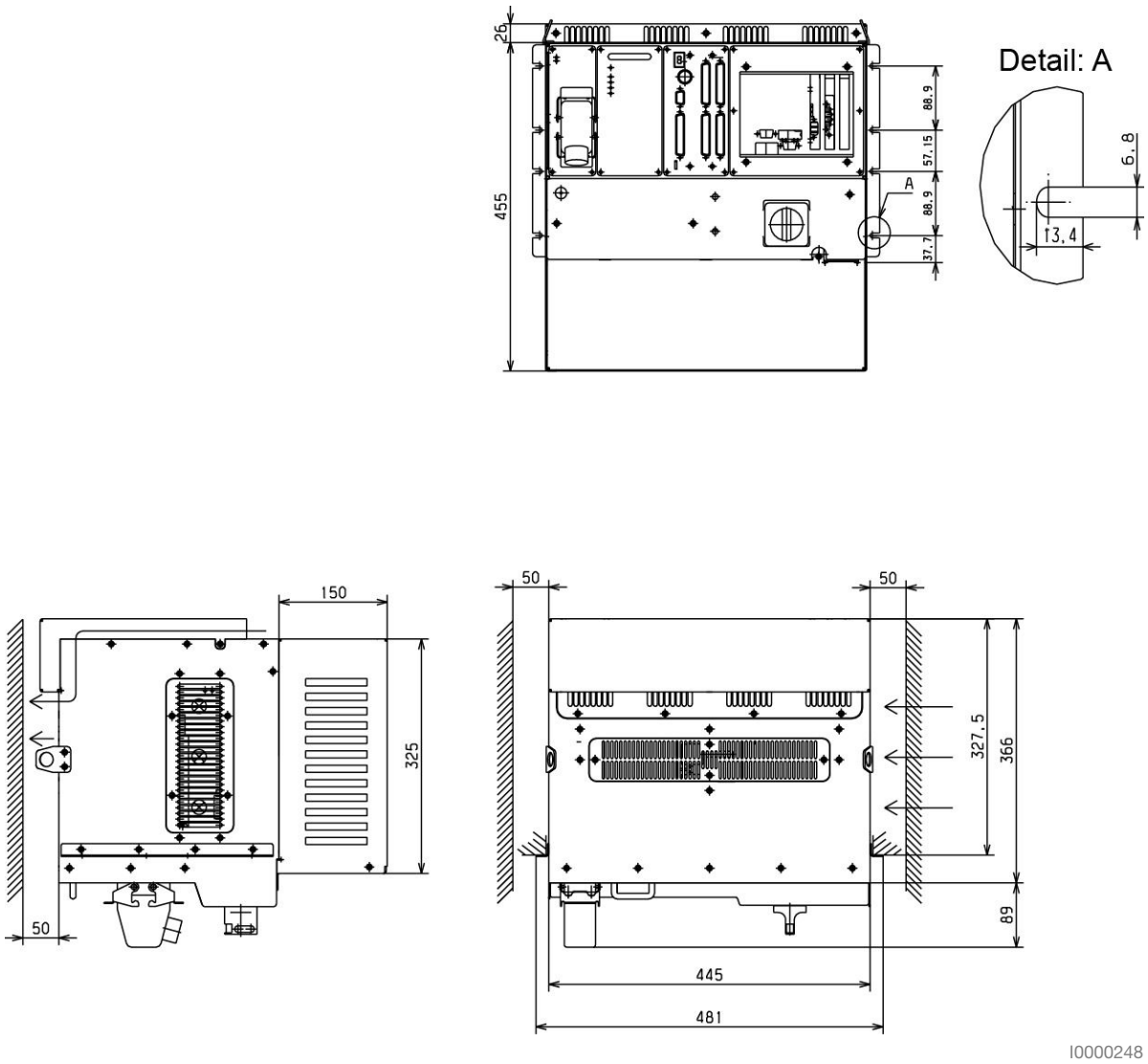
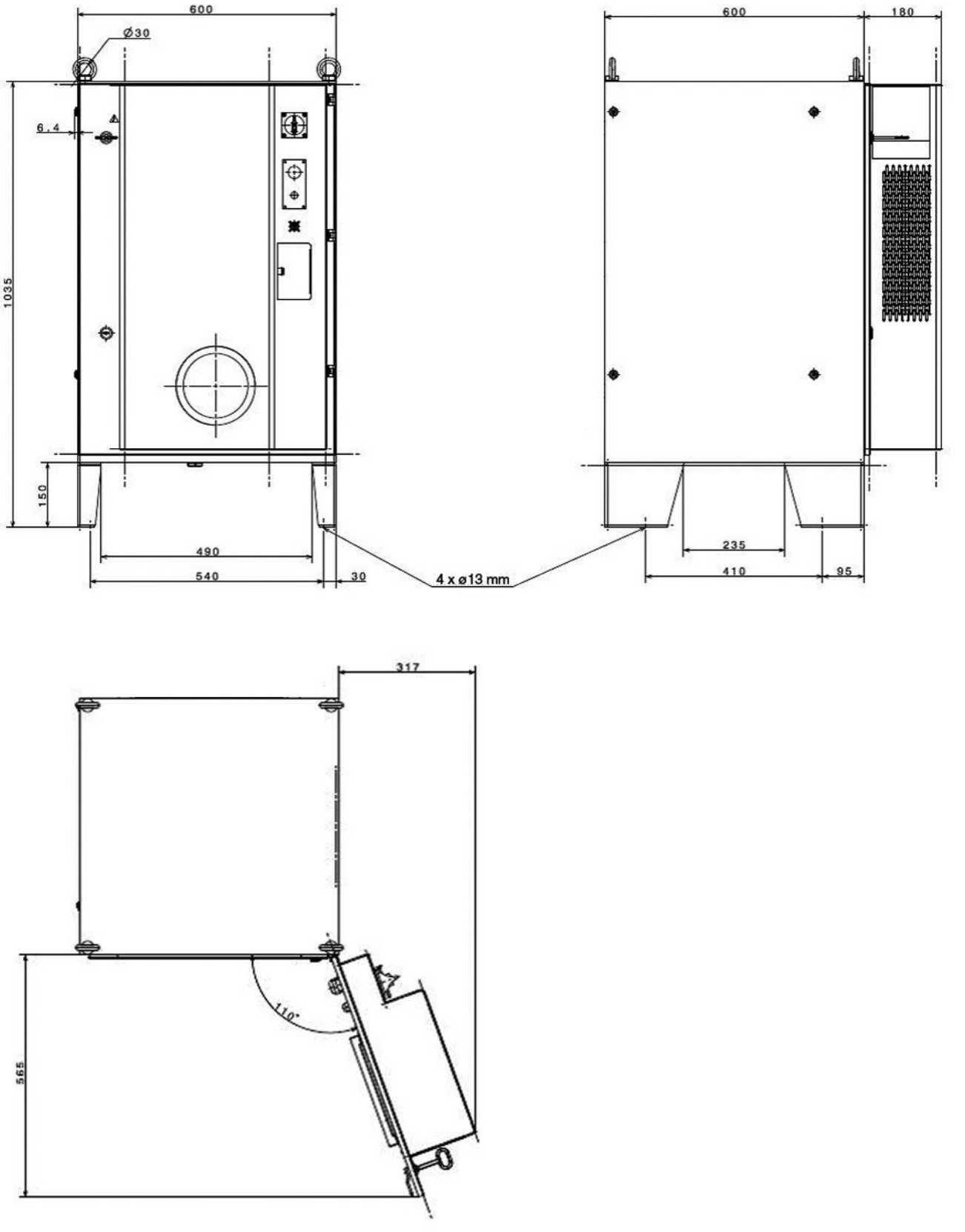


Figure 4.2

4.2.4.3 - Industrial chassis

M0000448.1



I0000214

Figure 4.3

When the industrial chassis is delivered together with the CS8C, the CS8C is already installed inside.

When the chassis is delivered afterwards, the assembly instructions are provided with the chassis. Refer to the document provided which specifies the mechanical assembly and the voltage configuration.

Ambient temperature: 35°C with air-air heat exchanger, 50°C with air conditioning unit.

4.2.5 - AIR CIRCULATION

M0000449.1

The airflow, which is directed upwards in the standard version, can be directed downwards as an option, except in the CS8C controller for the RX160, TP80, TX250.

The surface used for the installation must be horizontal and free from vibrations. If the CS8C controller is integrated in an industrial frame ensuring that the surrounding air is dust-free, the filter **(3)** must be removed.



I0000302

Figure 4.4

- Make sure that the controller is placed in such a way as to avoid obstructing the air inlets and outlets **(1) (2) (3) (4)** of the airflow path.
- Also make sure that the controller is placed in a location where the air can circulate freely (figures 4.2 and 4.4). If the controller overheats, this shortens the service life of the components and can lead to malfunctions.
- Take care to avoid damaging the interconnection cable when handling the controller.

4.2.6 - ACCESSIBILITY

M0000251.1

When designing the safety enclosure, it is necessary to ensure the accessibility of all the mechanical and electrical elements (robot, sensors, mechanical assemblies, etc.).

4.3 - UNPACKING AND HANDLING

4.3.1 - PACKAGING FOR THE "CS8C" CONTROLLER

M0000471.1

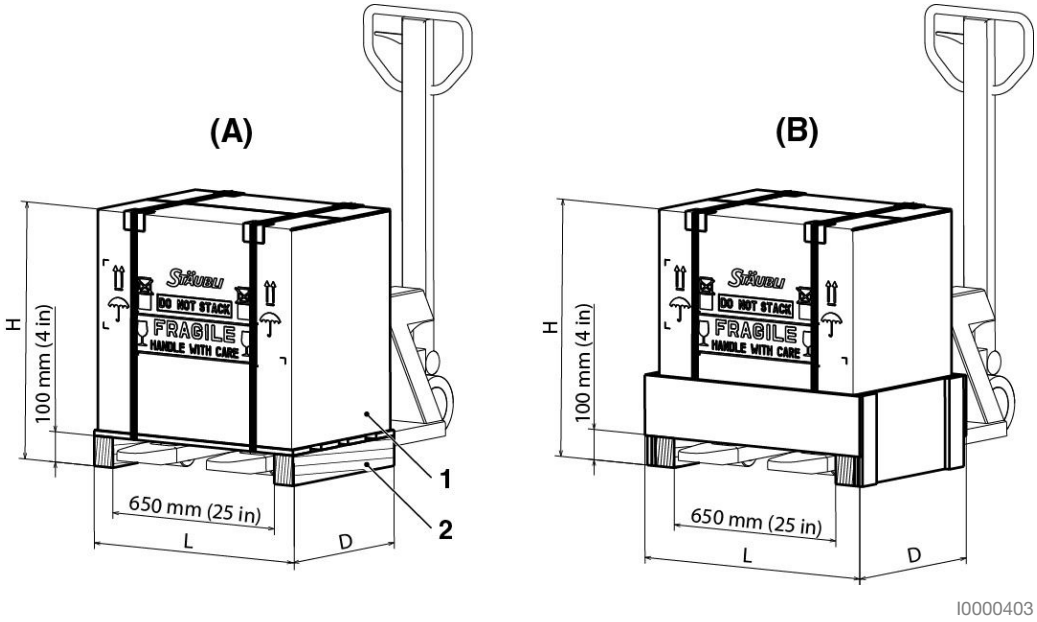


Figure 4.5

	CS8C 3P for TX, TS	CS8C SP for TX, TS	"CS8C" for RX160, TP80
Standard packaging **(A)** L x H x D Gross weight	800 x 660 x 570 mm 32 x 26 x 23 in 70 kg / 154 lb	800 x 660 x 570 mm 32 x 26 x 23 in 51 kg / 112 lb	800 x 850 x 570 mm 32 x 34 x 23 in 80 kg / 176 lb
International packaging **(B)** L x H x D Gross weight	870 x 660 x 600 mm 34 x 26 x 24 in 77 kg / 170 lb	870 x 660 x 600 mm 34 x 26 x 24 in 58 kg / 128 lb	870 x 850 x 600 mm 34 x 34 x 24 in 87 kg / 192 lb
Net weight	With transformer: 50 kg / 110 lb Without transformer: 31 kg / 69 lb	With transformer: 50 kg / 110 lb Without transformer: 31 kg / 69 lb	60 kg / 133 lb

4.3.1.1 - Handling of packing

M0000441.1

By pallet truck.



Do not stack.

4.3.1.2 - Unpacking and installing the controller

M0000431.1

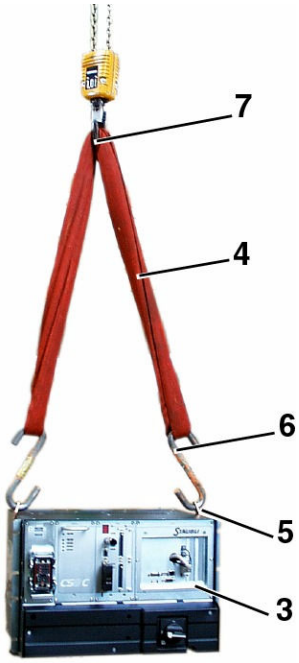


Figure 4.6

- Move the packing case **(1)** as near as possible to the installation site (see figure 4.5).
- Open the case.
- Take out the box containing the MCP and the upper wedges.
- Fit a sling **(4)** (fabric sling 200 kg / 442 lb) using hooks **(6)** between the lifting rings **(5)** on the controller **(3)** and the hook on the lifting tackle **(7)**. The hooks **(6)** must also stand up to 200 kg / 442 lb.
- Slowly lift the controller using the lifting tackle and set it down beside the pallet **(2)** (see figure 4.5).
- The lifting rings **(5)** on the controller are removable.
- Take out the documentation and side wedges.
- Take out the box containing the connecting cable.

If the controller has to be moved to another site, it shall be packed again in a similar way and protected against mechanical shocks and dust.

4.3.2 - PACKING FOR THE "CS8C" WITH INDUSTRIAL CABINET (IC)

M0000444.1

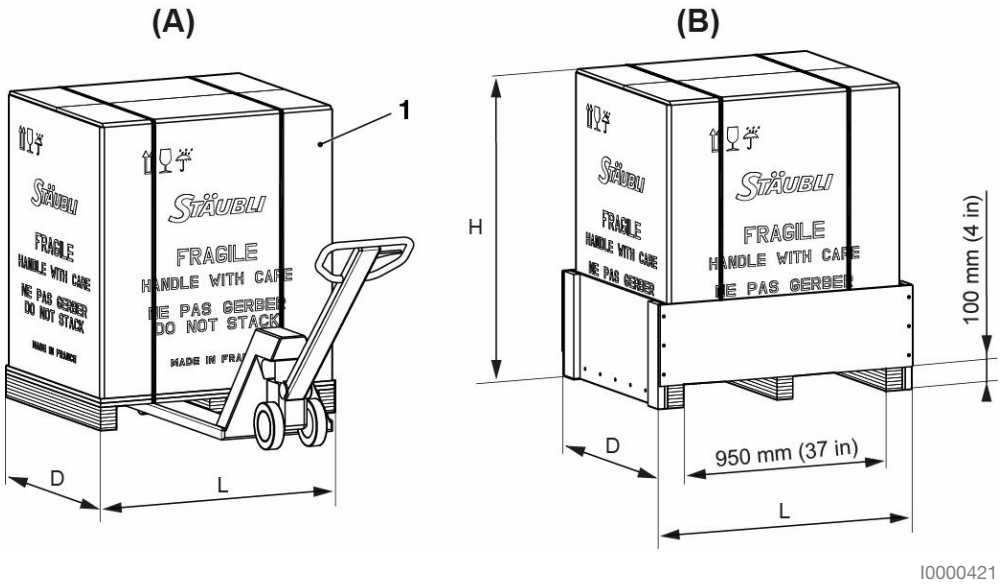


Figure 4.7

	"CS8C IC" for TX, TS	"CS8C IC" for RX160, TP80, TX250
Standard packaging **(A)** L x H x D Gross weight	1100 x 1320 x 950 mm 43 x 52 x 37 in With transformer: 232 kg / 511 lb Without transformer: 212 kg / 467 lb	1100 x 1320 x 950 mm 43 x 52 x 37 in 242 kg / 533 lb
International packaging **(B)** L x H x D Gross weight	1200 x 1320 x 970 mm 47 x 52 x 38 in With transformer: 247 kg / 544 lb Without transformer: 227 kg / 500 lb	1200 x 1320 x 970 mm 47 x 52 x 38 in 257 kg / 567 lb
Net weight	With transformer: 170 kg / 375 lb Without transformer: 150 kg / 331 lb	180 kg / 397 lb

4.3.2.1 - Handling of packing

M0000445.1

By pallet truck.



Do not stack.

4.3.2.2 - Unpacking and installing the controller

M0000443.1

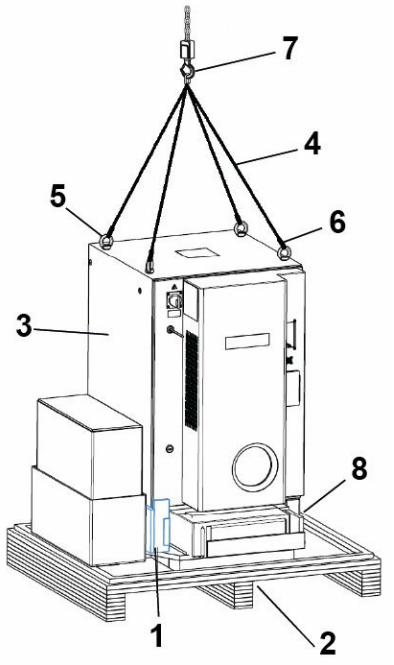


Figure 4.8

- Move the packing case as near as possible to the installation site.
- Open the box (see figure 4.7).
- Take out the upper wedge.
- Take out the box containing the MCP.
- Take out the box containing the connecting cable.
- Place 4 slings **(4)** (200 kg / 442 lb fabric slings) using hooks **(6)** between the lifting rings **(5)** on the cabinet **(3)** and the hook on the lifting tackle **(7)**. The hooks **(6)** must also stand up to 200 kg / 442 lb.
- Remove the 4 M10 bolts **(8)**.
- If present (international packaging), take out the wedge of the door (1).
- Slowly lift the controller using the lifting tackle and set it down beside the pallet **(2)**.
- The lifting rings **(5)** on the cabinet are removable.

If the controller has to be moved to another site, it shall be packed again in a similar way and protected against mechanical shocks and dust.

4.4 - FIXING THE MCP

4.4.1 - CHARACTERISTICS

M0000272.1

Protection class: IP54

The MCP must not be used in an explosive environment.

The MCP is made of plastic. It can be a source of electrostatic discharges to components located close by. This must be taken into account for components sensitive to electrostatic discharges in the machine.

4.4.2 - INSTALLATION

M0000427.1



I0000239

Figure 4.9

The MCP must be installed in accordance with the requirements of the installation. It must be accessible close to the workstation and must allow the arm to be seen from its location, outside the cell.

It is installed on the holder provided for the purpose. The holder must be fixed to an outside element of the cell (see figure 4.12). It must not be removable to prevent its use inside the cell. The fastening work must be carried out using the 8 x 12 mm oblong holesand screws of suitable size (see figure 4.12).

The MCP holder has two functions:

- To ensure that the MCP is easily accessible for the operator.
- To detect the presence of the MCP outside the cell to allow arm power on after an emergency stop.

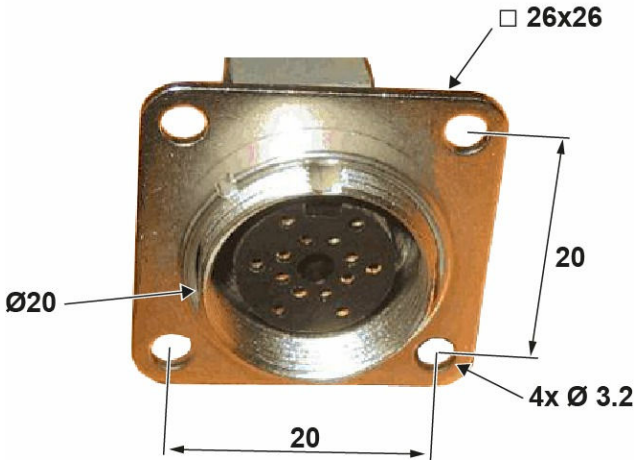
If MCP is not connected to the controller, it shall be removed from the cell or covered to avoid having a non-operating EStop push button. In this case, it has to be replaced with its dummy plug (J110). Provisions shall be made for storing or covering the unused MCP.

The MCP is provided with a 7 m cable. Extension cable can be used up to a limit of 30 m. This extension cable can be terminated either with a standard connector or a bulkhead connector to be installed on a panel.



I0000332

Figure 4.10 : Standard connector



I0000400

50 mm clearance space is required at the back of the panel to install this connector

Figure 4.11 : Panel mountable connector

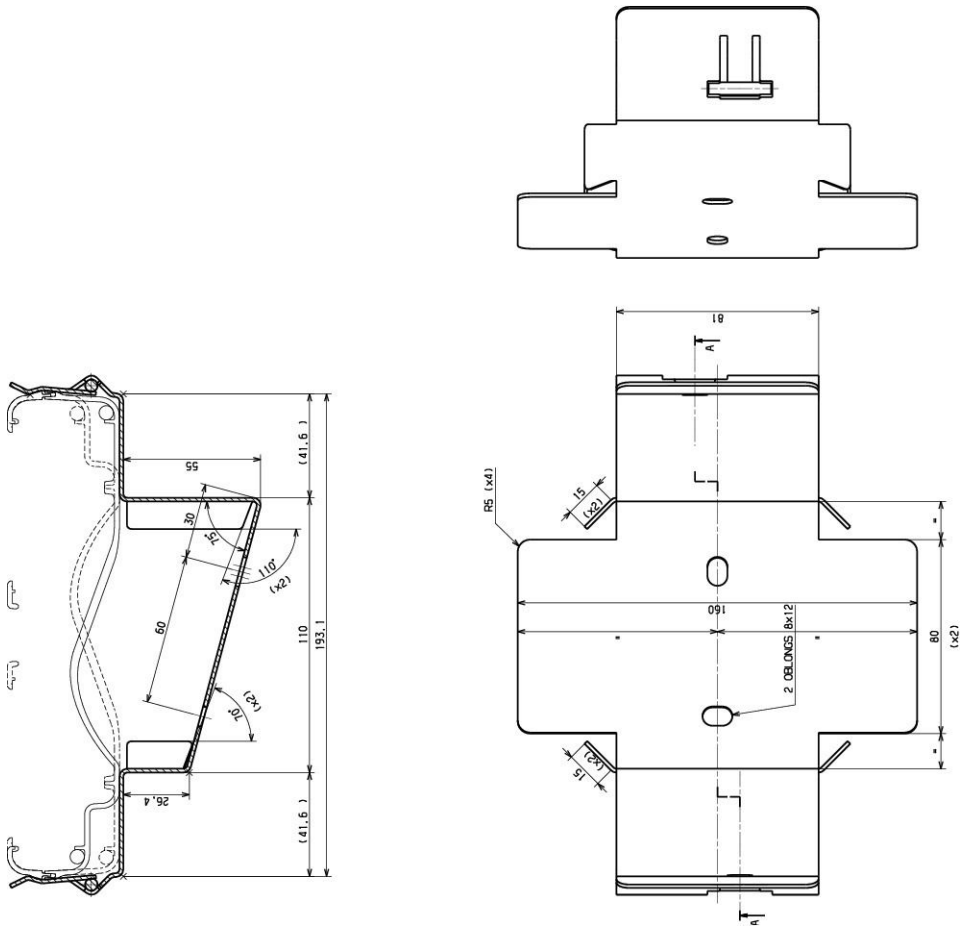


Figure 4.12 : Holder measurements

4.5 - FIXING THE WMS

M0000278.1



Figure 4.13

SAFETY

- The WMS (Working Mode Selector) must be installed outside of the safeguarded space of the cell in a location with good visibility of the possible risks. The Working Mode Selector shall not be accessible from within the safeguarded space.
- To comply with the UL requirements, it cannot be provided or modified by the user / integrator.

The WMS is designed to be installed on a panel with following dimensions:

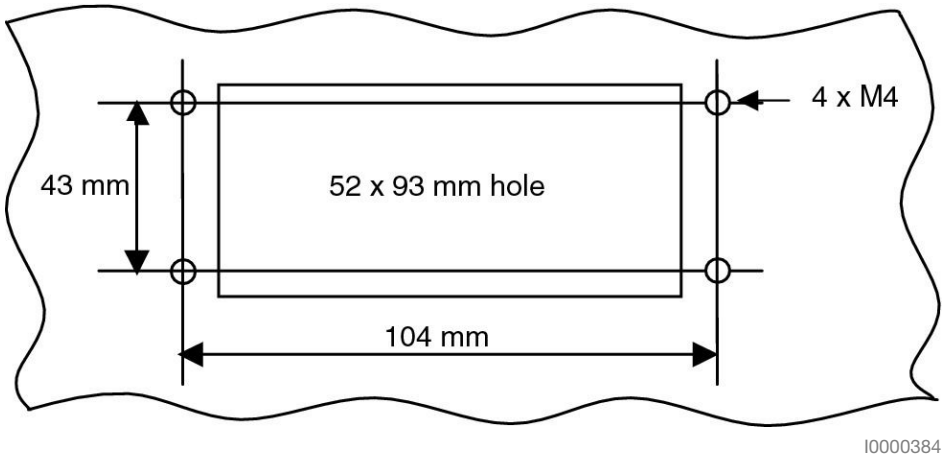


Figure 4.14

WMS overall dimensions:

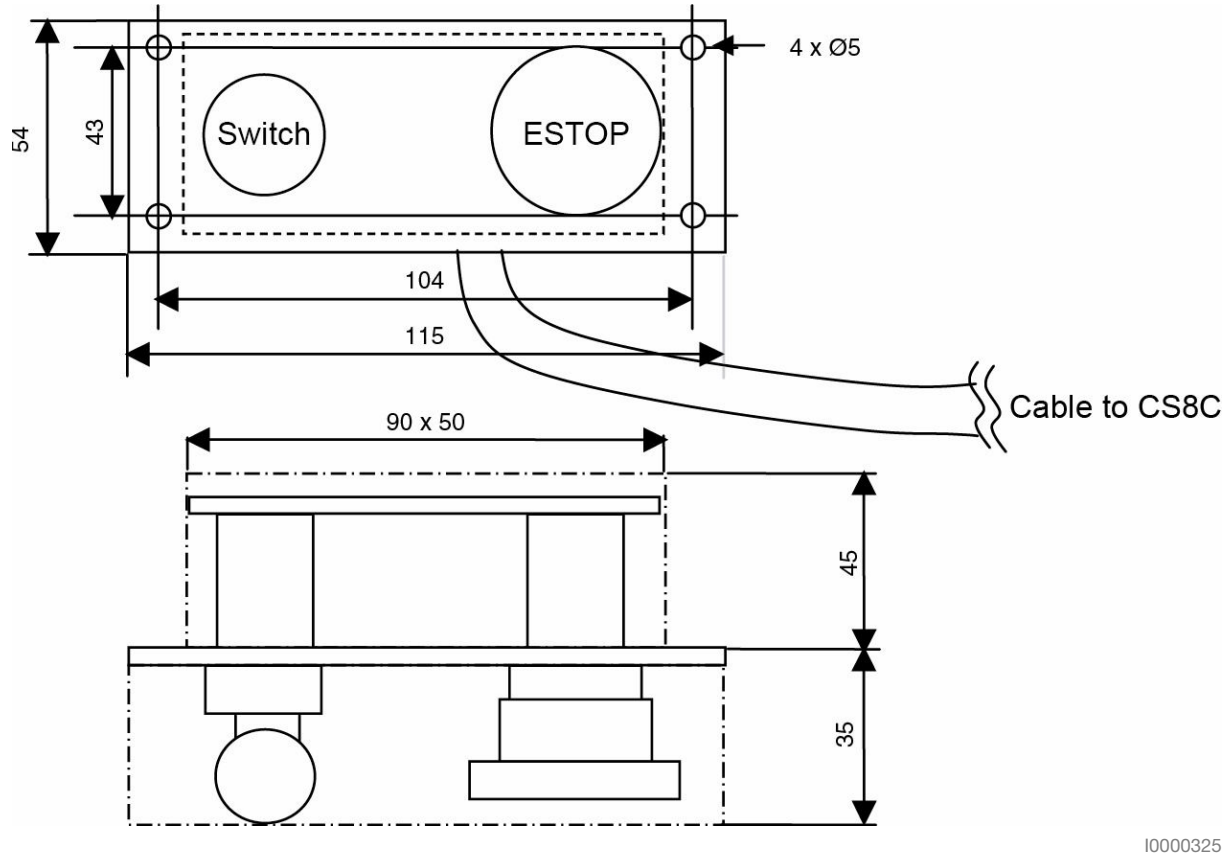


Figure 4.15

The cable between WMS and CS8C has a connector on both sides. It has to be connected to J113 on CS8C (RSI2 board) and to either of the connectors of the WMS. The second connector of WMS is provided to ease the integration (to be able to connect the WMS cable either on the side of WMS or the bottom the WMS) and shall remain empty (nothing connected to it).

SAFETY
The cable shall be protected from mechanical stress and installed in a metallic frame connected to the ground.

If WMS function is performed by an external equipment (i.e a PLC...), this equipment shall be at the same reference potential as the CS8C. A metallic braid is one way to ensure this same reference potential.

4.6 - CONNECTIONS

M0000217.1

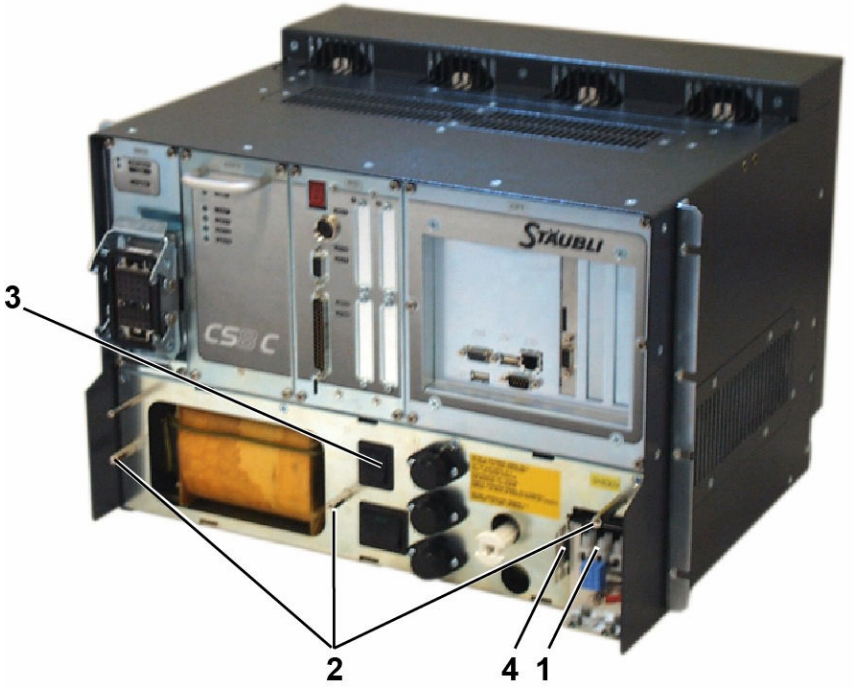
Service and maintenance on the robot has to be taken into account when wiring the controller: cables have to be long enough to be able to move components, cables should not obstruct air flow around the controller.

4.6.1 - CONNECTION TO THE MAINS POWER SUPPLY

M0000240.1

- Make sure that the voltage supplied corresponds to the voltage shown on the manufacturer's plate of the CS8C controller.
- Use a cable appropriately rated to the power mentioned on the identification plate, and protect the line accordingly.

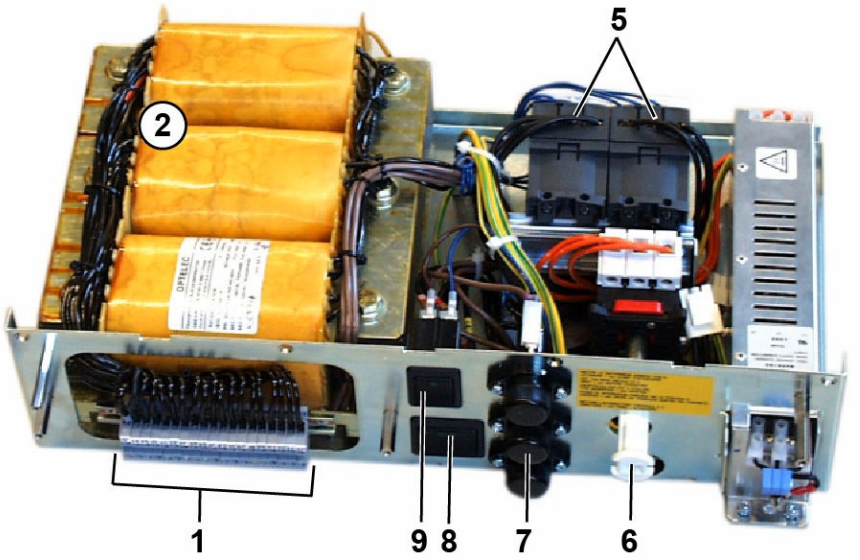
The mains input is connected via the terminal strip **(1)** that is underneath the cover. The cover can be removed after the fixing screws **(2)** have been taken out. The cable must be held in place by ties at **(4)**.



I0000338

Figure 4.16

If voltage configuration of the controller is not the good one, it can be changed on the terminal trip **(1)** of the transformer:



I0000198

Item	Description
1	Transformer terminal trip.
5	AP1 / AP2 contactors.
6	Main switch.
7	F1, F2, F3 fuses.
8	D2 circuit breaker.
9	D1 circuit breaker.

Figure 4.17

Three-phase voltage configuration:

200 V	208 V	230 V	400 V	440 V	480 V
7F 8F 9F	7E 8E 9E	7D 8D 9D	7C 8C 9C	7B 8B 9B	7A 8A 9A

Single-phase voltage configuration:

115 V	230 V
8G	8D

If voltage is changed, fuses F1, F2, F3 have to be adapted to the voltage:
- Fuses are 10 x 38 mm type, 500V for non UL controllers.
- For UL type controllers, replace fuses with UL type.

	Three-phase 400-480 V	Three-phase 200-230 V	Single phase 230 V	Single phase 115 V
TX40	4Am	6Am	10Am	16Am
TX60 - TS	4Am	8Am	10Am	16Am
TX90	6Am	12Am		
RX160 - TX250 - TP80	8Am	16Am		

- These fuses do not protect the mains power supply line which must be protected separately.
- Never replace these fuses with fuses of a higher rating or with different characteristics (see the "replacement parts" section).

Am means "slow-acting fuse" according to IEC 269-1.2.
AT means "slow-acting fuse" and AF "quick-acting fuse" according to IEC 127-2.

4.6.2 - CONNECTION BETWEEN THE ARM AND THE CONTROLLER

M0000218.1

The arm is connected to the controller via 2 unpluggable connectors on the arm and on the controller. The connectors must be locked in place to ensure correct connections and the cable has to be attached to the cell's frame to avoid having mechanical constraints on connectors.

This connection includes optical fibers. When the cable is disconnected, during maintenance for example, cover the ends to avoid soiling the ends of the optical fiber. In the event of soiling, the only cleaning product to be used is water. Never use alcohol.

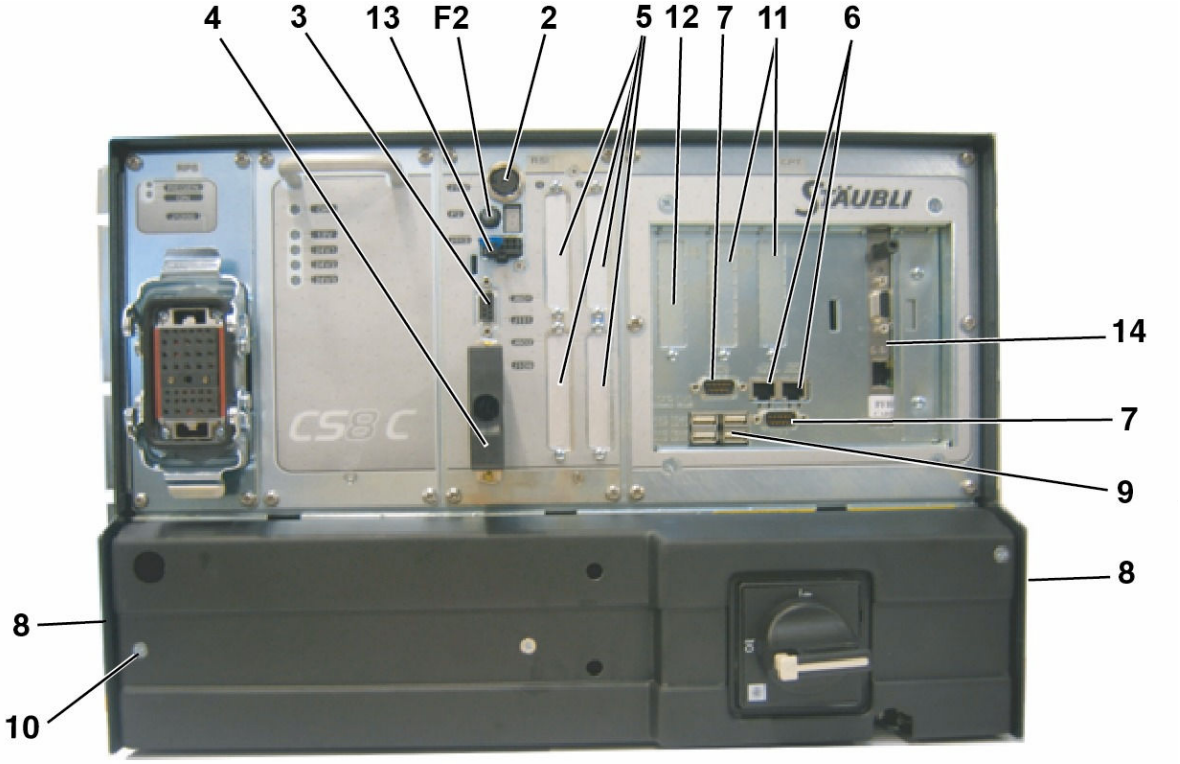
DANGER
Do not stand with your eyes directly opposite the optical fiber when it is lit, in order to avoid damage to the eyes.

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4.6.3 - CONNECTING THE SIGNALS

M0000216.1

The Input/Output signals are connected via the connectors located on the front panel of the controller.



I0000360

Item	Description
2	Connector for MCP
3	Fast inputs/outputs
4	Connection with cell (emergency stop, door, etc.)
5	Digital I/O options (BIO)
6	Ethernet links
7	Serial links
8	Fastenings
9	USB links
10	Connector for antistatic wrist wrap
11	Optional encoder input
12	CAN output for Scara robots
13	Connector for WMS front panel
14	Optional fieldbus board

Figure 4.18

The signals must be connected using shielded cables whose shield is grounded at the both ends. This is necessary both for the emergency stop signals (J109) and for the digital links (serial links, Ethernet, etc.). If screw terminal type connectors are preferred, an adaptor from Sub D to screw terminal is commercially available from several sources such as Phoenix Contact and others.

The CS8C is equipped with 4 USB ports on the front of the computer. It is recommended to use them with a USB stick directly connected on the board and not with an intermediate USB cable which can create EMC troubles. The recommended USB ports are J202 and J209.

The brackets used to hold the controller in place also provides protection from electrical noise. It is thus useful for the fastenings **(8)** to be linked to the ground circuit of the cell as a whole.

For TS robots, there are also Inputs/Outputs available on the arm.

4.6.4 - CABLE INLETS AND OUTLETS

M0000261.1

The connections for the CS8C controller are on the front panel. They must then be protected by backshells with good levels of mechanical strength and the cables have to be attached to the frame of the cell to avoid constraints on connectors.

DANGER

Make sure that the controller and the input/output signals are disconnected from the power supply before carrying out operations to connect or disconnect cables or modify the electrical configuration.

Grounded metal backshells provide improved protection from unwanted outside currents.

The cable layout must comply with a minimum bend radius for each type of cable. See the table below:

Bend radius in mm	
Standard interconnection cable	100
Flat interconnection cable	Static bend radius: 55 Dynamic bend radius: 110
hand I / O cable	50
MCP and WMS cables	50
Other	Depending on the cables used

During installation, the ends of the interconnection cable should be covered to protect them from dust. The only cleaning product that can be used is water. Never use alcohol.

Flat interconnection cable requires more space on controller side. The flat cable has to be installed in a chain to guaranty a good operation and dynamic constraints have to be applied to the cable itself, not the connectors.

Dynamic cable characteristics:

	Standard cable	Dynamic cable
Cable dimensions	ø 24 mm	52 x 13 mm
Static bend radius	100 mm	55 mm
Dynamic bend radius	350 mm	110 mm
Number of cycles	100 000	5 000 000
Max. speed	1 m/s	4 m/s
Max. accel	2 m/s^2	8 m/s^2

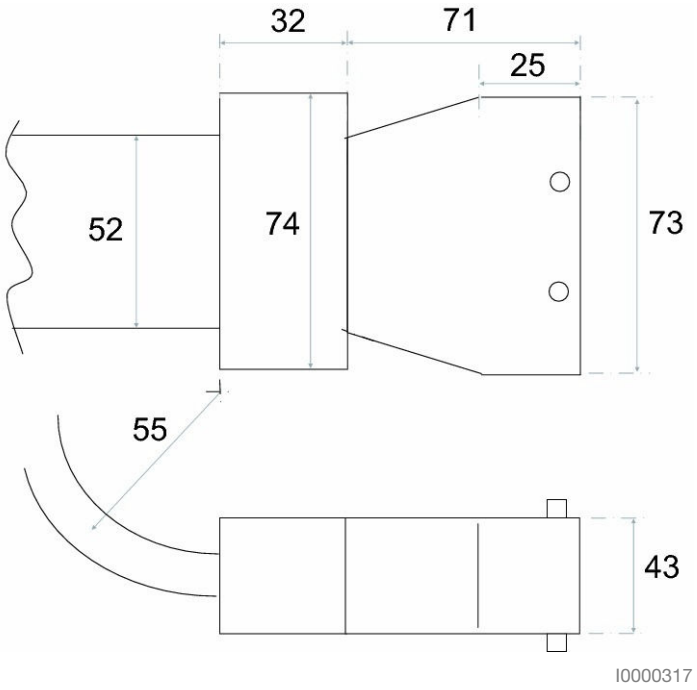


Figure 4.19

The length of the cables must be taken into account to facilitate maintenance.

Interconnection cable:

- Cable Ø: 25 mm
- Connector passage Ø: 90 mm

WMS cable:

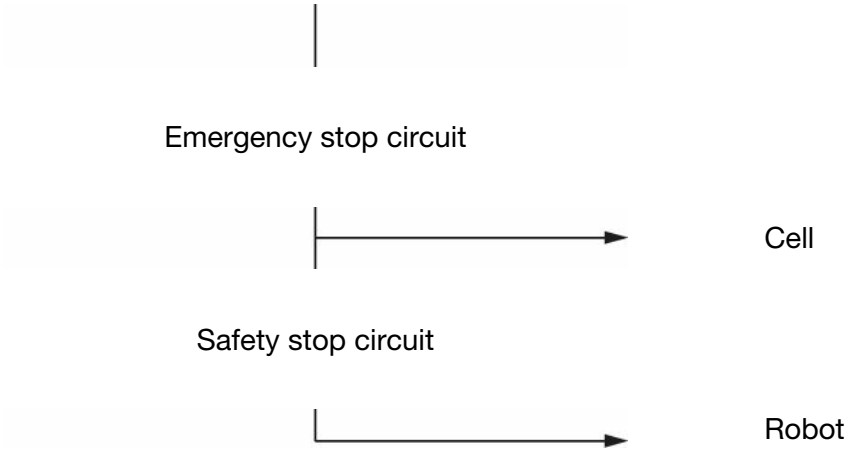
- Cable Ø: 7 mm
- Connector passage Ø: 25 mm

5 - INTEGRATION

5.1 - EMERGENCY AND SAFETY STOP CHANNELS

M0000274.1

In general, the "Emergency Stop Buttons" stop and remove power on the arm manipulator and all other equipments in the cell, where the "Safety Stop Buttons" stop and remove power on the arm manipulator only.



Stop conditions

Definition of stop categories (according to EN 60204-1):

- Category 0: Immediate power off on motors and brakes activation.
- Category 1: Controlled stop with arm power maintained followed by arm power off and brakes activation.
- Category 2: Controlled stop with arm power remaining on.

The different possibilities to stop the robot are indicated below:

Situation	Safety performance (see chapter 3.5)	Category 0	Category 1	Category 2	Comments
Move/Hold push button on MCP				√	Stop on path
stopMove instruction				√	Stop on path
DisablePower push button on MCP			√		Stop on path
DisablePower instruction			√		Stop on path
MCP enabling device (manual mode)	PId / SIL2		√		Stop on path
E-Stop push button and other safety stop functions	PId / SIL2		√		Stop on path
Door safety stop	PId / SIL2		√		Stop on path
Main power failure	PId / SIL2	√			Stop on brakes

Situation	Safety performance (see chapter 3.5)	Category 0	Category 1	Category 2	Comments
Envelop error		√			Stop with drive active braking and brakes
Electrical limit switch		√			Stop with drive active braking and brakes
Limited speed function (manual mode)	PId / SIL2	√			Stop with drive active braking and brakes
Hardware fault on drives at output level		√			The faulty axis is stopped with its brake, other axes are stopped with drive active braking and brakes
Hard stop		√			The concerned axis is stopped on hard stop and its brake; other axes are stopped with drive active braking and brakes



The emergency stop is not the normal method for stopping arm movements; it must only be used in a case of absolute necessity for a stop not provided in your application.

Stop on path: this is performed according to the deceleration parameter of the application's motion descriptor; for Category 1 stops, the stopping time is limited to a maximum value that depends on the robot model (refer to arm documentation).

Axis per axis controlled stop: on some error conditions listed above, the stop is not controlled at motion generator level but at drive's level with a controlled short circuit of the motor field windings.

Stop on brakes: with some hardware failures (i.e power part of the drive damaged), the stop of the concerned axis is performed with brakes only with a maximum stopping time depending on the robot model (refer to arm documentation).

5.1.1 - COMPOSITION OF THE EMERGENCY AND SAFETY STOP CHANNELS

M0000376.1

(see figures 5.1, 5.2)

The various elements forming the stop channel are as follows:

- The emergency stop (MCPES 1-2) on the MCP.
- The emergency stop (WMSES 1-2) from the WMS front panel.
- An emergency stop (UESA 1-2) to be wired according to the application.
- Two parallel channels corresponding to the automatic (COMP) and manual (MANU) modes, if controller configuration is "configuration 1" (see below) (see figure 5.1).

  Each of these channels has to be used according to the application. As a general rule, in the channel concerning the Automatic mode, there is an emergency stop (DOOR 1-2) triggered by the cell door. In the channel given over to the Manual mode, there is also an emergency stop or an authorization for specific operation in Manual mode (USER EN 1-2). These two emergency stop channels are specific to the application and depend to a great extent on the working mode selected.
- A (UESB 1-2) safety stop to be wired depending on the application.

All the contacts making up these various elements of the emergency and safety stop channels are duplicated.

The emergency stop is not the normal method for stopping the robot or disabling power on the arm.

The robot is delivered with a dummy plug connected to J109 (see figure 5.5).

When this connector is replaced with the final one, pay attention to the jumper between pins 18 and 37 (figure 5.2) which needs to be maintained, except if an external 24 V is required for EStop lines.

If the states of the EStops differ, a hardware fault is registered by the system and robot operation is not possible until the hardware fault is corrected, tested and acknowledged (see chapter 6.15).

The safety performance level of the emergency and safety stop channels is defined in chapter 3.5 with the stop and door safety functions.

Emergency stop output channels

Status of the emergency stop channels (ESOUT1 and ESOUT2) are available for the application using one of the controller configuration below (see figure 5.1):

- Either the status of MCPES, WMSES and UESA (configuration 1: the doors of the cell not included).
- Or the status of MCPES, WMSES, UESA and (USEREN or DOOR) (configuration 2: the doors of the cell are included). This is the default configuration when CS8C is delivered. To modify this configuration, refer to chapter 5.8.

The choice of position depends on the requirements of the application.

Status of these outputs must be the same: either both closed if there is no stop condition, or both open if there is a stop condition. Coherency verification shall be performed by the external device using them.

When the controller is switched off, the emergency stop status is not valid any more and the output is forced to the emergency stop state. For applications requiring the emergency stop system status to remain valid even when the controller is switched off, it is possible to power the emergency stop lines via an external 24 V supply provided between J109-37 and J109-19 (see diagrams below). The connection between J109-18 and J109-37 must be removed.

This mode of operation is only possible if controller is configured with "configuration 1" (see above).

When emergency stop lines are powered from the controller, it might be possible to have unstable EStop status at boot up sequence.

The safety performance level of the emergency output channels is defined in chapter 3.5.

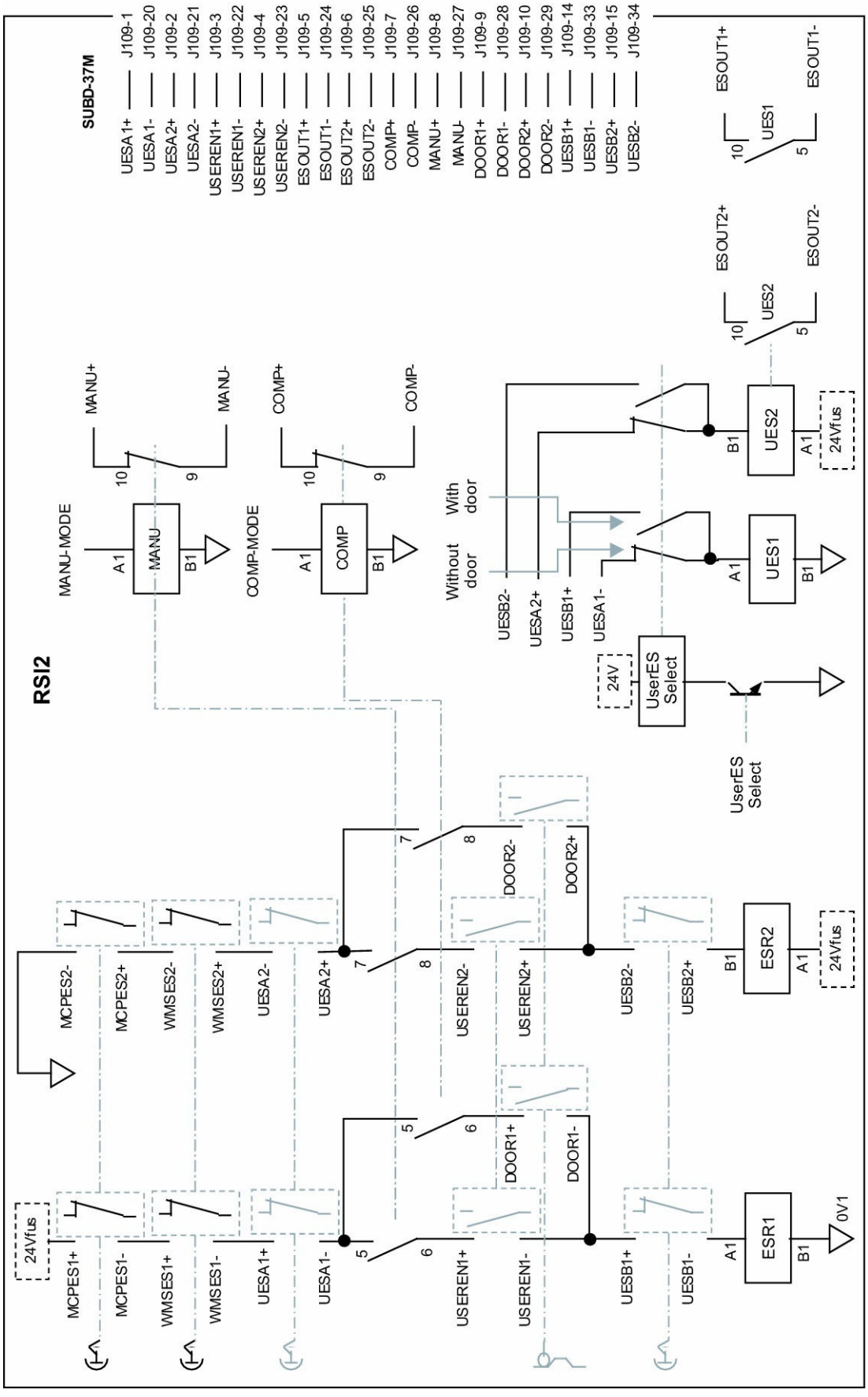
Display on the control panel of the MCP

The status of the stop channels is displayed on the control panel.



In this display, an active input ON shows that an emergency stop has been activated (channel in open position).

Emergency stop channels



I0000331

Figure 5.1

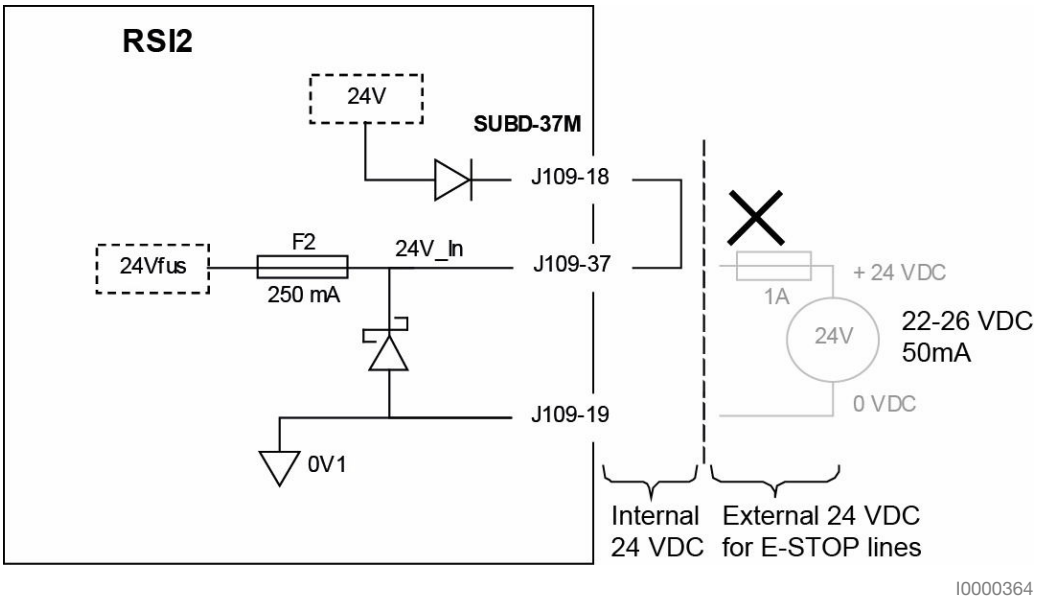

RSI2
24V
SUBD-37M
J109-18
F2
24V_In
24Vfus
250 mA
J109-37
J109-19
0V1
1A
+ 24 VDC
24V
22-26 VDC
50mA
0 VDC
Internal
24 VDC
External 24 VDC
for E-STOP lines
I0000364

Figure 5.2

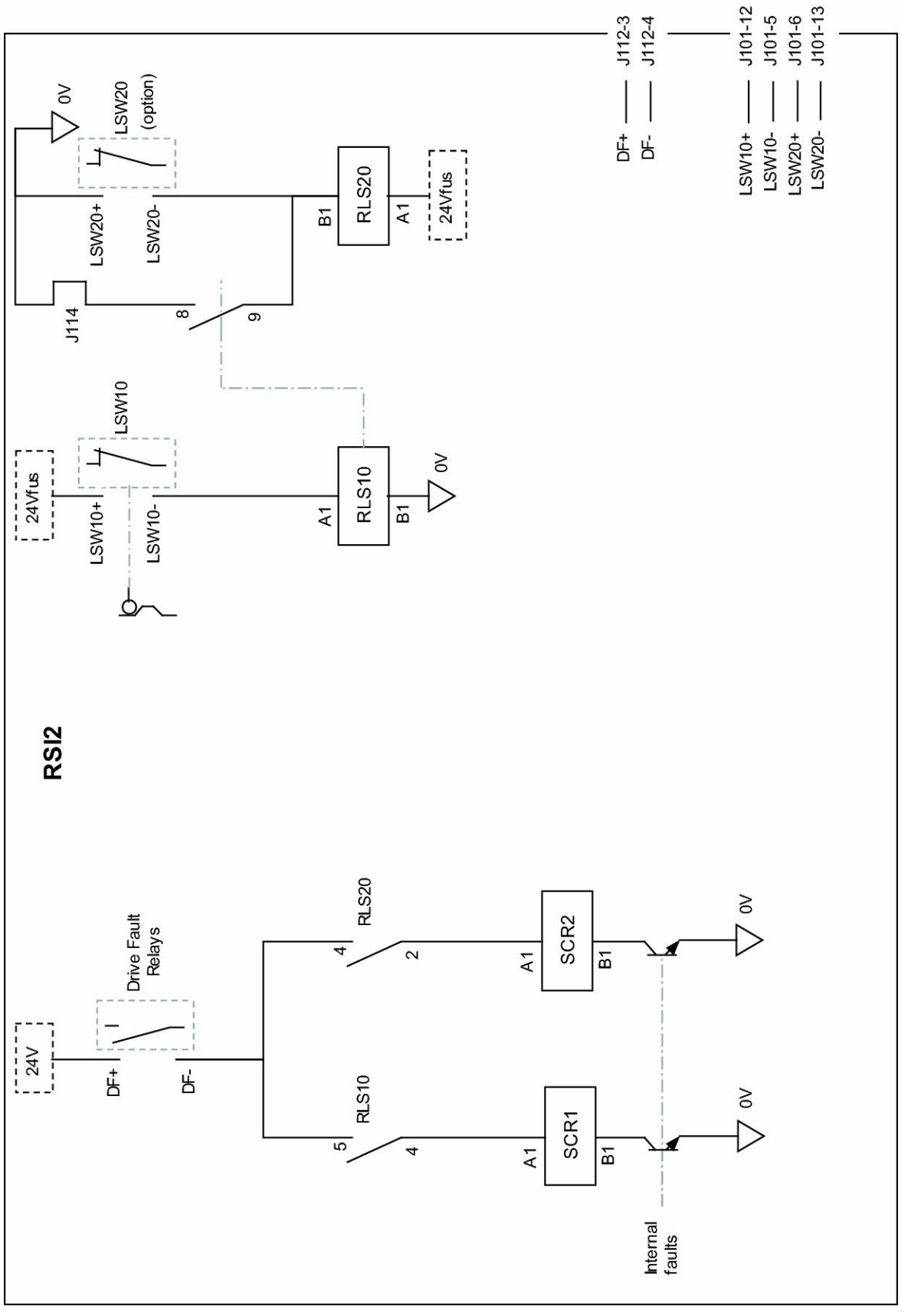

RSI2
24V
Drive Fault Relays
DF+
DF-
RLS20
RLS10
SCR2
SCR1
A1
B1
4
2
5
4
0V
Internal faults
LSW10
LSW10+
LSW10-
24Vfus
LSW20
(option)
LSW20+
LSW20-
J114
8
9
DF+
DF-
J112-3
J112-4
LSW10+
LSW10-
LSW20+
LSW20-
J101-12
J101-5
J101-6
J101-13
I0000202

Figure 5.3

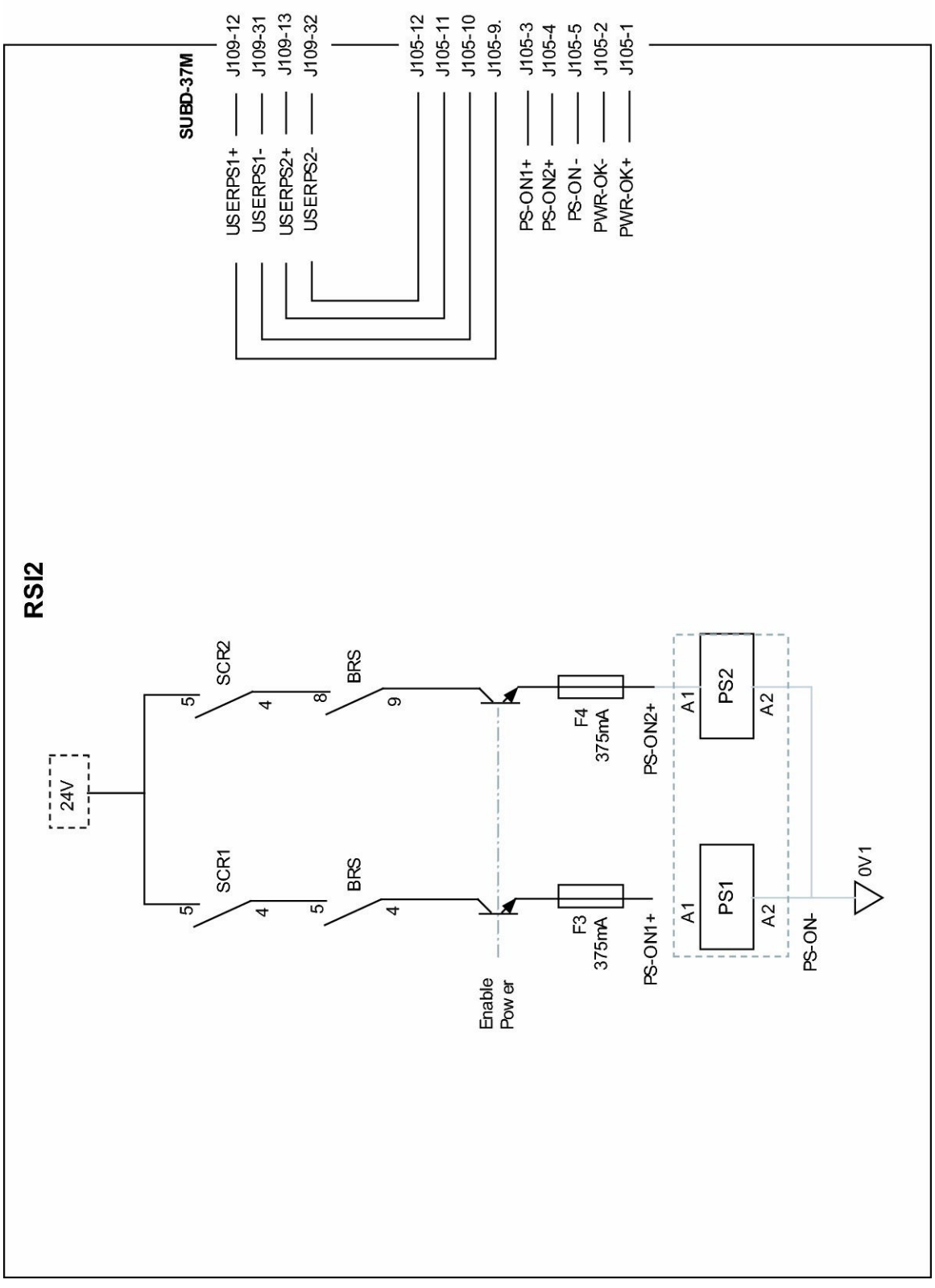

RSI2
SUBD-37M
USERPS1+
USERPS1-
USERPS2+
USERPS2-
PS-ON1+
PS-ON2+
PS-ON -
PWR-OK-
PWR-OK+
J109-12
J109-31
J109-13
J109-32
J105-12
J105-11
J105-10
J105-9.
J105-3
J105-4
J105-5
J105-2
J105-1
24V
SCR2
SCR1
BRS
BRS
F4
375mA
F3
375mA
PS-ON2+
PS-ON1+
A1
A2
PS2
PS1
0V1
PS-ON-
Enable
Power
I0000296

Figure 5.4

5.1.2 - CONNECTION WITH CELL

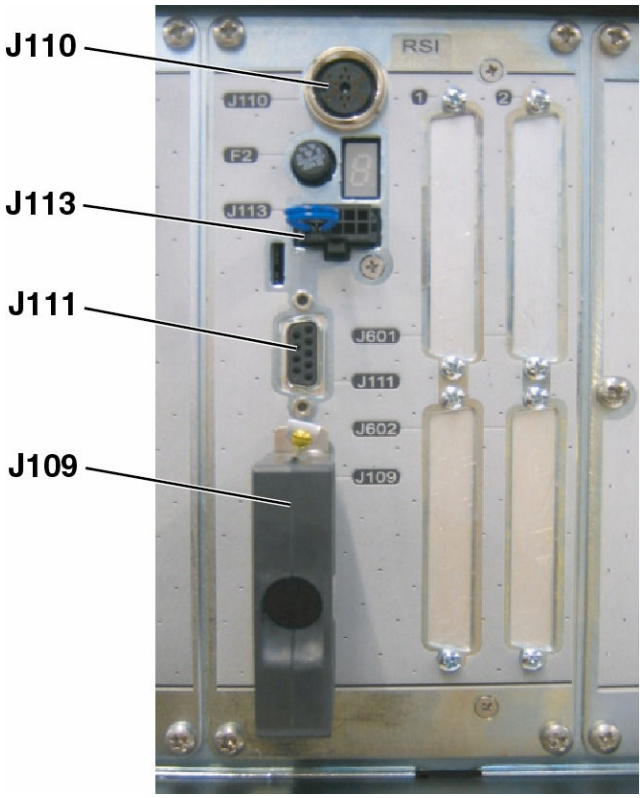
M0000275.1

Description of connection point

The RSI board is connected to the equipment in the cell via the J109 connector on the board on the front panel of the CS8C controller.

All the contacts to be connected up in the emergency stop channels must be duplicated dry contacts. An emergency stop button must activate two contacts at the same time, and the maximum authorized time lapse between the opening of the two contacts is 100 ms. If this time lapse is exceeded, an error message is displayed.

All the information supplied by the RSI board are in the form of dry contacts.



I0000392

Figure 5.5

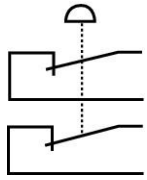
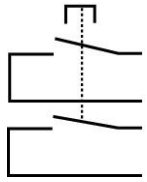
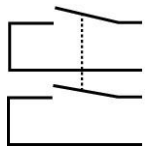
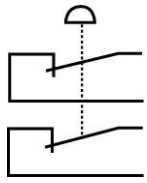
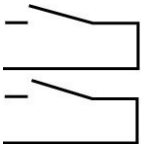
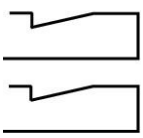
DANGER

Make sure that the controller and the input/output signals are disconnected from the power supply before carrying out operations to connect or disconnect cables or modify the electrical configuration.

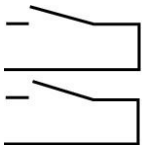
The CS8C controller is supplied with a "shorting connector" for J109 that can be used to power up the robot in manual mode only without wiring up the emergency stops. This connector is provided for diagnosis purposes only. It must be replaced by suitable wiring on the emergency stop circuits.

If screw terminal type connectors are preferred, an adaptor from Sub D to screw terminal is commercially available from several sources such as Phoenix Contact and others.

Connection point table

External electrical diagram	Name of electrical diagram	J109 (Sub-D37M)	Pin name	Internal electrical diagram	Switch open	Switch closed
	UESA 1-2 Emergency stop	1 20 2 21	UESA1+ UESA1- UESA2+ UESA2-		Emergency stop	Normal operation
	USER EN1-2 Emergency stop in manual mode	3 22 4 23	USEREN1+ USEREN1- USEREN2+ USEREN2-		Emergency stop in Manual mode	Normal operation
	DOOR1-2 Emergency stop in Auto mode	9 28 10 29	DOOR1+ DOOR1- DOOR2+ DOOR2-		Emergency stop in Auto mode	Normal operation
	UESB 1-2 Emergency stop	14 33 15 34	UESB1+ UESB1- UESB2+ UESB2-		Emergency stop	Normal operation
Note (1) Note (2)	ESOUT 1-2 Emergency stop channel status	5 24 6 25	ESOUT1+ ESOUT1- ESOUT2+ ESOUT2-		Emergency stop	Normal operation
Note (2)	COMP/MANU Working modes	7 26	COMP+ COMP-		Automatic mode	Automatic mode invalid
		8 27	MANU+ MANU-		Manual mode	Manual mode invalid

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External electrical diagram	Name of electrical diagram	J109 (Sub-D37M)	Pin name	Internal electrical diagram	Switch open	Switch closed
Note (2)	PS1 PS2 Arm power-up	12 31 13 32	USERPS1+ USERPS1- USERPS2+ USERPS2-		Arm not powered	Arm powered
	USER-IN X	16 35	USER-IN 1 + USER-IN 1 -			Note (3)
	USER-IN X	11 30	USER-IN 0 + USER-IN 0 -			Note (3)
Note (4)		18 19 37	24 V for EStop lines	Note (4)		

(1) This information is configurable (see chapter 5.1).
(2) Max. 48V AC/DC/ 0.5A.
(3) See figure 5.6.
(4) Refer to chapter 5.1 and figure 5.2.

5.2 - BASIC INPUTS/OUTPUTS

M0000408.1

To display the status of the inputs/outputs or to change their status, select the "I/O" branch in the control panel accessible via the main menu.

5.2.1 - USER-IN INPUTS

M0000409.1

2 inputs, User-in 1 and User-in 2, are available on the J109 connector.

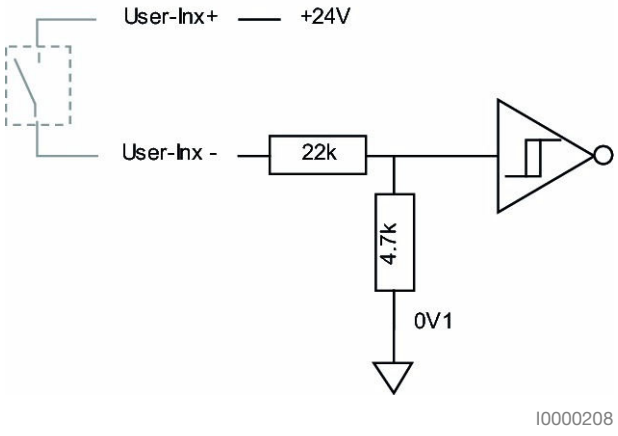


Figure 5.6

Characteristics

Operational voltage range	0 to 30 VDC
"Off" state voltage range	0 to 1 VDC
"On" state voltage range	4 to 30 VDC
Operational current range	0 to 240 µA
"Off" state current range	0 to 5 µA
"On" state current range	33 to 240 µA
Impedance	100 kΩ
Response time for equipment + software	6.5 ms max.

5.2.2 - FAST INPUTS/OUTPUTS

M0000425.1

2 inputs (Fast-In 0 and Fast-In 1) and 1 output (Fast-Out 0) are provided on the J111 connector (refer to Electrical diagrams manual for pinout).

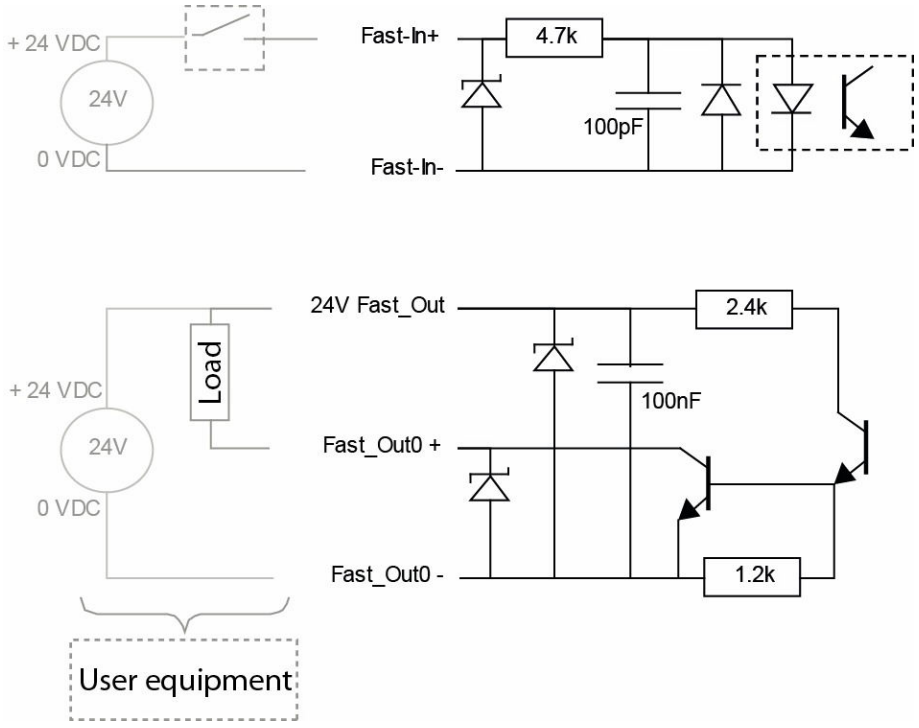


Figure 5.7

Characteristics

Inputs	
Operational voltage range	0 to 30 VDC
"Off" state voltage range	0 to 2 VDC
"On" state voltage range	6 to 30 VDC
Operational current range	0 to 9 mA
"Off" state current range	0 to 0.5 mA
"On" state current range	2 to 9 mA
Impedance	3.3 kΩ
Response time for equipment + software	50 µs

Outputs	
Power supply voltage range (24 V Fast-Out)	12 to 28 VDC
Current consumed	5.5 mA
Output voltage range	12 to 28 VDC
Range of use for current output	1 to 250 mA
Voltage drop at output when I = 250 mA	1.2 V max.
On state output resistance	1 Ω
Maximum leakage current in off state	5 µA
Response time for equipment + software	50 µs
Limit output current (overload)	2 A

5.2.3 - ENCODER INPUT

M0000284.1

With STARC2 board configuration, an optional DUAL ABZ board can be installed to provide 2 encoder inputs per board. The encoders to be used must be of the incremental type with 5V differential A, B, Z signals (RS422 type). The 5V power supply is provided by the DUAL ABZ board, its level of current is limited to 250mA per encoder. Each board has External Encoder Latch signals that can be used with the encoders; Latch signals will operate at voltages from 12 to 24 VDC. The connector used on the DUAL ABZ board is of the female SubD 25 point type.

These boards are installed in the computer (see figure 4.18, rep. (11)).

To install a new board:

- Extract and open computer part (CPT).
- Select board address with J324 jumper.
- Connect the board to STARC with cables provided with the board.

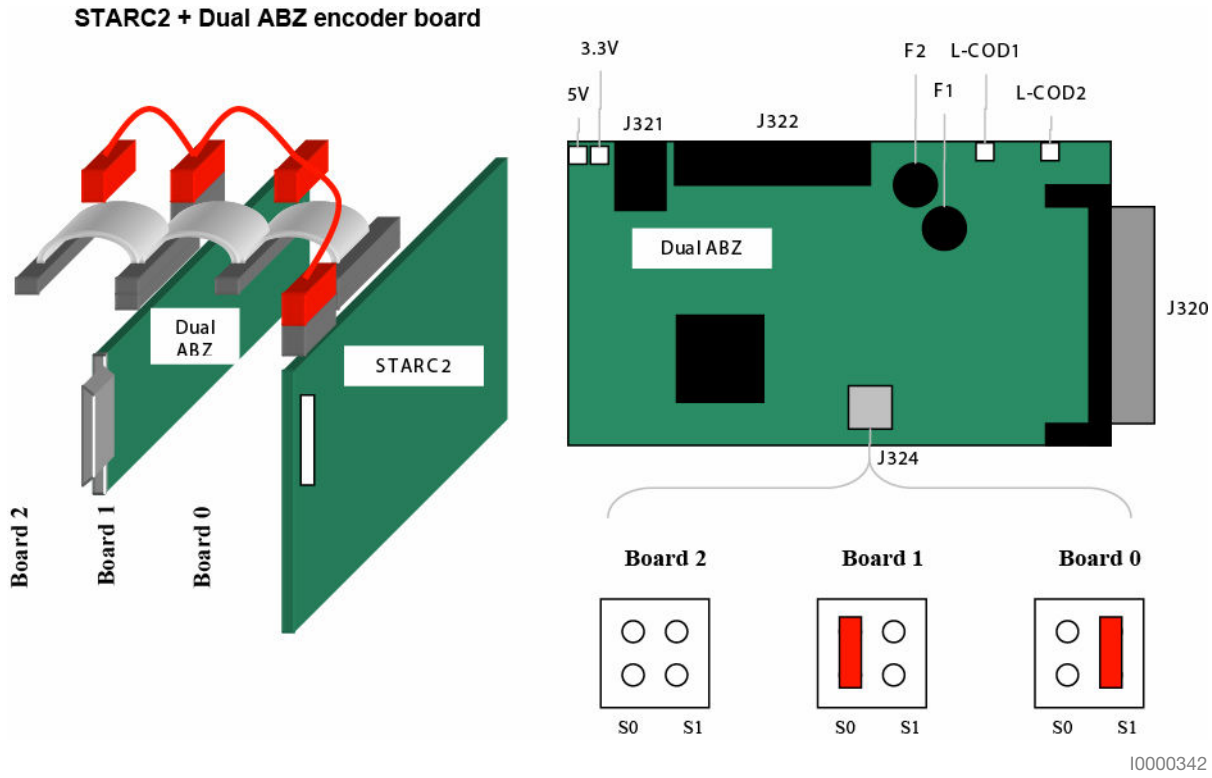


Figure 5.8

Each encoder has a corresponding set of digital and analog Inputs/Outputs. The name of these Inputs/Outputs is the same for all encoders except for the first and second digits that represent respectively the number of the optional board (0 to 1) and the number of the encoder on the board (0 or 1):
e01LatchSig is the latch signal of the second encoder input on the first optional board.

In this chapter, the Inputs/Outputs names are given for the first encoder input on the first board.

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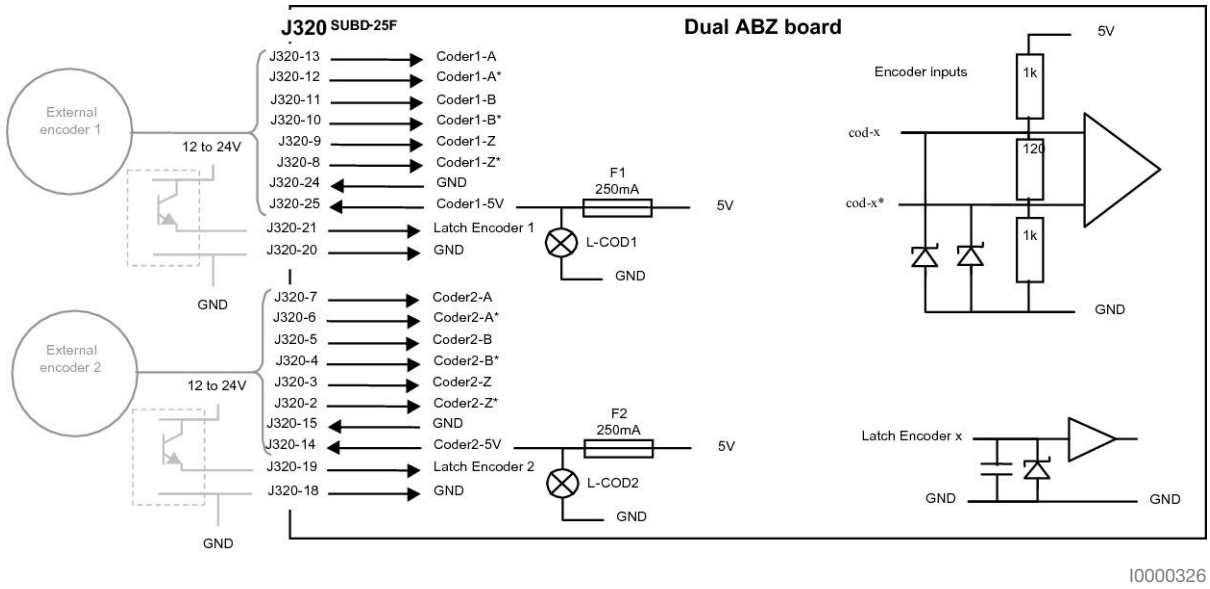


Figure 5.9

Recovery (preset)

Encoder recovery is used to define the zero position of the joint connected to the encoder. It is necessary to begin by configuring the encoder resolution using the e00Counts analog output (in encoder points per rotation, coded on an unsigned 16-bit integer). The resolution is saved by the system (encoder.cfx file).

When the encoder has moved by more than one turn, the system is able to detect its resolution that is then written to the e00CountsMes analog input.

There are 4 pulses per encoder point. A 1024-point (or "counts") encoder thus has a resolution of 4096 pulses per rotation.

Recovery procedure (preset):

- Stop the encoder at a reference position defined by the application.
- Write the current encoder position in the e00PrstPos output (Preset position: signed 32-bit integer).
- Activate the e00EnPrst (Enable preset) output.
- The current position e00CurrPos (Current position) then takes the value e00PrstPos, and the e00EnPrst Input/Output is deactivated.
- The recovery sequence is not saved by the system; it is necessary to repeat it each time the controller is powered up.

Position reading

The position reading is done with the e00CurrPos analog input. The position is in degrees, with an accuracy of 360/(4*e00Counts).

- The internal encoder position counter uses only 32 bits. When the encoder position reaches 2^31*360/(4*e00Counts) degrees, an overflow occurs and the position becomes -2^31*360/(4*e00Counts).
- No error is reported: The overflow must be managed by software programming, either by using a preset to avoid it, or by correcting the position with an offset of 2^32*360*4*e00Counts degrees.

The encoder position is updated every 4 ms. When a preset is done while the encoder is moving, it applies to the start of the current 4 ms time interval. The encoder position is updated only with the next time interval, where it is assigned the preset position plus the encoder movement of the last 4 ms. In that way, no encoder movement is lost with the preset.

Position capture (latching)

Position capture is used to record the encoder position on a rising signal for a fast input, and then read the position later.

Position capture procedure:

- Activate the e00EnLatch digital output.
- On the next rising or falling signal for the fast input e00LatchSig, the encoder position is recorded in the e00LatchPos analog input and the e00Latch digital input is activated to show that the capture has been executed. The e00EnLatch Input/Output is then automatically deactivated.

The precision of the capture is less than one microsecond. It is possible to cancel a capture request at any time by deactivating the e00EnLatch output.

- The position capture is made:
  - On the rising edge of the e00LatchSig signal if the e00LatchEdgFall digital output is set to False.
  - On the falling edge if the e00LatchEdgFall digital output is set to True.
- It is possible to filter out bounds on the latch signal by specifying a filter delay in milliseconds using the e00LatchFilter analog output: the latch is then effective only if the signal remains stable during the specified time. The latched position is then always the position at the rising or falling edge of the latch signal.

Errors

An encoder reading error is signalled by the e00HwErr digital input. If the encoder is rotating too fast, it is not possible for the controller to know for certain how many counts the encoder has missed. In this case, the e00OvsErr signal is activated. The maximum encoder speed is 7500 rpm.

When the encoder has moved by more than one turn, the system is able to detect its resolution that is then written to the e00CountsMes analog input. If this resolution does not match the specified resolution e00Counts, the encoder is in error and the e00CountsErr signal is activated.
The e00PowerErr digital output is activated when the encoder power supply is not correct.

5.2.4 - SYSTEM INPUTS

M0000268.1

The following inputs are accessible on the VAL 3 to detect various errors.

RSI board temperature

The CBT_TEMP input shows the temperature (°C) measured on the RSI board (in the controller). Correct controller operation can no longer be guaranteed if the temperature rises above 55°C. In this case, it is necessary to check the ventilation system on the CS8C controller (fans running, air inlets unobstructed).

Power supply for the controller

The SECTEUR_OK input is activated when the controller is powered up. When the power supply to the CS8C controller is cut off, the SECTEUR_OK signal is deactivated about 80 ms before the power is actually cut off. The SECTEUR_OK input may also be deactivated temporarily if the power supply voltage is too low.

Arm temperature thresholds

Temperature sensors are fitted in the arm to protect the motors and mechanical elements in the event of a temperature rise that is too great. If the temperature of a motor rises above 120°C, the arm stops immediately. An excessively high temperature on the DSI board (75°C) or the castings (100°C) stops the arm after a time lapse of about ten seconds.

The GLOBAL_PTC input signals an overtemperature on one of the arm sensors. The DSI_BOARD input shows an overtemperature on the DSI board.

On TX arms, the MOTOR_1_3_5, MOTOR_2_4_6 and CASTING inputs show an overtemperature on a motor and on the castings. Please refer to the EDS manual for details on the temperature sensors wiring.

On the TS arms, the MOTOR_1_3 and MOTOR_2_4 inputs show an overtemperature for a motor.

CPU board temperature

The CPU_TEMP input shows the temperature (°C) measured on the CPU board (in the controller). Correct controller operation can no longer be guaranteed if the temperature rises above 70°C. In this case, it is necessary to check the ventilation system on the CPU controller (fans running, air inlets unobstructed).

5.3 - CAN INPUTS/OUTPUTS OPTIONS

M0000461.1

The CS8C allows to drive a CAN bus Inputs/Outputs modules.

These Inputs/Outputs are driven from the controller via a dedicated CAN bus master module.

They are provided as two different options:

- The ARMIO module option is a compact form factor mounted inside the forearm of the TS SCARA.
- The REMIO module option is a rail form factor integrated in the paint cabinet.

Both Inputs/Outputs module option includes:

- 8 x 24V digital inputs,
- 8 x 24V digital outputs,
- 4 x ±10V analog inputs,
- 4 x 0/+10V analog outputs.

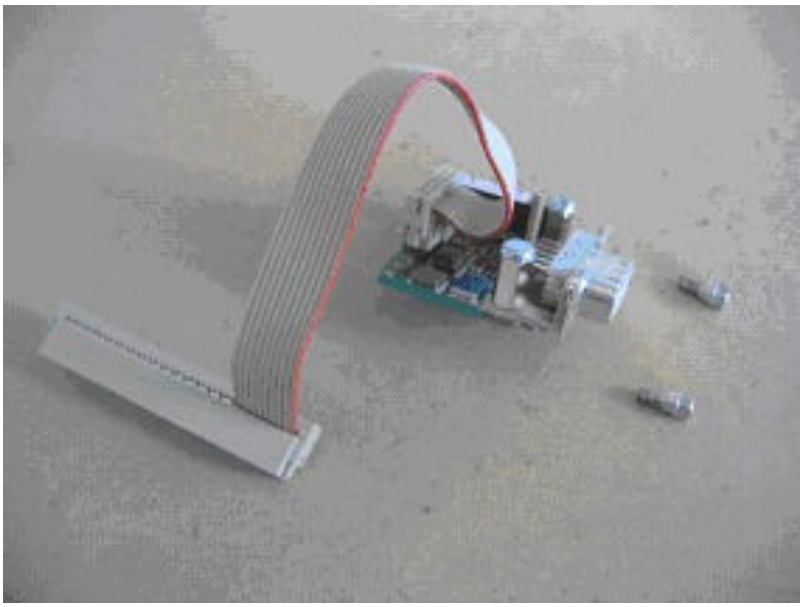
The information and operations on the CAN Inputs/Outputs are available in the "I/O" node of the SP1 Control Panel tree.

5.3.1 - INSTALLATION ON MM1 CPU

M0000276.1

The option includes:

- A CAN board to be installed in the CS8C computer module. The board is provided with a cable and mounting screws.
- A cable between CS8C and the base of the arm.
- An I/O board for the arm.



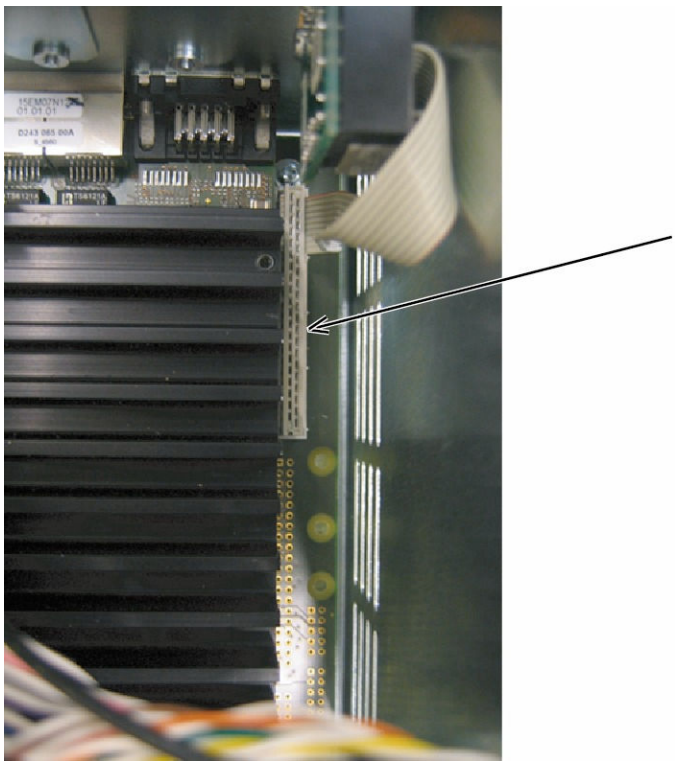
I0000407

Figure 5.10

5.3.1.1 - Installation of CAN board in computer

M0000369.1

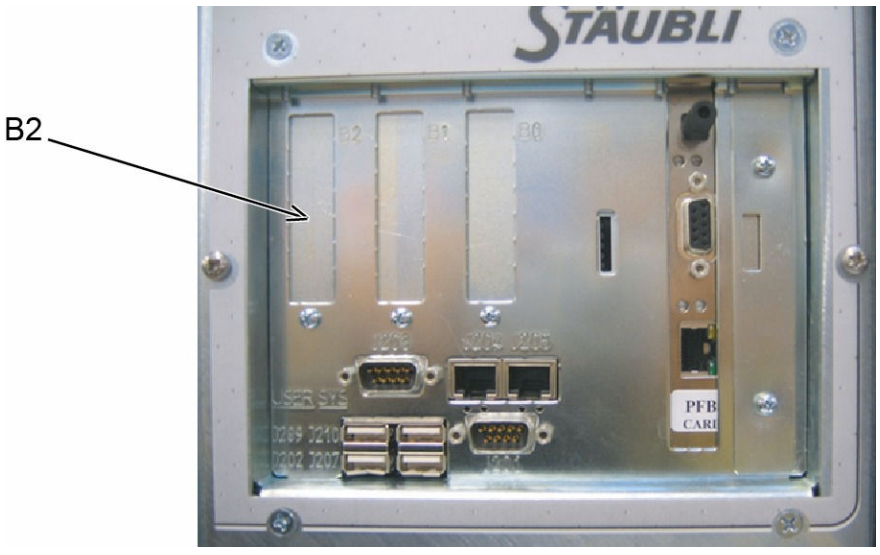
- Refer to chapter 8.8.2 to remove and open CS8C computer module.
- Plug the cable attached to the CAN board to CPU board.



I0000204

Figure 5.11

- Install the CAN board on B2 location of front plate using the 2 screws provided with the board.



I0000246

Figure 5.12

- Replace the cover and reinstall the CS8C computer module.

5.3.1.2 - Software setup

M0000277.1

The controller or emulator must be correctly configured to support the different possible CAN boards.

- The controller configuration file /usr/configs/controller.cfx (s6: /sys/configs/options.cfx) must contain:

  <Bool name="buildCio" value="True" />

- The cell configuration file /usr/configs/controller.cfx (s6: in /sys/configs/options.cfx) must contain:

  1) To enable the ARMIO board on the CAN bus:

  <String name="cioBoardConfig" value="ARMIO" />

  The Inputs/Outputs of the ARMIO board are then configured in the file /usr/configs/cio.cfx.

  The file /usr/configs/can.cfx, if any, must be removed.

  2) To enable the CAN bus with REMIO CAN board:

  <String name="cioBoardConfig" value="" /> (or any value)

  The CAN bus and the Inputs/Outputs are configured with the /usr/configs/can.cfx configuration file; original configuration is located in usr/configs/templates/can/remio.cfx.

  3) To enable the CAN bus with user-specific CAN boards:

  <String name="cioBoardConfig" value="" /> (or any value)

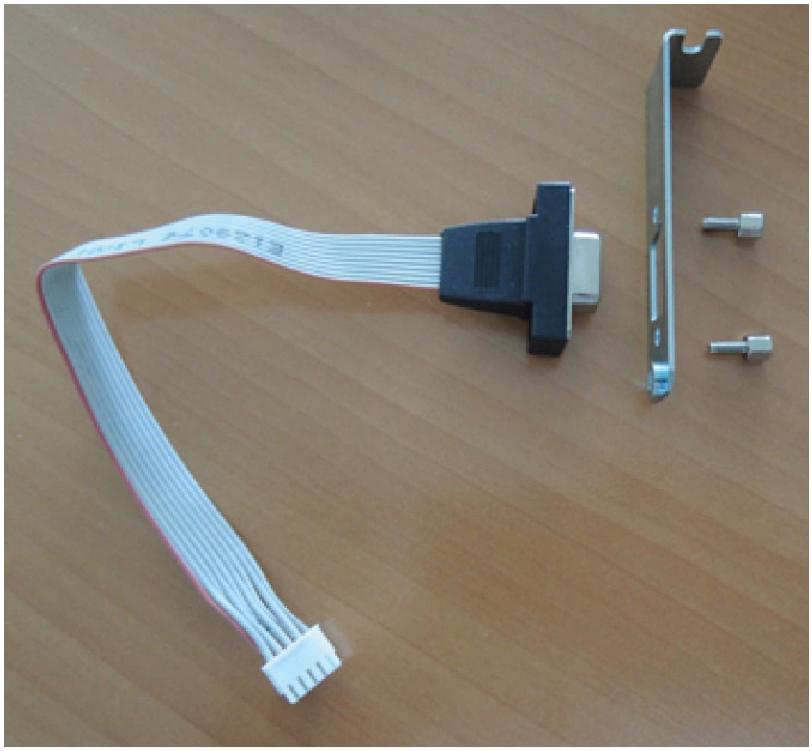
  The CAN bus can then be configured with the /usr/configs/can.cfx configuration file; examples are given in usr/configs/templates/can.

5.3.2 - INSTALLATION ON SCA01 CPU

M0005182.1

The option includes:

- A CAN board to be installed in the CS8C computer module. The cable is provided with a plate to be installed in front of the computer module.
- A cable between CS8C and the base of the arm.
- An I/O board for the arm.



I0005126

Figure 5.13

5.3.2.1 - Installation of CAN board in computer

M0005183.1

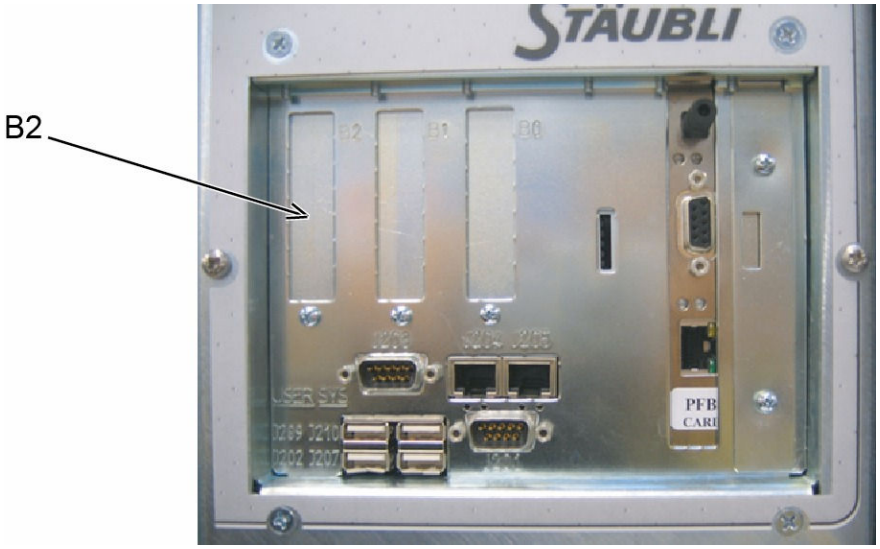
- Refer to chapter 8.8.2 to remove and open CS8C computer module.
- Plug the CAN cable on CPU board **(J212)**.



I0005127

Figure 5.14

- Install the CAN front plate on B2 location of the computer module.



I0000246

Figure 5.15

- Replace the cover and reinstall the CS8C computer module.

5.3.3 - TS SCARA ARMIO OPTION

5.3.3.1 - Installation of cable

M0000462.1

The CAN bus is connected to J211 on controller side and J1202 at the base of the arm. A user cable can be supplied as an option to take the intputs / outputs as close as possible to the tool flange through the ball screw.

5.3.3.2 - Installation of I/O board in the arm

M0000463.1

Remove the 2 side covers and the outer cover of the forearm. The board has to be attached by screws and connected to XB3 plug on the LPX5 board.

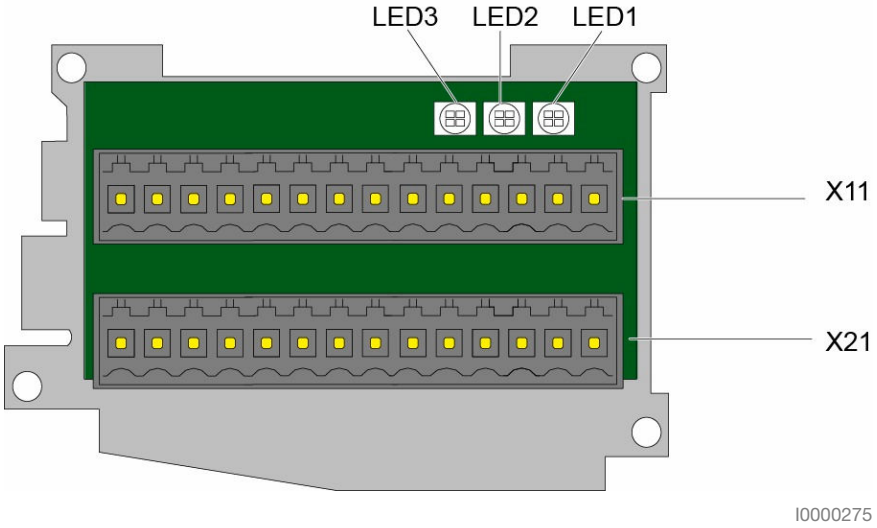


Figure 5.16

X11:

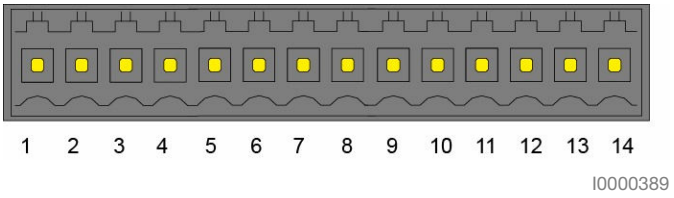


Figure 5.17

Pin	Correspondence
1	cAout 0
2	cAout 1
3	cDout0
4	cDout1
5	cDout2
6	cDout3
7	cDout4
8	cDout5
9	cDout6
10	cDout7
11	+24 VDC
12	0 V
13	cAout2
14	cAout3

X21:

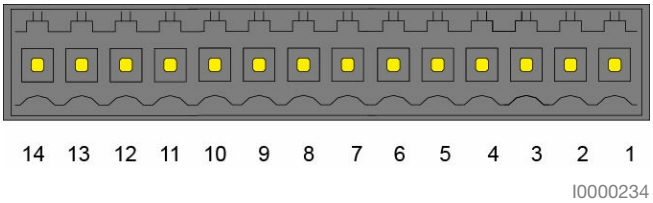


Figure 5.18

Pin	Correspondence
14	cDin0
13	cDin1
12	cDin2
11	cDin3
10	cDin4
9	cDin5
8	cDin6
7	cDin7
6	cAin3
5	0 V analog
4	cAin2
3	cAin1
2	0 V analog
1	cAin0

Characteristics:

- Digital inputs:
  - Nominal voltage: 24 VDC (minimum 20 VDC, maximum 28 VDC).
  - Voltage for logical 0: 0 to 11 VDC.
    Voltage for logical 1: 16 to 28 VDC.
  - Input current: 11 mA maximum.
  - Response time (hardware and software): 6 ms
- Digital ouputs:
  - Nominal voltage: 24 VDC (minimum 20 VDC, maximum 28 VDC).
  - Max. current per output: 0.5 A.
  - Max. current for all outputs: 2 A.
  - Response time (hardware and software): 6 ms maximum.
- Analog inputs:
  - Input voltage: ± 10 V.
  - Resolution: 78 mV.
  - Accuracy: 5 %.
  - Response time: 6 ms.

- Analog outputs:
  - Output voltage: 0/+10 V.
  - Max. current: 20 mA.
  - Resolution: 2.4 mV.
  - Accuracy: 5 %.
  - Response time: 6 ms.

Hardware configuration:

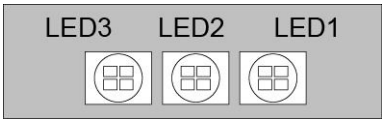


Figure 5.19

	Light emitting diodes LED Display	**Meaning**
LED1	Off	No power supply connected
	Green	Logic power supply OK
	Red	Overload on logic power supply
LED2	Off	No power supply connected
	Green	OK power supply
	Red	Overload on outputs
LED3	Red	CAN bus not operational (not connected or bad cable)
	Flashing red	ID node not valid
	Fast flashing red	Module in passive error status (communication error)
	Green	Module in operational status
	Flashing green	Module in pre-operational status (not connected or bad cable)

The ARMIO board has configuration switches whose positions must be as follows:
- 1, 7, 8, 9 = on
- 2, 3, 4, 5, 6, 10 = off (see figure 5.20)
- Switches 1-5: node number
- Switches 6-7: baud rate, 500k
- Switch 8: 120 ohm termination on
- Switch 9: Can Rho
- Switch 10: unused

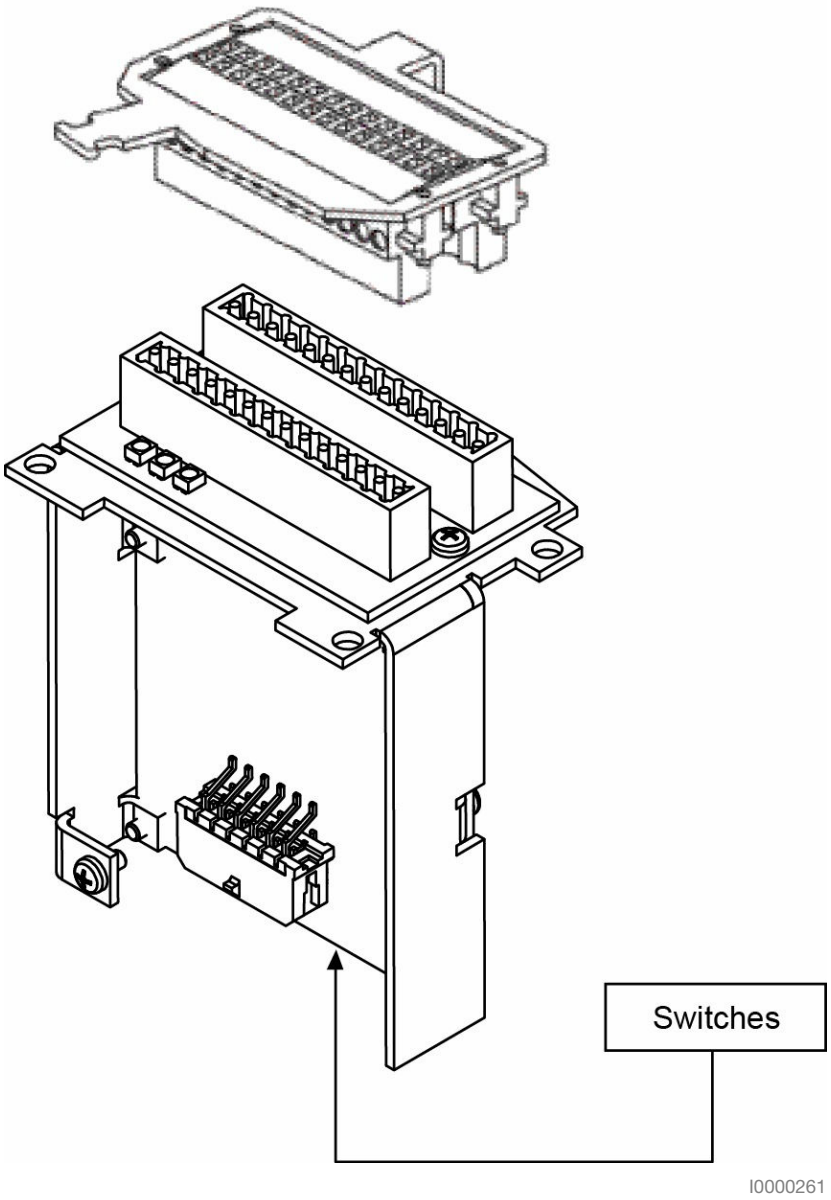


Figure 5.20

5.3.3.3 - Connections on the tool interface

M0000270.1

The user cable is connected on the forearm to X11 and X21 before leading to the tool interface. Once the cable has been installed, connect the sockets X14.1, X14.2 and X14.3 as shown in the table below.

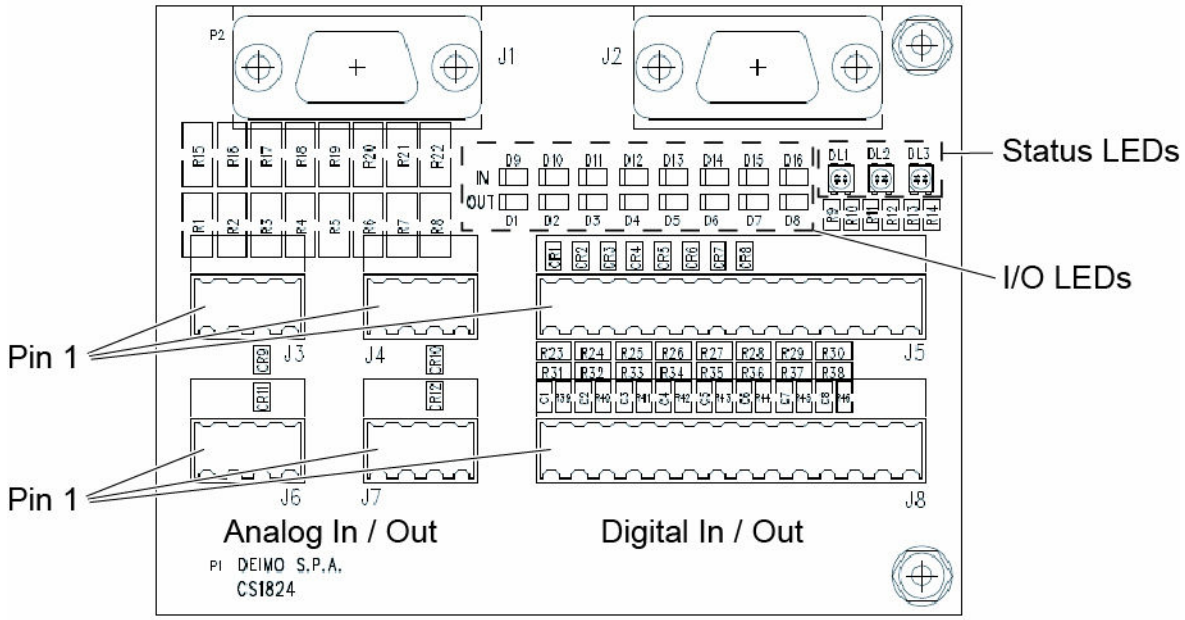
Socket	Pin	Colour	Socket	Pin	Function
X21	14	Grey	X14.1	1	cDin0
	13	White, green	X14.1	2	cDin1
	12	White, yellow	X14.1	3	cDin2
	11	White, brown	X14.1	4	cDin3
	10	White, orange	X14.1	5	cDin4
	9	Orange	X14.3	1	cDin5
	8	Yellow	X14.3	2	cDin6
	7	Green	X14.3	3	cDin7
X11	3	White	X14.2	1	cDout0
	4	White, black	X14.2	2	cDout1
	5	White, violet	X14.2	3	cDout2
	6	Blue	X14.2	4	cDout3
	7	Black	X14.2	5	cDout4
	8	White, grey	X14.2	6	cDout5
	9	White, blue	X14.2	7	cDout6
	10	White, red	X14.2	8	cDout7
X11	11	Red	X14.1	6	+24 V
	12	Violet	X14.3	4	0 V
	12	Brown	X14.1	7	0 V
X21	5	Green, yellow	PE	-	

5.3.4 - REMIO OPTION

M0000469.1

REMIO board has the same electrical characteristics than ARMIO with another form factor for the connectors.

Pin out for the inputs / outputs



I0000333

Figure 5.21

J5	Inputs	J8	Outputs
1	cDin0	1	cDout0
2	cDin1	2	cDout1
3	cDin2	3	cDout2
4	cDin3	4	cDout3
5	cDin4	5	cDout4
6	cDin5	6	cDout5
7	cDin6	7	cDout6
8	cDin7	8	cDout7
9	0V_In	9	+24V_Power
10	0V_In	10	+24V_Power
11	0V_In	11	+24V_Power
12	0V_In	12	+24V_Power
13	+24V_Power_In	13	+24V_Power_In
14	0V	14	0V

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J3	Signals	J4	Signals	J6	Signals	J7	Signals
1	+24V_Power	1	+24V_Power	1	+24V_Power	1	+24V_Power
2	cAout0	2	cAout1	2	cAout2	2	cAout3
3	cAin0	3	cAin1	3	cAin2	3	cAin3
4	0V	4	0V	4	0V	4	0V

J1	Signals	J2	Signals
1	Term_1	1	Term_1
2	CAN L	2	CAN L
3	0V	3	0V
4	NC	4	NC
5	GND	5	GND
6	0V	6	0V
7	CAN H	7	CAN H
8	Term_2	8	Term_2
9	+24V_L	9	+24V_L

The board shall be powered by a 24 VDC supply between +24_PowerIn and 0 V (minimum 20 VDC, maximum 28 VDC, 4 A maximum). These 4 A are shared between analog and digital I/O's.

The digital inputs / outputs are connected as shown below:

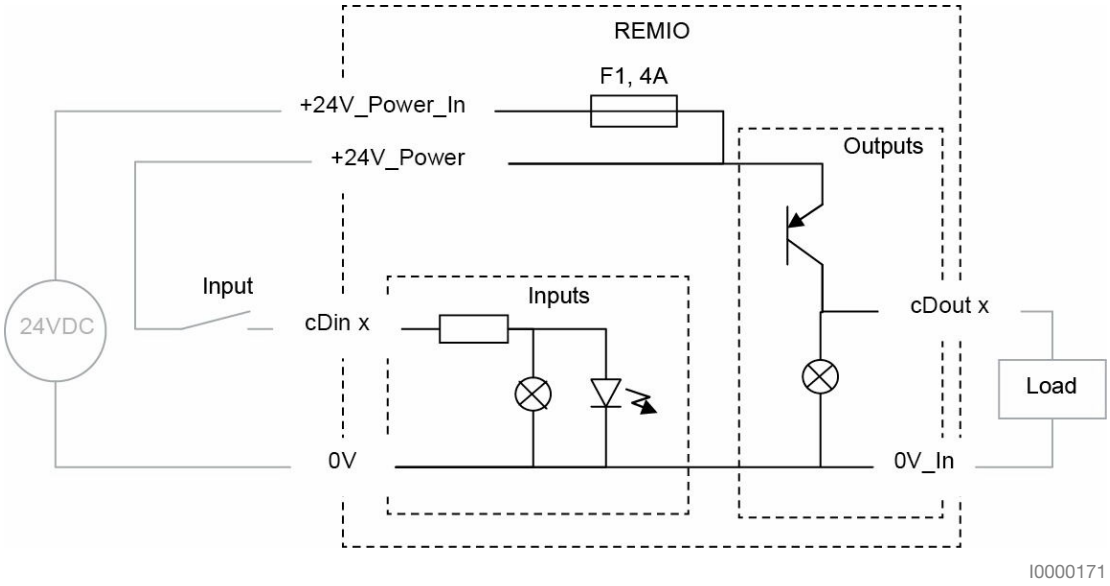


Figure 5.22

The analog inputs / outputs are connected as shown below:

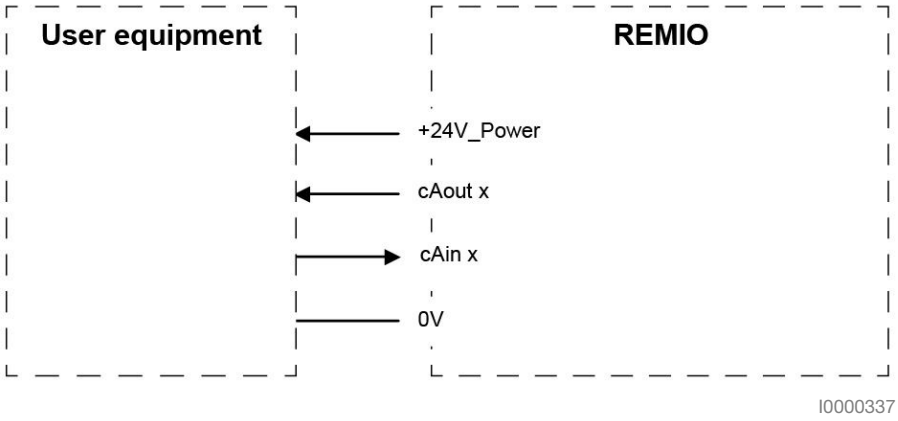


Figure 5.23

The REMIO can provide a 24 VDC to the user equipment within the limit of 4 A, including all I/O consumption (digital and analog).

Analog inputs and outputs are referenced to the 0 V pin.

The board configuration is as follows; it must not be changed:

SW1 configuration	
Node ID	1
Baud Rate	500 k
CAN termination	On
CAN protocol	CANOpen

1	2	3	4	5	6	7	8	9	10
X						X	X		X
	X	X	X	X	X			X	
Node ID					Baud Rate		Termination	Protocol	Not used

Diagnostics

LEDS are present on the board (see figure 5.21).

The input statuses are shown by LEDS D9-D15 for inputs cDin 0-7.

The output statuses are shown by LEDS D0-D18 for outputs cDout 0-7.

The board status is shown by 3 LEDS DL1, DL2, DL3:

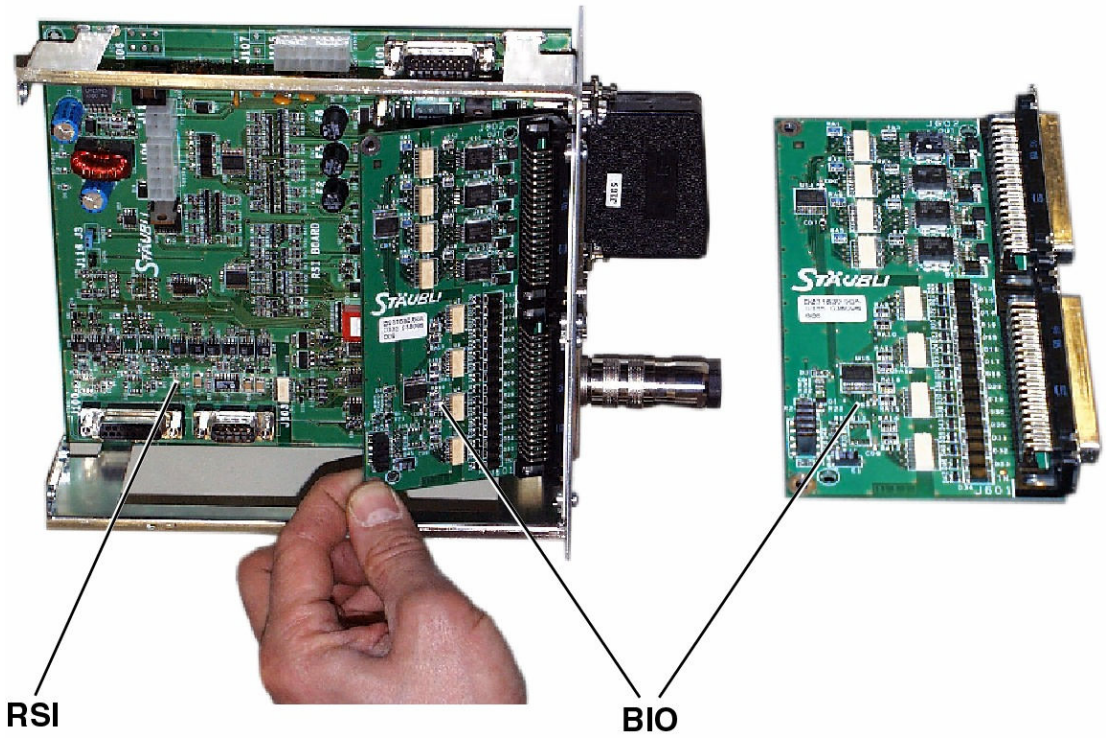
LED		Meaning
DL3	Red	Overload on the 24V_Logic power supply
	Green	24V_Logic present
DL2	Red	Overload on the 24V_Power power supply
	Green	24V_Power present
DL1	Red	CAN bus not operational (wiring problem or no communication with the controller)
	Flashing red	Wrong board ID (see SW1, switch 1-5)
	Fast flashing red	Communication errors for the CAN bus
	Green	Communication bus operating
	Flashing green	Communication bus being initialized

During the CS8C start-up phase, the normal status for the three LEDS is DL2 and DL3 = green and DL1 = flashing green. At the end of the CS8C start-up sequence, the 3 LEDS are green.

5.4 - DIGITAL BIO INPUT/OUTPUT BOARD (OPTION)

M0000285.1

BIO 16I / 16O board description



I0000226

Figure 5.24

The kit is made up of a BIO board to be mounted on the RSI board. Up to 2 BIO boards can be used.

The BIO board is fitted with:

- 16 optocoupler inputs.

  The signals are numbered from 0 to 15 on each board and they correspond to inputs 0 to 15 and then 16 to 31.
- 16 optocoupler outputs protected against overcurrents.

  The signals are numbered from 0 to 15 on each board and they correspond to outputs 0 to 15 and then 16 to 31.



The Inputs/Outputs have to be powered by a rectified, filtered external power source (not supplied).

Wiring the I / 0s

Wiring is described in the "Electrical Diagrams" manual.

Characteristics of the BIO Inputs

Each input channel is made up of an input and a corresponding return wire.

The connector pin out is given in the "Electrical diagrams" manual.

Parameter	Value
Operational voltage range	0 to 24 VDC
"Off" state voltage range	0 to 3 VDC
"On" state voltage range	11 to 24 VDC
Typical threshold voltage	Vin = 8 VDC
Operational current range	0 to 6 mA
"Off" state current range	0 to 0.5 mA
"On" state current range	2 to 6 mA
Typical threshold current	2.5 mA
Impedance (Vin / Iin)	3.9 kΩ minimum
Current at Vin = 24 VDC	Iin ≤ 6 mA
Equipment and software response time	15 ms max.
Insulation voltage / Leakage current	2,5 kV / 4 mm



The characteristics of the input current are given for informational purposes only.

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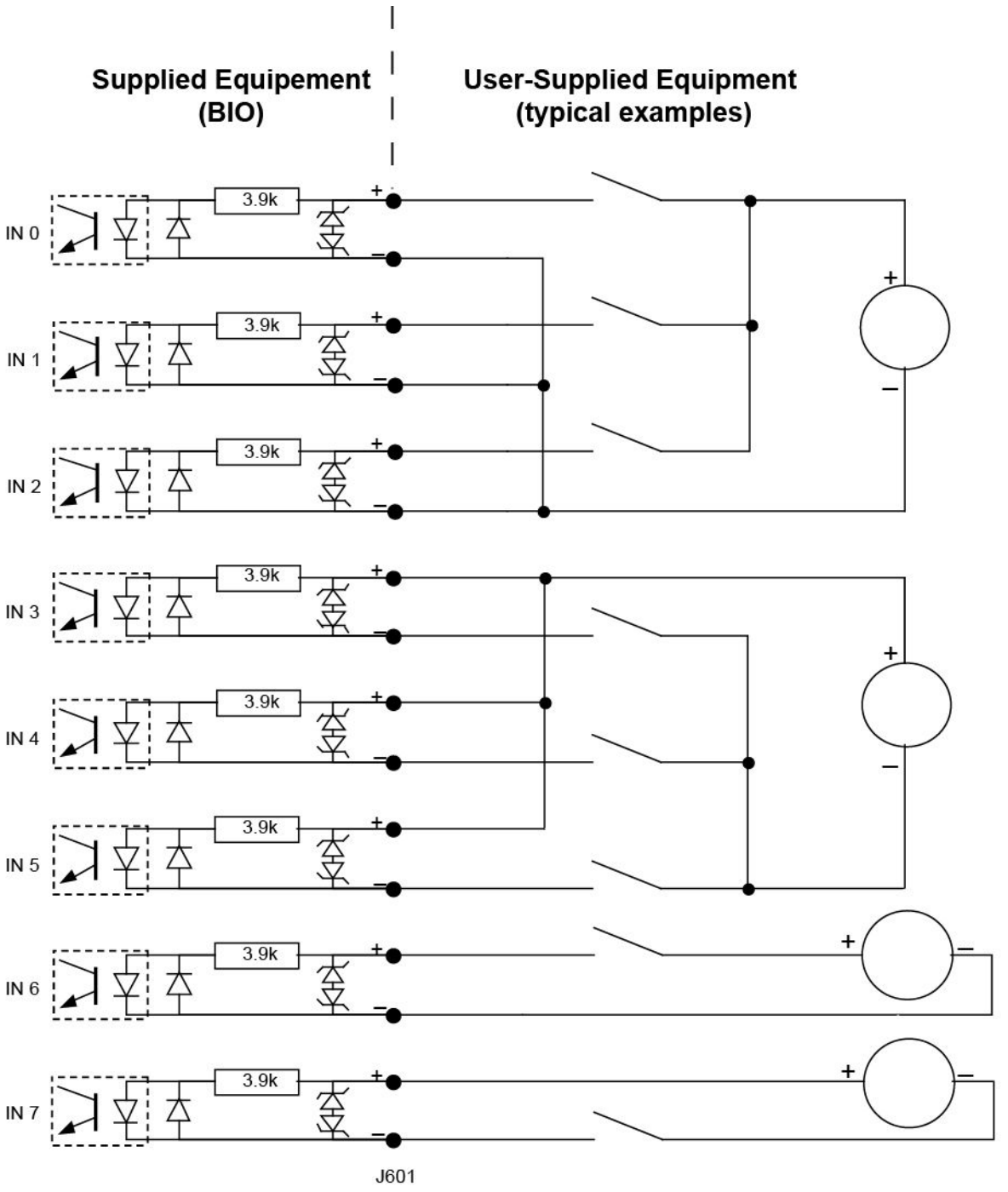


Figure 5.25 : "BIO" board - Inputs

The inputs are numbered from 0 to n on each Input/Output board.

Characteristics of the BIO outputs

The 16 outputs are installed in 4 groups of 4. Each group is electrically isolated from the other groups and optically isolated from the controller. The 4 outputs of each group have a shared return connection. The outputs are protected from overvoltages and inverse voltages.

The connector pin out is given in the "Electrical diagrams" manual.

Parameter	Value
Power supply voltage range	10 VDC < Vsup < 30 VDC
Low voltage stop	5 VDC < Vusd < 8 VDC
Earth current	Ig < 60 mA
Functional current per channel	I_{out} < 700 mA, protected against short circuits
On state resistance (Iout = 0.5 A)	Ron < 0.32 W @ 85 °C (Ron = 0,4 W @ 125 °C)
Output off leakage current	Iout < 25 µA
Equipment and software response time	15 ms maxi
Output cut-off voltage on inductive charge (Iout = 0.5 A, L = 1 mA)	(Vsup - 65) < Vdemag < (Vsup - 45)
Maximum DC short circuit current	0.7 A < Ilim < 2.5 A
Peak short circuit current	Iovpk < 4 A
Insulation voltage / Leakage current	2.5 kV / 4 mm

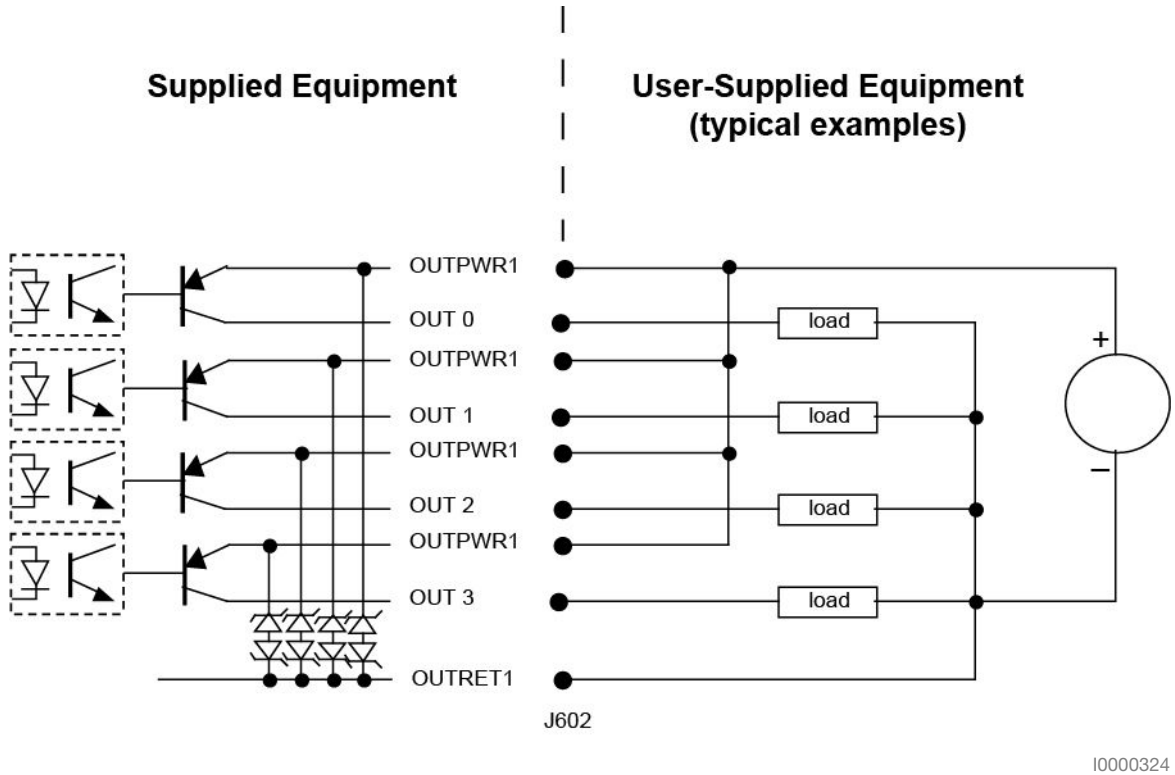


Figure 5.26 : "BIO" board - Outputs

The outputs are numbered from 0 to n on each input/output board.

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Installation of the BIO option:

- Before removing or inserting a board, disconnect the installation from the power supply, following the procedure.
- Furthermore, electronic boards must only be handled after an antistatic work area has been created. To do this, the After Sales Support technician (or the customer) must use an antistatic mat connected to the earth and to the cabinet, as well as wearing the antistatic bracelet supplied with the controller.

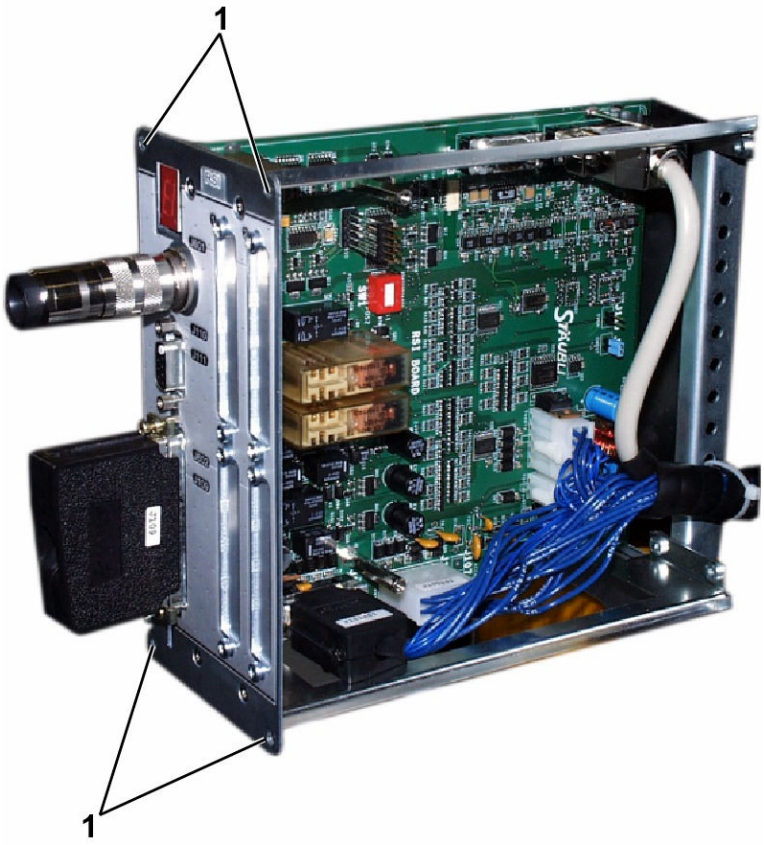
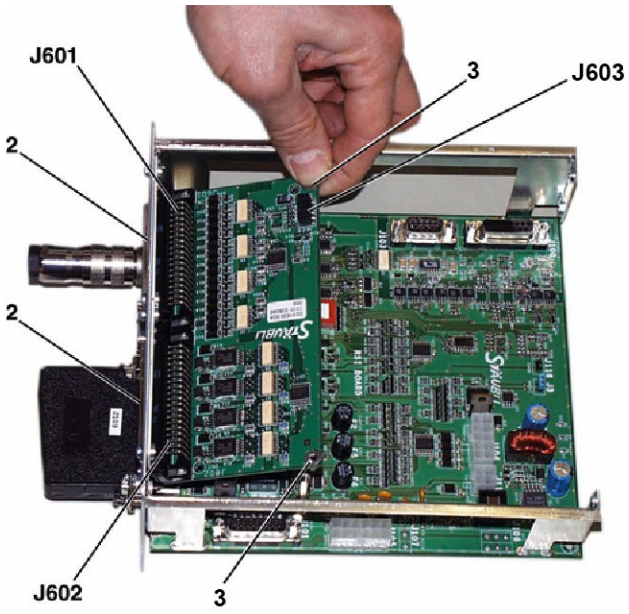


Figure 5.27

- Remove the 4 fastening screws **(1)** from the RSI board and take it out of the controller.



I0000267

Figure 5.28

Installing the first board:

- Insert the **(J601)** and **(J602)** connectors on the BIO board through the holes cut in the front panel of the RSI **(2)** board and then plug the BIO board into the **(J603)** connector.
- Secure the BIO board using the locks on the **(J601)** and **(J602)** connectors and the 2 fastening screws **(3)**.

Installing the second board:

- The second board is installed in accordance with the same principle, using the accessories supplied with the kit (spacers, connector).
- Replace the RSI board.
- The presence of the BIO board is detected automatically by the CS8C cabinet on start-up. The "Control panel" application enables you to see that the board is in place and check the status of its Inputs/Outputs.

To display the status of the Inputs/Outputs or to program them, select the "I/O" branch in the control panel accessible via the main menu.

5.5 - FIELD BUS

5.5.1 - MOLEX MASTER AND SLAVE FIELD BUSES

5.5.1.1 - Characteristics

M0000246.1

There are various types of field busses (DeviceNet, Profibus, CANopen, EtherNet/IP, Profinet, Modbus-TCP).

Each of these options consists of a PCI format board located in the computer and a CD-ROM for installation of the PC configuration software program.

The elements making up the bus as such are not supplied.

The characteristics and wiring for the field bus are specific to the equipment selected. See the constructor's recommendations, especially for the terminal resistances at the end of the lines.

Wiring is described in the "Electrical Diagrams" manual.

The Profibus board requires a straight connector on the field bus cable. It is not possible to fit a 90° right angle connector.

5.5.1.2 - Configuration

M0000273.1

The controller is configured for a field bus using a PC ApplicomIO software application supplied on a specific CD-ROM.

This tool enables you to configure the field bus board on the one hand, and to generate Inputs/Outputs names in the ConfigTag.xml file on the other hand. This file is automatically copied in the controller with ApplicomIO 4.0 or latter, it must be copied by Ftp in the /usr/applicom/io directory with earlier releases. If this file is present on start-up, the Inputs/Outputs of the field bus are displayed on the control panel and can be used in a VAL 3 application.

The Ethernet configuration procedure using the ApplicomIO console software is as follows:

1) Preparation of the configuration (Controller)
- The field bus board cannot be configured while it is operating. If the field bus board has already been configured in the controller, press the "Init" menu on the MCP (Control Panel > I/O > Fieldbus) and confirm (Restart the s6: controller). The field bus board is stopped and ready to be reconfigured.
- Give the controller an IP address and check that it is accessible via the network from the PC on which the ApplicomIO software has been installed.

2) Remote creation of a configuration (ApplicomIO)
- Create a new field bus configuration (File > Configuration Manager > New).
- Enter the name of the new configuration.
- Select "On remote computer (LAN TCP/IP)","IP address" and enter the IP address for the controller. Leave the port on 5001, with the "Automatic transfer" menu selected.

3) Defining the field bus configuration (ApplicomIO)
- Configure the board via the network (Description > Add board).
- Detect the equipment ("Network detection" tab and then Network > Read network configuration).
- Insert the equipment and configure the corresponding Inputs/Outputs.
- The analog Inputs/Outputs can have an unsigned format (default configuration) or a signed one. The min/max value and processing configurations are supported. The "off time" value is ignored. During writing on analog outputs, the linear transformation $y = a.x + b$ is applied and then the result is saturated with the min/max values, and finally sent to the field bus. During reading of an analog Input or Output, the value read from the field bus is first saturated with the min/max values, and then the linear transformation $x = (y-b) / a$ is applied.
- Save the configuration (File > Save).

4) Controller configuration (ApplicomIO)
- Create the XML configuration file (File > Export > items -> XML).
- Initialize the board and download the configuration (File > Download in flash).

5) Checking the configuration of the controller
- Restart the controller.
- Check the field bus board Inputs/Outputs (Control Panel > I/O > Fieldbus).

5.5.1.3 - Diagnosis

M0000252.1

Field bus errors are shown by messages on the MCP, which can also be accessible using the "events logger " application. These errors start with the word "FIELDBUS", followed by a controller diagnosis, the identification of the board, the equipment and the channel concerned, and the Status (Applicom diagnosis).

The controller diagnoses are:

Status	Definition
InitSoftware #Status	/usr/applicom/io/configTag.xml file missing.
BuildItem-#name	The #name item has not been created (incorrect name, already used, or insufficient memory).
Write #Board #Equip. #Channel #Status	Writing error on the field bus.
Read #Board #Equip. #Channel #Status	Reading error on the field bus.
EquipmentStatus #Board #Equip. #Status	Problem with an item of equipment on the field bus.
RefreshIn #Board #Status	Error during the board input update phase.
RefreshOut #Board #Status	Error during the board output update phase.
InitBoard #Board #Status	Error during initialization of the board controller driver. This error is always present for board 2, if there is only one field bus board.
ExitBoard	Error during reinitialization of the controller driver for the board.
BuildPort	Error during construction of a controller Input/Output port: The /usr/applicom/io/ConfigTag.xml file must contain contradictory information.
BoardId #Board Status=1	The OEM board number is not valid. Only boards purchased from Stäubli are accepted.
ConfigBoard #Status	Board initialization error. Check the .ply configuration file, and the board type configuration (Compact PCI "CPCI" boards are not accepted).
NetworkStatus #Board #Status	Problem with the field bus.
Version	Identification of the board version, its BIOS and the playerIO.

The ApplicomIO diagnoses are:

Status	Definition
0	No anomaly detected. The function has been executed correctly.
1	**Function unknown** The function requested is not accepted.
2	**Address error** The address of the variable requested is incorrect.
3	**Data error** MODBUS: Incoherency in the frame content.

Status	Definition
4	**Data inaccessible** MODBUS: The physical address does not exist, the module does not exist, or the data are protected. CANOPEN: ■ For reading / writing objects in SDO, the equipment refuses access to the object requested: - The object does not exist. - The object is read or write protected. - The number of bytes written exceeds the size of the object. ■ For sending or receiving a CAN message: - The message cannot be sent. - The receiving COB-ID is already used by the board or is not valid. DEVICENET: The remote equipment is in error status. Check its status.
9	MODBUS: A MODBUS customer is trying to modify a write protected data item.
10	Negative layer 2 acknowledgement from the equipment (NACK). PROFIBUS: UE (User Error), error in the remote equipment.
11	Negative layer 2 acknowledgement from the equipment (NACK). PROFIBUS: RR (Remote Ressource), insufficient resources in the remote equipment or initialization parameters not valid.
13	Negative layer 2 acknowledgement from the equipment (NACK). PROFIBUS: RDL (Response FDL/FMA1/2 Data Low), remote equipment resources insufficient to deal with the data received, low priority response.
14	Negative layer 2 acknowledgement from the equipment (NACK). PROFIBUS: RDH (Response FDL/FMA1/2 Data High), remote equipment resources insufficient to deal with the data received, high priority response.
32	**Incorrect parameter sent to the function** Wrong number of variables.
33	**Response time exceeded** The equipment does not respond. Check its status and wiring. MODBUS: Equipment configured but not connected to the network. ■ Wiring problem, the CPU does not execute the communication blocks, connection not declared or incorrectly declared in the CPU. ■ Incorrect IP address for the equipment or the gateway. ■ Check that the Ethernet frame format configured in the remote equipment is "ETHERNET II". DEVICENET: ■ The DeviceNet master has no equipment to monitor in its configuration. ■ The slave has not been configured by the master during the initialization phase. ■ The slave has not been reached by the master during the time lapse defined by the master during the initialization phase. PROFIBUS: The Input/Output configuration for the master does not correspond to the Input/Output configuration for the slave.
34	**Physical fault on the line** DEVICENET: No 24V supply detected. The CAN component of the applicom interface is "bus off". Check the wiring and the Baud Rate for the network.
35	Data not available for cyclic reading.

Status	Definition
36	**Equipment not configured** Define the equipment configuration using applicomIO Console and reinitialize the field bus board.
40	Writing or reading attempt deferred by a task, although the maximum number of tasks that can use the deferred mode at the same time has already been reached.
41	Writing or reading attempt made although the deferred request register is full.
42	Attempt to transfer a request deferred although the deferred request register is empty.
46	Board number not configured, or master / customer function attempting to use a channel with a master / slave configuration, or the other way round.
47	The field bus board is invalid or incorrectly initialized by the IO_Init function.
49	**Time delay fault for adding to the queue file** MODBUS: It was not possible to send the request due to a lack of resources (no communication channel available). This period corresponds to 4 times the value of the time-out for the current requests. Increase the time-out value, or the maximum number of simultaneous requests for the equipment in question.
51	Driver system problem.
53	DEVICENET: Synchronization problem on the line. The master DeviceNet is offline (no power supply detected or the CAN component of the applicom interface set to "Bus Off"). Check the wiring and the Baud Rate for the network.
55	**Response time exceeded-Message lost** Check the equipment status. MODBUS: Queuing time exceeds the value of the "time-out for requests being processed", connection made, question acknowledged but no response. DEVICENET: The equipment has accepted the connection but has not responded to the request.
59	■ Protection key missing from the applicom interface ■ Use of applicom functions without prior initialization
63	Communication error on the serial port.
65	Connection refused. DEVICENET: The connection to the master DeviceNet is in progress or refused by the equipment.
66	Insufficient memory for the applicom interface. Insufficient resources for a further connection.
70	MODBUS: Connection closed by the equipment following a communication problem. Wiring problem, CPU off, the CPU is not executing the communication blocks. The equipment does not accept this messaging system. Check the equipment status. DEVICENET: Connection over. MAC ID duplication detected on the DeviceNet network. Modify the MAC ID of the master DeviceNet.
79	Profile incompatible. The equipment does not correspond to the configuration. Check the equipment and the bandwidth of the connections.
93	Driver not accessible.
97	Operating mode not accepted.
99	The applicom interface is already in use.

Status	Definition
255	Local reading buffer not initialized by the IO_RefreshInput function.

5.5.2 - ETHERNET BASED FIELDBUS

5.5.2.1 - Characteristics

M0000465.1

The CIFX50 Hilscher board supports Ethernet based field buses.

This card is used to connect the controller as a slave device for robot control under uniVAL configuration or for software IO exchanges under VAL3 configuration.

The table below shows the compatibilities of theses functionalities according to the SRC releases.

	EtherCAT	Powerlink	Sercos3
uniVAL	>= SRC s7.1	>= SRC s7.3	>= SRC 7.0
Software IO exchanges	>= SRC s7.3	>= SRC 7.4	Not supported

5.5.2.2 - Configuration

M0000451.1

uniVAL configuration

Refer to uniVAL user manual.

Slave IO configuration

When it is use for software IO exchanges under VAL3 configuration, the card must be declared with tags in the controller configuration file.

For VAL3 < s7.4

The configuration file is /usr/configs/hilscher.cfx.

<String name="fieldbusIO" value="coe"> => CanOverEtherCAT

For VAL3 >= s7.4

The configuration file is /usr/configs/hilscher.cfx.

<Hilscher fieldbus="coe"> ⟶ Use CanOverEtherCAT

<Hilscher fieldbus="powerlink"> ⟶ Use powerlink

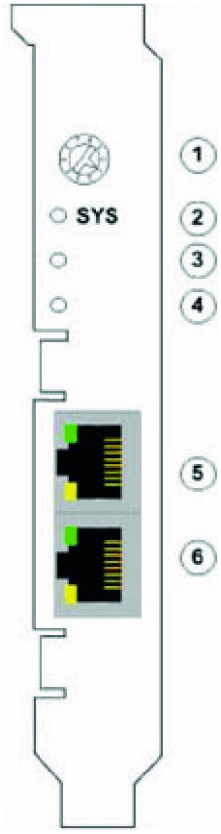
Default value: "coe"

<Hilscher hideOtherBoards="false"> ⟶ All controller IO boards are accessible from the fieldbus master.

<Hilscher hideOtherBoards="true"> ⟶ Only the IO boards defined by the user are accessible from the fieldbus master.

Default value: "false"

Refer to SRS VAL3 Studio physical IO tool for more details.



I0000227

Item	Description
1	Rotary switch slot number (unused)
2	Led SYS
3	Led COM 0 (Designation and meaning depends by protocol)
4	Led COM 1 (Designation and meaning depends by protocol)
5	Ethernet Interface (Channel 0)
6	Ethernet Interface (Channel 1)

Figure 5.29 : Front plate X 50-RE or CIFX 50E-RE

5.5.2.3 - Diagnosis

LED Descriptions

The meaning of the communication status and the RJ45 LEDs depends on the protocol configured on the controller.

LED naming in the device drawing		LED naming by protocol		
		EtherCAT - Slave	**Powerlink**	**SERCOS III - Slave**
SYS (System status) (Yellow/Green)		**SYS**	**SYS**	**SYS**
COM 0 (Communication status) (Red/Green) (Orange = Red/Green Simultaneously)		**RUN** (Run)	**BS** (Bus status)	**S3** (Status/Error)
COM 1 (Communication status) (Red/Green)		**ERR** (Error)	**BE** (Bus error)	**-**
RJ45	(Green)	**L/A IN** (Link/Activity input)	**L/A** (Link/Activity)	**L/A** (Link/Activity)
	(Yellow)	**-**	**-**	**-**
RJ45	(Green)	**L/A OUT** (Link/Activity output)	**L/A** (Link/Activity)	**L/A** (Link/Activity)
	(Yellow)	-	-	-

System LED

The subsequent table describes the meaning of the system LED.

LED	Colour	State	Meaning
"SYS"	**Duo LED Yellow/Green**		
	(Yellow)	Static	Bootloader netX (= roomloader) is waiting for the second stage Bootloader
	(Yellow/Green)	Blinking Green/Yellow	Second stage bootloader is waiting for firmware
	(Green)	ON	Operating system running
	(OFF)	OFF	Power supply for the device is missing or hardware defect

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EtherCAT Slave

The subsequent table describes the meaning of the LEDs Communication Interface when the configured protocol on the controller is EtherCAT.

LED	Colour	State	Meaning
RUN Name in the device drawing: **COM 0**	**Duo LED Red/Green**		
	(OFF)	OFF	**INIT:** The device is in state INIT
	(Green)	Blinking	**PRE-OPERATIONAL:** The device is in state PRE-OPERATIONAL
	(Green)	Single flash	**SAFE-OPERATIONAL:** The device is in state SAFE-OPERATIONAL
	(Green)	ON	**OPERATIONAL:** The device is in state OPERATIONAL
ERR Name in the device drawing **"COM 1"**	**Duo LED Red/Green**		
	(OFF)	OFF	**No error:** The EtherCAT communication of the device is in working condition
	(Red)	Blinking	**Invalid Configuration:** General configuration error (Example: State change commanded by master is impossible due to register or object settings).
	(Red)	Single flash	**Unsolicited State Change:** Slave device application has changed the EtherCAT state autonomously: Parameter "Change" in the AL status register is set to 0x01:change/error (Example: Synchronisation Error, device enters Safe-Operational automatically).
	(Red)	Double flash	**Application Watchdog Timeout:** An Application Watchdog Timeout has occurred (Example: Sync Manager Watchdog Timeout).
	(Red)	ON	**PDI Watchdog Timeout:** A PDI Watchdog Timeout has occurred (Example: Application controller is not responding any more).
L/A IN/ RJ45 Ch0 - **L/A OUT/** RJ45 Ch1	**LED Green**		
	(Green)	ON	A link is established
	(Green)	Flashing	The device sends/receives Ethernet frames
	(OFF)	OFF	No link established
RJ45 Ch0 RJ45 Ch1	**LED Yellow**		
	(Yellow)	-	-

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Powerlink

The subsequent table describes the meaning of the LEDs Communication Interface when the configured protocol on the controller is Powerlink.

LED	Colour	State	Meaning
BS Name in the device drawing: **COM 0**	**Duo LED Red/Green**		
	 (OFF)	OFF	Slave initializing
	 (Green)	Flickering	Slave is in a Basic Ethernet state
		Single flash	Slave is in Pre-Operational 1
		Double flash	Slave is in Pre-Operational 1
		Triple flash	Slave is in ReadyToOperate
		ON	Slave is Operational
		Blinking	Slave is Stopped
BE Name in the device drawing **COM 1**	**Duo LED Red/Green**		
	 (OFF)	OFF	Slave has no error
	 (Red)	ON	Slave has detected an error
"L/A/" RJ45 Ch0 & Ch1	**LED Green**		
	 (Green)	ON	**Link:** A connection to the Ethernet exist
	 (Green)	Flashing	**Activity:** The device sends/receives Ethernet frames
	 (OFF)	OFF	The device has no connection to the Ethernet
RJ45 Ch0 & Ch1	**LED Yellow**		
	-	-	The LED is not used

SERCOS III

The subsequent table describes the meaning of the LEDs Communication Interface when the configured protocol on the controller is **"SERCOS III"**.

LED	Colour	State	Meaning
"STA" Name in the device drawing: **"COM 0"**	**Duo LED Red/Green/Orange** (Orange = Red/Green Simultaneously)		
	 (Green)	ON	**CP4:** Communication phase 4, Normal operation, No error
	 (Green)	Flashing (4 Hz)	**Loopback:** The network sate has changed from "fast-forward" to "loopback"
	 (Red/ Green)	Flashing (4 Hz) the LED flashes at least for 2 seconds from red to green	**Communication error:** Depends on IDN S-0-1003 (for details refer to SERCOS III Slave Protocol API.pdf on the product DVD). Shows how long the master may in the communication phases CP3 and CP4 not received Master SYNC telegrams.
	 (Red)	ON	**SIII C1D:** Error detected according to SERCOS III Cass 1 diagnosis.
	 (Orange)	ON	**CP0...CP3:** Communication phase 0 to communication phase 3.
	 (Orange)	Flashing (4 Hz)	**Identification:** Bit 15 in the slave device control that indicates remote address allocation or configuration errors between master and slaves (for details refer to SERCOS III Slave Protocol API.pdf on the product DVD).
	 (OFF)	OFF	No SERCOS III communication
Name in the device drawing **"COM 1"**	**Duo LED Red/Green**		
	-	-	The LED is not used
"L/A/" RJ45 Ch0 & Ch1	**LED Green**		
	 (Green)	ON	**Link:** A connection to the Ethernet exist
	 (Green)	Flashing	**Activity:** The device sends/receives Ethernet frames
	 (OFF)	OFF	The device has no connection to the Ethernet
RJ45 Ch0 & Ch1	**LED Yellow**		
	-	-	The LED is not used

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5.6 - ETHERNET LINK

5.6.1 - CONFIGURATION

M0000262.1

The CS8C has 2 Ethernet ports, J204 and J205. The IP address of each of these ports can be modified via the control panel. The modification takes effect immediately. On delivery, the first port is configured with the address 192.168.0.254 (mask 255.255.255.0) and the second one with the address 172.31.0.1 (mask 255.255.0.0).

It is also possible to get automatically an IP address from the network (with DHCP protocol).



- The DHCP protocol may not always assign the same IP address to the controller. The DHCP mechanism should therefore not be used when the IP address of the controller is used by other peripheral equipment.
- The two Ethernet ports must correspond to different sub-networks. Two IP addresses on the same sub-network are not supported.

The CS8C controller can reach other Ethernet sub-networks, through gateways configurable from the Control Panel.

Each gateway is defined with:

- The IP address of the device used as gateway. The address must belong to the same sub-network as the CS8C controller.
- The IP address of the sub-network to reach. A null address "0.0.0.0" can be used to define a default gateway, to reach all sub-networks that are not handled by a specific gateway.

5.6.2 - FTP PROTOCOL

M0000263.1

The CS8C controller is an Ftp server that enables file exchanges over the Ethernet. All that is required is to define the IP address of the CS8C controller to enable it to be accessed via Ftp, and use a network login and password corresponding to a user profile as defined on the CS8C. The reading and writing access rights depend on the login user profile selected (see chapter 5.8.3).

A free client Ftp software application is supplied on the controller CD-ROM.

5.6.3 - MODBUS TCP PROTOCOL

M0000250.1

The CS8C controller can be configured to exchange Inputs/Outputs via the Ethernet using the Modbus Tcp protocol. In this case, the CS8C controller is considered as a Modbus Server.

The CS8C controller is configured for Modbus Tcp using a Stäubli Robotics Suite tool (Tools > Modbus IO Config). This tool is used to generate a modbus.xml file that has to be saved in the /usr/applicom/modbus directory of the CS8C emulator. Then the file has to be transferred to the CS8C controller using the Transfer Manager tool of Stäubli Robotics Suite. If this file is present on start-up, the corresponding Modbus Tcp Inputs/Outputs are displayed on the control panel and are directly accessible via a VAL 3 application.

The "MODBUS TCP/IP" Client software option for your PC provides access to Modbus Inputs/Outputs declared on CS8C from PC software using OPC Client, ActiveX, DLL interface (f.i Visual Basic, Delphi, Visual C++, LabView, ...). A trial version of the DirectLink PC Network Interfaces software can be downloaded from the online Technical Database (http://www.staubli.com).

Configuration

The configuration procedure for the Modbus Tcp Inputs/Outputs using the Stäubli Robotics Suite Modbus IO Config tool is as follows:

- Create a new configuration ("New") or open an existing configuration ("Open").
- Define the name, type, size and access for each item.
- Save the configuration ("General parameters", "Save" tab).
- Copy the file to the controller using Stäubli Robotics Suite Transfer Manager.
- Restart the controller.
- Check the modbus Inputs/Outputs (Control panel > I/O > Modbus).

Diagnosis

Modbus Tcp errors are shown by messages on the MCP, which can also be displayed using the "events logger" application. These errors begin with the word "MODBUS" followed by a CS8C diagnosis, identification of the channel concerned, and the Status (applicom diagnosis).

The CS8C diagnoses are:

Status	Definition
InitLib #Status	/usr/applicom/modbus/modbus.xml file missing
BuildItem #name	The #name item has not been created (incorrect name, already used, or insufficient memory).
Write #Channel #Status	Writing error
Read #Channel #Status	Reading error
StartServer	It has not been possible to start the Modbus server. The modbus.xml file must be invalid.
StopServer	It has not been possible to stop the Modbus server

The diagnoses given by the Status are the same as those given for the Modbus field bus (see chapter 5.5).

5.6.4 - ETHERNET SOCKETS (TCP)

M0000483.1

The CS8C controller can be configured to communicate via the Ethernet using sockets (TCP or UDP). The CS8C controller accepts up to 40 sockets simultaneously, in client mode or server mode. The Ethernet sockets are configured via the "Control panel" application (Control panel > I/O / Socket).

The parameters of a server socket are:

- The connection port between 0 and 65535.

  The ports between 0 and 1000 are reserved.

- The maximum number of simultaneous clients.
- The time lapse prior to triggering of an error message (maximum time lapse reached in reading or for a connection). A zero value suppresses the time lapse control.
- The end of string character.
- The enabling/disabling of the Nagle optimization algorithm.

Details of the last parameters are given in the VAL 3 reference manual (SIO type).

The parameters of a client socket are the same, with the extra element of the address of the server socket to be reached. A "Test" menu can be used to test the connection with the server.

A server socket is activated ("opened") in the CS8C each time a VAL 3 program uses it, and deactivated ("closed") when the last customer logs off. When the maximum number of clients has been reached for a server socket, other clients attempting to log on are accepted but the communication is interrupted immediately by the server.



If no VAL 3 program has accessed the server sockets in the CS8C, they are not activated, and all attempts by a customer to connect will remain fruitless. In particular, the "Test" menu used on a CS8C controller to test a server socket on another CS8C controller produces an error message if no VAL 3 application is running on that controller.

5.7 - SERIAL PORT

M0000258.1

Two serial ports are available on the CS8C controller (J203, COM1 and J201, COM2) to exchange data between a VAL 3 application and an equipment item in the cell.

The serial links are configured via the Input/Output display on the Control panel.

The parameters that can be configured in the Series link are:

- The transmission speed (from 110 to 115200 bauds).
- The number of data bits (from 5 to 8).
- The number of stop bits (1 or 2).
- Parity (even, odd or no parity).
- For J201 (COM2), the RS232/RS422 configuration. Default configuration is RS232.
- The flow control (none / hardware) (not available with RS422 configuration on COM2).
- The time lapse prior to triggering of an error message (maximum reading time lapse). A zero value suppresses the time lapse control.
- The end of string character.

Details of the last two parameters are given in the VAL 3 reference manual (SIO type).



- Check the configuration of the Series link to the external equipment before connecting to the electricity supply.
- When the controller is powered up, characters are sent via COM1 (start-up information for BIOS, etc.) and they may disrupt operation of equipment connected to J203. This point must be taken into account in the application.

5.8 - SOFTWARE CONFIGURATIONS

M0000245.1

The software configuration can be used to modify certain characteristics of the controller, program user profiles to limit access to certain functions, and program Inputs/Outputs to enhance CS8C integration in the cell.

5.8.1 - CONFIGURATION OF THE CONTROLLER CHARACTERISTICS

M0000254.1

The Control panel application gives the various system characteristics. The serial number of the controller is displayed with the list of installed software and hardware versions.

Certain characteristics can be modified (depending on the login user profile selected):

- The software limits linked to the arm or the cell. The software limits linked to the arm must correspond to the mechanical limit switches.
- The unit of length (millimeter or inch).
- The language (see hereafter).
- The date and time, fixed or from SNTP time server.
- The address and IP mask of each Ethernet port (fixed or from DHCP).
- The list of gateways to reach other Ethernet sub-network.
- The current user profile and the user profile when the CS8C is started.
- The port number for different system Ethernet socket servers (Stäubli Robotics Suite remote maintenance, telnet, ApplicomIO fieldbus console, Stäubli Robotics Suite 3D Studio).
- The maximum Cartesian speed in manual mode (up to 250 mm/s).
- The status of the emergency stop channels (ESOUT1 and ESOUT2) with or without DOOR signal (see chapter 5.1.1).



- The configuration of some controller characteristics (such as software limits, unit of length, maximum Cartesian speed, emergency chain) may impact the safety of the robot installation. The default user profile prevents the change of these characteristics (see chapter 5.8.3).
- These configurations do not have the safety performance level required in the ISO 10218-1:2011 standard for safety control systems. They are provided for device protection and not for functional safety in the cell.



Date and time are maintained with a capacitor whose guaranteed backup time is 3 days. When discharged, date and time come back to January 1998.

Language configuration:

The controller is delivered with some predefined translations (English, German, French, Italian, Spanish, Chinese, Japanese, Korean, etc.). Each translation is defined in a .cfx XML file with Unicode UTF8 encoding, located in the /sys/configs/resources directory. The file itself consists in a set of string definitions, such as:

<String name="invalidBinaryOperator" value="Invalid binary operator for these types" /> where the name attribute is a translation identifier (to be kept unchanged), and the value attribute is the corresponding translation. The help attribute, if any, defines the help message associated with the main text.

It is possible to remove/add/modify a translation by removing/creating/modifying the corresponding .cfx file. Missing texts in a language definition file are replaced with the default English translation, therefore a user-defined translation file can be used on a newer VAL 3 version: Only new texts will not be translated correctly.

The Unicode UTF8 format of the language definition file makes it possible to use any Unicode character or symbol in it. However, the correct display of the characters on MCP depends on the installed fonts on the MCP. Currently, ASCII, Turkish, Czech, Hungarian, Polish, Scandinavian, Chinese, Japanese and Korean characters are supported.

5.8.2 - CONFIGURATION OF THE ARM CHARACTERISTICS

M0000435.1

The arm characteristics are displayed in the Control Panel. The serial number of the arm is displayed with the list of installed software and hardware versions. This list also details the "Arm" version that shows:

- The arm type with its mechanical version, such as TX90-S1.
- The arm tuning version, such as R2.
- The arm mounting (floor / wall or ceiling).
- Possibly arm mechanical options.

The controller is delivered with the configuration of the arm delivered with it. If another arm is attached to the controller, the arm characteristics, defined in the /usr/configs/arm.cfx configuration file, should be updated: This can be done with the "Exp." export and "Imp." import buttons from the calibration menu.

If some arm characteristics must be updated, you need to edit the arm configuration file /usr/configs/arm.cfx that is exported with the "Exp." export button from the calibration menu.



- The configuration of some arm characteristics (such as software limits) may impact the safety of the robot installation. The default user profile prevents the change of these characteristics (see chapter 5.8.3).
- These configurations do not have the safety performance level required in the ISO 10218-1:2011 standard for safety control systems. They are provided for device protection and not for functional safety in the cell.

Arm mounting configuration

The arm mounting configuration is used by the controller for gravity compensation. It can be updated by changing the value of the "mount" parameter in the arm.cfx file:

<String name="mount" value="floor" />

The value can be set to "floor" (floor mounting), "ceiling" (ceiling mounting), or "wall" (any mounting). When the wall mounting is defined, the controller uses the gravity vector configured in the /usr/configs/cell.cfx file, and displayed in history utility at boot time.



- For a wall mouting, the coordinates of the gravity vector in World must be correct !
- Default value (gravity along the X direction of World) may not fit the reality !
- Ceiling and wall mounting are not possible for all models of arm. Refer to robot arm documentation.

Arm dynamic parameters setting

From SRC s7.6, it is possible to define the actual loads moved by the arm.

Thanks to the flexibility compensation, this improves the static accuracy for the arms with the “absolute calibration” option.

More generally, this also improves the behavior of the arm when the application requires very high dynamics, since the torque feed-forwards are more accurate when the actual load is declared.

Principles:

The loads are defined with a weight and inertia. For each joint of the arm, three sets of parameters exist:

- The "system" parameters. Defined in the system.zfx file, they describe the arm load itself. They cannot be read or modified by the user.
- The "cell" parameters. They define the static loads that are mounted on the arm. For instance a gripper or equipment mounted on the forearm is static loads. They are stored in the cell.cfx and initialized once during the controller startup. If no load is defined in the cell.cfx, the system uses a default value. This default value is cleared as soon as a payload is declared by the VAL 3 application. Note that with SRC s7.6, the behaviour is slightly different: the payload declared with the instruction setPayload() is added to the default payload.
- The "user" parameters represent the robot payload: Typically the parts that are picked and placed during the application. The payload can be changed dynamically from the VAL 3 application (refer to the VAL 3 Reference Manual). It is cleared at the system startup but not when stopping/unloading an application.

The controller dynamic model uses the combination of all the above parameters.

This feature is not supported by all the arms. This can be checked either by the VAL 3 application (VAL 3 Reference Manual) or by checking the error.log file where the robot actual payload is logged at system boot: if no payload information is found in the file, this feature is not supported by the arm.

Arm static load declaration

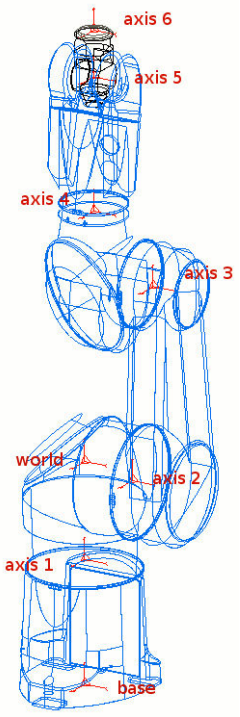


Figure 5.30

The static loads can be defined for each joint of the arm.

They are individually defined and referenced regarding the joints reference frames as:

- The mass (kg).
- The location of the gravity of the load center (x, y, z in millimeters).
- The inertia 3x3 matrix (in kg/m^2).

Please contact your local support for more information regarding the reference frames position on a particular arm.

These parameters must be set in the /usr/config/cell.cfx file as showed below.

The example below shows how to declare 2 static loads on a TX90 arm.

- A 2.5 kg tool effector located on the flange with a 100 mm offset in Z direction (no inertia).
- A 10 kg static load fixed on the forearm (joint 1) and located with 150 mm offset in X direction and 300 mm offset in Z direction (no inertia).

Warning: In the cell.cfx, the joint are identified from the index 0 (0, 1, 2, 3, 4, 5 for a 6 axis arm).

```
<CellDynamicParams>
<VectCart name="gravity" x="0" y="0" z="-9.81" />
<Payload>
<Float name="mass" value="2.5" />
<VectCart name="gravityCenter" x="0" y="0" z="100" />
<Inertia>
<Value line="0" column="0" >0</Value>
<Value line="0" column="1" >0</Value>
<Value line="0" column="2" >0</Value>
<Value line="1" column="0" >0</Value>
<Value line="1" column="1" >0</Value>
<Value line="1" column="2" >0</Value>
<Value line="2" column="0" >0</Value>
<Value line="2" column="1" >0</Value>
<Value line="2" column="2" >0</Value>
</Inertia>
</Payload>
<AxisInertia>
<Uint name="axis" value="1" />
<Float name="mass" value="10" />
<VectCart name="gravityCenter" x="150" y="0" z="300" />
<Inertia>
<Value line="0" column="0" >0</Value>
<Value line="0" column="1" >0</Value>
<Value line="0" column="2" >0</Value>
<Value line="1" column="0" >0</Value>
<Value line="1" column="1" >0</Value>
<Value line="1" column="2" >0</Value>
<Value line="2" column="0" >0</Value>
<Value line="2" column="1" >0</Value>
<Value line="2" column="2" >0</Value>
</Inertia>
</AxisInertia>
</CellDynamicParams>
```

5.8.3 - CONFIGURATION OF THE USER PROFILES

M0000379.1

The user profiles are configured using the Stäubli Robotics Suite tool. Each profile is defined by a configuration file that must be installed on the CS8C under /usr/configs/profiles. There is no limit to the number of profiles.

Profiles are selected via the control panel or using the Shift-User keyboard shortcut.

The setProfile() instruction can also be used to change the user profile in a VAL 3 program.

The file name determines the corresponding profile name. An user profile configuration is defined using the following key words:

Key word	Description
password	Profile password.
connectionPassword	Profile password for network connections (Ftp and remote maintenance).
writeAccess	Writing access to the VAL 3 applications and the CS8C configuration (using the MCP or via Ftp). If writing access is activated, reading access is also activated.
readAccess	Reading access to VAL 3 applications via the MCP or Ftp (opening, display without modification, exporting controller data, connection for the remote maintenance tool).
armWriteAccess	Writing access to the specific data for the controller and arm (software limits, adjustment offsets, user marks, safety chain, maximum Cartesian speed).
recovery	Access to the recovery from the calibration menu.
ioWriteAccess	Activation of the "On"/"Off" menus for the inputs/outputs in manual mode. Does not prevent use of keys **(1)**, **(2)** and **(3)**.
123KeysControl	Access to programming for keys **(1)**, **(2)** and **(3)**.
manualMode	Access to manual working mode (disabled when WMS is used).
localMode	Access to local working mode (disabled when WMS is used).
remoteMode	Access to remote working mode (disabled when WMS is used).
monitorSpeed	Access to the MCP monitor speed.
powerButton	Activation of the power button in remote mode (only power cut-off remains possible, manual connection to the power supply remains prohibited).
moveHoldKey	Activation of the Move/Hold button in remote mode.
stopKey	Activation of the "Stop" button for the VAL 3 application.
menuKey	Activation of the "Menu" browsing button in the MCP interface.
hardwareFaultAck	Activation of the ’Ack’ menu in the MCP Control Panel (see chapter 6.15)
Integrator	true: Hardware safety faults don't need a repair evidence but only an acknowledge operation to allow arm power. false (default): Hardware safety faults required the repair evidence to allow arm power (see chapter 6.15).

The controller is supplied with 2 profiles:

- "default" (with empty profile and network passwords) providing complete access except for arm configuration
- "maintenance" (with "spec_cal" as the login and network password) that provides complete access

Of course, the profiles can be adapted or deleted. If no profiles are defined, the default profile does not have any access restrictions. This means that the network connections must use the "default" log-in with an empty password.



- The configuration of some controller and arm characteristics may impact the safety of the robot installation. The default user profile prevents the change of these characteristics.
- The configuration of the user profiles must be carefully designed so that only qualified personnel may modify characteristics with potential safety impact (mainly writeAccess and armWriteAccess key words).
- These configurations do not have the safety performance level required in the ISO 10218-1:2011 standard for safety control systems. They are provided for device protection and not for functional safety in the cell.
- Using the integrator flag set to true implies to accept the use of the system in reduced safety condition. Thus it must be used only by trained people and limited to dedicated robot operations during integration phase.

  For this reason, the integrator flag is not available from the default flags in Stäubli Robotics Suite profile editor. Adding this flag is only possible by editing the file /usr/configs/profiles.cfx.

  If the integrator flag is included in a profile set as “starting profile”, the password is required once to make the integrator flag active after the start of the controller.

  Warning: To prevent the end user to use the robot in this configuration, the integrator must insure that no profile including the integrator flag is delivered with the robot.

5.8.4 - CONFIGURATION OF SYSTEM INPUTS/OUTPUTS

M0000255.1

System inputs/outputs are configured in the /usr/configs/iomap.cfx (s6:iomap.cf) file.

Each CS8C controller is supplied with an example /usr/configs/templates/iomap/iomap.cfx (s6:/usr/configs/iomapExample.cf) in which the configurations are commented out. To activate a configuration, it is necessary to:

- Copy /usr/configs/templates/iomap/iomap.cfx to /usr/configs/iomap.cfx.
- Complete the ’value’ parameter of the system inputs/outputs to be configured with the physical address of an input/output. The physical address of an input/output can be copied from the VAL 3 Studio application of Stäubli Robotics Suite (Tools/Physical IOs menu).
  - Rename the iomapExample.cf file as iomap.cf.
  - Remove the "//" comments in front of the key words to be configured and replace the description after the "=" by the name of an input/output. For example: enablePower = usrIn0.
- Restart the controller.

Any configuration errors in the iomap.cfx (s6: iomap.cf) file are listed in the events logger on start-up.

Configuration of system inputs

Certain controller functions require a user signal for which the default wiring can be reprogrammed.

Key word	Description	Type	Default wiring
estopAcknowledge	Acknowledgement of the emergency stop in manual mode.	Digital input	Internal signal concerning detection of the MCP on its holder
	The emergency stop is acknowledged if this signal is activated when the arm is powered on. **DANGER** Current standards require that power must be switched on from outside the cell after an emergency stop. This digital input must thus be linked to an item of equipment outside the cell.		
enablePower	Enable or disable arm power in remote mode with signal rising edge (see chapter 6.6.3).	Digital input	No wiring
hardwareFaultAcknowledge	Acknowledgement of a hardware fault (see chapter 6.15).	Digital input	No wiring

When the MCP is replaced by its shorting plug, it is possible to simulate pressing certain keys using inputs:

Key word	Description	Type
remoteMonitorSpeed	Monitor speed selection [0, 100]	Analog input
remoteMoveHold	Replacement of the Move/Hold button	Digital input

Access to the system statuses and signals

The status signals for the safety system are set to "On" when an emergency stop has been activated.

The wiring of the safety system makes it impossible to know the state of each signal when several EStop signals are activated. The software then returns the last known state.

Key word	Description	Type
limitSwitch	Signal that a limit switch has been reached on a joint	Digital output
driveFault	Variable speed controller fault signal	
initSwitch	Software command signal opening the safety system	
watchdog	Fault signal from the watchdog on the RSI board	
fuse24V	Status of the 24V supply at fuse F2	
estopMCP	Emergency stop signal from the MCP	
estopWMS	Emergency stop signal from the WMS	
estopUser1-2	Emergency stop signal from UESA on the safety system	
estopUser3-4	Emergency stop signal from UESB on the safety system	
userEnable	Emergency stop signal from USER EN on the safety system  The USER EN signal is not valid in local or remote mode. Its state is then not updated.	
door	Emergency stop signal from DOOR on the safety system  The DOOR signal is not valid in manual mode. Its state is then not updated.	
brakeSelect	Signal from the joint selector at the base of the arm	
brakeRelease	Signal requesting brake release at the base of the arm	
deadman	Status of the MCP enabling device	
park	Signal concerning detection of the MCP on its holder	
power	Arm power status	
dummyPlug	Signal that the MCP is replaced with a shorting plug	
hardwareFault	Signal that a hardware fault impacting safety was detected	
temperature	Temperature (°C) measured on the RSI board	Analog output
popup	Messages displayed on the MCP	Serial port

Configuration for the "remoteMCP" option

This option can be used to make a full simulation of the MCP when it is replaced by its shorting plug. It thus enables an MCP OEM to be connected to the CS8C controller.

This option can also be used to replace only the WMS box with signals coming from a fieldbus or any other IO device. The settings for the "remoteWMS" option are explained below.



The "remoteMCP" option must be used with great care to meet the requirements of the safety standards in force. In particular:

- A mutual supervision software mechanism must be set up between the MCP OEM and the CS8C. It stops the robot as soon as the MCP OEM switches to fault status and checks that the status of the robot on the MCP OEM is correct.
- The performance level of the safety MCP enabling device function (chapter 3.5.5) can not be defined by Stäubli any more. This function has to be implemented by the OEM, for instance by connecting the outputs of the OEM enabling device on the cell emergency chain.

Key word	Description	Type
remoteTestMode	Signal for activation of the test mode	Digital input (remote WMS settings)
remoteManualMode	Signal for activation of the manual mode	
remoteLocalMode	Signal for activation of the local mode	
remoteRemote-Mode	Signal for activation of the remote mode	
remoteEnablePower	Signal for manual connection to the power supply	Digital input
remoteDeadman1	Signal from the enabling device (1/2)	
remoteDeadman2	Signal from the enabling device (2/2)	
remotePark1	Signal showing presence on the holder (1/2)	
remotePark2	Signal showing presence on the holder (2/2)	
remoteJogJoint-Mode	Signal showing activation of the "Joint" mode for manual movement	
remoteJogFrame-Mode	Signal showing activation of the "Frame" mode for manual movement	
remoteJogTool-Mode	Signal showing activation of the "Tool" mode for manual movement	
remoteJogUser-Mode	Signal showing activation of the "User" mode for manual movement	
remoteJogMove1	Speed of manual movement along the 1 or X axis	Analog input [-100, +100]
remoteJogMove2	Speed of manual movement along the 2 or Y axis	
remoteJogMove3	Speed of manual movement along the 3 or Z axis	
remoteJogMove4	Speed of manual movement along the 4 axis or in RX rotation	
remoteJogMove5	Speed of manual movement along the 5 axis or in RY rotation	
remoteJogMove6	Speed of manual movement along the 6 axis or in RZ rotation	
remoteSpeedLimit	Signal for incrementation of the maximum speed authorized in test mode	Digital input

5.9 - SYSTEM EVENTS

M0000477.1

All events detected by the system are stored in a logger file. The logger mechanism uses 2 text files: The /log/errors.log hosts actual history of the system events. When this file is full, it is saved in the backup file /log/errors.old and cleared to get the next events. Only one backup exists.

The system events can also be read from the SP1 using the logger option from the main menu. From this page, the 'Exp' function keys copy the current log file in the specified location. Note that additionally to the current logger information, this command adds a full dump of the system state.

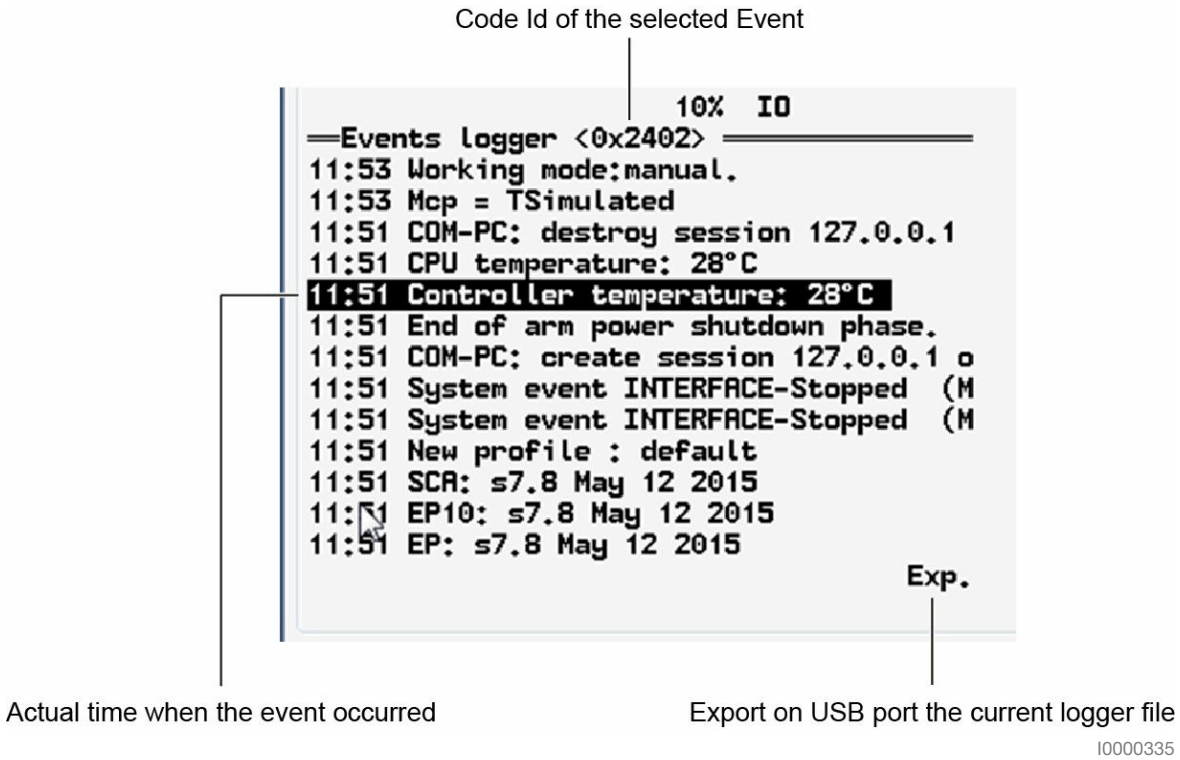


Figure 5.31

Each system event comes with the actual time when it occurred and since SRC s7.6, an unique identifier that appears in hexadecimal format (0x000) in the logger file, the SP1 Events logger and in the teach pendant popups.

5.9.1 - FORMAT OF THE LOGGER FILE

M0000478.1

In the logger file, the system events are stored with the following format:

Before SRC 7.8

[date and time]: <event ID> Event text

For instance:

[JUL 06 2005 23:28:44]: <0x2401> Configuration Version: c1.030

From SRC 7.8

[date and time]: <event ID> Event text {Internal Code}

For instance:

[JUL 06 2005 23:28:44]: <0x2401> Configuration Version: c1.030 {6ddfbaea}

5.9.2 - SIZE OF THE LOGGER FILE

M0000479.1

From SRC 7.8, the size of the logger file is increased by setting the new parameter 'loggerSize' in controller.cfx (unit is bytes).

- The default value of loggerSize is 1 MBytes (10485576 bytes) the same than in the previous SRC releases.
- loggerSize cannot be set to a value less than the default value.
- The maximum value allowed for loggerSize is 9 MBytes for MM1 CPU and 2 MBytes for others CPU models.

See below an example to set the logger size at the maximum value on a MM1 CPU:

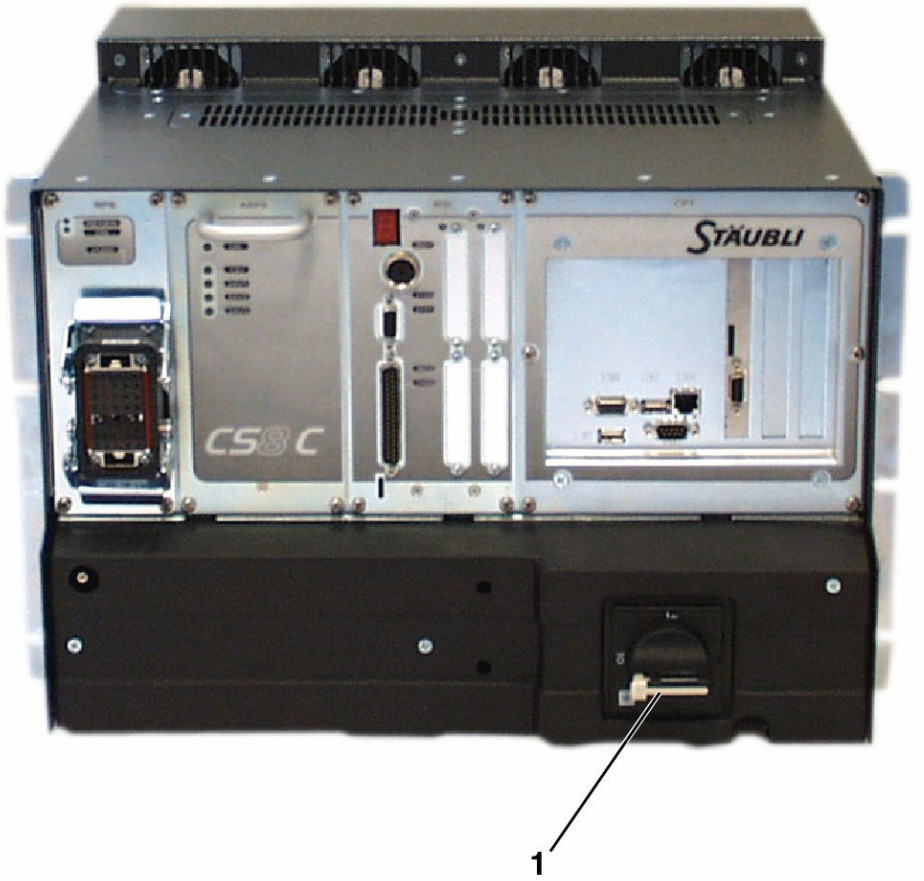
<Uint name="loggerSize" value="9437184" />

Remark: Note that because loggerSize is a new parameter in the controller.cfx file, this parameter will be lost in case of SRC downgrade to a release that doesn't manage this parameter.

6 - OPERATION

6.1 - POWERING UP THE CONTROLLER

M0000244.1



I0000345

Figure 6.1

To start the controller, set the switch **(1)** to position "1".

DANGER
Before carrying out any work on the components inside the controller, it is essential to set the master switch **(1)** to position "0".

See the Safety chapter 3.3.1 for the procedure to isolate the system from the electrical power supply.

DANGER
When the main switch **(1)** is on "0", the cables and the filter located upstream from the switch remain alive.

6.1.1 - SP1 SCREEN AT BOOT

M0000249.1

When the controller is switch on, the MCP screen and all the LEDs flash on and off.

Next, a "Stäubli Robotics Controls" message is displayed on the screen, and then the main menu is shown after about 2 minutes.



I0000306

Figure 6.2

6.2 - PRESENTATION OF THE MCP

6.2.1 - GENERAL PRESENTATION

M0000283.1



I0000262

Figure 6.3

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Working mode (1)

The working mode selected on the WMS operator interface is displayed around the button located opposite the working mode icons. The selected mode is also displayed on the WMS front panel.

Button turning on arm power (2)

This illuminated button enables you to enable or disable arm power. When the green indicator light comes on steady, this shows that arm power is on. In manual mode, the enabling device **(11)** must be kept pressed to enable arm power.

Emergency stop (3)

The emergency stop must only be used in the event of absolute necessity for an unforeseen stop in your application.

Movement keys (4)

These keys are active in manual mode and they enable you to generate arm movements per axis or using Cartesian coordinates, depending on the movement mode selected (see chapters 6.3 and 6.6.1).

LEDs and Luminous indicator keys

DANGER

- When the MCP is powered on all the LED **(L)** flash to enable you to check that they are working correctly.
- If a LED is found to be faulty, the MCP must be replaced for safety reasons.



Figure 6.4

Movement mode selection keys (5)

Joint | Frame | Tool | Point



When the arm power is on and in manual mode, each of these 4 keys enable you to select the desired movement mode (Joint, Frame, Tool or Point). The indicator light associated with the key shows the current mode.

Speed adjustment key (6)



I0000313

This key enables you to vary the speed within the limits imposed by the movement mode. It can be deactivated, depending on the current user profile (see chapter 5.8.3).

The speed is shown in the MCP status display bar.

The speeds vary in accordance with predefined values (with a factor of about 2 each time the key is pressed). The current speed is incremented or decremented by 1%, by pressing Shift at the same time as the speed key.

Function keys (7)

These are used to select the menus displayed above them.

Alphanumerical keys (8)

These keys are used to enter the data for your application.

Interface and navigation keys (9)



I0000263

The functions of these keys are described in chapters 6.2.3 and 6.2.4.

Application control keys (10)



I0000353

These keys are used to start or stop an application and to enable arm movements. The functions of these keys are set out in chapter 6.2.2.

Enabling device (11)

This button has three positions, and it states are:

- Open when the button is not pressed.
- Closed in the intermediate position.
- Open in the fully depressed position, which corresponds to tenseness in the user. These contacts stay open until the button is released.

This button can be used to authorize connecting arm power in manual mode (see chapters 6.3, 6.6.1), but only when it is in intermediate position. The 2 other positions prevent the arm from being powered up or cut off the power supply if the arm is under power in manual mode. In automatic mode, the position of the button is not taken into account.

The pendant is designed to enable the button to be pressed by right-handed operators (holding the MCP one way up) or by left-handed operators (holding the MCP the other way).

Digital output activation keys (12)



I0000200

In manual modes, these keys change the status of the digital outputs that are associated with them (see chapter 6.2.4).

Jog keys (13)



I0000289

These keys are activated in manual mode and they enable you to generate arm movements, per axis or using Cartesian coordinates, depending on the movement mode selected (Joint, Frame, Tool), with one hand (see chapter 6.7).

6.2.2 - CONTROL KEYS

M0000420.1



Stop key
This key stops the current application. Depending on the user profile, it may be inactive (see chapter 5.8.3).



Run key
This key enables you to start an application.



Move / Hold key

- In manual mode, the arm movements are allowed when the Move / Hold key is pressed. The arm stops on the programmed trajectory as soon as the key is released.
- In local and remote modes, the movements can be stopped and the robot set to pause mode, by pressing the Move / Hold key. Press the key again to reactivate the movements.
- In remote mode, the Move / Hold key may be inactive depending on the user profile (see chapter 5.8.3).

In manual and local modes, the robot is always in pause phase when the arm power is switched on. In remote mode, arm movements are authorized as soon as it is powered.

DANGER
When the green light is on and an application has been started, the robot arm can carry out movements at any time.



Shift + User keys
When these two keys are pressed together, the user profile change page is displayed.

6.2.3 - INFORMATION AND HELP KEYS

M0000236.1

Online help



By pressing the Help key, you can access online help at any time. In the "online help" mode, the function keys are desactivated. On the other hand, by pressing the function keys, you can call up a display window explaining the associated function.



To exit the "online help" mode and reactivate the function keys, press the Help key or the Esc key again.





Menu key
This key enables you to return to the main menu. Depending on the user profile, it may be inactive (see chapter 5.8.3).



User key
By pressing this key, you can call up the VAL 3 user display page (e.g. if the entry icon ? appears).

(s7): 

User + Pg up keys
By pressing both these keys at the same time, you can switch to the previous VAL 3 user display page.

(s7): 

User + Pg dn keys
By pressing both these keys at the same time, you can switch to the next VAL 3 user display page.

(s7): 

User + Shift + Pg up keys
By pressing these keys at the same time, you can switch to the first VAL 3 user display page.

(s7): 

User + Shift + Pg dn keys
By pressing these keys at the same time, you can switch to the last VAL 3 user display page.

Pick list

The pick list enables you to access the element you are looking for directly in all the lists shown on the display unit.

All you have to do is use the keyboard to enter the first letter or letters of the name of the element you are looking for. The selector moves to the first element whose initial characters correspond to the search criteria.



This key enables you to move on to the following element corresponding to the search criteria.

6.2.4 - NAVIGATION KEYS

M0000237.1



Arrow keys
Besides the classic browsing functions, these keys have some other functions that are specific to the CS8C controller.



"End" key
Expands an element when it has been contracted (preceded by the "+" sign).



Home key
Contracts an element that has been expanded (preceded by the "-" sign).



Shift lock key
Provides access to the second function of each key (except for $ and \).



Shift + End keys
By pressing both these keys at the same time, you can move to the end of the list.



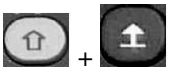
Shift + Home keys
By pressing both these keys at the same time, you can select the first item on the list.



Shift + Pg up keys
By pressing both these keys at the same time, you can move up one page in the display.



Shift + Pg dn keys
By pressing both these keys at the same time, you can move down one page in the display.



Shift + Shift Lock keys
By pressing both these keys at the same time, you can access capital letters and the characters $ and \.



The full stop "." and the comma "," can be accessed in normal mode and in capital mode.



Esc key
Cancels the entry and restores the initial value in the box, or exits the current page.



Return key
Starts the action associated with the element selected.
Enables you to modify the box selected (see chapter 6.2.5).
Validates the box being modified.



Tab key
Enables you to switch quickly from one box to another.



Backspace key
This key has the classic function of deleting the character to the left of the cursor.



Keys activating the digital outputs "1", "2", or "3"
In manual modes, these keys change the status of the digital outputs that are associated with them.

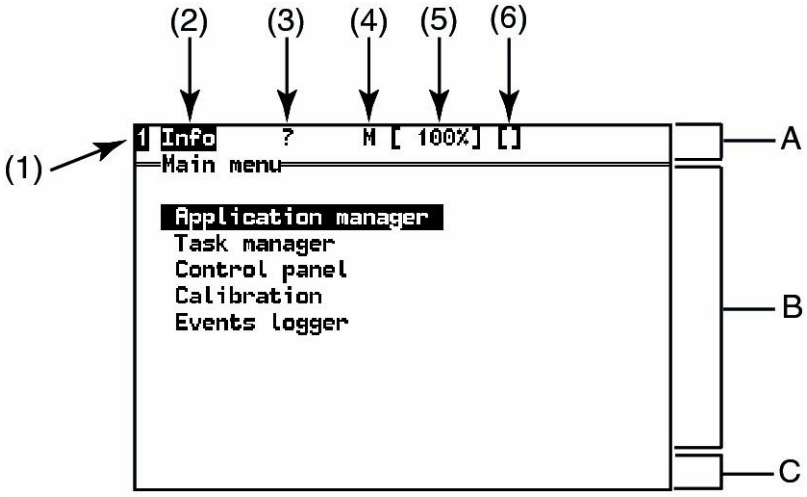
The keys can be associated with the digital output using the control panel on the Input/Output display.

To allocate a key for a digital output, select the output in the list of Inputs/Outputs on the control panel and then press the "Shift" key and the "1", "2", or "3" key at the same time. This operation may be inactive, depending on the user profile (see chapter 5.8.3).

6.2.5 - MCP DISPLAY

M0000256.1

The display is made up of three areas (A, B, C):



I0000230

Figure 6.5

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Status bar

The status bar **(A)** gives the following information whatever the current navigation status:

- System connection and activity indicator **(1)**. (s7) A blinking number indicates the number of network connections from PC running the Stäubli Robotics Suite. These remote users may interact with the controller at any time. A spinning line indicates that the controller is busy. The interface will not respond until the controller has completed its task.
- Indicator showing the presence of new information messages **(2)**. Its presence shows that one or more new information messages have been stored in the events logger. This indicator always flashes and it remains active until the user has consulted the information.
- Entry indicator **(3)**. It flashes when a VAL 3 application is awaiting an operator entry in the application page. It stays active as long as the application under consideration is active and until the entry is made.
- Arm movement status indicator **(4)**. 'M' indicates arm movement, 'S' indicates that a current motion has been stopped by the "Move / Hold" press button or by the "stopMove" VAL 3 instruction, '@' indicates, for manual movements, that the arm has reached its target position. The indicator is blank when there is no pending move.
- Arm movement speed indicator **(5)**. It is applicable to all the movements (manual and programmed). The brackets [ ] are displayed when the operator's profile does not allow changing the monitor speed.
- (s7) Input Output lock indicator **(6)**. The blinking square brackets [ ] indicates that some input or output has been locked: the VAL 3 application does no have access to these hardware inputs or outputs any more. They are now under the control of the operator, from the Control panel.

The work page

- The work page **(B)** is the part of the display screen located between the status bar and the menu sector. This page is used to exchange all the information concerning the current application (display, information windows, entries). The work page always has a title located on the line just below the status bar (see figure 6.5).

Menus

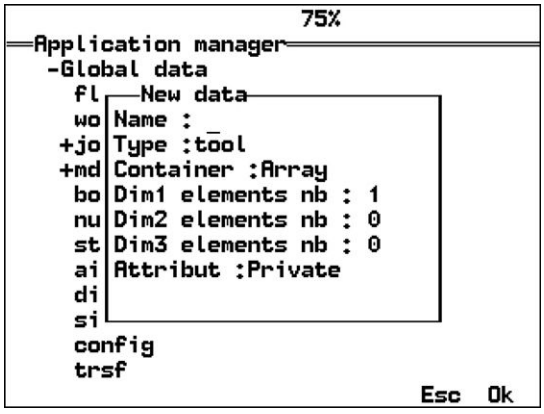
The menus **(C)** enable you to carry out a specific action for the element selected or the navigation page (see figure 6.5). To trigger the action, press the key located below the corresponding label.



For ergonomic reasons and for certain interface elements, a default menu is defined (action associated with the menu most often used). This action can be triggered by pressing the Return or End keys, or with the corresponding menu key.

Entry boxes

Entry boxes are areas on the display via which the user communicates information to the system when the system requires it.



I0000315

Figure 6.6



Press the Return key; this makes the cursor visible.

Modify the box (enter the information)

Validate the entry by pressing Return or cancel the modifications by pressing Esc.



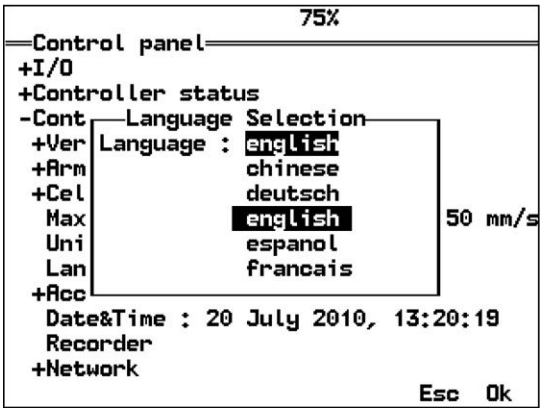
Press the OK key on the menu to validate the display as shown, as a whole.



The backspace key enables you to delete the character to the left of the cursor.

Drop down lists

When the information required by the system consists of a selection from a predefined list, the interface proposes a selection of all the possible values in the list.



I0000394

Figure 6.7



Press the Return key to display the list.





Use the Pg up / Pg dn keys or the lexical search to move around the list.



Validate the entry by pressing Return or cancel the modifications by pressing Esc.
Use the OK key in the pop-up menu to validate the entries displayed.

6.3 - ARM POWER-UP

M0000279.1



DANGER

Before powering up the arm, make sure that the cell is completely free from obstructions and that there is no one within the work area of the robot. When power is switched on, the robot arm is liable to follow unforeseen trajectories.



DANGER

Each time the arm is powered on, keep one hand close to the "Emergency stop" button in order to be able to press it as quickly as possible in the event of a problem.



I0000363

Figure 6.8

Under normal operating conditions, the operating method is as follows:

- Select the Manual Mode from the WMS front panel. The selected mode is displayed on WMS front panel and on the MCP **(1)** (see chapter 6.2.1).
- Press button **(3)** to switch the arm power on. This action is taken into account if the enabling device **(2)** has been put into its middle position in the last 15 seconds.

  If it is not possible to switch the arm power on, because the enabling device was pressed more than 15 seconds previously, it is necessary to release it and then press it again.

DANGER
Wherever possible, the manual mode of operation shall be performed with all people outside of the safeguarded space. When it is necessary to operate in manual mode inside the cell, the operator must keep in mind the inherent residual risks. Typical residual risks inside the cell are listed in chapter 3.5.2.

The button indicator light flashes for a few seconds and then comes on steadily; this means that the arm power has now been switched on and it is ready to carry out movements.

Press the button **(3)** again to switch the arm power off and apply the brakes. It is then necessary to repeat the procedure for switching the arm power on to make movements possible.

In manual mode, the power is also cut off if the enabling device is released, or if WMS key position is changed (working mode changed).

6.4 - EMERGENCY STOP

M0000220.1

The emergency stop is not the normal method for stopping arm movements; it must only be used in a case of absolute necessity for a stop not provided for in your application.

When the emergency stop button is pressed, the arm is stopped, the brakes are engaged after the stop and then the power is removed from the arm. The other equipment in the cell is also cut off from the power supply (depending on the cell wiring).

Restarting

DANGER
When the robot is restarted, all persons are prohibited from remaining in the isolation area in which the arm moves.

Once those concerned have made certain that safety conditions have been restored, the arm power procedure can be carried out using the MCP.

- Release the emergency stop button by turning it clockwise 1/4 turn.
- In manual mode, the MCP must be placed on its holder, fixed permanently outside the cell (a contact inside the MCP enables this operation to be checked).
- Restore power to the arm in accordance with the standard procedure, using the arm power button on the MCP (and the enabling device in manual mode).

DANGER
If the MCP is not connected to the controller, it must not be left near the cell, because its emergency stop button no longer works.

6.5 - CALIBRATION, ADJUSTMENT, RECOVERY

M0000428.1

Each time an adjustment or recovery procedure is done, the calibration of the arm has to be checked carefully to verify that the robot is able to move within its expected angular range and not outside that range. This verification has to be done at slow speed.

6.5.1 - DEFINITION

M0000265.1

Stäubli arms are calibrated in the factory, to determine the specific "zero" arm position with maximum precision. Calibration quality is essential for arm accuracy, i.e. its ability to respect the required Cartesian positions.

If drive elements (motor, encoder) have been replaced, or in the event of mechanical slippage due to a shock, the specific "zero" arm position can be displaced on one or more joints: it is then necessary to adjust the joints to restore the arm's original precision.

If one or more axes have been displaced, there are simple procedures for resetting them, using pre-established reference positions. If more than two joints have been displaced, or if no reference positions are available, it is not possible to adjust the arm correctly and it is necessary to carry out a full readjustment procedure.

It is important to provide for an adjustment procedure in the cell and define the associated reference positions beforehand.

The arm position at any given moment is shown by:

- The position measured for each motor (encoder),
- The "zero" calibration offsets.

  These offsets are stored in the respective encoders of each motor. A backup is saved in the arm.cfx file of the controller, a copy of which is supplied on the CdRom delivered with each arm.

During start-up, if a zero offset from a motor encoder is different from the one in the arm.cfx file (f.i. replacement of the arm linked to the controller), it is necessary to determine which offset is correct. This is done in the recovery operation.

The motor encoder also stores a phase offset for the encoder-motor combination. In the same way as for the zero offsets, the motor phase offsets are saved in the arm.cfx file of the controller. During start-up, if a motor phase offset on a motor encoder is different from the one in the arm.cfx file, it is necessary to state which offset is correct, using the recovery operation.

6.5.2 - RECOVERY PROCEDURE

M0000266.1

The recovery application is accessed via the main menu on the MCP.

This is used to update arm or controller data if an inconsistency is detected between the data on a motor encoder and the data in arm.cfx file of the controller.

The procedure consists of determining which set of data is correct, using a series of questions:

- If you have simply replaced the arm linked to the controller, the arm.cfx file of the controller corresponds to the former arm. It is possible, but not recommended, to use the recovery menu to update it:



The arm.cfx file contains other specific data for the arm (modified software limits, factory marks, user marks) that are not updated during this operation. When replacing an arm, it is preferable to recover the arm.cfx file for the new arm and install it on the controller ("Imp" Import menu for the calibration application).

- If you have replaced a motor or an encoder, the maintenance procedure you followed should have updated the offsets on the motor encoder and in the arm.cfx file. The recovery menu thus invites you to repeat the procedure to deal with the problem.
- If you have not changed anything on the arm or the controller, the problem may be due to a faulty motor encoder (data loss) or a corrupt arm.cfx file on the controller. Check the contents of the arm.cfx file ("Exp" Export menu in the recovery application) and compare it with the original file. If it is corrupted, restore it from a backup, or using the recovery menu to simulate an arm replacement. If the file is correct, the fault may be the motor encoder, which can be reprogrammed using the recovery menu.



The motor phase offsets for the encoders are critical, and if incorrect they can make a motor uncontrollable. Never update an encoder if you are not sure of the data.

6.5.3 - ADJUSTMENT PROCEDURE

M0000382.1

Before adjusting one or more joints on the robot, it is necessary to set the joints to a reference position beforehand, with the highest possible level of precision. The exact coordinates of the reference position must be determined in advance.

Sequence of operations:

- Select the reference position (press Return and then use the arrows).
- If the reference position has not been entered beforehand ("Here" menu), it is possible to enter its coordinates manually ("Edit" menu).
- Use the arrows to select the desired joint.
- Press 'Adj.' to carry out the adjustment process.
- Check that the arm has been reset correctly by moving to a known point at low speed.

Once the axis has been reset, you will be asked to save the new adjustment offsets for the arm on an external device.

You can reset all axes at the same time by using the arrow key to select the reference position instead of a joint.

6.6 - WORKING MODES

M0000418.1

Selection of Working Mode is made from WMS front panel. The 3-position keyswitch allows to choose one of the 3 modes (local, remote or manual). The selected mode is indicated on a light on the WMS front panel and on the MCP.

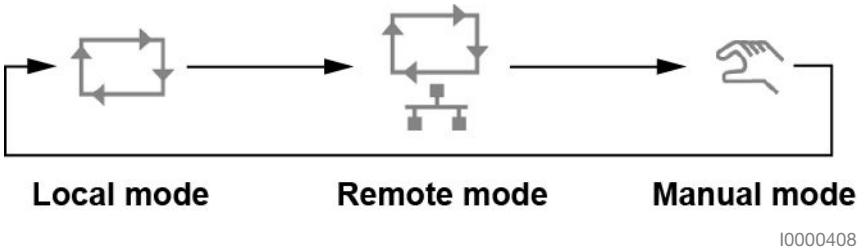


Figure 6.9

Monitor speed management changes from SRC 7.8.

In manual mode, the monitor speed is frequently 100%, and the robot is limited to 250 mm/s.

When switching to Local mode, the high monitor speed results in very fast move, which may surprise the operator: It is usually safer to start slowly, and increase the speed manually up to the desired value. This is even more important during maintenance when safety protective devices are disabled.

For this reason from SRC 7.8 the system automatically reduce to 10 % the value of the monitor speed when the working mode is switched from Manual to Local or Remote.

Please note that this automatic reduction of the monitor speed is not done when the teach pendant is not present and replaced by the dummy plug.

For some cells a reduction of monitor speed is however not desirable: the monitor speed reduction value has to be configurable. This can be done setting the configuration data maxStartMonitorSpeed in the cell.cfx (the default value is 10).

Note that if maxStartMonitorSpeed is set to 100, the controller has the same behavior than with previous SRC releases.

An upgrade of SRC to 7.8 will initialize maxStartMonitorSpeed to 10%.

6.6.1 - MANUAL MODE

M0000280.1

The manual mode is required in the following cases:

- Manual arm movements.

  The operator controls the arm movements using the MCP.
- Test/fine tuning of an application.

  In this case, it is a program that determines the arm movements.

The manual mode enables the robot to be moved at low speed (250 mm/s maximum) in accordance with the applicable standards listed in chapter 3.1.

- Wherever possible, the manual mode of operation shall be performed with all people outside of the safeguarded space.
- When it is necessary to operate in manual mode inside the cell, the operator must keep in mind the inherent residual risks. Typical residual risks inside the cell are listed in chapter 3.5.2.

Jog interface is made using the movement keys (see chapter 6.7).

Program movements are only carried out if the Move / Hold key has been pressed. The movements stop as soon as the key is released (see chapter 6.2.2).

Power is only maintained on the arm if the enabling device is kept pressed down to its intermediate position (see chapter 6.2.1).

When the manual mode is selected, the other modes are disabled and the movements cannot be initiated via outside equipment.

6.6.2 - LOCAL MODE

M0000281.1

The local mode enables the robot to be moved without any human intervention at the maximum speed defined for the application. The movements are the result of a scenario written in a program.

DANGER
When the robot is in local mode, all persons are prohibited from remaining inside the isolation area in which the arm moves.

Monitor speed management changes from SRC 7.8.

In manual mode, the monitor speed is frequently 100%, and the robot is limited to 250 mm/s.

When switching to Local mode, the high monitor speed results in very fast move, which may surprise the operator: It is usually safer to start slowly, and increase the speed manually up to the desired value. This is even more important during maintenance when safety protective devices are disabled.

For this reason from SRC 7.8 the system automatically reduce to 10 % the value of the monitor speed when the working mode is switched from Manual to Local or Remote.

Please note that this automatic reduction of the monitor speed is not done when the teach pendant is not present and replaced by the dummy plug.

For some cells a reduction of monitor speed is however not desirable: the monitor speed reduction value has to be configurable. This can be done setting the configuration data maxStartMonitorSpeed in the cell.cfx (the default value is 10).

Note that if maxStartMonitorSpeed is set to 100, the controller has the same behavior than with previous SRC releases.

An upgrade of SRC to 7.8 will initialize maxStartMonitorSpeed to 10%.

The robot is only operational when the following conditions are combined:

- The arm is powered up.
- A movement application has been loaded in the memory and is being executed.



The movement order is given via the MCP using the Move / Hold key (see chapter 6.2.2). The arm movements are controlled exclusively by the application.



Only the operator can only stop or start the movement and adjust the running speed using the "+/- " key.

6.6.3 - REMOTE MODE

M0000282.1

The remote mode operates in a similar way to the local mode.

The differences are as follows:

- The arm is powered on via an external system (controller, external MCP) on a digital input system signal, or using the enablePower instruction (see the VAL 3 language reference manual).
- The Move / Hold movement order can be generated automatically as soon as the arm power has been turned on (see the autoConnectMove instruction in the VAL 3 language reference manual).
- The Move / Hold button may be deactivated depending on the user profile.
- The power cut-off button may be deactivated depending on the user profile (see chapter 5.8.3).

The enablePower system signal must be configured beforehand to enable it to be linked to a digital input (see chapter 5.8.4).

- To connect the arm to the power supply, turn on the enablePower signal for 200 ms
- Turn on the enablePower signal again to cut off the power supply to the arm

DANGER

When the robot is in remote mode, all persons are prohibited from remaining inside the isolation area in which the arm moves.
When remote enablePower is used, the external PLC shall take into account the error messages delivered by the contoller to avoid periodic enablePower retries which could damage the controller.

DANGER

When driven in remote mode, the enable power sequence shall take into account the status of the controller and shall not be repeated if it is not taken into account the first time.

6.7 - JOG INTERFACE

6.7.1 - PRESENTATION

M0000222.1

The jog interface is a utility dedicated to the manual control of robot movements and the teaching of robot positions.

DANGER

- Wherever possible, the manual mode of operation shall be performed with all people outside of the safeguarded space.
- When it is necessary to operate in manual mode inside the cell, the operator must keep in mind the inherent residual risks. Typical residual risks inside the cell are listed in chapter 3.5.2.

When arm power is disabled, the jog interface can be accessed by pressing a movement mode selection key (Joint, Frame, Tool or Point), whatever the current robot working mode. When arm power is enabled, the jog interface can be accessed with the same keys, but only in manual mode.

The jog interface displays the context of the movement under manual control. This context is important to avoid unexpected movements. Therefore when the jog interface is exited, the movement mode is also reset automatically. It is however possible to exit the jog interface while keeping the movement mode active by pressing Shift-Esc.



I0000207

Figure 6.10

- To select manual mode, turn the 3 position keyswitch to the appropriate position. The selected mode is indicated both on WMS front panel and on MCP **(1)**.



Manual mode icon

- Carry out the procedures for switching on power in manual mode (see chapter 6.3).
- Select the movement mode (Joint, Frame, Tool, or Point) ; the corresponding indicator light **(4)** comes on.
- Press one of the movement keys (5 or 6) or, in Point mode, press the Move / Hold key.

In manual mode, the speed of movement is limited to 250 mm/s in accordance with the applicable standards listed in chapter 3.1.

When one of the movement modes is selected, the jog interface page is shown automatically on the MCP display (see figure 6.11).

To exit the page, press the Esc key. To return to the page, select a movement mode (Joint, Frame, Tool, Point).

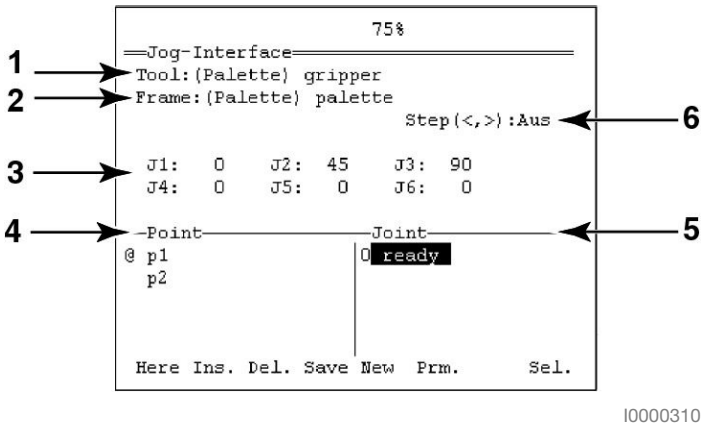


Figure 6.11

Description of the jog interface page

(see figure 6.11)

Whatever the movement mode selected, this page enables you to see:

- The current tool **(1)**.

  The Cartesian speed is limited at 250 mmps at the Tool Center Point (TCP) and at the robot flange. If the tool is not correctly defined, the Cartesian speed limit at TCP may not be correct.

- The current frame **(2)**.
- The current position depending on the movement mode selected **(3)**.
- In the Joint, Frame and Tool modes, a list of the Cartesian points for the current frame **(4)**.
- In the Point mode, a list of all the Cartesian points for the application **(4)**.
- A list of the points for the joints **(5)**.
- For joint positions and cartesian points, a position marker may preceed the name of the data:
  - '@' indicates that the robot is exactly at the position.
  - '~' indicates that the robot is near the position.
  - '0' indicates that the position is null.
  - '!' indicates that the position is not reachable with the current tool.
- The step mode **(6)**, and, if enabled, the current step size. When the step mode is active, each manual movement is a step of the specified size (in mm or degree). The step size can be easily modified by using the < and > keys. It is also possible to move of several steps in one movement by pressing quickly several times the jog key. By pressing 3 times the jog key, the robot performs a step of 3 times the initial step distance. The display of the step size is updated with each key strike, and is reset to its default value when the movement is completed.

Menus

(see figure 6.11)

- Sel. (Selection) menu

  This menu displays a window in which it is possible to select the current tool and the current frame in the application data list.

- Par. (parameters) menu

  This menu is used to edit the manual movement parameters (step mode and step size).

- Here menu

  This menu is used for teaching points. By pressing the key, you can modify the position of the point selected. A confirmation window is then used to validate the selection.

- New menu

  This is used to create a new data or Cartesian variable. A confirmation window enables you to give the variable a name and validate it.
- Save menu

  Used to save the application.
- Ins. & Del. Menu

  Enables you to insert a new element in a table, delete an element from the table, or delete a variable.

6.7.2 - MOVEMENT IN JOINT MODE (JOINT)

M0000219.1

After the arm has been powered up, press the Joint button **(1)** on the MCP. The corresponding indicator light comes on.

The keys **(3)** enable you to carry out movements in joint (Joint) mode for the various axes (1, 2, 3, etc.). These movements are carried out in the positive direction (set of keys with the "+" sign) or the negative direction (set of keys with the "-" sign). It is possible to move several axes at once. Only the indicator light for the last joint key pressed (+ or -) and the minijog indicator light then come on.



If one of the SEL keys on the minijog **(4)** is pressed, the joint number selected changes and the corresponding indicator light comes on in the set of keys **(3)**.



If one of the "+/-" keys on the minijog is pressed, the joint selected moves.



I0000297

Figure 6.12

6.7.3 - MOVEMENT IN CARTESIAN MODE (FRAME, TOOL)

M0000460.1

After the arm has been powered up, press the Frame button **(1)** on the MCP. The corresponding indicator light comes on.



Figure 6.13

By pressing the keys in the set of movement keys **(2)** or one of the Sel keys on the minijog **(5)**, it is possible to carry out movements along the three axis of the current coordinate system (Frame as default setting). These movements are carried out in the positive direction (set of keys with the "+" sign) or the negative direction (set of keys with the "-" sign).

The movements can be made in translation and rotation: (see figures 6.14, 6.15, 6.16)

- In Translation (X, Y, Z keys):

  A movement in the direction of the X axis is called X+, and a movement in the opposite direction to the X axis is called X- (the same is the case for the Y and Z axes).
- In Rotation (RX, RY and RZ keys):

  Rotation around the X axis in the direction of X+ is called RX+, and a rotation in the opposite direction from the X axis is called RX- (the same is the case for the Y and Z axes).

Specific case (TS arms):

RZ rotation is only possible if the Z axis of the current position coincides with the Z axis of the World mark. RX and RY rotations are without any effect.

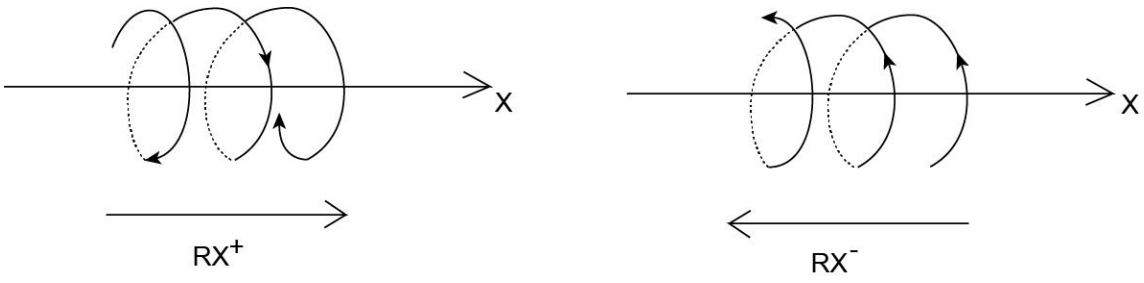


Figure 6.14

If the Tool key **(3)** has been pressed, the movements are made parallel to the axis of the current tool (Flange as default setting).

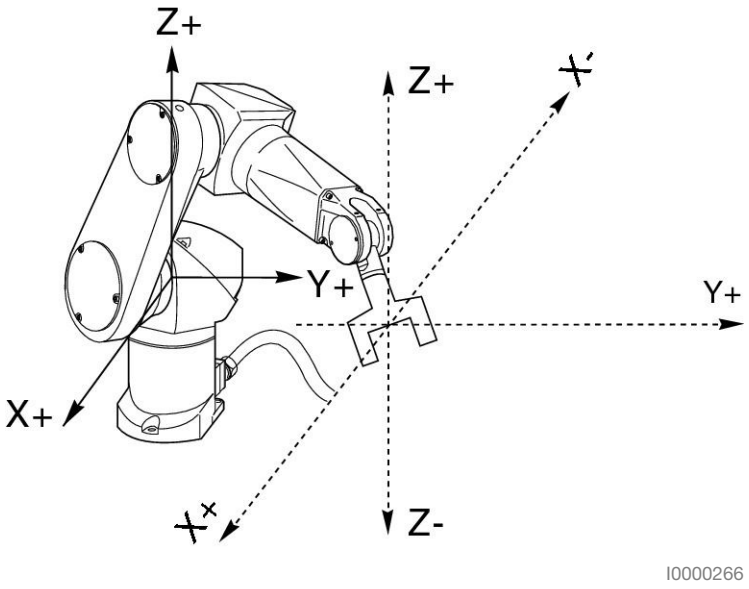


Figure 6.15

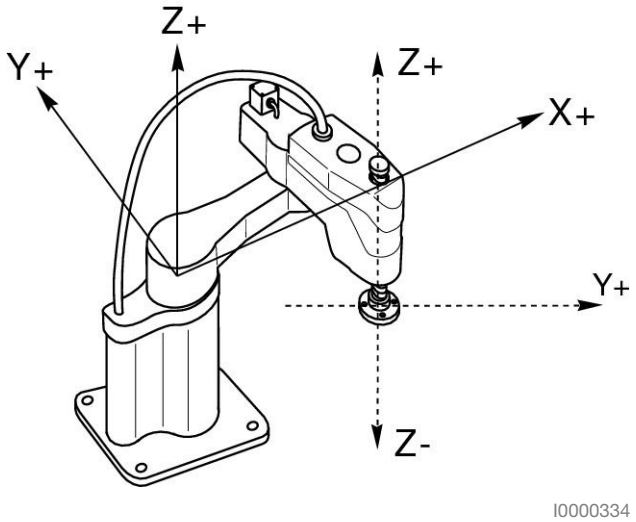
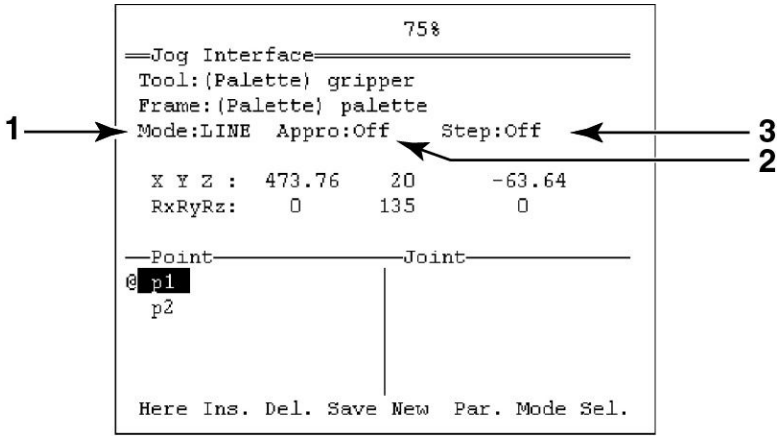


Figure 6.16

6.7.4 - MOVEMENTS IN POINT MODE

M0000269.1



I0000185

Figure 6.17

The POINT mode is used for movements to a point of the application. To display the points of an application, you will first be asked to select a tool in that application.

When the POINT movement mode is selected, the display shows:

- The active movement mode **(1)**:
  - Line mode, the movements towards the target are made in a straight line.
  - Joint mode, the movements are made from point to point.
  - Align mode, the Z axis of the tool is lined up with the closest axis to the current marker. The end of the tool rotates without translation.
- The approach parameters **(2)**. The approach can be inactive OFF or active ON.
- The current step mode **(3)**, and, if enabled, the current step size. When the step mode is active, each manual movement is a step of the specified size (in mm or degree). The step size can be easily modified by using the < and > keys.

Further menu keys are dealt with in Point mode:

- Mode key

  This key enables you to switch successively from Joint mode to Line mode, on to Align mode and then back to Joint mode.
- Par. (parameters) key

  This key calls up a window in which it is possible to configure the approach and the step parameters for the movement. The approach can be specified along X, Y, Z, RX, RY and RZ.

The minijog indicator light **(5)** and that of the last axis selected **(2)** remain off in this mode. Nothing happens when the keys of the minijog **(5)** or the movement keys **(2)** are pressed (see figure 6.13).

6.8 - STARTING AN APPLICATION

M0000224.1

Monitor speed management changes from SRC 7.8.

In manual mode, the monitor speed is frequently 100%, and the robot is limited to 250 mm/s.

When switching to Local mode, the high monitor speed results in very fast move, which may surprise the operator: It is usually safer to start slowly, and increase the speed manually up to the desired value. This is even more important during maintenance when safety protective devices are disabled.

For this reason from SRC 7.8 the system automatically reduce to 10 % the value of the monitor speed when the working mode is switched from Manual to Local or Remote.

Please note that this automatic reduction of the monitor speed is not done when the teach pendant is not present and replaced by the dummy plug.

For some cells a reduction of monitor speed is however not desirable: the monitor speed reduction value has to be configurable. This can be done setting the configuration data maxStartMonitorSpeed in the cell.cfx (the default value is 10).

Note that if maxStartMonitorSpeed is set to 100, the controller has the same behavior than with previous SRC releases.

An upgrade of SRC to 7.8 will initialize maxStartMonitorSpeed to 10%.

6.8.1 - STARTING UP IN LOCAL MODE

M0000225.1

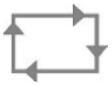
The local mode is the mode most commonly used in production.

DANGER
In local mode, the robot arm makes high speed movements. These movements can be dangerous. Always comply with the safety standards recommended for robot use and inform operators about the dangers faced.

DANGER
If an application has been configured in "Automatic start" mode, it starts as soon as the controller is powered up.

Once the application has been opened:

- Turn the 3-position keyswitch to the appropriate position. The selected mode is indicated both on WMS front panel and on MCP **(1)**.



Local mode icon

- Switch the arm power on by pressing the arm power button (see chapter 6.3).

  When the powering up process has been completed, the button comes on steadily.
- To start the application, press the Run key.



Run key

- Command the movement by pressing the Move / Hold key



Move / Hold key

6.8.2 - STARTING UP IN MANUAL MODE

M0000238.1

Once the application has been opened (see chapter 6.10.1):

- Turn the 3-position keyswitch to the appropriate position. The selected mode is indicated both on WMS front panel and on MCP **(1)**.



Manual mode icon

- Keep the enabling device in its intermediate position (see chapter 6.3).
- Switch the arm power on by pressing the arm power button.
- To start the application, press the Run key.



Run key

- Command the movement by pressing the Move / Hold key



Move / Hold key

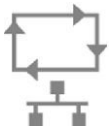
6.8.3 - STARTING UP IN REMOTE MODE

M0000226.1

In remote mode, the arm is powered up by an external system (external MCP, automatic controller) via a dedicated digital input (see chapter 6.6.3). The arm can also be powered up using the EnablePower instruction (see the Reference Manual for the VAL 3 language).

Once the application has been opened (see chapter 6.10.1):

- Turn the 3-position keyswitch to the appropriate position. The selected mode is indicated both on WMS front panel and on MCP **(1)**.



Remote mode icon

- To start the application, press the Run key.



Run key

6.9 - STOPPING MOVEMENTS

M0000227.1

When the robot is carrying out programmed movements, it is always possible to stop them, using the MCP. Depending on the stop mode selected by the user, the system can adopt several types of behaviour.

6.9.1 - STOPPING/STARTING MOVEMENTS USING THE MOVE / HOLD KEY

M0000248.1



Move / Hold key

- In manual mode, the arm movements are activated when the Move / Hold key is pressed. As soon as the key is released, the arm stops immediately on the programmed trajectory.
- In local and remote modes, the movements can be stopped and the robot set to pause mode, by pressing the Move / Hold key. Press the key again to reactivate the movements.
- In remote mode, the Move / Hold key may be inactive depending on the user profile (see chapter 5.8.3).

Movements on restarting

When programmed movements are stopped by pressing the Move / Hold key or during an arm power failure,the system memorizes a stopping point.

When movements are restarted, the arm goes back to the stopping point (if needed) using point to point movement, at a speed limited to 250 mm/s.

The restarting movement is commanded by pressing the Move / Hold key in local and manual modes. The restarting movement can be automatic in "Remote" mode.

The Move / Hold key does not stop the current application, it simply suspends the arm movements. The robot is then in pause mode.

Stopping using the powering up/ power switch-off button

When switching off the power (see chapter 6.3), the movements are first suspended as with the Move / Hold key, and then the system applies the brakes and cuts off the power supply to the arm. To restart movements, follow the particular procedure for the working mode selected (see chapter 6.3).

The arm power can be switched off via the MCP in all the working modes.

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6.9.2 - STOPPING/STARTING MOVEMENTS USING THE STOP KEY

M0000239.1



Stop key

To stop the current application, press the Stop key and validate using the OK key in the pop-up menu. This key may be inactive depending on the user profile (see chapter 5.8.3).

To restart, press the Run key; the controller restarts the application from the beginning.



Run key

Depending on the type of application, the arm may continue to move until it has finished the current cycle.

6.9.3 - STOPPING MOVEMENTS VIA THE EMERGENCY STOP BUTTON

M0000228.1

During an emergency stop, the standard imposes a fast cut-off of the arm power supply, which may lead to a higher deceleration than that obtained using the power supply cut-off button. This means that the emergency stop must not be used as a normal method for stopping the robot or switching off the power supply to the arm.

Once those concerned have made certain that safety conditions have been restored, the arm power procedure can be carried out using the MCP (see chapter 6.4).

6.10 - VAL 3 APPLICATION MANAGER

6.10.1 - OPERATIONS CONCERNING APPLICATIONS

M0000229.1

Changes to applications are best done on PC with the VAL 3 Studio tool of Stäubli Robotics Suite. They are also possible on the MCP as described below.

The application manager can be accessed via the main menu.

In the application manager, the New menu enables you to create a new application and to give its storage location (hard disk, USB key).

An application contains (see figure 6.18):

- The libraries **(1)**,

  The list of library identifiers defined in the application is given here.
- (s7): The user types **(2)**,

  The list of user types defined for the application is given here.
- The global variables **(3)**,

  The global datas for the application are classified according to type (see the Reference Manual for the VAL 3 language). The Tool type variables are to be found under the Flange variable. The Frame, Point type variables are to be found under the World variable.
- The programs **(4)**.

  - The Start program that is called up by the system when the application starts up.
  - The Stop program that is called up by the system when the application stops.

  These two programs do not have any parameters, and they cannot be deleted or renamed.

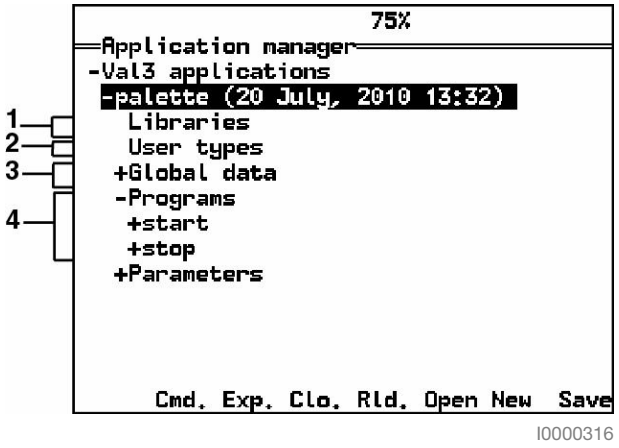


Figure 6.18

Opening an existing application

- Press the Open key in the pop-up menu.
- Select the application.
- Press the Ok key in the pop-up menu.

Closing an application

This action is only possible in the applications opening page.

- Select the application and then press the Clo. key in the pop-up menu.

  If there are modifications under way that have not been backed up, a confirmation window can be used to record or delete the modifications.

Deleting an application

This action is only possible via the opening page of the application.

- Press the Del. key in the pop-up menu.

  This operation is irreversible; it is preceded by a confirmation window.

Reloading an application

When the files on disk of an opened application have been modified through the network, it is possible to quickly update the application in memory by using the menu Rld. (Reload). This action is equivalent to the closing / re-opening of the application.

Opening mode for an application

Behaviour for applications when the controller is started up can be configured in the application selection window.

Three types of behaviour can be selected by pressing the Mode contextual menu key:

- Manual: No action on the application.
- Autoload: The application is opened automatically.
- Autostart: The application is opened and then started up automatically.



It is not possible to apply the "Autostart" mode to more than one application at a time.

Saving a VAL 3 application

The Save pop-up menu enables you to save the VAL 3 application in full in its original location. If the application uses libraries, the pop-up menu proposes to save the application alone (Yes/No) or the application and all its libraries recursively (Y2A/N2A:Yes to all/No to all).

All the information is saved as soon as the key has been pressed:

- The global datas and their current value.
- The application programs.
- The configuration data for the application.

During the saving process, the system is not available and the system activity indicator is displayed on the status bar.

Exporting a VAL 3 application

To save the application under another name or in another storage location, use the Exp. menu.

Online commands

The Cmd. menu provides access to an online command enabling you to display the variable values (using the "?") and execute a VAL 3 instruction line.



The use of an online command during execution of a program may modify its behaviour.

6.10.2 - MODIFYING AN APPLICATION

M0000230.1

Changes on applications are best done on PC with the VAL 3 Studio tool of Stäubli Robotics Suite. They are also possible on the MCP as described below.

Library, user type, data, program creation and edition

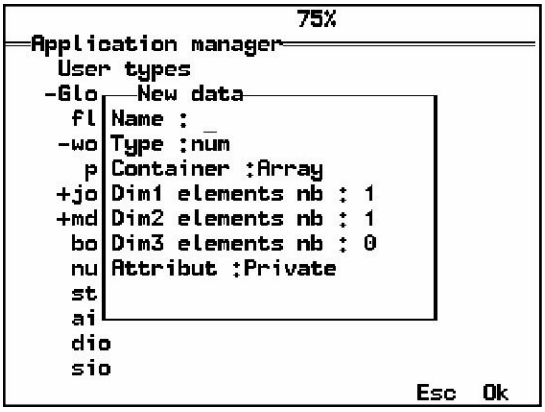
- To declare a new library, go to the library node of the application, press the Add key in the menu, then select the library to load and enter the corresponding identifier for the application.
- (s7): To declare a new user type, go to the user type node of the application, press the Add key in the menu, select the application to use as type definition, and enter the corresponding type name for the application.
- To create a new data, select the type of variable and press the New key in the menu.
- To add a frame or a point in a frame, go to this frame and press the New key in the menu.
- To add a tool, go to its reference tool and press the New key in the menu.

The interface window to create a new data is shown below.

- You must enter the name of the new variable in the first line.
- (s7): You can switch between an array and a collection container.
- (s7): When an array container is selected, you can decide for the size of each dimension. By default, a one dimension array is created with a single element. Enter a size greater than 0 on dimensions 2 or 3 to ask for a two dimensions or a three dimensions array.

The size of an array can be easily modified later, but not its number of dimensions.

- You can switch between public and private data on the last line.



I0000308

Figure 6.19

For further information on modifying programs, consult the Reference Manual for the VAL 3 language

It is not possible to delete a variable, a program, a point or a tool if they are used in a program. It is not possible to add a parameter or a local variable to a program if the application is active.

Program editor

The VAL 3 program editor can be accessed via the list of programs in the application manager ; it enables you to modify programs (to insert, delete, or modify an instruction).

Each instruction added to the editor is checked. If it is not valid, an error message is displayed and the instruction is refused. In the event of an error, see the reference manual for VAL 3 language to correct the instruction concerned.

The editor proposes a program tree, which means that the instructions set up (if, while, for) are displayed in the form of a node that can be expanded or contracted.

Example:

Collapsed	Expanded
-if nNb>12 +switch nNb else putln("Error: too many items") endIf	-if nNb>12 -switch nNb -case 5 break +case 7 +default endSwitch else putln("Error: too many items") endIf
Figure 6.20	**Figure 6.21**

The editor makes sure at all times that the program is coherent. This means that when a compound statement (if, while, for …) is deleted, the corresponding instruction is deleted.

Example:

<table>
<tr>
<td rowspan="2"> 

Figure 6.22</td>
<td>Removal of if</td>
<td> 

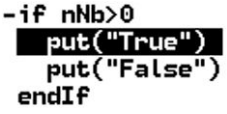
Figure 6.23</td>
</tr>
<tr>
<td>Removal of else</td>
<td> 

Figure 6.24</td>
</tr>
</table>

It is possible to mark instructions in order to globalize certain actions (copy, delete).

- If a composite statement is selected, all the instructions between the start and the end are also selected.
- Deselection of a composite statement deselects only that statement.

This means, for example, that to mark all the instructions contained between a "while" and an "endWhile", all that has to be done is to mark " While" twice.

Example:

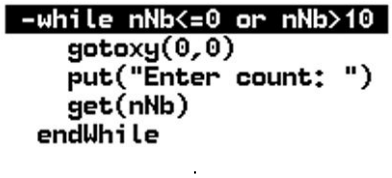
<table>
<tr>
<td></td>
<td>Selection of "While" or "endWhile".</td>
<td></td>
</tr>
<tr>
<td> 

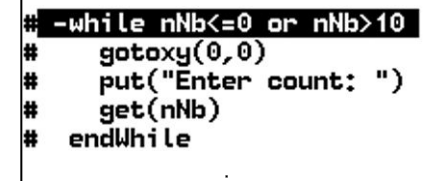
Figure 6.25</td>
<td>All the instructions are selected and the selection goes after "endWhile"</td>
<td> 

Figure 6.26</td>
</tr>
</table>

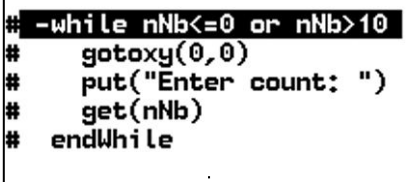
<table>
<tr>
<td></td>
<td>Deselection
of "While" or "endWhile".</td>
<td></td>
</tr>
<tr>
<td> 

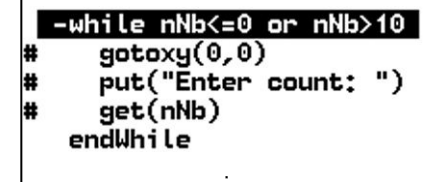
Figure 6.27</td>
<td>Automatic deselection
of "While" and "endWhile"</td>
<td> 

Figure 6.28</td>
</tr>
</table>

There is a clipboard enabling you to copy instructions. Make sure that the instructions are valid in the program in which they are pasted. Be careful with the local variables!

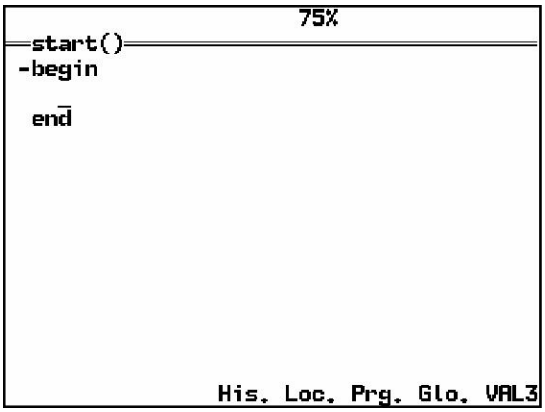
When adding or modifying an instruction, a list of menus makes it easier to enter (see figure 6.29).

- His. enables you to select an instruction in the list of the last 20 instructions entered.
- Loc. enables you to search for or create a local variable or a parameter.
- Prg. enables you to select or create a new program.
- Glo. enables you to select or create a global variable.

- VAL 3 enables you to select an instruction in the list of VAL 3 instructions.

It is possible to insert a breaking point in an instruction (see chapter 6.10.3).

To exit, press the Esc key.



I0000350

Figure 6.29

6.10.3 - DEBUGGING AN APPLICATION

M0000383.1

s7: Debugging applications is best done on PC with the VAL 3 Studio online debugger tool of Stäubli Robotics Suite. It is also possible on MCP as described below.

The VAL 3 task manager (accessible via the main menu) provides access to a debugging program that supplies the tools necessary to fine-tune tasks.

To access the debugging program, select a task and press the Dbg key in the pop-up menu. When the debugging session starts, the task selected is suspended immediately and a display page is opened (see figure 6.30).

In the debugging page, the ">" program pointer **(1)** shows the next instruction to be carried out.

- The Bpts menu enables you to associate a break point **(2)** with an instruction, via a window, or to delete a break point that has already been positioned. The stopping point is shown by the * icon opposite the instruction concerned.
- The ; menu enables you to add the selected instruction as a comment.
- The Data menu provides access to the variables for the instruction selected.
- The ->* menu enables you to move the task pointer **(1)**; this action does not execute any instructions.
- The {}* menu activates the step-by-step mode. In this mode, the program is executed as it is shown in the display (per display line).
- The {*} menu activates the detailed step-by-step mode. In this mode, each step is equal to a VAL 3 instruction and the sets of instructions (if, while…) are expanded.
- The Rsm./Sus. menu enables you to suspend and restart execution of the task without exiting the debugging program.
- The Save menu enables you to save the application.

To exit the debugging page, press the **"Esc"** key.

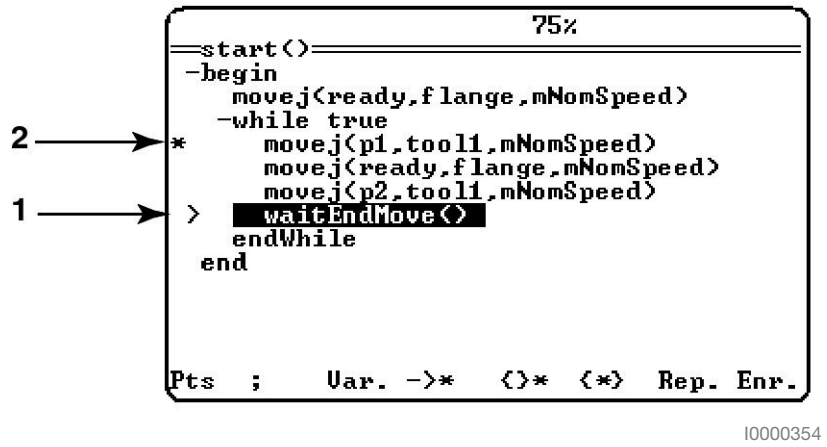


Figure 6.30

6.10.4 - METHODS FOR DIAGNOSIS

M0000231.1

System events

System events can be shown in several ways in the MCP display:

- **By showing a window with an explanatory message in natural language concerning the system error or event.**
- By a history, which provides access to the last 100 system events classified in chronological order with their date and time. The messages are stored in the events logger that is accessible via the main menu.
- By showing the "Info" indicator in the status bar (see figure 6.5). This shows that a new event has been added to the history without opening a window to warn the user. The "Info" indicator is removed when the user has consulted the events list.
- By showing the "?" indicator in the status bar (see figure 6.5). This shows that a VAL 3 application is awaiting an operator entry in the application page. It stays active as long as the application under consideration is active and until the entry is made.

Task status

The VAL 3 task manager is accessible via the main menu and it can be used to display a list of tasks and their status (for further information concerning errors, press the Help key). (s7) The current line of the selected VAL 3 task is displayed at the bottom of the screen. It shows where a task is stopped, without having to enter the debugger.

Input/output status

To display the Input/Output statuses, select the "I/O" branch in the control panel accessible via the main menu (see chapter 6.2.5).

This branch is used to display the status of the Inputs/Outputs of the controller boards defined in the system (RSI, BIO boards, field buses, Modbus TCP).

Indicator lights for system boards

To find out the meanings of the indicator lights for the RSI board and the other system boards (see chapter 6.2.5).

Examples of frequent events and means of diagnosis

Event	Diagnosis
 **In spite of starting up the application using the "Run" key, the robot does not move.**	■ Check the indicator light associated with the "Move / Hold" key. ■ Check that the task has not been suspended using a break point (see chapter 6.10.3). ■ Check that the task is not in error (the Task manager is accessible via the main menu). ■ Check that the application is not in waiting mode (? icon in the status bar, see chapter 6.2.5).
The application has started, but nothing is shown on the LED display.	The information shown for a VAL 3 program can only be seen via the user page (see chapter 6.2.5). See also the "userPage" instruction in the Reference Manual for the VAL 3 language.
A page is displayed automatically each time a movement mode is selected.	This jog interface management page is indeed displayed each time the movement mode is changed (see chapter 6.6). To return to the previous page, press the"Esc" key.
The fact of pressing the "Rsm." (Resume) pull-down menu key for a task does not have any effect.	■ Check whether the task is in error status. ■ Check whether the task has been stopped at a break point. If this is the case, delete the break point or use the Rsm. menu via the debugging program (see chapter 6.10.3).
The robot does not go to the right place	If the robot does not carry out the movements as programmed when the "Move / Hold" key is pressed: ■ Check whether the controller is in "Point" manual mode (associated indicator light on); in this case the robot moves to the last point selected for a jog interface. To return to the programmed movements, exit the "Point" mode (see chapter 6.6).
 **When an application is stopped using the "Stop" key, the "Run" indicator light stays on.**	There must be a task still being debugged. ■ Exit the debugging program, and then kill the task in the task manager.
 **When the "1" or "2" keys are pressed, the solenoid valves are not switched.**	■ Check that these keys have not been assigned to other outputs.

6.11 - TEACHING FRAMES

M0000232.1

This chapter constitutes a frame teaching procedure using the three-point method.

This method enables you to define the orientation of the new frame in a precise way by recording three points:

- The frame origin (O),
- A point (Ox) located on the X axis of the frame on the positive x side,
- A point (Oxy) on the plane formed by the X and Y axis on the positive y side.

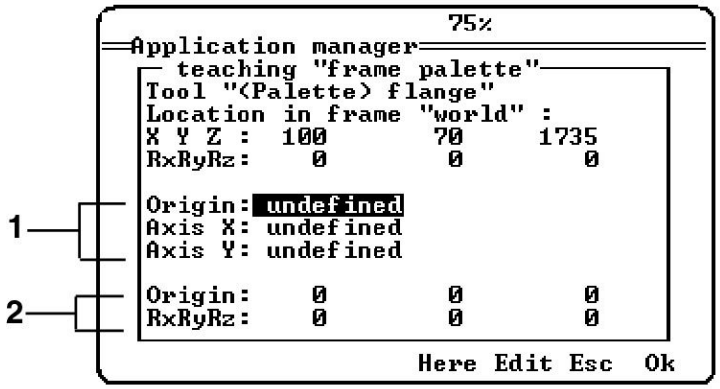
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The method used is as follows:

- After creating a new Frame type data, use the Teac key in the menu to call up a "Teaching" display (see figure 6.31).
- Position the point of the robot tool at the desired point of origin using jog interfaces and press the Here key in the pop-up menu.
- Repeat the operation for "X axis" and "Y axis" and then validate the orientation of the new frame **(1)**.
- Save the modifications.

The frame coordinates are displayed in the box **(2)**.

The point values and marker orientations can be modified using the Edit menu.



I0000196

Figure 6.31

6.12 - TEACHING POINTS

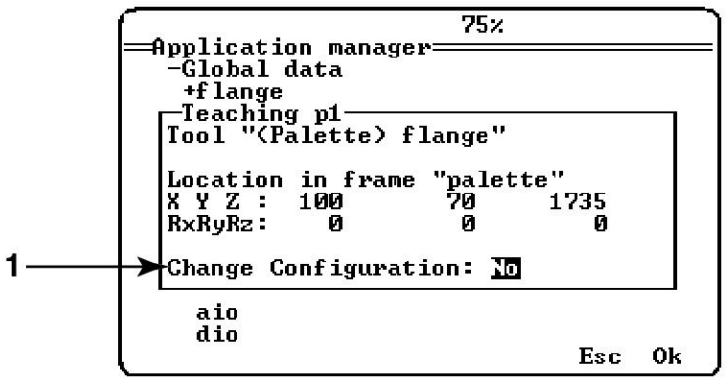
M0000233.1

Teaching points, the method used is as follows:

- Create a new data of "point" type.
- Use the Here pop-up menu to call up the "Teaching p1" display (see figure 6.32).
- Move the tool to the location and position where the teaching is to be done.
- Validate the position using the "Ok" key in the pop-up menu.
- Save the modifications.

The "Configuration" box **(1)** can be filled in two ways:

- "No": The point configuration remains unchanged.
- "Yes": The point is taught with the current arm configuration (see the "Reference Manual for VAL 3 language").



I0000209

Figure 6.32

6.13 - MOTION DESCRIPTOR EDITOR

M0000410.1

When editing a motion descriptor in the application manager, a simplified view is displayed where only speed and blending type can be modified. The ">>" menu gives access to the advanced interface where all motion descriptor parameters can be modified.

A change in the velocity parameter of the simplified interface modifies both the joint velocity and joint acceleration/deceleration, so that the arm behavior remains harmonious. The accel and decel parameters should indeed be roughly the square of the vel parameter. For instance, a velocity of 120% = 1.2 is best adapted with accel and decel of 1.2 x 1.2 = 1.44 = 144%. Higher values for accel and decel give a more aggressive, but shakier arm behaviour.

When the accel and decel parameters are modified in the advanced interface, this relation with the velocity parameter may not be effective any more. In that case, when the simplified interface is displayed with the "<<" menu, the speed parameter is displayed as "User". It can still be edited to recover the default relation between joint velocity and acceleration/deceleration.

6.14 - CONTROLLER BACKUP

M0000367.1



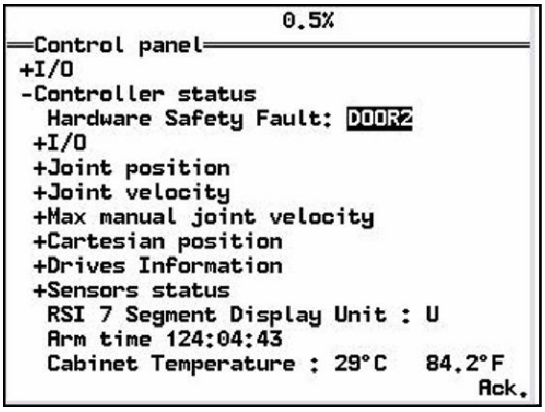
It is important to maintain an up-to-date controller backup for an easier and faster maintenance. The backup should include controller and arm specific configuration data, controller firmware, safety application program and user application.

The controller backup can be done either from "Stäubli Robot Studio" or directly from the controller by using a USB key.

6.15 - SAFETY FAULT ACKNOWLEDGEMENT

M0000378.1

After a fault is detected that may impact the safety level of the controller or the cell, robot operation is not allowed until the fault is acknowledged. Hardware faults on the EStop signals must be corrected and tested before acknowledge. The controller keeps track of the detected safety faults and, for hardware faults, checks automatically for evidence that the fault has been solved. After the hardware fault is repaired, it is tested by reproducing the initial fault conditions, then it is required to acknowledge the correction to resume robot operation. The safety fault acknowledgment can be done from the Controller status section of the Control Panel display (see figure 6.33).



I0000365

Figure 6.33

- The 'Info' menu is displayed when the system is waiting for the evidence that the hardware fault has been repaired.
- The 'Ack.' menu is displayed when the system is waiting for an acknowledge of the safety fault.

The wiring of the safety system makes it impossible to know the state of each signal when several EStop signals are activated. Hardware faults on EStop signals that are currently not visible are marked with brackets []. Deactivate the different EStop signals to enable the hardware check again.

The acknowledge of the safety fault can also be done:

- by simultaneously pressing the MCP keys 'A', 'C' and 'K'.
- from an external device (see chapter 5.8.4).

Acknowledge from the MCP requires the adequate access rights defined in the user profile (see chapter 5.8.3). Acknowledge from an external device can be done whatever the user profile configuration.

7 - PC UTILITIES

7.1 - STÄUBLI ROBOTICS SUITE (SRS)

M0000259.1

PC Software Suite for robotics application development and maintenance.

Basic functions

- Easy file transfer between the robot controller and the development PC.
- Make full backups of the entire contents of the robot system to a remote PC for complete systems backup, either manually or automatically as a scheduled task.
- Visualization of robot arms only in 3D scene.
- Execution / modification of VAL 3 programs in emulated controller.
- Single window approach, fully customizable layout.

Development Studio

- VAL 3 program editor with high end functionalities such as online debugging, auto completion, intellisense, snippets and much more.
- Possibility to import CAD data (STEP, IGES, STL, VRML).
- Possibility to build models based on standard primitives (cubes, spheres, etc...).
- Direct link between 3D scene and program editor (for geometric data creation/edition, etc...).
- Full multi-robot simulation capabilities (robot reach, realistic cycle times through the individual clocking of each controller in the cell).
- Each simulated robot can be driven by a real or emulated controller.
- High performance collision detection (highlighting of colliding objects, minimum distance verification).

Maintenance Studio

- Remote Access provides full access to the robot system from a remote location. You can control the system from your remote PC and get a live visual feedback of any operation performed on the teach pendant by the operator.
- 3D scene in visualization mode with the ability to load a 3D robot cell previously created in Development Studio.

7.2 - FTP ACCESS FROM A PC

M0000480.1

Most of the network exchanges between a PC and a controller can be easily done with the Transfer Manager tool of Stäubli Robotics Suite.

This chapter describes how to access to the flash disk of the controller with a simple FTP connection.

7.2.1 - FTP CLIENT

M0000380.1

A FTP client is accessible using Windows (95, 98, NT, 2000, XP). To set up a connection:

- Open an online command session and enter: ftp w.x.y.z (w x y z corresponds to the IP address of the controller) (see chapter 5.6).
- Then when the "User" prompt is displayed, enter the name of a user profile, and when the password prompt is displayed, enter the network password for the profile (see chapter 5.8.3).
- Go to the USR reader using the cd /usr command and then carry out the desired actions.



The freeware "Ftp surfer" is supplied with the robot CD ROM.

7.2.2 - IP ADDRESS CONFIGURATION

M0000234.1

The IP address of the controller is configured via the control panel. This is accessible via the main menu. The IP address is in the controller node (see chapter 5.6).

7.2.3 - FUNCTIONS VIA FTP

M0000423.1

All the information available for the user is on the disk called "usr".

Updating the configuration file:

Using a FTP client.

Connect up to the controller and go to the /usr/configs directory.

The files enabling login user configuration of the controller are as follows:

- "arm.cfx" contains the configuration concerning the arm (recovery, marks). This file is only accessible for a backup prior to maintenance.
- "cell.cfx" contains the configuration linked to the cell (language, max. Cartesian speed, etc...).
- "network.cfx" contains the network configuration of the controller (IP address, system TCP ports, gateways...).
- "controller.cfx (s6: controller.cf)" contains the controller configuration.
- bio.cfx, bio2.cfx, mio.cfx, sio.cfx, can.cfx, encoder.cfx, cio.cfx, asi.cfx contain the input/output configurations.

DANGER
All ill-considered modifications made to the configuration can lead to bodily injury or serious material damage.

7.3 - FTP ACCESS TO A PC

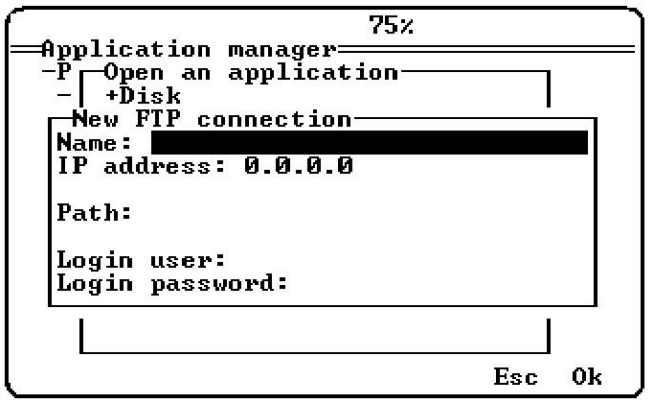
M0000235.1

This function enables centralization of the VAL 3 applications on a PC (backup on a CD, just one storage location, etc).

To do so, it is necessary for an FTP server to be executed on the PC to share a directory containing the VAL 3 applications. Stäubli supplies the free Cesar FTP server enabling this functionality but cannot be held responsible in the event of incorrect use of the software. To install the server, see chapter 7.3.1.

Configuration of an FTP node on the MCP

On the home page of the application, press the key of the "Ftp" pop-up menu to move on to the next page:



I0000188

Figure 7.1

- The name enables you to give the FTP connection a name.
- The IP address corresponds to the IP address of the PC on which the FTP server is executed.
- The path corresponds to the shared directory on the FTP server. With Cesar FTP, if the shared directory is c:\temp\VAL 3\apps, enter only \apps.

- The user corresponds to the user name created on the FTP server
- The login password must correspond to the FTP user password (Be careful with upper and lower case letters).

Once the parameters have been entered, validate the page using "OK". In the applications opening page, there is now a new node corresponding to the FTP server that has just been created. The applications that can be seen in this node are used in the same way as local VAL 3 applications.

To enable a backup of the applications, the Ftp connection must be conserved. However, it is not necessary to run the application.

7.3.1 - INSTALLING CESAR FTP

M0000411.1

This chapter does not set out to explain the FTP connections, it simply supplies a guide to installing the software and creating the login user account.

- Execute the CesarFTP.exe file to be found on the robot CD-ROM and follow the installation instructions.
- Once the software has been installed, create a new user.
- In the user creation window, add the sharing of the directory containing the VAL 3 applications (File Access Right button). To add it, drag and drop the directory onto the user.
- Select the default directory by right clicking and then "set as default".
- Close the files window, and validate the new login user.

For further information, see the software documentation.

8 - MAINTENANCE

8.1 - HOW TO USE THIS MANUAL?

8.1.1 - SAFETY RECOMMENDATIONS

M0000286.1

The recommendations for safety are found in the "Safety" chapter 3. They have to be read and understood. In case of doubt or incomprehension, contact the Stäubli technical support. Refer to the service organization label inside the controller.

Even if it is not specified at every step of trouble shooting, each component change or component disconnection has to be made with the main switch of the CS8C in the 0 position (off).

Some of the diagnostic steps require the power to be turned on or off. Do not forget to turn it off before changing components.

8.1.2 - REQUESTED LEVEL

M0000287.1

Service personnel have to have the experience to be able to perform electrical and mechanical interventions. Refer to the regulations currently in force in the country concerned.

8.1.3 - INTERVENTION LEVELS

M0000288.1

The different troubleshooting steps represented below use 3 intervention levels:

- Level 1 (by default): Operations that can be carried out by a maintenance technician without specific Stäubli training.
- Level 2: Operations that can be carried out by a maintenance technician who has undergone specific Stäubli training.
- Level 3: Operations that must be carried out by the Stäubli Customer Support.

8.1.4 - SERVICE METHODOLOGY

M0000289.1

The following chapters give a general methodology to trouble shoot the robot. It is based first on visual indicators provided by the controller (led, display) and on indications provided on MCP (popup messages, statuses).

For each point, it is assumed that previous ones have been checked and operate properly.

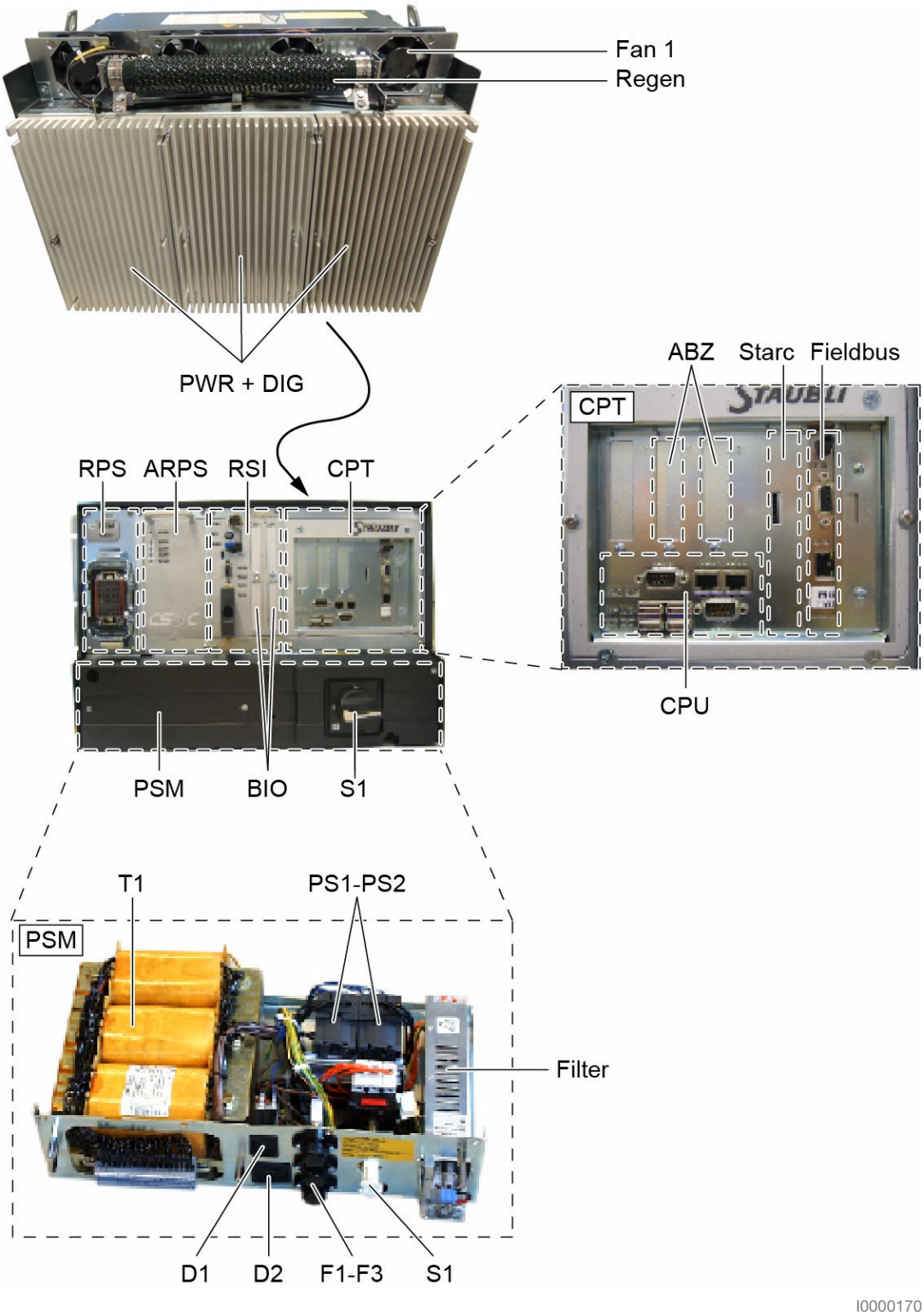
8.1.5 - ELECTRICAL DRAWINGS

M0000290.1

Electrical drawings provided in this manual are for information only. The electrical drawings of the robot is provided in a separate manual.

8.2 - COMPONENT LOCATION

M0000447.1



I0000170

Figure 8.1

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8.3 - SAFETY

M0000291.1

To disconnect the system from the power supply, set the master switch (S1) located on the front of the controller to 0. Before doing so, you must stop the arm motion and switch off arm power supply.



DANGER

- Disconnect all the electrical and pneumatic power supplies before carrying out any work on the controller or the arm. Wait for at least 1 minutes before starting to work.
- Lockout and tagout equipment before servicing.

See the Safety chapter 3.3.1 for isolation of the system.



- Use an antistatic wrist strap and an antistatic mat connected to the controller for all work involving handling of boards or components.
- Take all the necessary precautions as set out in chapter 3.4.3 to avoid the risk of electrostatic charges.
- During maintenance and/or diagnostic operations, if parts are replaced or exchanged between different systems, make sure that they are fully compatible (hardware and software compatibility). Check, at low speed, that the robot is operating correctly, especially for calibration.



After a change of a safety-related component, the corresponding safety functions must be validated:

- The RSI board (safety stops functions, enabling device).
- The PS1/PS2 contactors (motors power supply).
- The ARPS (brakes power supply).
- The MCP (emergency stop and enabling device).

8.4 - INPUT VOLTAGE

8.4.1 - DESCRIPTION

M0000293.1

The following components are located in PSM (Power Supply Module) located at the bottom of the cabinet.

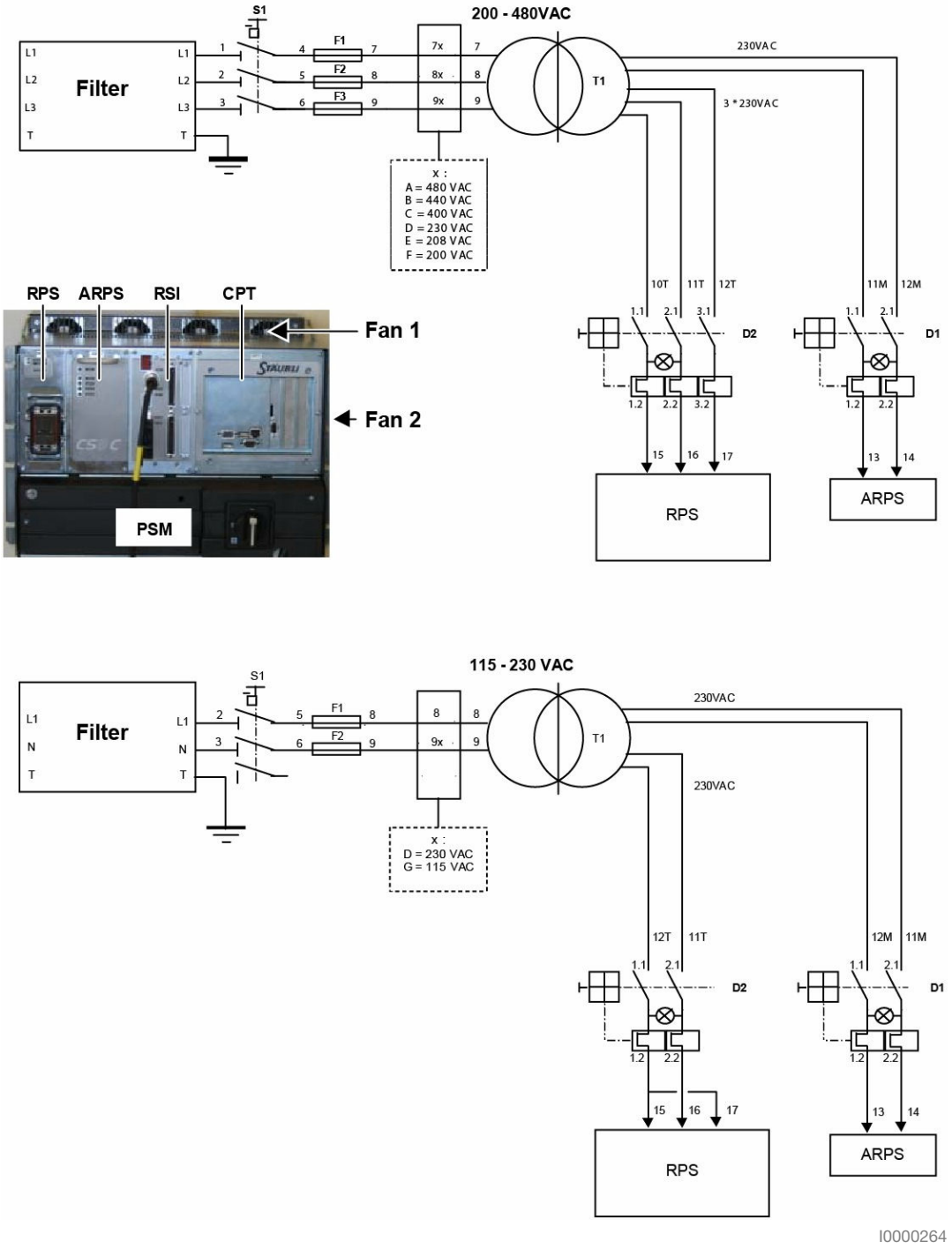


Figure 8.2

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8.4.2 - ACCESS

M0000294.1

Remove 3 screws **(4)** to remove cover.



Figure 8.3

To access the components of the PSM, remove the screws **(1)** and pull it forward.

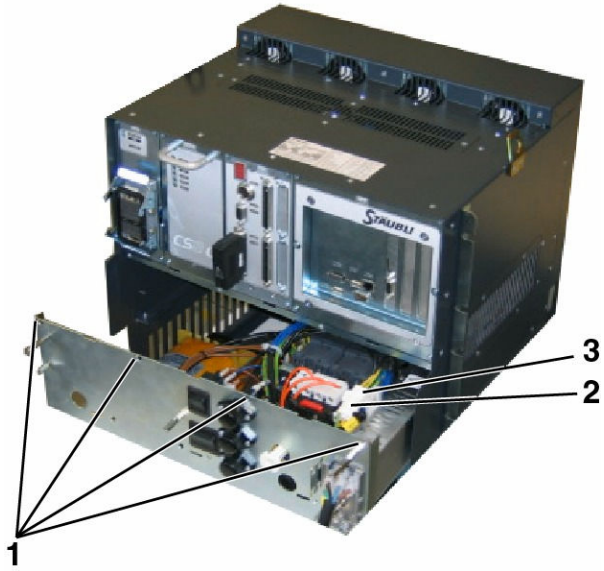


Figure 8.4 : CS8C TX/TS

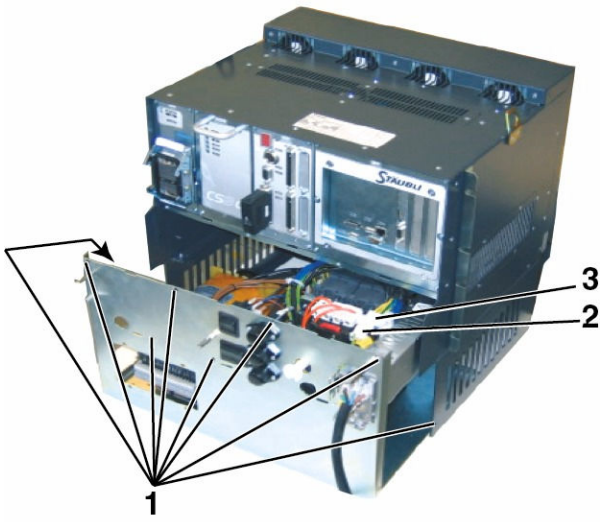


Figure 8.5 : CS8C RX160/TP80/TX250

- To completely remove the PSM, disconnect connectors **(2)** and **(3)**.
- The PSM is heavy; take all necessary precautions to avoid dropping it and to avoid making efforts in an incorrect position.

8.4.3 - TROUBLE SHOOTING

8.4.3.1 - Case 1

M0000531.1

Problem:

All lights of ARPS are off.

Solution:

- Check that main switch S1 is on position 1 and that input voltage is provided to CS8C (external line).
- Switch off CS8C.
- Turn off main switch S1 to position 0.
- Check input fuses (F1, F2, F3):
  - Fuses are 10 x 38 mm type, 500V for non-UL controllers.
  - For UL type controllers, replace fuses with UL type.

	Three-phase 400-480 V	Three-phase 200-230 V	Single phase 230 V	Single phase 115 V
TX40	4Am	6Am	10Am	16Am
TX60 - TS	4Am	8Am	10Am	16Am
TX90	6Am	12Am		
RX160, TX250, TP80	8Am	16Am		

- These fuses do not protect the mains power supply line which must be protected separately.
- Never replace these fuses with fuses of a higher rating or with different characteristics (see the "replacement parts" section).

Am means "slow-acting fuse" according to IEC 269-1.2.
AT means "slow-acting fuse" and AF "quick-acting fuse" according to IEC 127-2.

- Check D1 circuit breaker.

8.4.3.2 - Case 2

M0000539.1

Problem:

D1 circuit breaker light is off.

Solution:

- Check that D1 circuit breaker is on position 1.
  - If D1 circuit breaker does not remain on position 1, change ARPS and / or D1 circuit breaker.
- Check input fuses (F1, F2, F3):
  - Fuses are 10 x 38 mm type, 500V for non-UL controllers.
- Check voltages at the input of the controller (L1, L2, L3).
- Check voltages after the transformer. At this point, voltages are 230 VAC ±10% for all the input voltages.
- Change D1 circuit breaker.

To change D1 or D2 circuit breakers, remove PSM.

DANGER

- Main switch S1 should be in off position AND main power to the controller has to be disconnected.
- Orange wires inside PSM indicate that dangerous voltage remains even if S1 is off.

Remove 3 screws **(4)** to remove cover.



Figure 8.6

To access the components of the PSM, remove the screws **(1)** and pull it forward.

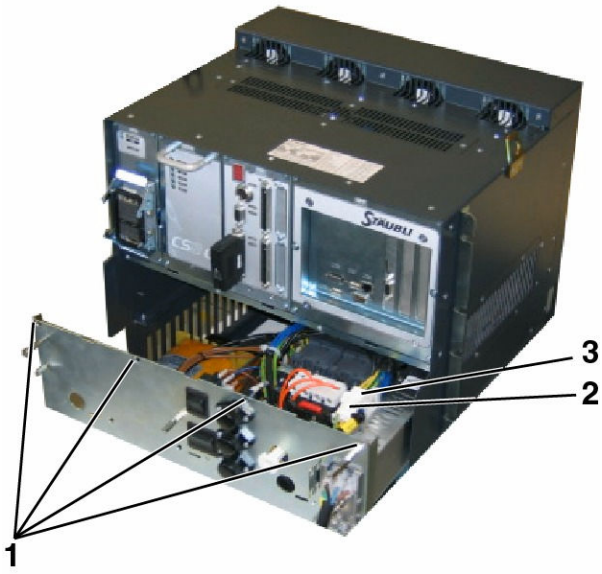


Figure 8.7 : "CS8C TX/TS"

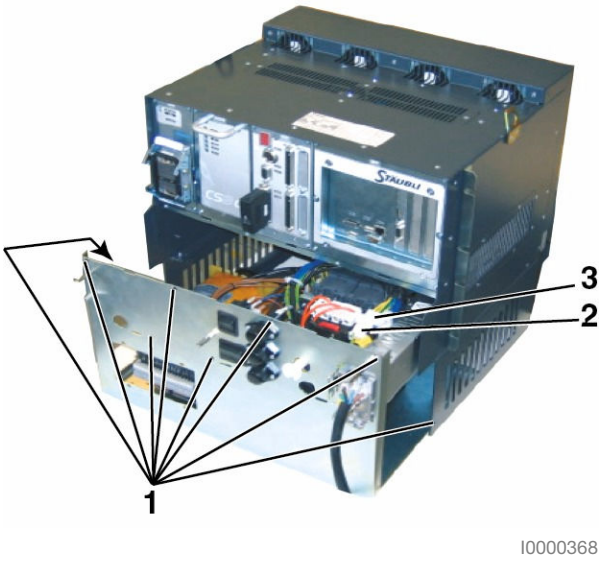
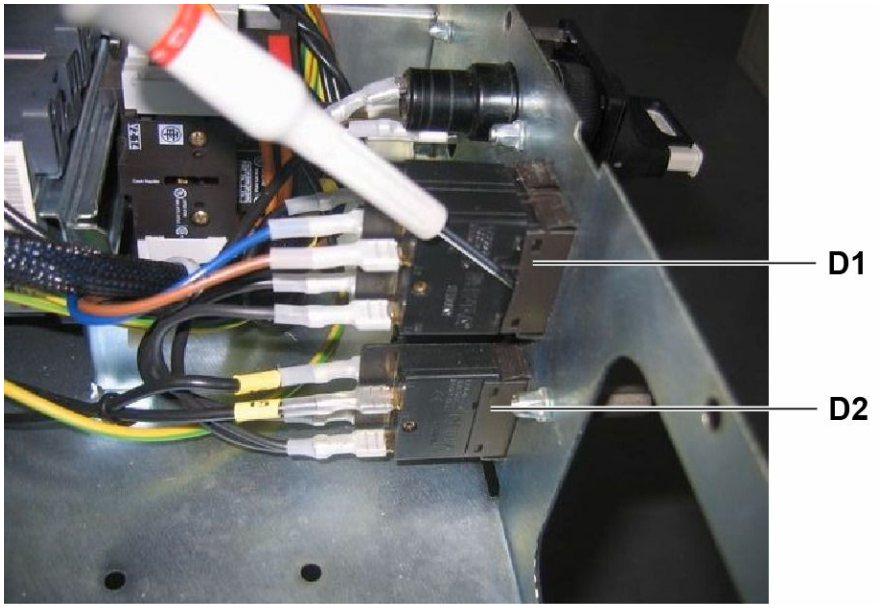


Figure 8.8 : "CS8C RX160/TP80/TX250"

- To remove it completely, disconnect connectors **(2)** and **(3)**.
- The PSM is heavy; take all necessary precautions to avoid dropping it and to avoid making efforts in an incorrect position.

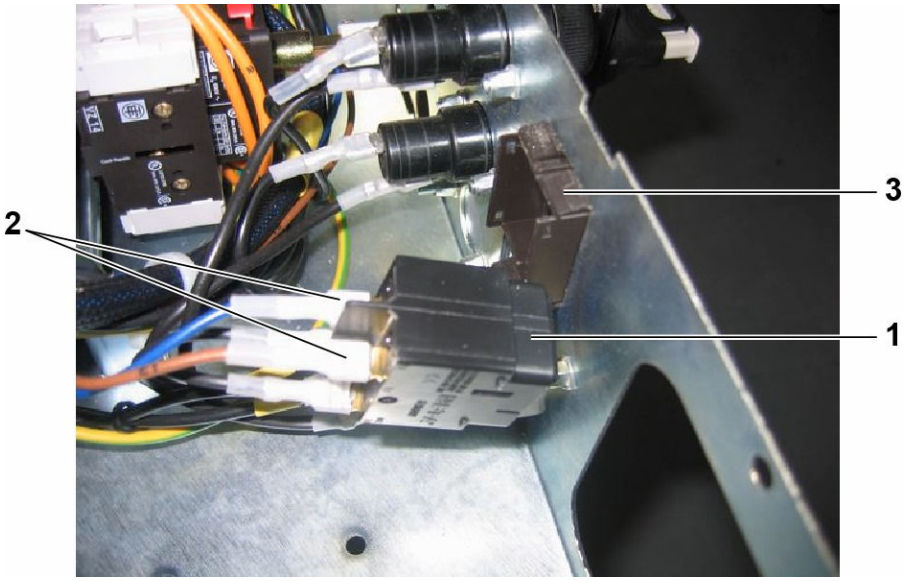
The frame of circuit breaker can be separated from the front face with a screw driver allowing it to be pulled back.



I0000224

Figure 8.9

Push on **(1)** to separate it from the front face **(3)** and disconnect the wires **(2)**.



I0000314

Figure 8.10

8.4.3.3 - Case 3

M0000540.1

Problem:

D2 circuit breaker light is off.

Solution:

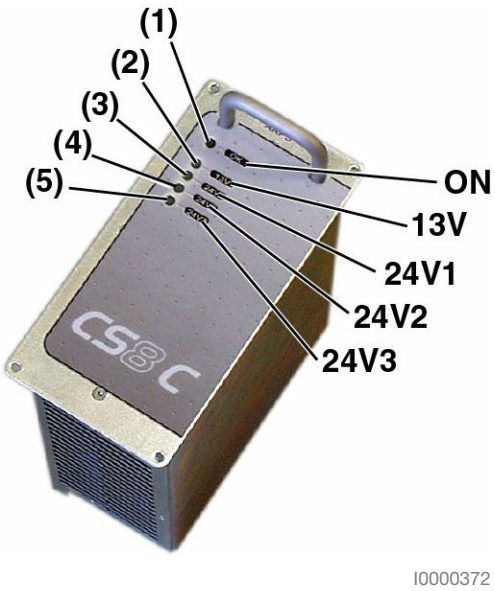
- Check that D2 circuit breaker is on position 1.
  - If D2 circuit breaker does not remain on position 1, change D2 circuit breaker.
- Check input fuses (F1, F2, F3):
  - Fuses are 10 x 38 mm type, 500V for non-UL controllers.
- Check voltages at the input of the controller (L1, L2, L3).
- Check voltages after the transformer. At this point, voltages are 230 VAC ±10% for all the input voltages.
- Change D2 circuit breaker.

8.5 - ARPS AUXILIARY ROBOT POWER SUPPLY

8.5.1 - DESCRIPTION

M0000413.1

ARPS is powered with 230 VAC from the D1 circuit breaker. Its outputs are overload protected, which means that output voltages are automatically set to 0 when there is a short circuit in the components it supplies.



I0000372

Figure 8.11

Normal status, arm power off:

Green LEDs	Status
ON	**ON**
13V	**ON**
24V1	**ON**
24V2	OFF
24V3	**ON**

Normal status, arm power on:

Green LEDs	Status
ON	**ON**
13V	**ON**
24V1	**ON**
24V2	**ON**
24V3	**ON**

8.5.2 - ACCESS

M0000296.1

- Turn off main switch S1 to position 0.
- Remove the 4 screws **(1)** and pull the ARPS board.

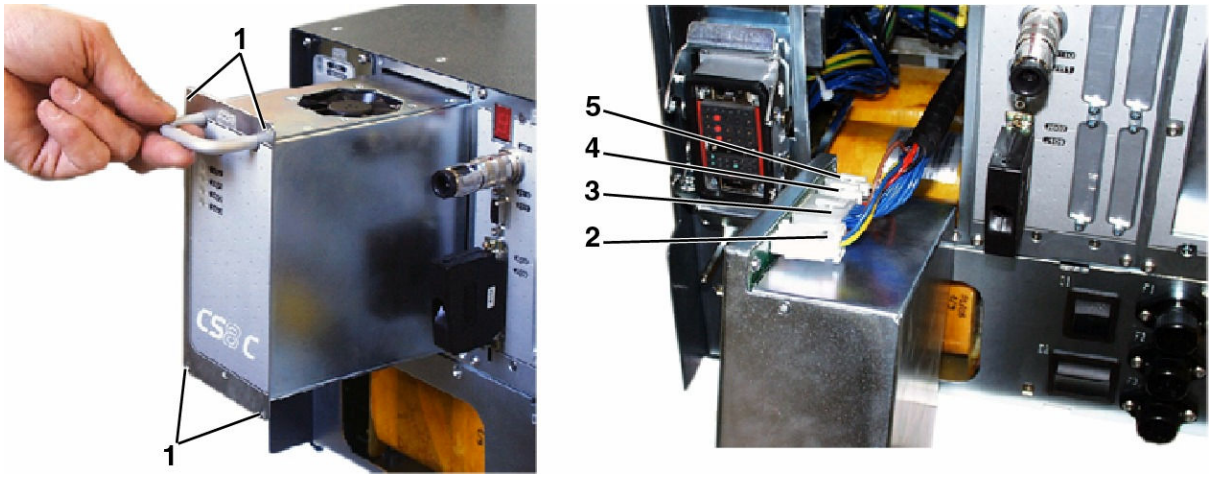


Figure 8.12

8.5.3 - TEST POINTS

M0000297.1

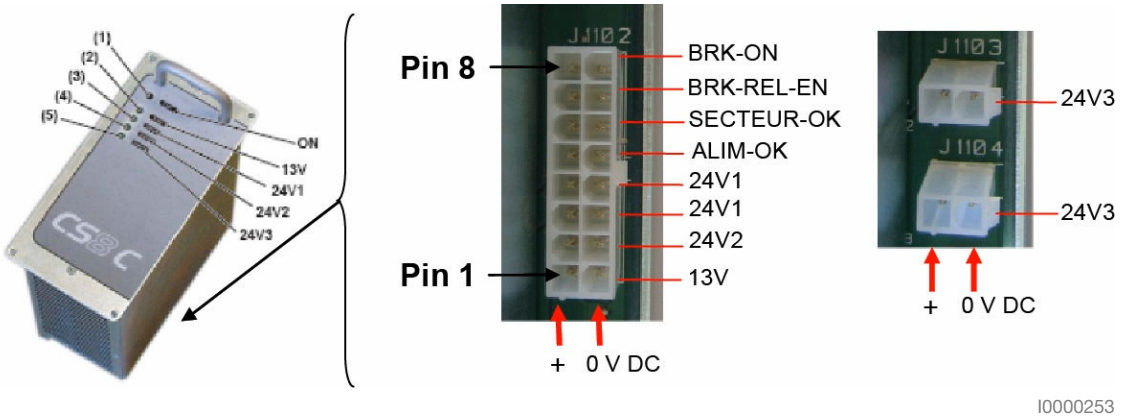


Figure 8.13

8.5.4 - TROUBLE SHOOTING

8.5.4.1 - Case 1

M0000489.1

Problem:

ON indicator remains off.

Solution:

- Refer to chapter 8.4 to check input voltages:
  - Check that main switch S1 is on position 1.
  - Check fuses (F1, F2, F3).
  - Check D1 circuit breaker.
- Change the ARPS board.

8.5.4.2 - Case 2

M0000491.1

Problem:

13V, 24V1 or 24V3 light remains off.

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Solution: Step 1

- Unplug J1102 **(3)**, J1103 **(4)**, J1104 **(5)** connector at ARPS outputs (see figure 8.12):
  - If the indicator lights remain off, change ARPS.
  - If the indicator lights come on again, there is a short circuit on the corresponding outputs.

Solution: Step 2

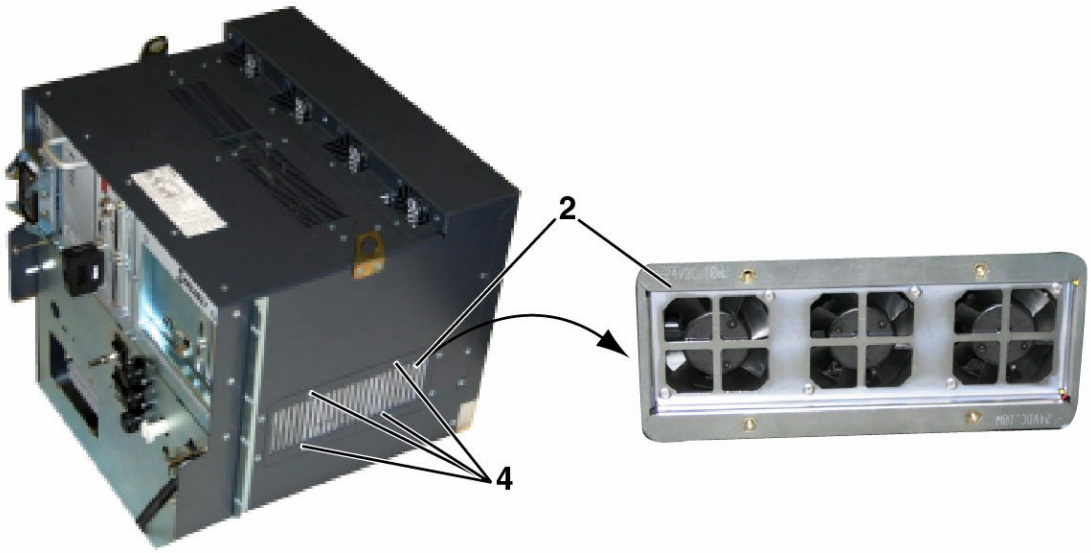
- Plug only J1104 (24V3) for internal fans (fan 2 and fan 3):
  - If 24V3 light goes off, check wiring and fans.
  - Change the defective part.

Access to fans

DANGER

- The cover **(2)**, the resistor **(8)** and the amplifiers **(7)** may be very hot, especially in the event of a ventilation malfunction.
- This disassembly operation provides access to the regeneration resistor powered at 400V during normal operation. It is essential to cut off all power supplies before carrying out this operation. Wait for at least 1 minutes before starting to work.

The fan2 group **(2)** is accessed by removing the screws **(4)**.



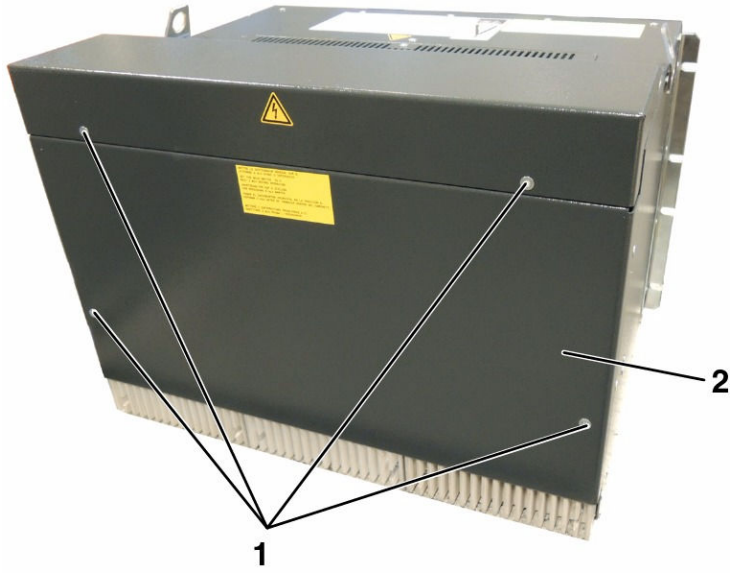
I0000270

Figure 8.14

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The fan 3 **(5)** for the RPS power supply can be accessed by removing the amplifier for joints 3-6:

- Unscrew the screws **(1)** to remove back covers **(2)**.



I0000277

Figure 8.15



I0000187

Figure 8.16

Unscrew the screws **(3)** to remove the amplifier for joints 3-6 **(4)** and access to the fan **(5)**.

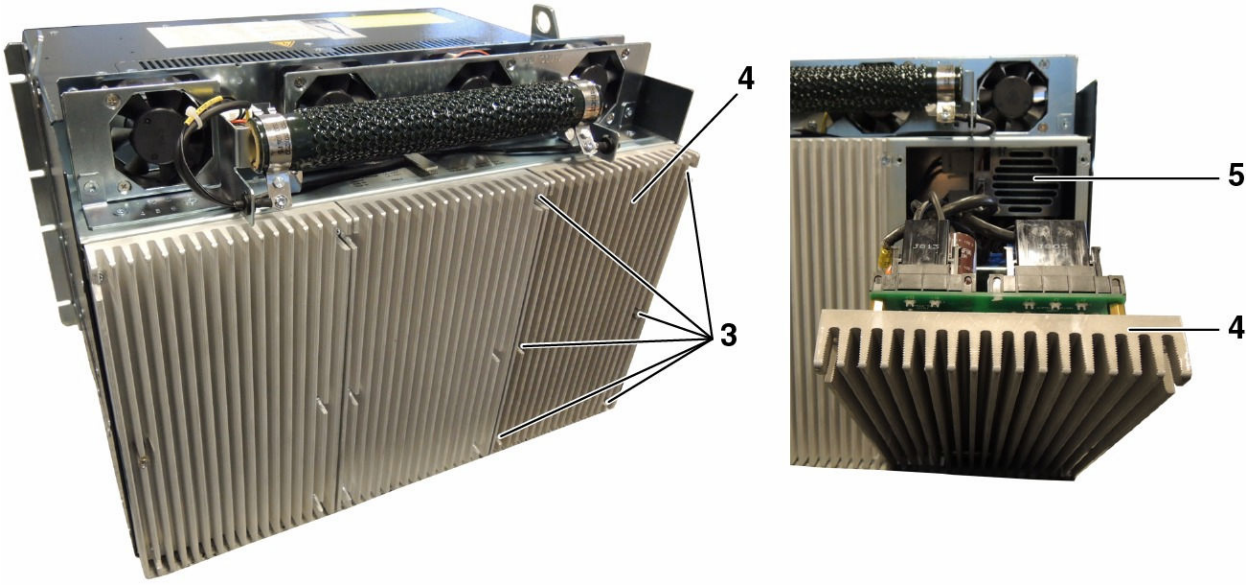


Figure 8.17

If fan needs to be changed, unscrew part of the regeneration assembly kit to access to the screws **(6)** of the fan.

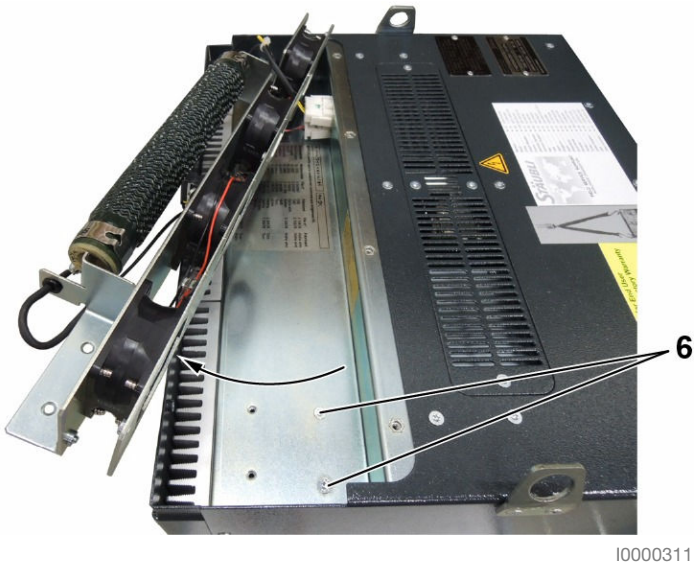
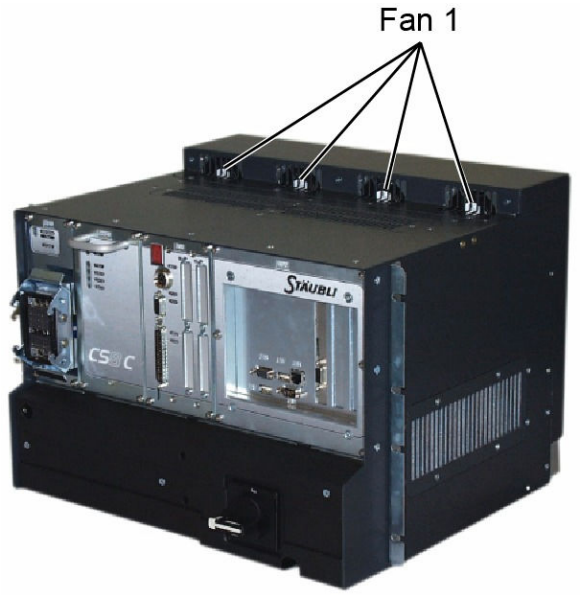


Figure 8.18

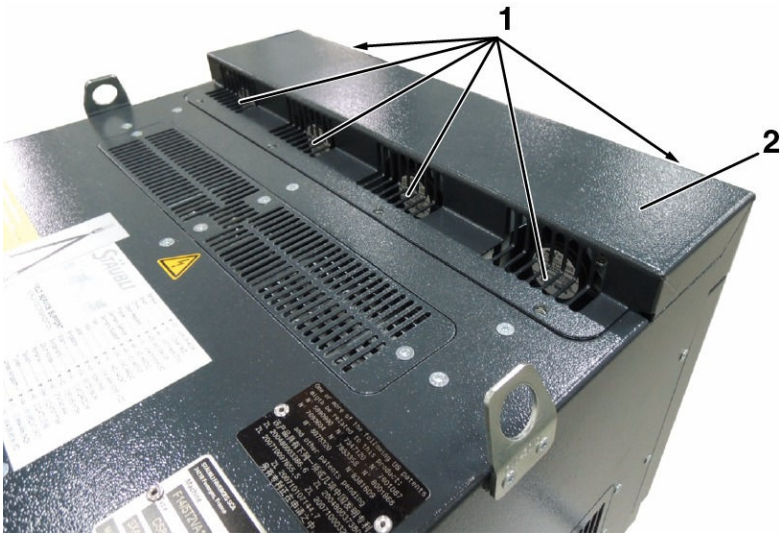
Solution: Step 3

- Plug only J1103 (24V3 for fan 1 on top of CS8C).
- If 24V3 light goes off, check wiring and fans.
- Change the defective part.



I0000254

Figure 8.19



I0000393

Figure 8.20

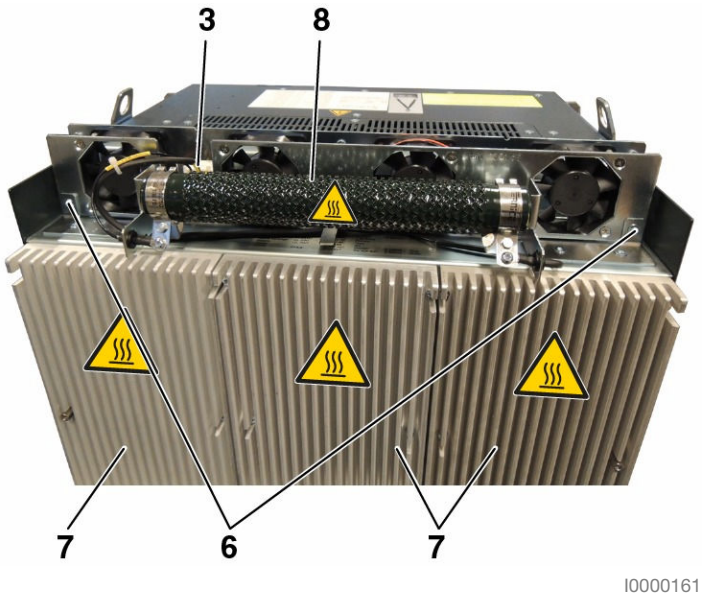


Figure 8.21

DANGER

- The cover **(2)** (see figure 8.20), the resistor **(8)** and the amplifiers **(7)** may be very hot, especially in the event of a ventilation malfunction.
- This disassembly operation provides access to the regeneration resistor powered at 400 V during normal operation. It is essential to cut off all power supplies before carrying out this operation. Wait for at least 1 minutes before starting to work.

- Remove the 6 screws **(1)** (see figure 8.20).
- Remove the cover **(2)** (see figure 8.20).
- The 4 fans can be accessed by removing the screws **(6)** and the connectors J1301 and J1303 **(3)**.

Advanced information

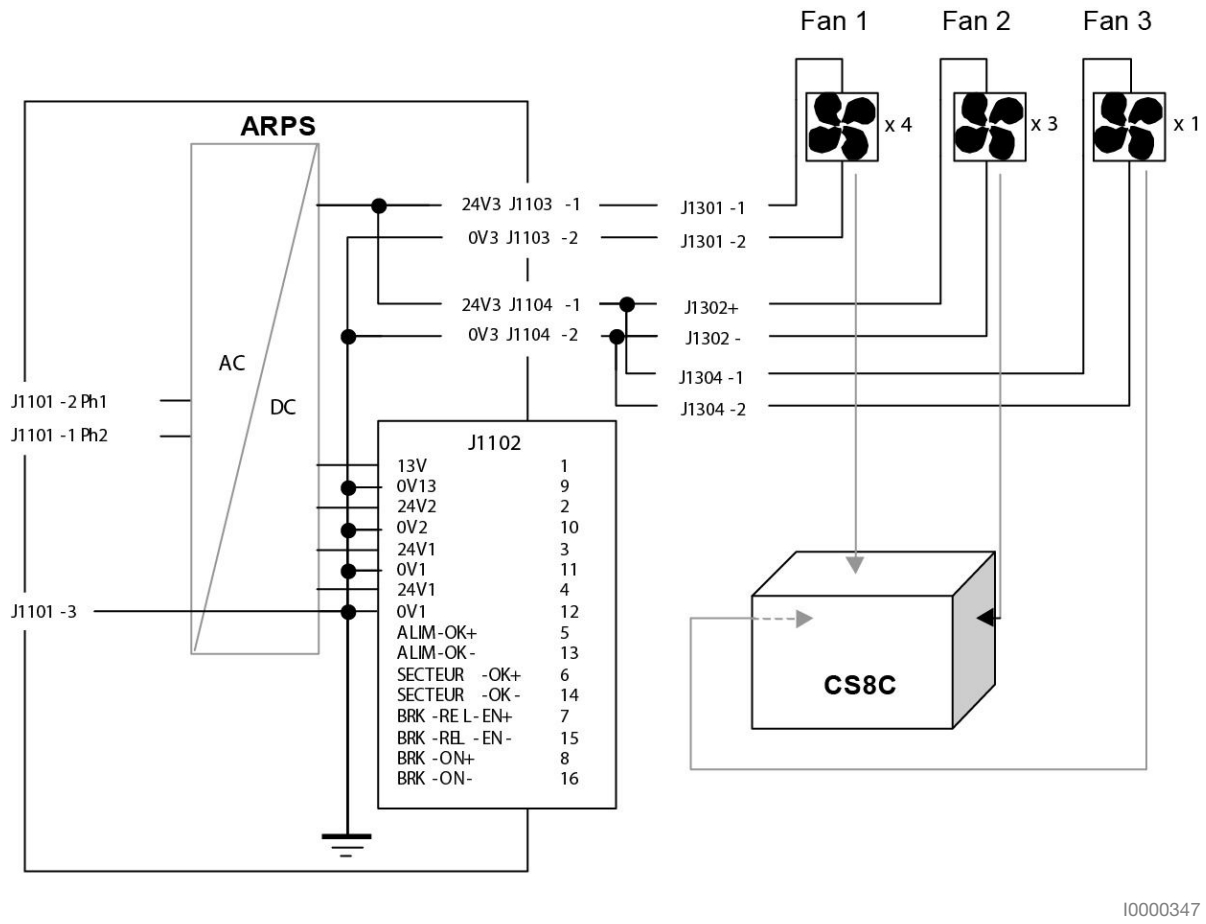


Figure 8.22

Solution: Step 4

- Plug only J1102 (24V1 for RSI, drives and CPT ; 24V2 for brakes ; 13V for DSI inside arm).
  - If 13V light goes off, repeat the same operation with the interconnection cable disconnected.
    - If 13V light comes on, there is a short circuit either in the cable or in the DSI in the arm. Change the defective part.
    - If 13V light remains off, the short circuit is inside CS8C. Check wiring from J1102 and J1200: short circuit, damaged wire ...

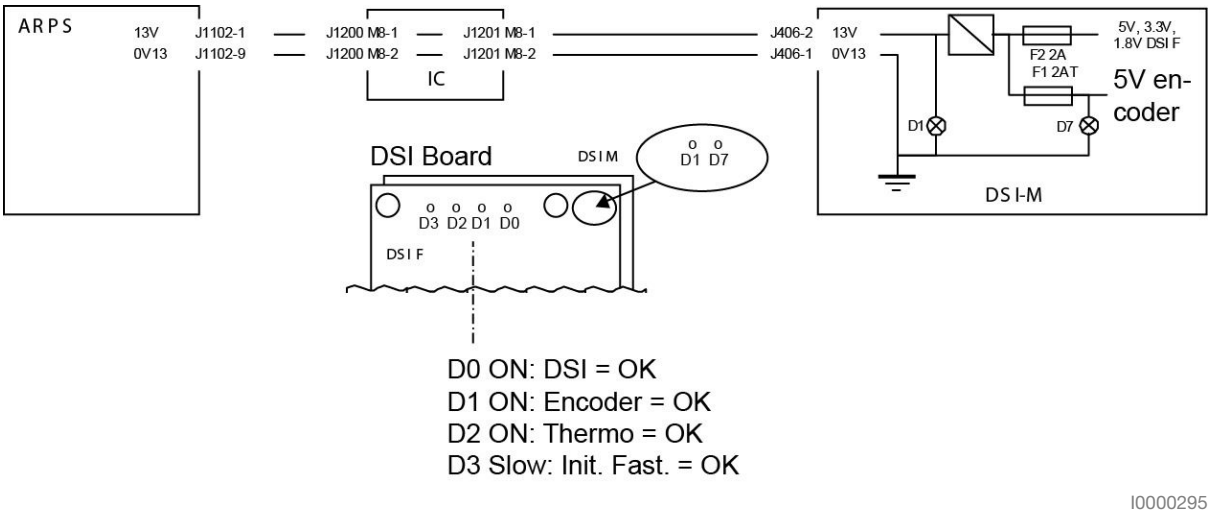


Figure 8.23

Solution: Step 5

- Plug only J1102 (24V1 for RSI, drives and CPT ; 24V2 for brakes ; 13V for DSI inside arm).
  - If 24V1 light goes off, repeat the same operation with powered components disconnected one by one: CPU, RSI, drives in order to differentiate the defective component or wiring.

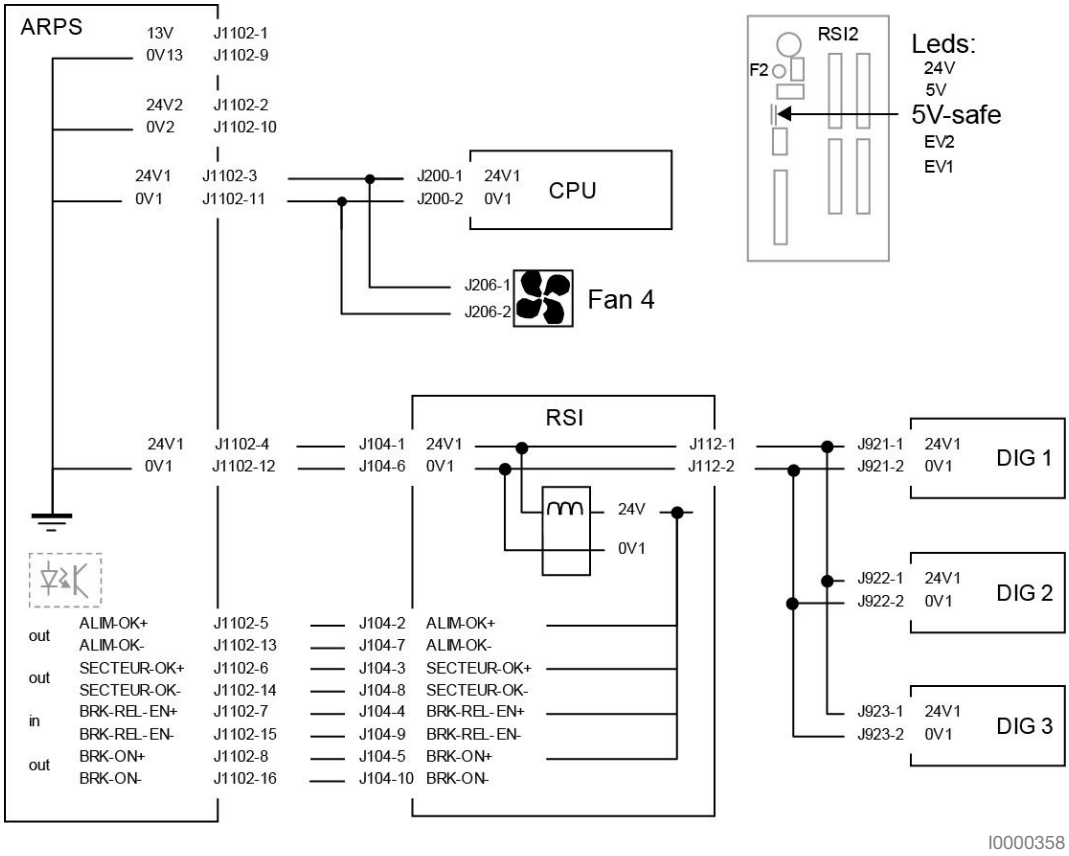


Figure 8.24

8.5.4.3 - Case 3

M0000515.1

Problem:

24V2 light remains off when enabling power on arm.

Solution: Step 1

- Brake command signal (BRK_REL_EN) not received by ARPS.
- Brake command is not taken into account by ARPS.

Refer to chapter 8.5.7, BRK-x signals.

Solution: Step 2

- Short circuit on brakes or wiring.

  Check if all brakes are operating in Manual Brake Release mode:

DANGER

- Refer to chapter 3.
- Pay attention to risks related to the size of the robot, the size of the payload, and so on when using Brake Release mode.

- If 24V2 light goes off when a specific brake is released, check this particular brake and its wiring from the robot base to the brake.
- If 24V2 light goes off whatever is the selected brake, check brake release board in the robot base and the wiring from ARPS to the arm.

Changing a brake requires an intervention level 2 or 3 depending if brake is integrated or not in the motor.



Figure 8.25

On 4-axis robots, 24V2 voltage is connected to LPX2 board (refer to electrical drawings).

8.5.4.4 - Advanced information

M0000415.1

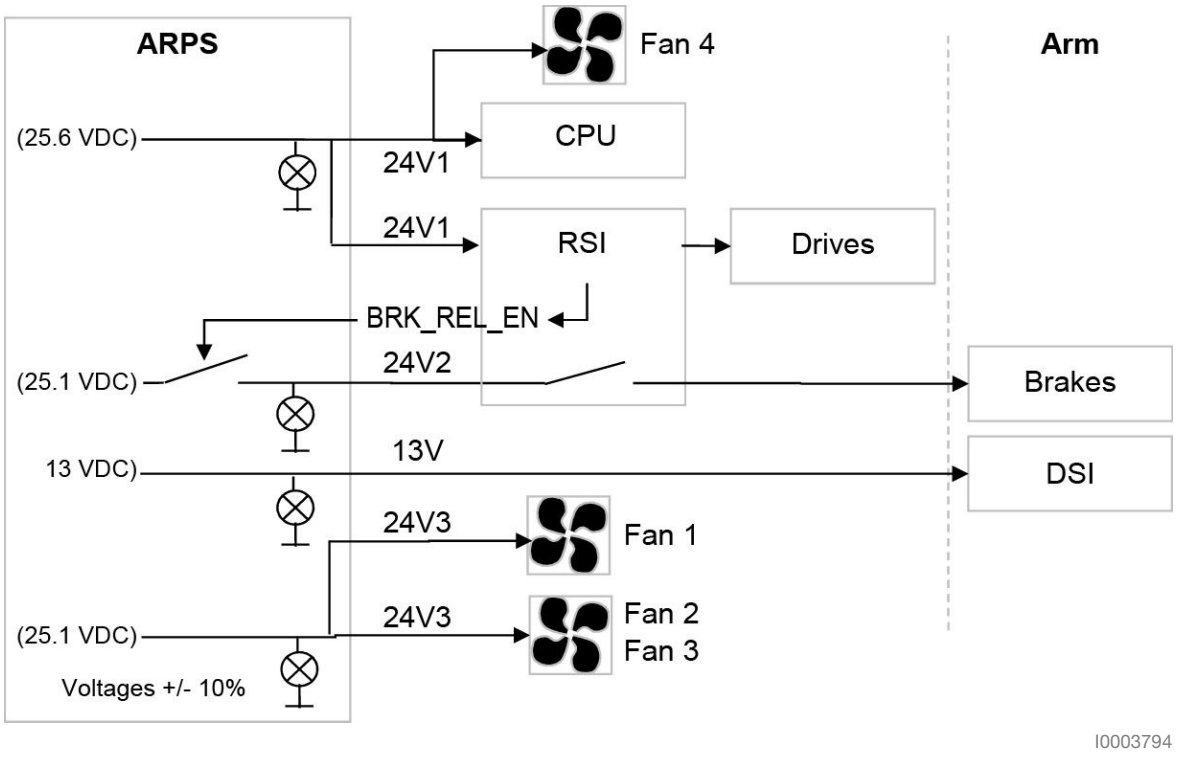


Figure 8.26

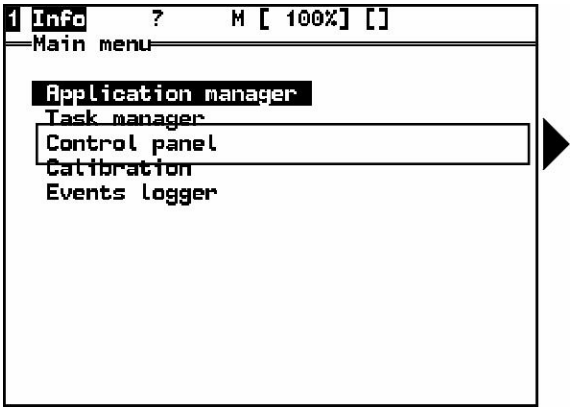
There are no brakes on TP80 robot. 24V2 voltage is used to drive a pump for axis 1 cooling system (refer to electrical drawings).

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8.5.5 - ALIM_OK SIGNALS

8.5.5.1 - Description

M0000381.1



- Screen display:
  - Controller status
    - I/O
      - SystemIO
        - Inputs

        ...

        28 ALIM_OK

        ...

Figure 8.27

	Status	Signal from ARPS to RSI
28 ALIM_OK	ON	Normal status
	OFF	Fault

8.5.5.2 - Access

M0000424.1

See chapter 8.5.2.

8.5.5.3 - Trouble shooting

M0000535.1

Problem:

ALIM_OK status = OFF: There is a fault detected by ARPS and corresponding output is set to 0 V.

Solution:

1) It can be a short circuit on one output (see chapter 8.5.4).

2) Or it is an internal fault: Change the ARPS board.

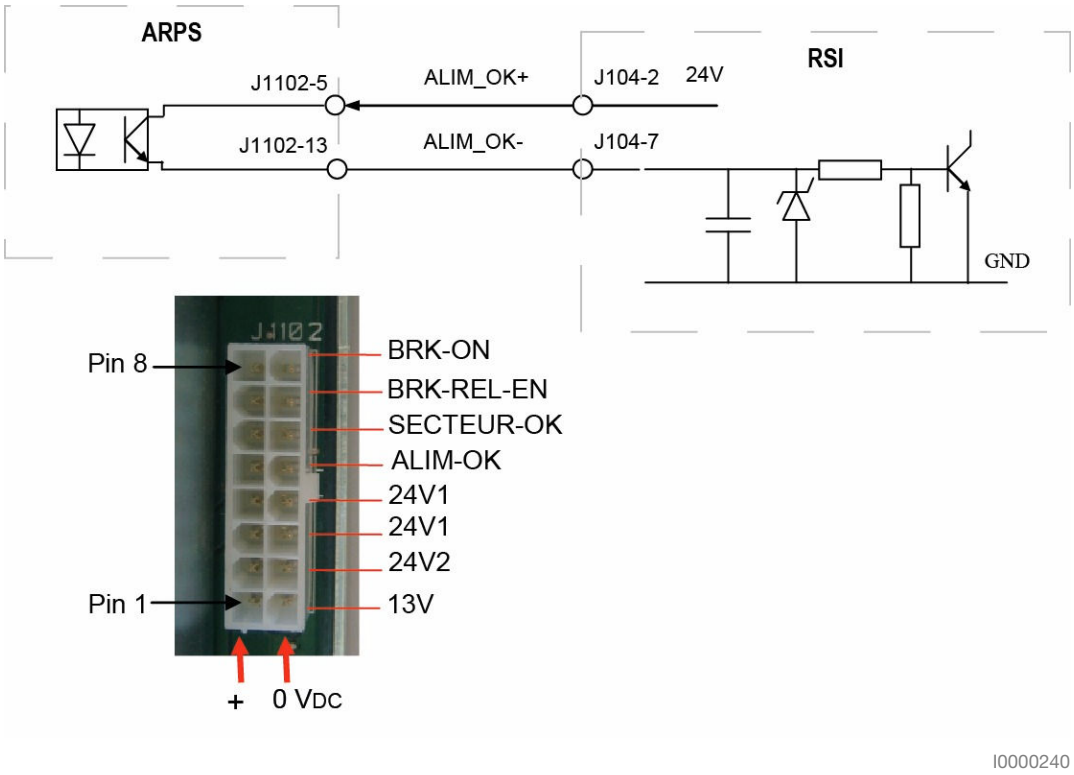
8.5.5.4 - Advanced information

M0000303.1

This output is a 24 VDC signal.

When there is no fault (status in control panel = ON) ALIM_OK signal is set to 24 V.

When there is a fault (status in control panel = OFF) ALIM_OK signal is set to 0 V.



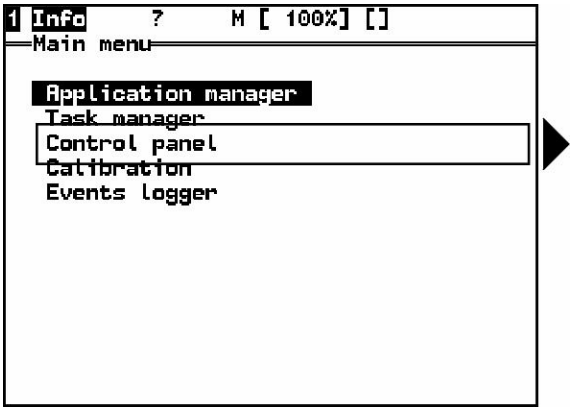
I0000240

Figure 8.28

8.5.6 - SECTEUR_OK SIGNALS

8.5.6.1 - Description

M0000305.1



- Screen display:
  - I/O
    - Controller
      - SystemIO
        - Inputs

          ...

          2 SECTEUR_OK

Figure 8.29

	Status	Signal from ARPS to RSI
2 SECTEUR_OK	ON	Normal status

This signal provides the status of input voltage of ARPS.

Normal status: Input voltage of ARPS is correct (>190 VAC).

Status = OFF: There is a power supply failure detected by ARPS.

Failure means that input voltage is <190 VAC for more than 20 ms.

In that case, 24V1 output is maintained for ~100 ms, 13V output is maintained for ~ 300 ms, there is no voltage backup on 24V2 and 24V3 outputs.

8.5.6.2 - Access

M0000424.1

See chapter 8.5.2.

8.5.6.3 - Trouble shooting

M0000536.1

Problem:

SECTEUR_OK status = OFF: As this condition shut off the controller, this status is recorded in the error logger.

Solution:

- Check input voltage to the ARPS.
- Check main voltage quality. A voltage drop can be caused by bad connections in the line or by too much of a power draw by the other equipment on the same line.

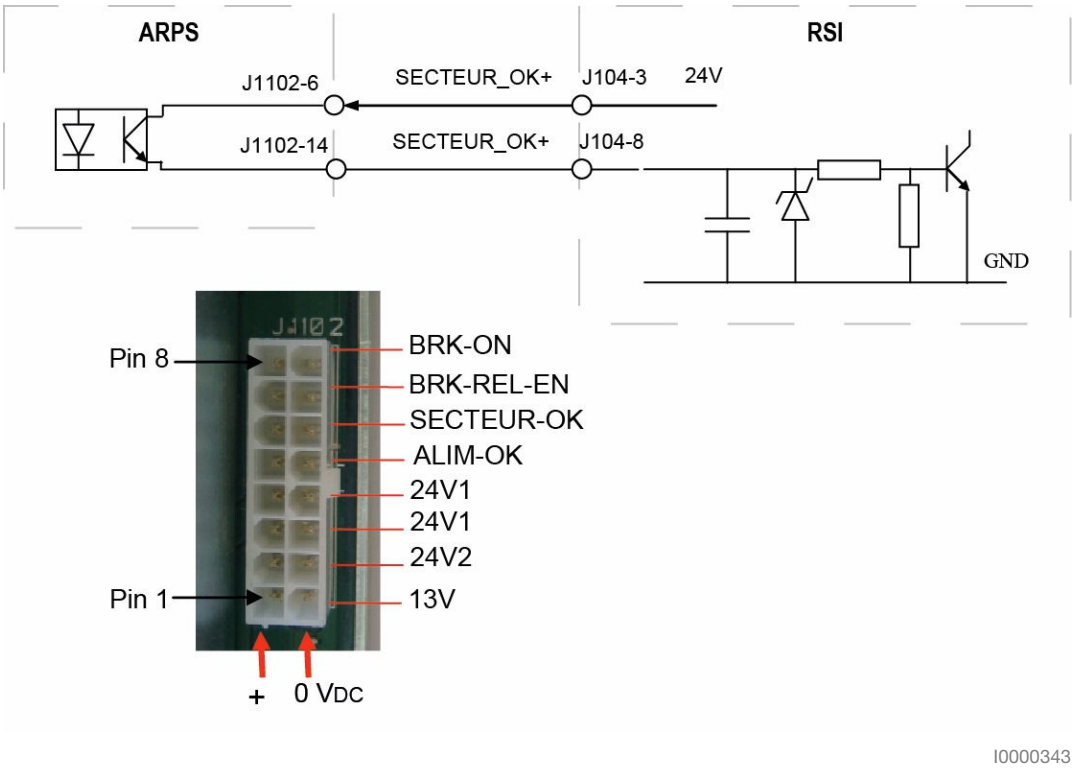
8.5.6.4 - Advanced information

M0000307.1

This output is a 24 VDC signal.

When there is no fault (status in control panel = ON) SECTEUR_OK signal is set to 24 V.

When there is a fault (status in control panel = OFF) SECTEUR_OK signal is set to 0 V.



I0000343

Figure 8.30

8.5.7 - BRK_X SIGNALS

8.5.7.1 - Description

M0000309.1

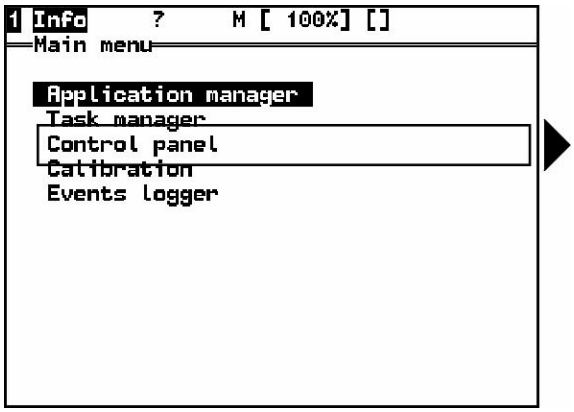


Figure 8.31

- Screen display:
  - Controller status
    - I/O
      - SystemIO
        - Inputs

          ...

          26 BRK_REL_EN

          ...
        - Outputs

          ...

          5 BRK_EN

          ...

	Arm Off	Arm On
26 BRK_REL_EN	OFF	**ON**
5 BRK_EN	OFF	**ON**

These signals provide status of command and feedback signals for the brakes. Brakes are off when arm power is disabled. Brakes are on when arm power is enabled.

These signals operate the same way in Manual Brake Release mode which is easier to test.

DANGER

- Refer to chapter 3.
- Pay attention to risks related to the size of the robot, the size of the payload, and so on when using Brake Release mode.

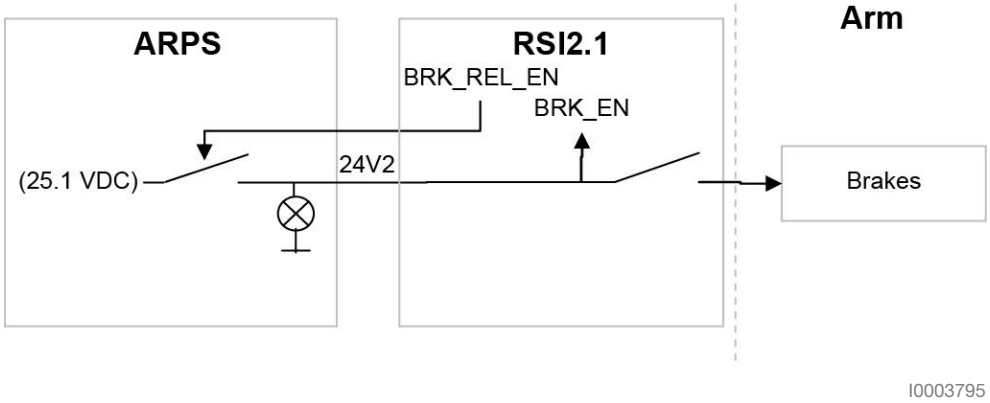


Figure 8.32

8.5.7.2 - Access

M0000424.1

See chapter 8.5.2.

8.5.7.3 - Trouble shooting

8.5.7.3.1 - Case 1

M0000537.1

Problem:

When trying to enable power on arm, 24V2 led on front of ARPS remains off. BRK_REL_EN remains off: no brake command issued to ARPS.

Solution:

- Change the RSI board.

8.5.7.3.2 - Case 2

M0000538.1

Problem:

When trying to enable power on arm, 24V2 led on front of ARPS remains off. BRK_REL_EN is on.

Solution:

- Change ARPS.

8.5.7.3.3 - Case 3

M0004432.1

Problem:

When trying to enable power on arm, 24V2 led on front of ARPS becomes on but brakes are not released.

Solution:

- The redundancy command on RSI is off:
  - Check wiring between ARPS and RSI.
  - Change the RSI board.

8.5.7.3.4 - Case 4 M0004433.1

Problem:

When trying to enable power on arm, 24V2 led on front of ARPS is on but brakes are not released.

Solution:

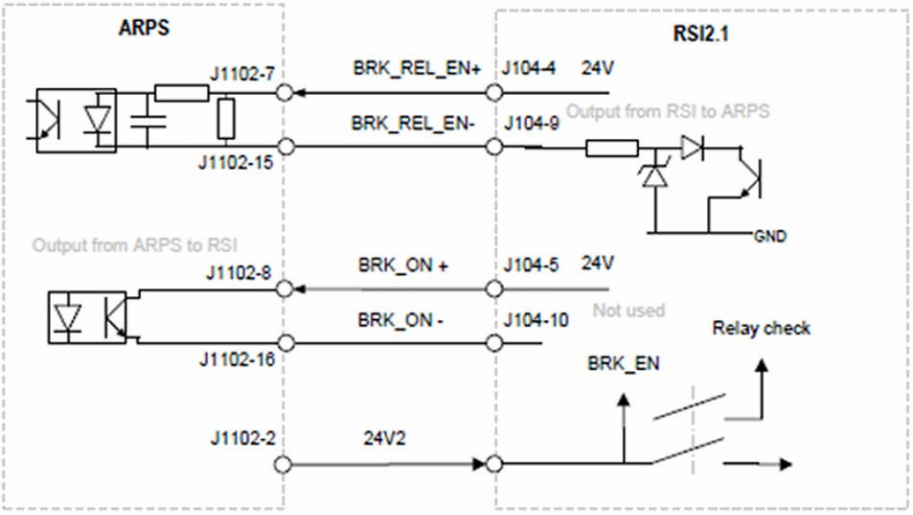
- The RSI has detected that ARPS provides a permanent 24V and consequently keep the redundancy command off: Change the ARPS board.

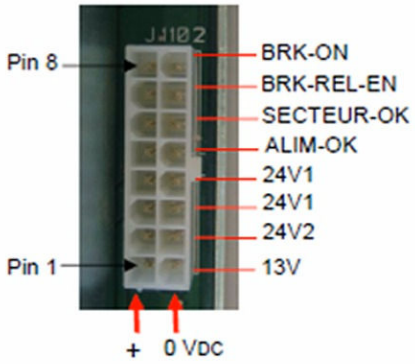
8.5.7.4 - Advanced information M0000312.1

Input and output are 24 VDC signals.

BRK_REL_EN is active (status in control panel = ON) when BRK_REL_EN signal is set to 0V.

In that case, 24V2 goes to 24V and its level is checked by RSI (BRK_EN signal) which authorize to close the relay contact.





I0003796

Figure 8.33

8.6 - RPS POWER SUPPLY

8.6.1 - DESCRIPTION

M0000314.1

RPS is the 325 VDC power supply for the drives. It is powered through D2 circuit breaker and contacts on the PS1, PS2 relays closed by the RSI board when arm is powered on. It has an overvoltage protection mechanism called regeneration.

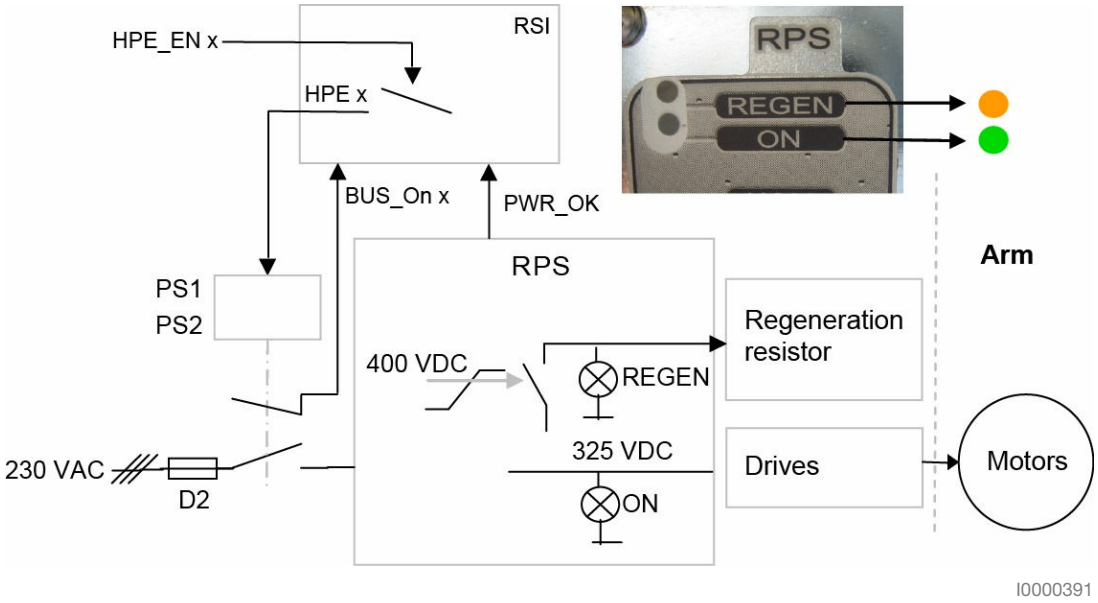


Figure 8.34

Normal operation:

Status	ON light	REGEN light
Arm not powered	OFF	OFF
Arm powered on	**ON**	OFF
Arm powered, motion with high deceleration	**ON**	**ON**

8.6.2 - ACCESS

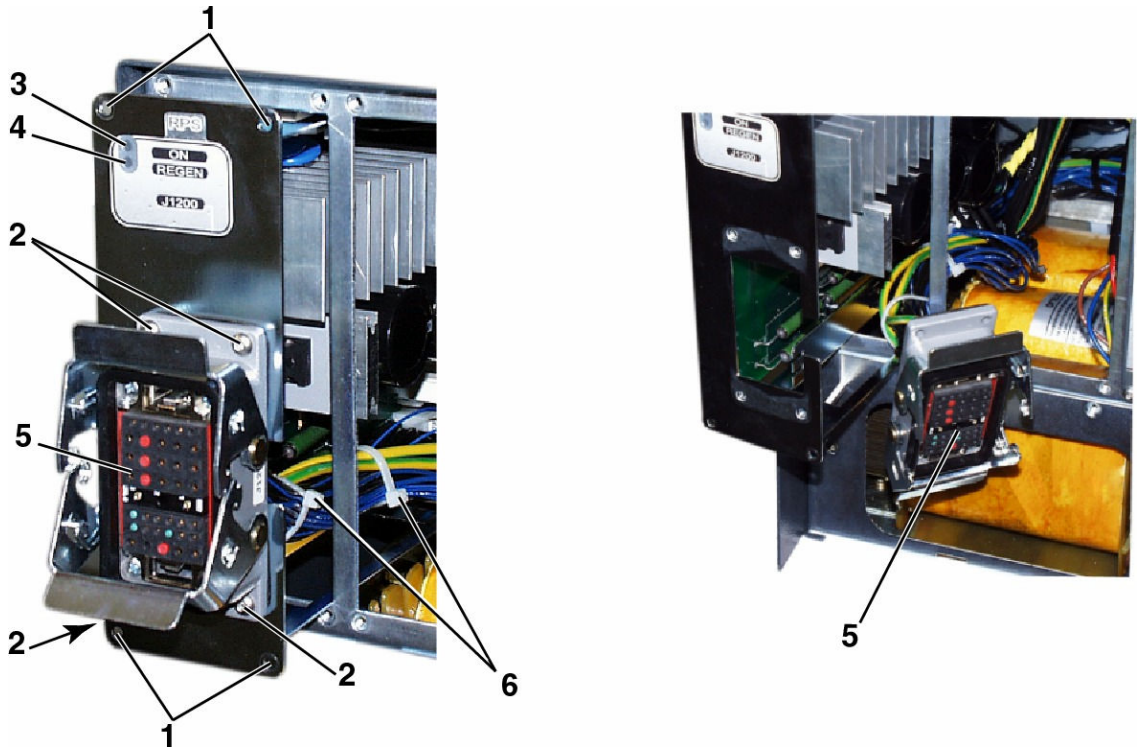
M0000364.1

- When the power supply to the arm is cut off, the output voltage is still present even when the indicator light goes off.
- Wait for at least 1 minutes before starting to work.

8.6.2.1 - "CS8C" for TX or TS robots

M0000429.1

- Set aside the ARPS power supply (see chapter 8.5.2).
- Remove the 4 screws **(1)** holding the RPS325 power supply.
- Remove the 4 screws **(2)** holding the J1200 **(5)** connector.
- Release the connector **(5)** and the cable clamps **(6)**.



I0000278

Figure 8.35

- When replacing the RPS325 power supply, make sure the wire **(12)** is correctly placed, in the same position as on the original power supply.

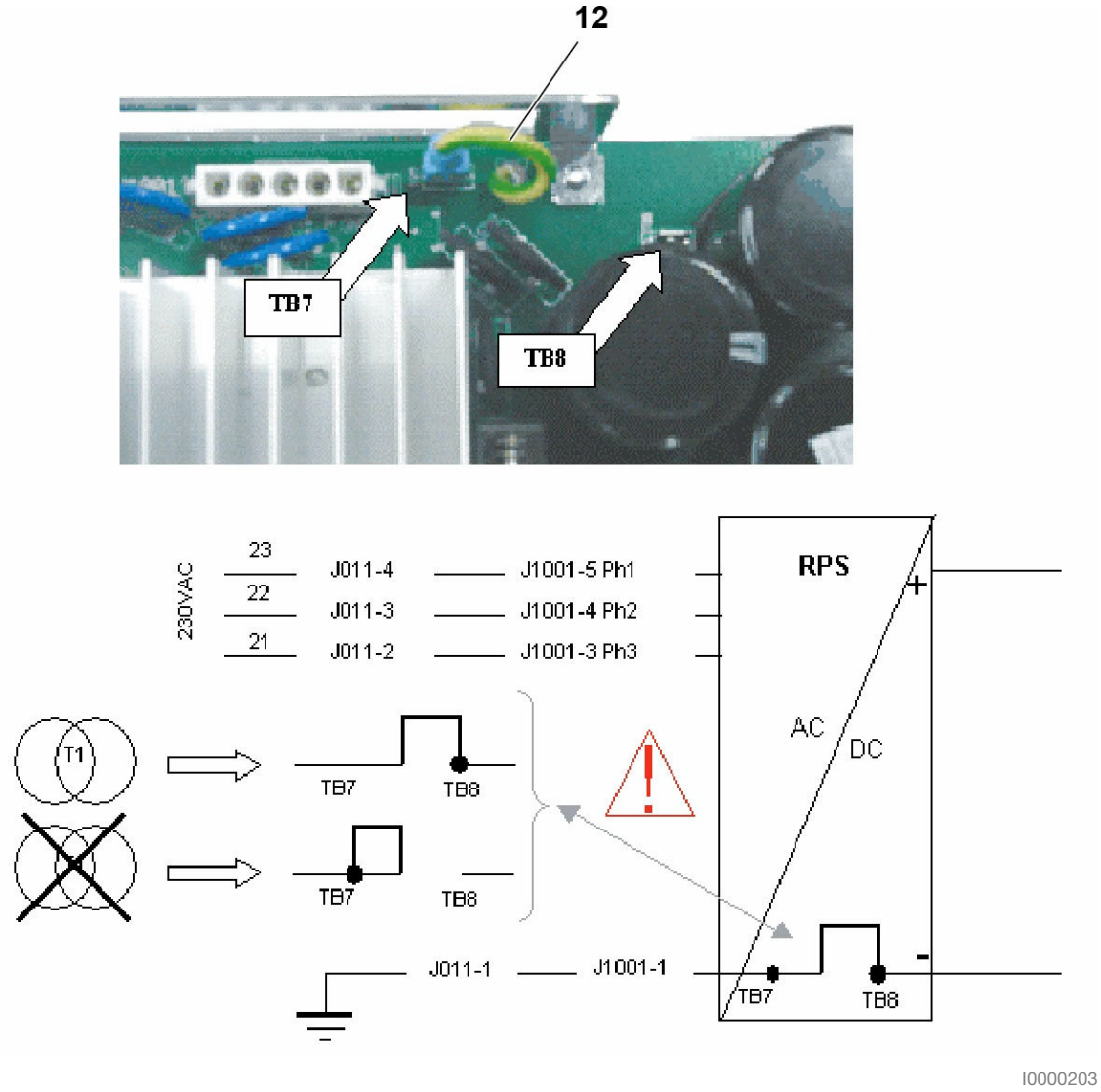
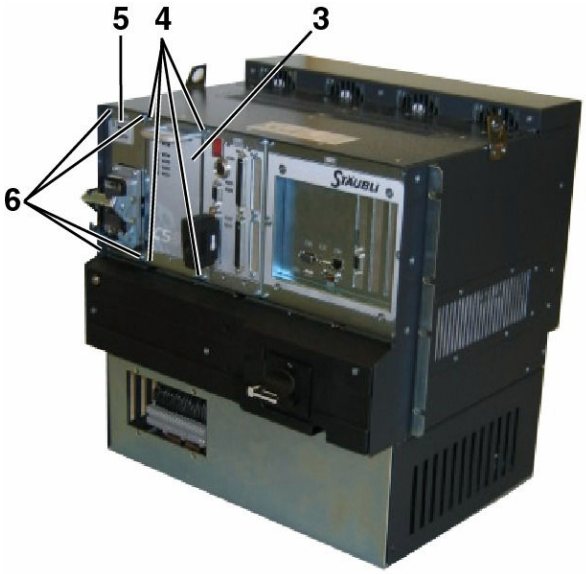


Figure 8.36

8.6.2.2 - CS8C for RX160, TP80, TX250 robots

M0000416.1

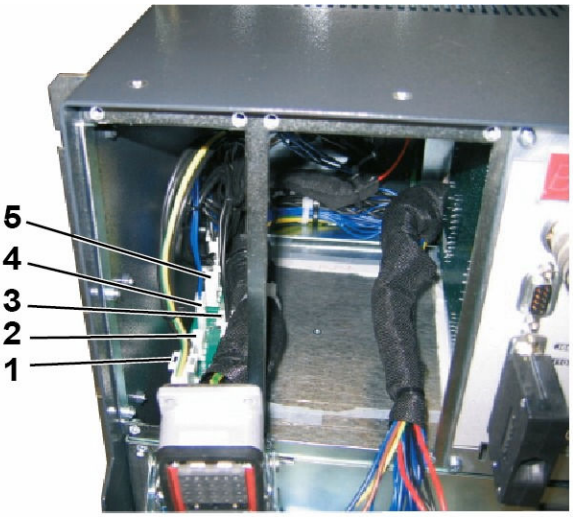
- To access the connectors, remove the ARPS **(3)** power supply wires held in place by 4 screws **(4)** (see chapter 8.5.2) and the metal plate **(5)** holding the socket and kept in place by 4 screws **(6)**.



I0000268

Figure 8.37

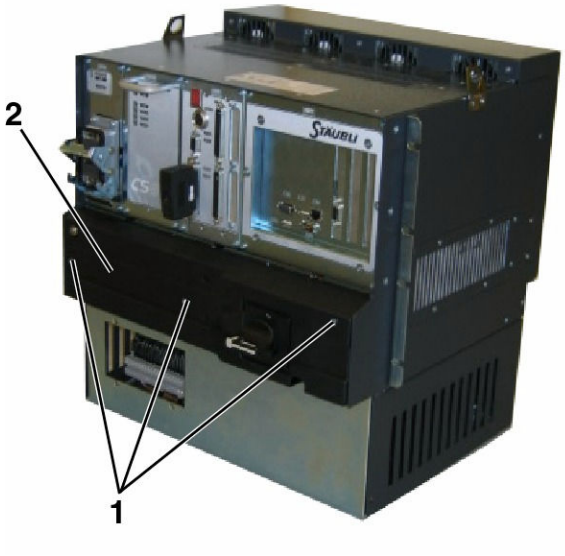
- Disconnect the connectors J1001 **(1)**, J1002 **(2)**, J1003 **(3)**, J1004 **(4)**, J1005 **(5)**.



I0000356

Figure 8.38

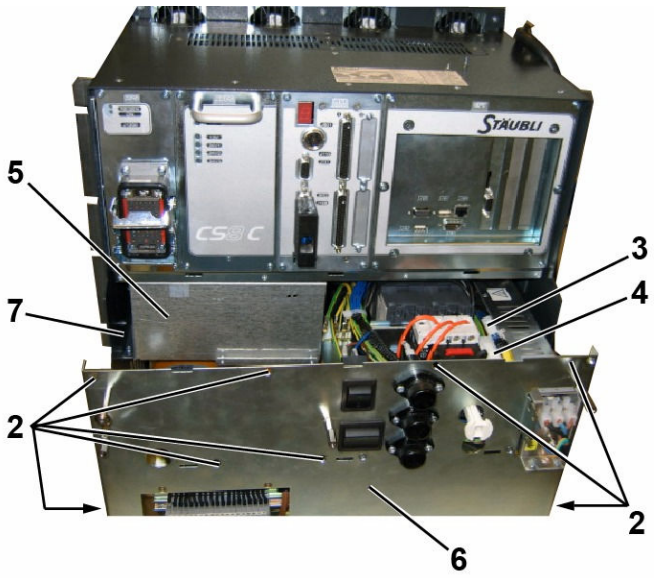
- Remove the plastic cover **(2)** held in place by 3 screws **(1)**.



I0000197

Figure 8.39

- Remove the 8 screws **(2)** holding the PSM **(6)** power module in place.



I0000213

Figure 8.40

The PSM is heavy; take all necessary precautions to avoid dropping it and to avoid making efforts in an incorrect position.

- Pull the PSM **(6)** forward to take it out. To remove it completely, disconnect connectors J010 **(3)** and J011 **(4)**.
- To access the RPS **(5)** power supply, lift it up in front to extract it from the fixing point **(7)**, and then pull it to free it from the rear lug.

8.6.3 - TROUBLE SHOOTING

8.6.3.1 - Case 1

M0000517.1

Problem:

ON indicator light remains off when arm is powered.

Solution: Step 1

- Check D2 circuit breaker inside PSM:
  - If it is off, turn it on. It may be due to a temporary overload on power part.
  - If it comes back to off position, change D2.
  - If its light remains off when it is turned on:
    - Check fuses F1, F2, F3.
    - Change D2.

Solution: Step 2

- Check the operation of PS1/PS2: Output command signals and feedback information can be checked from MCP Control Panel and RSI 7-segments error codes.

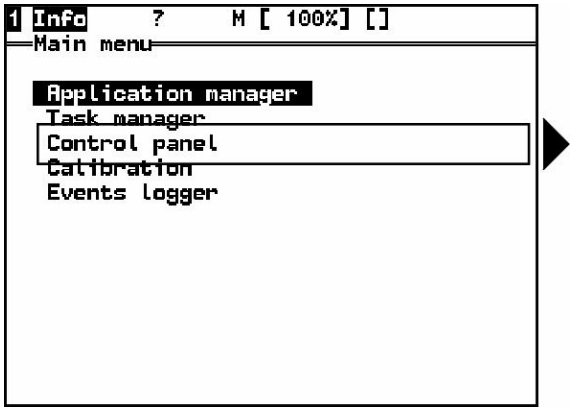


Figure 8.41

- Screen display:
  - Controller status
    - I/O
      - SystemIO
        - Inputs

          ...

          4 HPE2 = On

          5 HPE1 = On (Step 2)

          ...

          17 BUS_ON1 = On

          18 BUS_ON2 = On (Step 3)
        - Outputs

          0 HPE1_EN = On

          1 HPE2_EN = On (Step 1)

          ...

Step 2:
Enable power command applied to PS1 and PS2 relays
(RSI error codes o, P)

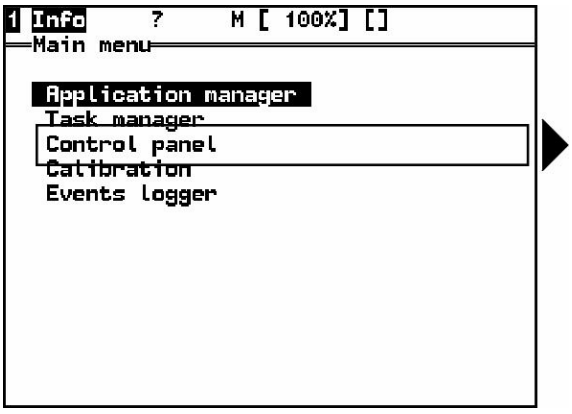
Step 3:
Feedback signals from PS1 and PS2 relays
(RSI error codes L, n)

Step 1:
Command signals to enable power on arm
(RSI error codes i, J)

HPEx_EN	HPEx	BUS_ONx	Action
Arm is Off			
OFF	OFF	OFF	Normal status when arm is Off
ON	OFF	-	Change the **"RSI"** board
OFF	**ON**	-	Change the **"RSI"** board
OFF	OFF	**ON**	Bad connection in the wiring between **"RSI"** and PS1/PS2 or PS1/PS2 defective
Arm enable power sequence requested			
OFF	OFF	OFF	Change the **"RSI"** board
ON	OFF	OFF	Change the **"RSI"** board
ON	**ON**	OFF	PS1/PS2 relays are not operating or there is a bad connection in the wiring between **"RSI"** and PS1/PS2
Arm is On			
ON	**ON**	**ON**	Correct status when arm is powered on

Solution: Step 3

- Check PWR_OK signal from MCP Control Panel.
- If PWR_OK signal is off, change RPS (PWR_OK becomes Off when output voltage of RPS is below 260VDC).



- Screen display:
  - Controller status
    - I/O
      - SystemIO
        - Inputs

          ...

          27 PWR_OK=

Figure 8.42

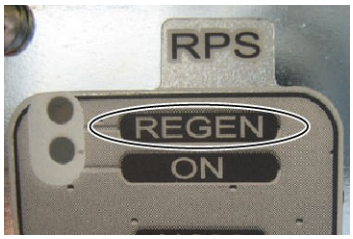
	Normal	Fault
27 PWR_OK=	**ON**	OFF

8.6.3.2 - Case 2

M0000518.1

Problem:

REGEN indicator light (yellow) remains on.



I0000238

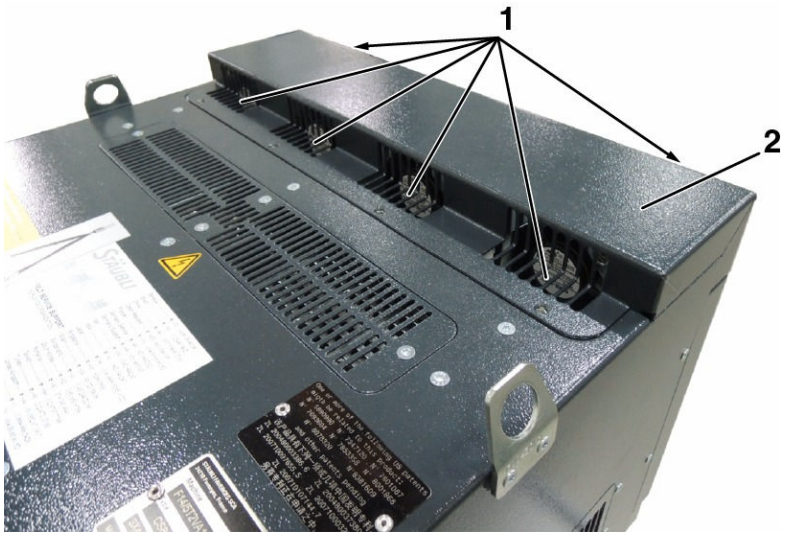
Solution:

- Change RPS and check that regeneration resistor has not been damaged.

Regeneration resistor is located at the back of CS8C.

Acces to the regeneration resistor:

- At the back of cabinet, remove the top cover **(2)** which is maintained with 6 screws **(1)**.



I0000393

Figure 8.43

DANGER

- The cover **(2)**, the resistor **(8)** and the amplifiers **(7)** may be very hot, especially in the event of a ventilation malfunction (see figure 8.21).
- This disassembly operation provides access to the regeneration resistor powered at 400 V during normal operation. It is essential to cut off all power supplies before carrying out this operation. Wait for at least 1 minutes before starting to work.

The regeneration resistor is good if its resistance is 22 ohm (±15%).

If it needs to be changed, remove 7 screws **(6)** and disconnect the wires **(9)**.

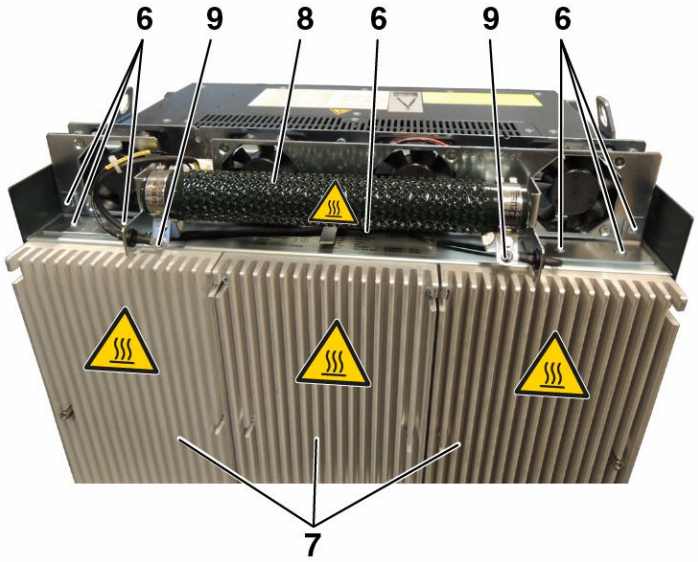


Figure 8.44

8.6.3.3 - Case 3

M0000532.1

Problem:

Not able to move the arm due to the error message "... under voltage ..." or " ... envelop error ...". The DC bus voltage applied to the amplifiers is too low or power requested from amplifier is too high.

Solution: Step 1

- Check that main voltage is correct according to identification plate of the controller.

  In case of 3-phase, it is important to verify voltages on the 3 phases.
- Check fuses F1, F3.
- Check D2 circuit breaker.

Undervoltage level = 50VDC, Overvoltage level = 440VDC.

Solution: Step 2

- If error occurs only when arm is accelerating very high:
  - Check that payload conforms to the arm specifications.
  - Reduce speed and/or acceleration parameters in the application program.

Solution: Step 3

- Check in manual brake release mode that there are no damages to the mechanics.

DANGER

- Refer to chapter 3.
- Pay attention to risks related to the size of the robot, the size of the payload, and so on when using Brake Release mode.

8.7 - RSI

M0000516.1

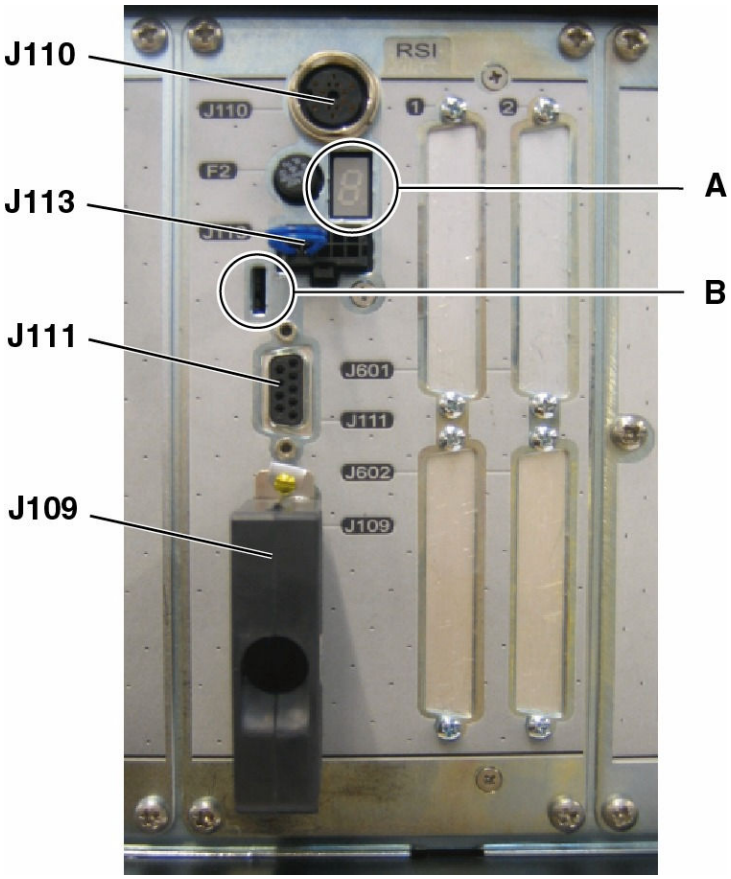
First CS8C controller generations were equipped with RSI1 board which changed to RSI2 from 2007. "RSI" applies for both RSI1 and RSI2 where RSI1 or RSI2 applies specifically to a dedicated board version.

8.7.1 - DESCRIPTION

M0000319.1

RSI board manages all hardware signals for safe operation of the robot.

RSI2 board front view:



I0000247

Item	Description
A	A 7-segment display indicates status of internal signals (see chapter 8.7.4).
B	Power status is indicated with green lights: D49: 24 V power supply D50: 5 V internal power supply D52: 5 V power supply from STARC2 board D74: Valve 2 through F5 fuse D73: Valve 1 through F6 fuse

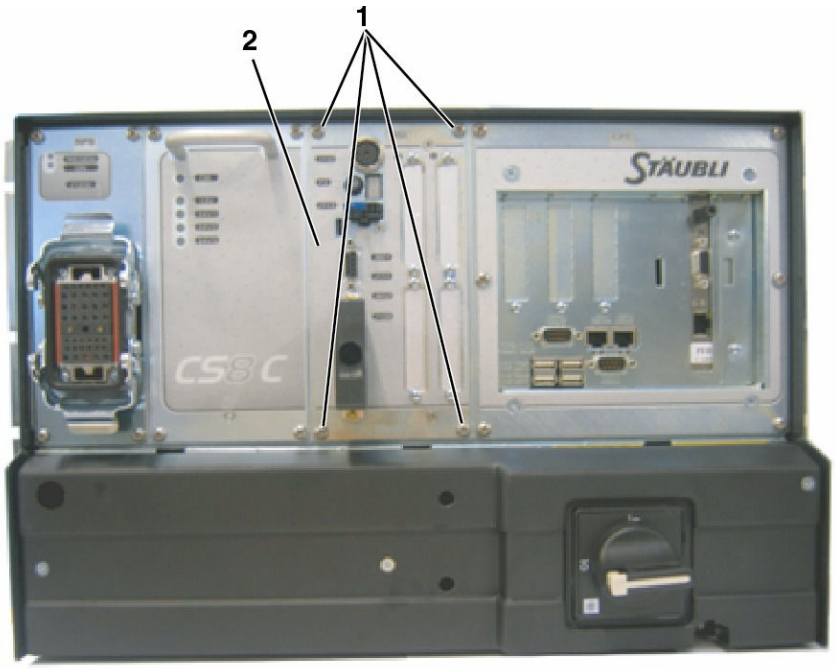
Refer to chapter 8.7.3 for trouble shooting.

Figure 8.45

8.7.2 - ACCESS

M0000320.1

Remove the 4 screws **(1)** and pull the RSI **(2)** board.



I0000276

Figure 8.46

8.7.3 - TROUBLE SHOOTING

8.7.3.1 - Case 1

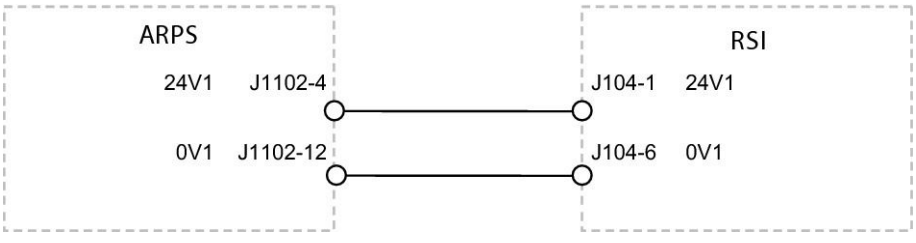
M0000519.1

Problem:

D49 remains off (24V1 from ARPS).

Solution:

- Check ARPS (see chapter 8.5).
- Check wiring between ARPS J1102 and RSI J104.



I0000179

Figure 8.47

- Change the RSI board.

8.7.3.2 - Case 2

M0000520.1

Problem:

D52 remains off (5V from STARC).

Solution:

- Check 5V on STARC (see chapter 8.8).
- Check wiring between STARC J302 and RSI J100.
- Change the RSI board.

8.7.3.3 - Case 3

M0000521.1

Problem:

D52 is on and D50 remains off.

Solution:

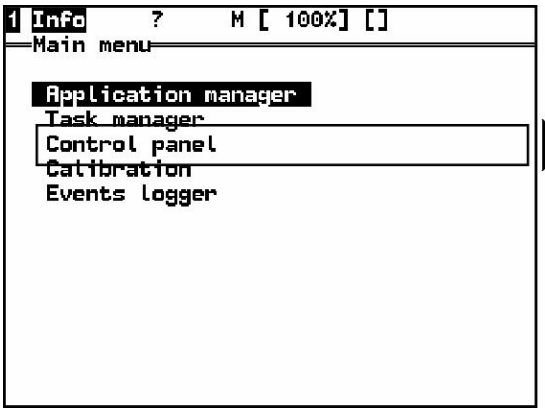
- Refer to STARC chapter 8.8 if there is an error message from STARC (STARC has the possibility to shut off 5V on RSI in case of major error).
- Check wiring between STARC J302 and RSI J100.
- Change the RSI board.

8.7.3.4 - Case 4

M0000522.1

Problem:

D73 (EV1) or D74 (EV2) remains Off when corresponding I/O on MCP is set to On.



- Screen display:
  - I/O
    - Controller
      - UserIO

        + Inputs

        - Outputs

        ..0 valve = on

        ..1 valve = on

Figure 8.48

0 valve = on	D73 = ON
1 valve = on	D74 = ON

Solution:

- Check F5 and F6 fuses on RSI. If fuses are blown, check that there is no short circuit in the wiring from RSI to the arm and no short circuit in the valves.
- Change the RSI board.

8.7.3.5 - Advanced information

M0000421.1

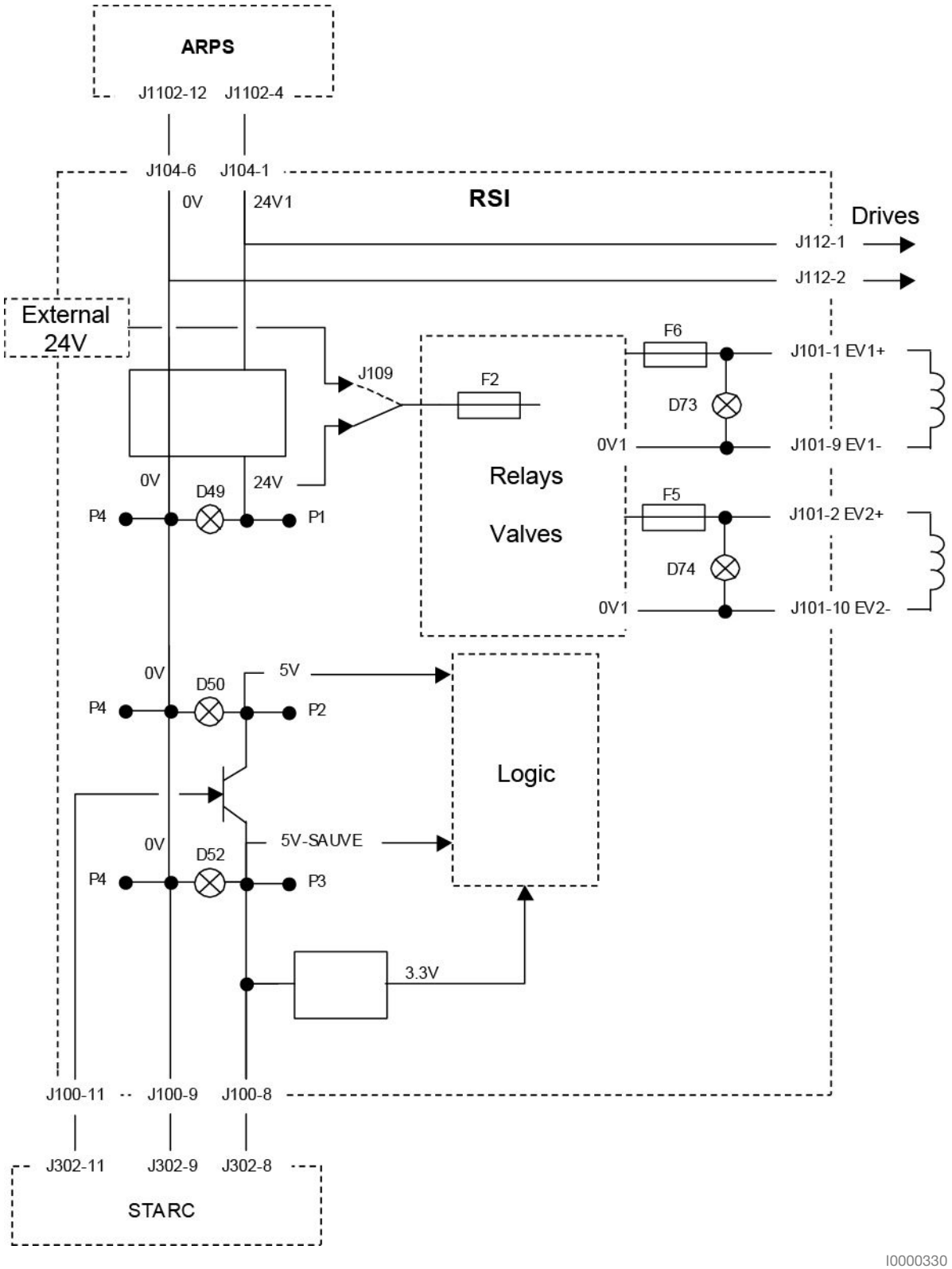


Figure 8.49

8.7.3.6 - Case 5

M0000523.1

Problem:

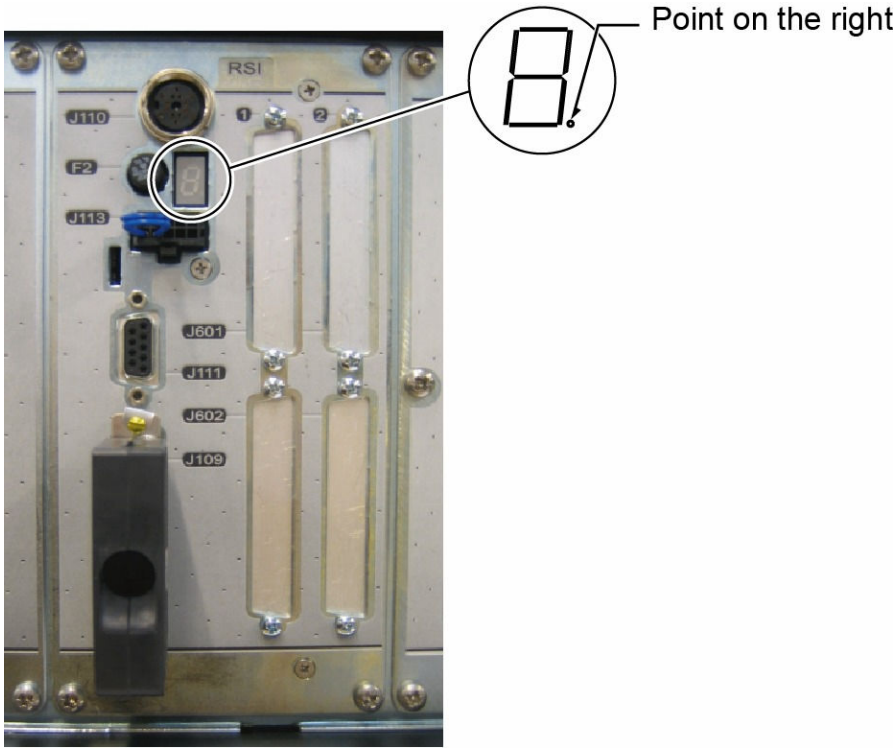
RSI display is off, no indication.

Solution:

- If D52 is off: (see chapters 8.4, 8.5 and 8.6).
- If D49, D50 and D52 are On:
  - Check that STARC board is operating (see chapter 8.8).
  - Change it if necessary.
  - Check wiring between STARC J302 and RSI J100.
  - Change the RSI board.

8.7.4 - ERROR CODES ON 7-SEGMENT DISPLAY

M0000433.1



I0000373

Figure 8.50

	Chapter			Chapter			Chapter
E - **E** flashing	8.7.4.1		8 **(8)**	8.7.4.14		U **(U)**	8.7.4.22
r - **r** flashing	8.7.4.2		9 **(9)**	8.7.4.14		y **(y)**	8.7.4.22
0 **(0)**	8.7.4.3		A **(A)**	8.7.4.15		i **(i)**	8.7.4.23
5 - 5 + right point flashing	8.7.4.4		a **(a)**	8.7.4.15		J **(J)**	8.7.4.24
i - **(i)**	8.7.4.5		b **(b)**	8.7.4.16		L **(L)**	8.7.4.25
0 **(0)**	8.7.4.6		C **(C)**	8.7.4.16		n **(n)**	8.7.4.25
1 **(1)**	8.7.4.7		c **(c)**	8.7.4.17		o **(o)**	8.7.4.26
2 **(2)**	8.7.4.8		d **(d)**	8.7.4.17		P **(P)**	8.7.4.26
3 **(3)**	8.7.4.9		E **(e)**	8.7.4.18		r **(r)**	8.7.4.27
4 **(4)**	8.7.4.10		E **(E)**	8.7.4.19		t **(t)**	8.7.4.28
5 **(5)**	8.7.4.11		F **(F)**	8.7.4.20		u **(u)**	8.7.4.29
6 **(6)**	8.7.4.12		H **(H)**	8.7.4.20			
7 **(7)**	8.7.4.13		h **(h)**	8.7.4.21			

When switching on controller, the RSI display goes through intermediate steps where the 7-segment and right point are alternatively flashing.

Display appearance	Status
E - **E** flashing	The RSI board is waiting to be synchronised with the STARC board
r - **r** flashing	The STARC board is waiting to be synchronised with the computer
5 - 5 + right point flashing	Waiting for the "Safety" task to start
- **(i)**	Waiting for the power supply to be switched on

This sequence lasts approximately 2 minutes.

Trouble shooting when display remains on intermediate error codes with 7-segment and point alternatively flashing.

8.7.4.1 - Case 1

M0000529.1

Problem:

E - **E** flashing The RSI board is waiting to be synchronised with the STARC board.

Solution:

- If it remains on this status:
  - Check STARC board (see chapter 8.8).
  - If STARC is OK:
    - Check wiring between STARC J302 and RSI J100.
    - Change the RSI board.

8.7.4.2 - Case 2

M0000524.1

Problem:

r - **r** flashing The STARC board is waiting to be synchronised with the computer.

Solution:

If it remains on this status, the issue is related to STARC board (see chapter 8.8).

8.7.4.3 - Case 3

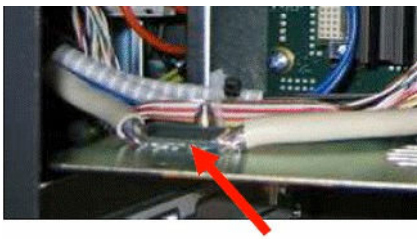
M0000525.1

Problem:

0 **(0)** NACK I2C error: Communication problem between the STARC and RSI boards.

Solution:

- Check wiring between STARC J302 and RSI J100. If this error occurs when arm is powered, pay attention to the correct shielding of this cable.



I0000250

Figure 8.51 : View of computer card cage

- This error can also be created from external noisy signals coming to the RSI board: EStop lines, I/O signals ...

8.7.4.4 - Case 4

M0000526.1

Problem:

5 - 5 + right point flashing Waiting for the "Safety" task to start.

Solution:

- If it remains on this status, change CPU board.

Changing a CPU board requires an intervention level 2 or 3 to set up parameters of the new CPU (level 2 if CPU is preconfigured, level 3 if not).

8.7.4.5 - Case 5

M0000330.1

When initialisation ends up properly, display is:

. - **(i)** Waiting for the power supply to be switched on.

From this step, left point is flashing fast, which means the board is properly inilialized. In all above cases, the point is flashing fast.

8.7.4.6 - Case 6

M0000530.1

Problem:

0 **"(0)"** Regeneration thermoswitch: Contact J1303, 3-4 open.

Solution:

- RSI1 case: Either J117 on RSI or J1303, 3-4 at the back of CS8C are defective (bad contacts) (the regeneration thermoswitch is not implemented).
- RSI2 case: This input does not exist.

8.7.4.7 - Case 7

M0000533.1

Problem:

1 **(1)** RSI board power supply problem: J104, 1-6 or internal power supplies.

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Solution:

RSI is powered from ARPS 24V1 output.

- If 24V1 led on ARPS is Off:
  - See chapter 8.5.
- If 24V1 led on ARPS is On:
  - Check wiring between ARPS J1102 and RSI J104.

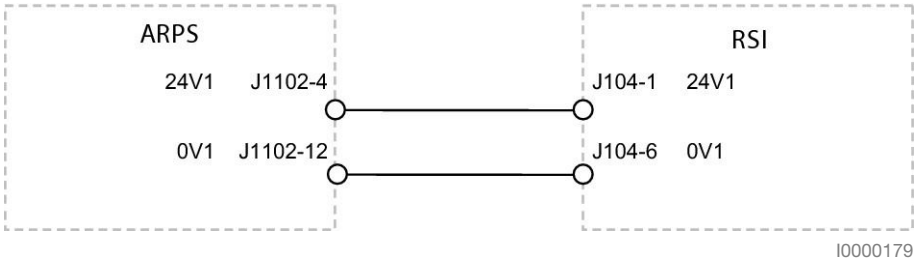


Figure 8.52

- If wiring is OK, change RSI.

8.7.4.8 - Case 8

M0000527.1

Problem:

2 **"(2)"** Watch Dog open: Computer failure.

Solution:

- A major failure has been detected on the CPU:
  - Reboot controller.
- If this error code remains:
  - Change the CPU board.

Changing a CPU board requires an intervention level 2 or 3 to set up parameters of the new CPU (level 2 if CPU is preconfigured, level 3 if not).

8.7.4.9 - Case 9

M0000528.1

Problem:

3 **"(3)"** Electric stop activated in the arm: J101, 5-12

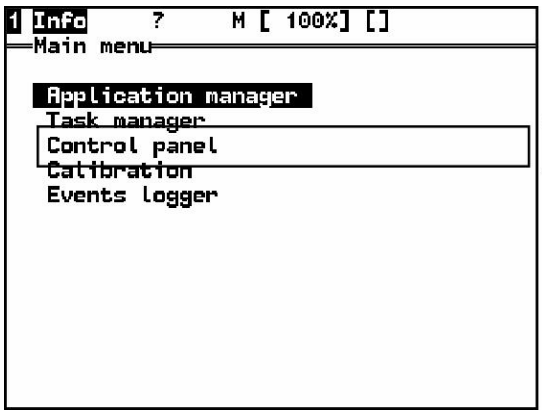
Solution:

An open electric limit switch condition has been detected.

Information 1:
The presence and number of electrical limit switches depends on arm type (see figure 8.54).

Information 2:
With RSI1 board, limit switch circuit is powered from 24V1.
With RSI2 board, limit switch circuit is powered from 24Vfus.

- Check first F2 fuse on front of RSI2 (refer to error code #7).
- Check that arm is inside the defined range through MCP Control Panel information.



- Screen display:
  - Controller status

    ...
  - Joint position

    J1: xx J2: xx J3: xx

    J4: xx J5: xx J6: xx

    ...

Figure 8.53

The maximum angular range is defined in the robot product characteristics.

- If arm is inside its predefined range, check wiring inside the arm.

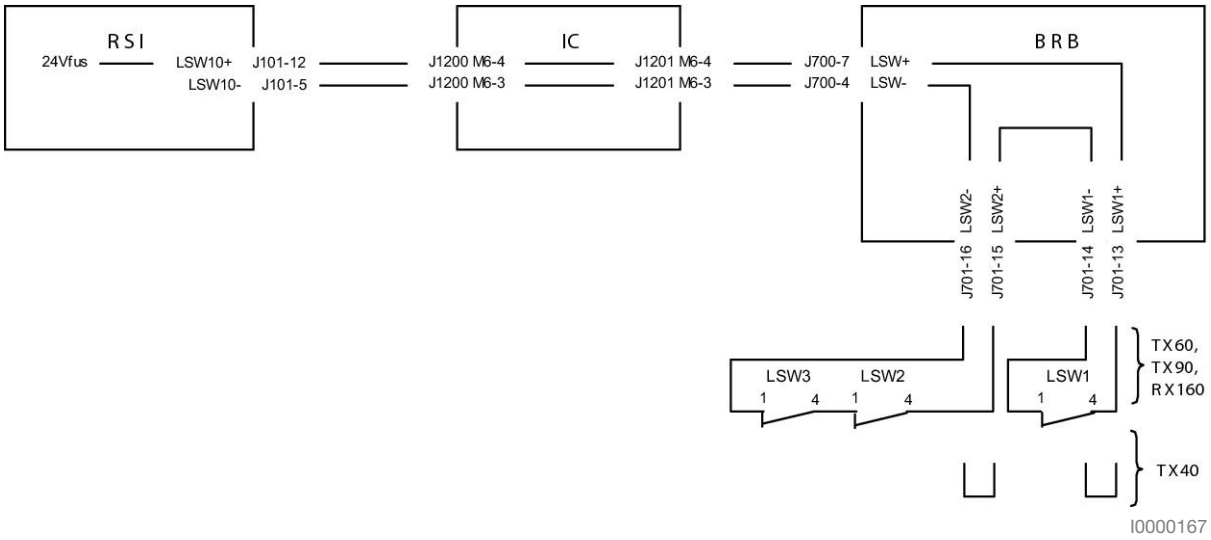


Figure 8.54

- Check wiring from RSI to the arm base (mainly the interconnection cable).
- If wiring is OK, change RSI board.

8.7.4.10 - Case 10

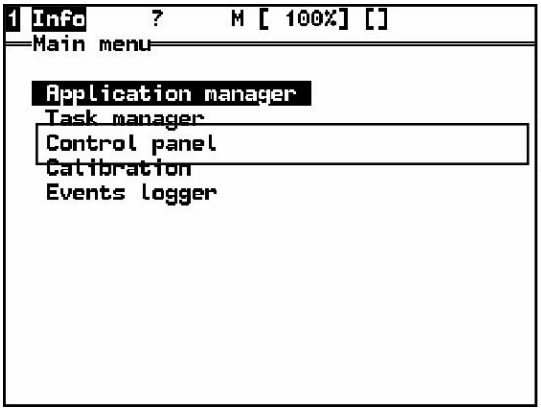
M0000534.1

Problem:

Ч "(4)" Amplifier fault: Contact DF open on J112, 3-4. The events history can be used to find the number of the amplifier concerned, and it provides further information on the fault. The control panel can also be used to display the status of the various variable speed drives.

Solution:

- Before checking faults on drives, check first that STARC board is operating properly (see chapter 8.8).
- Check on MCP Control Panel which drive is faulty and change the drive.



- Screen display:
  - Controller status

    ...
  - Drives information

    Drive 1: Disabled

    Drive 2: Error

    ...

    Drive 6: Disabled

Figure 8.55

- If there is no fault on drives, check wiring between drives and RSI.

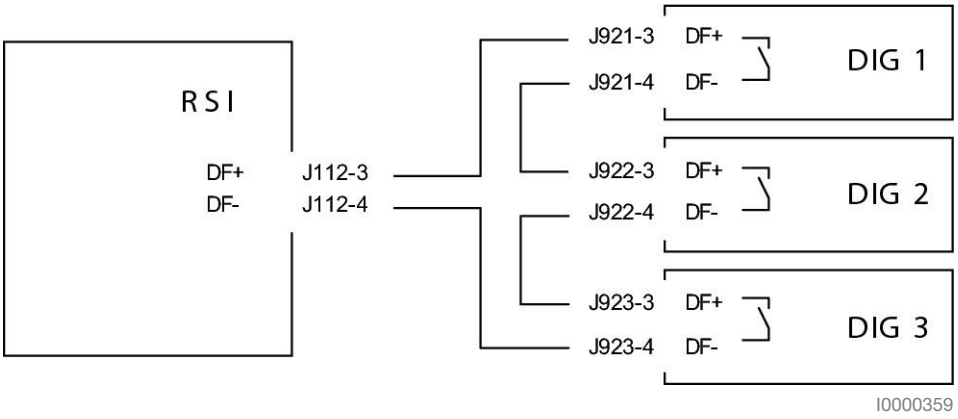


Figure 8.56

8.7.4.11 - Case 11

M0000514.1

Problem:

5 (5) ShortR-EN command not activated: The safety management task is not operational (computer). This error is displayed if the software has detected a blocking fault on start-up. The controller thus has no working mode, and the events history provides details on the fault detected (invalid arm configuration, etc.).

Solution:

- Reboot controller.
- If error remains, check events history in the error logger.

This signal can be activated from the CPU in case of malfunction of brakes. If it is the case:

- Check ARPS and RSI, especially the brake command signals (see chapter 8.5).

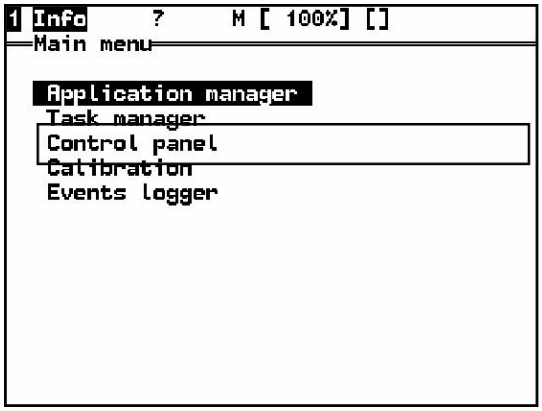


Figure 8.57

- Screen display:
  - Controller status
    - I/O
      - SystemIO
        - Inputs

          ...

          26 BRK_REL_EN

          ...
        - Outputs

          ...

          5 BRK_EN

          ...

	Arm Off	Arm On
26 BRK_REL_EN	OFF	**ON**
5 BRK_EN	OFF	**ON**

- If there is no error related to the brakes, check STARC and CPU boards (see chapter 8.8).

8.7.4.12 - Case 12

M0000513.1

Problem:

6 **"(6)"** The ShortR-EN signal is not taken into account by the RSI board.

Solution:

- Check the error logger of the controller. If errors starting with "STARC-Safety..." appear, there is an issue with software safety controls. Please contact the Stäubli Robotics Customer Support.

8.7.4.13 - Case 13

M0000512.1

Problem:

7 **"(7)"** Fuse F2 unserviceable or no 24V supply on J109-37.

Solution:

Internal voltage (24Vfus) for EStop lines is off.

- If EStop lines are powered from CS8C:
  - Check that there is no short circuit between EStop lines.
  - Change F2 fuse.

- If EStop lines are powered from an external device:
  - Check that external 24 V is provided between J109-19 (0 V) and J109-37 (+24 V).
  - Check that there is no short circuit between EStop lines.
  - Change F2 fuse.

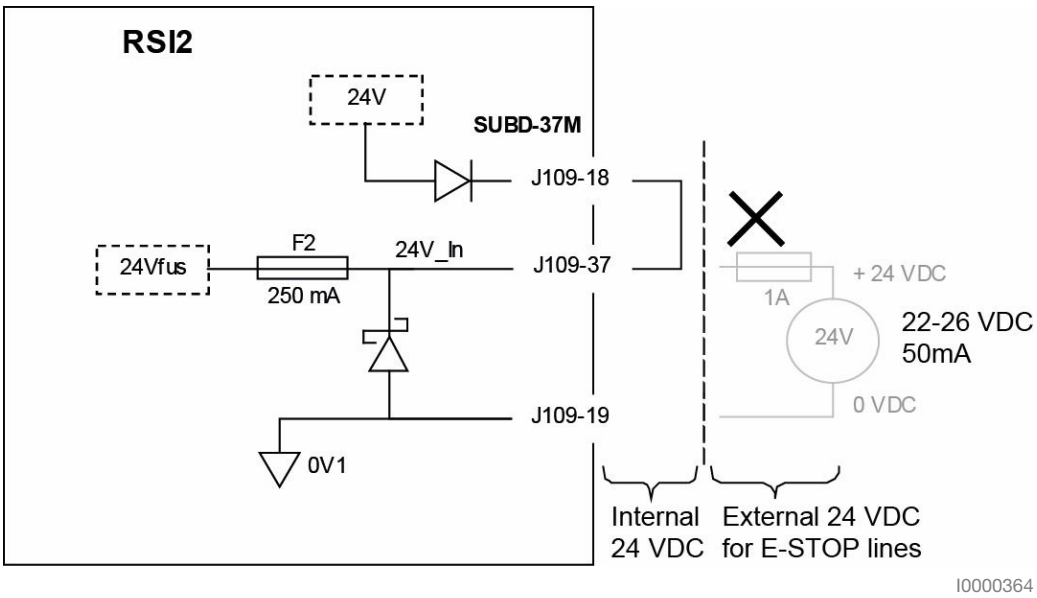


Figure 8.58

8.7.4.14 - Case 14

M0000511.1

Problem:

8 **"(8)"** MCP emergency stop 1 activated: Contact MCPES1, J110.
S-J or WMS emergency stop activated: Contact WMSES1, J113, 1-2.

9 **"(9)"** MCP emergency stop 2 activated: Contact MCPES2, J110.
T-U or WMS emergency stop activated: Contact WMSES2, J113, 6-7.

Solution:

There is one EStop contact open on MCP (Manual Control Pendant) or on WMS (Working Mode Selection) devices.

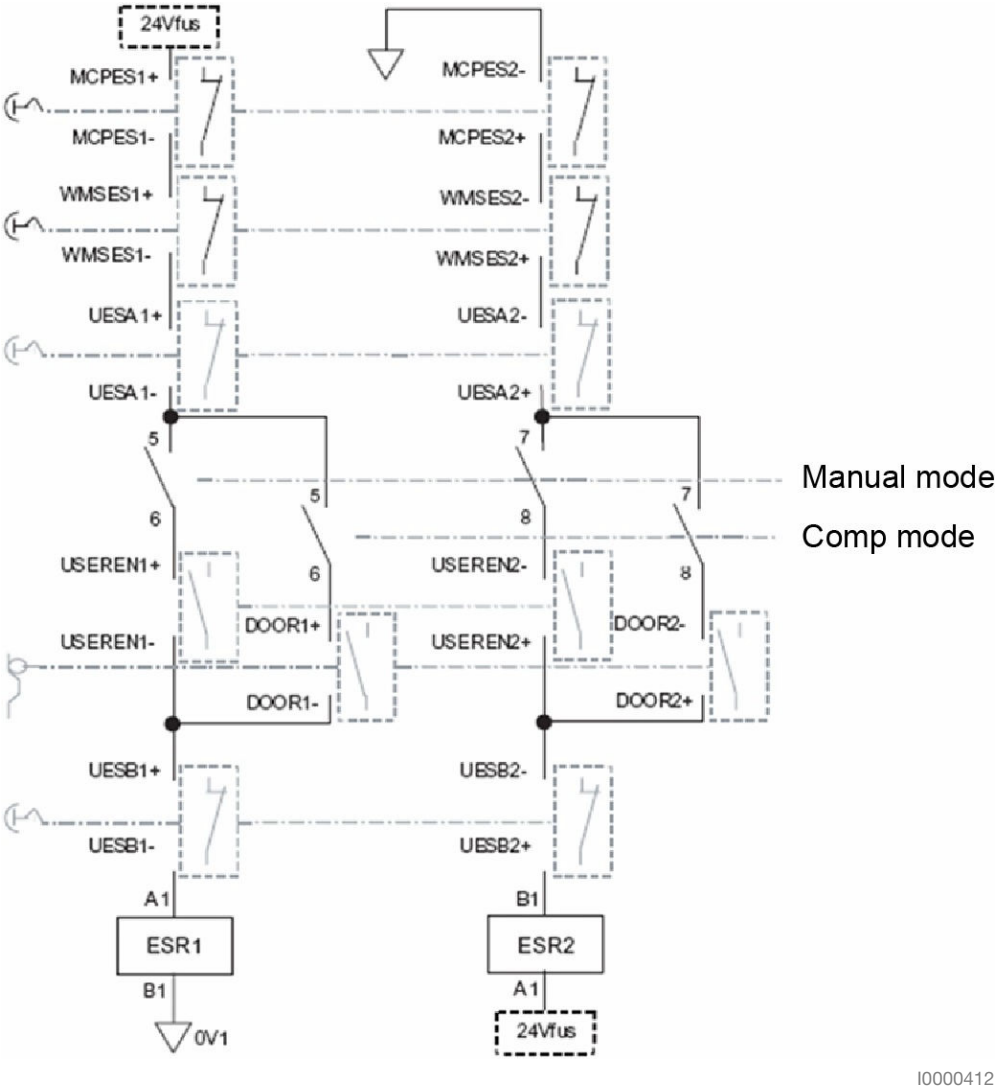


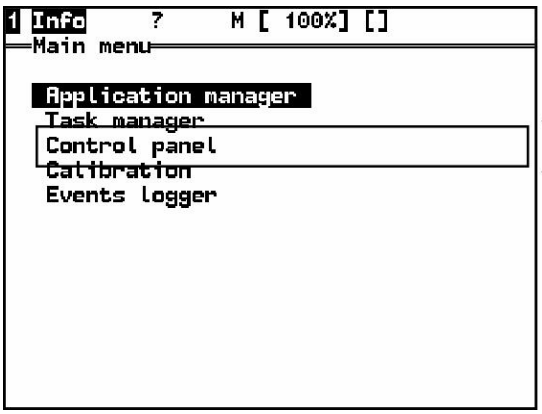
Figure 8.59

- MCP is connected on J110:
  - Check that MCP connector is properly plugged and screwed on J110.
  - Check that MCP EStop is released:
    - If error remains, disconnect MCP and replace it with J110 dummy plug provided with CS8C:
      - If error disappears, change MCP.
      - If error remains, change RSI.

DANGER

If only one EStop contact is operating, the robot will be stopped but unsafely. It is mandatory to repair the EStop line to get both contacts operating: Robot operation is not possible until the hardware fault is repaired (see chapter 6.15).

Status of MCP EStop push button can be obtained from MCP Control Panel:



- Screen display:
  - Controller status
    - I/O
      - SystemIO
        - Inputs

          ...

          20 MCPES1=

          21 MCPES2=

Figure 8.60

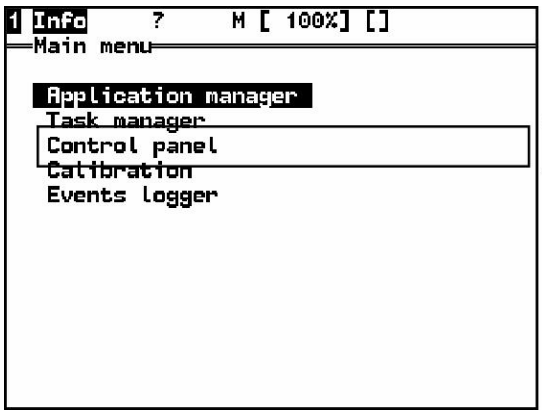
	EStop activated	EStop released
20 MCPES1=	**ON**	OFF
21 MCPES2=	**ON**	OFF

- WMS is connected on J113:
  - Check that WMS connector is properly plugged on J113.
  - Check that WMS EStop is released:
    - If error remains, disconnect WMS and replace it with J113 dummy plug provided with CS8C:
      - If error disappears, change WMS or its cable.
      - If error remains, change RSI.

DANGER

If only one EStop contact is operating, the robot will be stopped but unsafely. It is mandatory to repair the EStop line to get both contacts operating: Robot operation is not possible until the hardware fault is repaired (see chapter 6.15).

Status of WMS EStop push button can be obtained from MCP Control Panel:



- Screen display:
  - Controller status
    - I/O
      - SystemIO
        - Inputs

          ...

          39 WMSES1=

          42 WMSES2=

Figure 8.61

	EStop activated	EStop released
39 WMSES1=	**ON**	OFF
42 WMSES2=	**ON**	OFF

8.7.4.15 - Case 15

M0000510.1

Problem:

A (A) User emergency stop A1 activated: Contact UESA1, J109, 1-20.

a (a) User emergency stop A2 activated: Contact UESA2, J109, 2-21.

Solution:

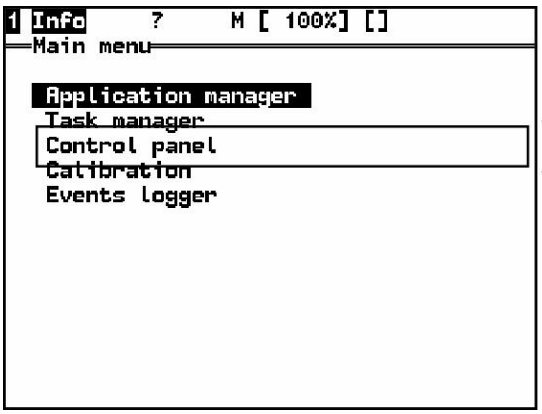
This EStop is coming from the cell and is connected to RSI J109.

- Check wiring from this EStop to J109 1-20 and 2-21.

DANGER

If only one EStop contact is operating, the robot will be stopped but unsafely. It is mandatory to repair the EStop line to get both contacts operating: Robot operation is not possible until the hardware fault is repaired (see chapter 6.15).

Status of UESA EStop push button can be obtained from MCP Control Panel:



- Screen display:
  - Controller status
    - I/O
      - SystemIO
        - Inputs

          ...

          22 UESA1=

          23 UESA2=

Figure 8.62

	EStop activated	EStop released
22 UESA1=	**ON**	OFF
23 UESA2=	**ON**	OFF

8.7.4.16 - Case 16

M0000509.1

Problem:

b **"(b)"** User emergency stop UEN1/DOOR1 activated: Contact USEREN1/DOOR1, J109, 3-22/9-28.

C **"(C)"** User emergency stop UEN2/DOOR2 activated: Contact USEREN2/DOOR2, J109, 4-23/10-29.

Solution:

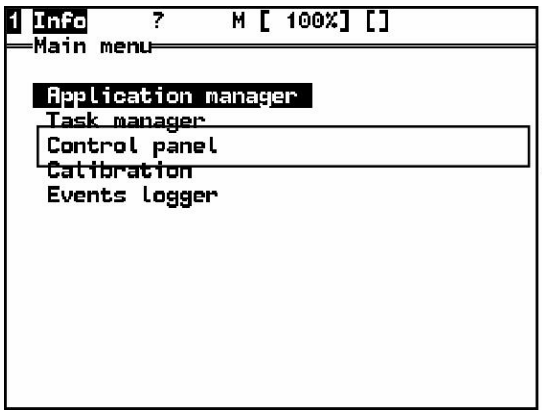
This EStop is coming from the cell and is connected to RSI J109. It is DOOR contact when robot is in COMP mode (automatic mode); it is USEREN contact when robot is in Manual mode.

- Check wiring from these EStop to J109 3-22, 4-23, 9-28 and 10-29.

DANGER

If only one EStop contact is operating, the robot will be stopped but unsafely. It is mandatory to repair the EStop line to get both contacts operating: Robot operation is not possible until the hardware fault is repaired (see chapter 6.15).

Status of UEN and DOOR can be obtained from MCP Control Panel:



- Screen display:
  - Controller status
    - I/O
      - SystemIO
        - Inputs

          ...

          0 UEN2-DOOR2=

          1 UEN1-DOOR1=

Figure 8.63

	Abnormal condition	Ready to operate
0 UEN2-DOOR2=	**ON**	OFF
1 UEN1-DOOR1=	**ON**	OFF

8.7.4.17 - Case 17

M0000508.1

Problem:

c (c) User emergency stop B1 activated: Contact UESB1, J109, 14-33.

d (d) User emergency stop B2 activated: Contact UESB2, J109, 15-34.

Solution:

This EStop is coming from the cell and is connected to RSI J109.

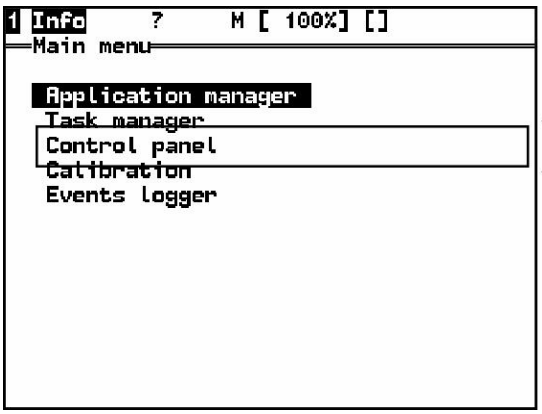
- Check wiring from this EStop to J109 14-33 and 15-34.

DANGER

If only one EStop contact is operating, the robot will be stopped but unsafely. It is mandatory to repair the EStop line to get both contacts operating: Robot operation is not possible until the hardware fault is repaired (see chapter 6.15).

Status of UESB EStop push button can be obtained from MCP Control Panel:

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- Screen display:
  - Controller status
    - I/O
      - SystemIO
        - Inputs

          ...

          24 UESB1=

          25 UESB2=

Figure 8.64

	EStop activated	EStop released
24 UESB1=	**ON**	OFF
25 UESB2=	**ON**	OFF

8.7.4.18 - Case 18

M0000507.1

Problem:

"(e)" Manual brake control selected: The rotary switch at the base of the arm is not set to 0

Solution:

- Check that rotary switch at the base of the arm is in position 0.
- If error remains, shut-off CS8C, disconnect J1201 plug at the base of the arm and check arm wiring with an ohm-meter (if the arm is equiped with a remote brake release box, this box shall be connected). Refer to electrical drawing set manual.

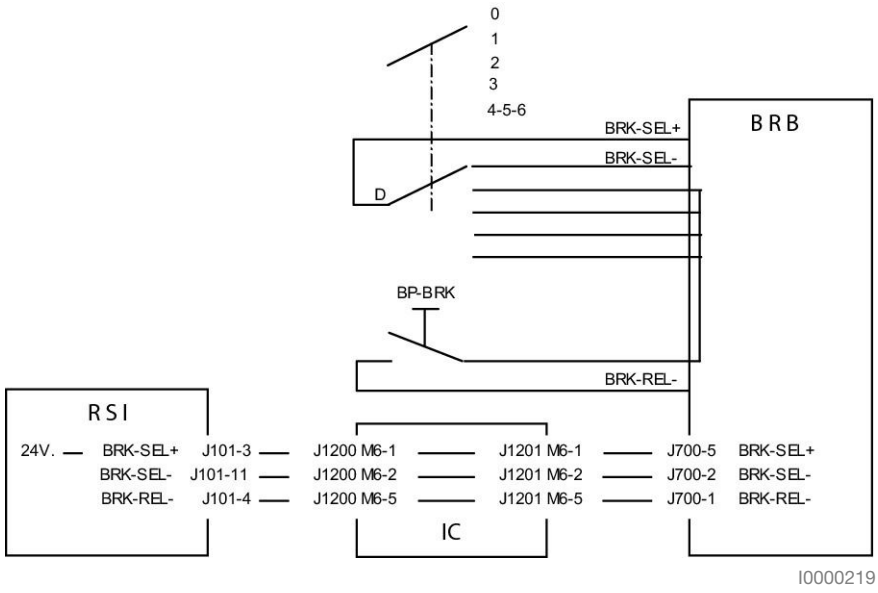


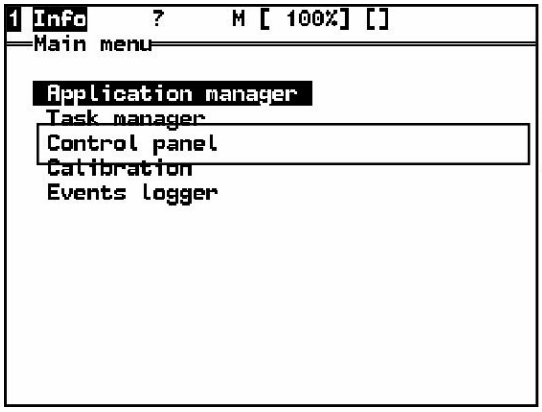
Figure 8.65

Test between	Rotary switch	
	Position 0	Other position
M6-1 and M6-2	Close	Open
M6-1 and M6-5	Open	Closed when push button pressed

- If the above check-up is not correct, change arm harness.
- If it is correct, repeat the same test from J200 (controller side) with J1201 connected on the arm.
- If the above check-up is not correct, change controller-to-arm cable.
- If it is correct, change RSI.
- For RX160, TP80 and TX250 arms, the manual brake release function is always located on a remote brake release box and the ohmeter test is not possible: replace the remote brake release box with another one. If error remains, change RSI.

Changing arm harness requires an intervention level 3.

Status of manual brake release switch can be obtained from MCP Control Panel.



- Screen display:
  - Controller status
    - I/O
      - SystemIO
        - Inputs

        ...

        10 BRK_SEL =

        ...

        15 BRK_REL =

Figure 8.66

	Position 0	Other position	Brake release
10 BRK_SEL =	OFF	**ON**	**ON**
15 BRK_REL =	OFF	OFF	**ON**
(RSI display)	**(i)**	**(e)**	**(E)**

8.7.4.19 - Case 19

M0000506.1

Problem:

"(E)" Manual brake control activated: One of the brakes has been released manually.

Solution:

- Refer to the case above.

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8.7.4.20 - Case 20

M0000505.1

Problem:

F "(F)" Manual brake control channel 1: The BRS 4-8 contact is not closed.

H "(H)" Manual brake control channel 2: The BRS 9-13 contact is not closed.

Solution:

- If manual brake release circuit is correct (see above), change RSI.

8.7.4.21 - Case 21

M0000504.1

Problem:

h **"(h)"** MCP: Enabling device or parking contact not activated.

Solution:

- Check that enabling device on MCP is correctly pressed or that MCP is properly inserted on its cradle.
- If OK, change MCP.

8.7.4.22 - Case 22

M0000503.1

Problem:

U **"(U)"** EStop 1 order memorised: An emergency stop on line 1 has been recorded by the RSI board. This fault is reset the next time the arm is powered up.

y **(y)** EStop 2 order memorised: An emergency stop on line 2 has been recorded by the RSI board. This fault is reset the next time the arm is powered up.

Solution:

- If this error code remains when next enable power sequence is initiated, change RSI.

DANGER

To ensure that the system is operating properly after this error has occurred, it is necessary to perform several EStop sequences with arm power enabled and robot stopped.

8.7.4.23 - Case 23

M0000345.1

When the robot is waiting for an enable power sequence, the display is:

' "(i)"

The following error codes are related to an enable power sequence.

8.7.4.24 - Case 24

M0000502.1

Problem:

ı (i) Power-up command 1 not activated by the computer.

ȷ (J) Power-up command 2 not activated by the computer.

Solution:

- Change the RSI board.

8.7.4.25 - Case 25

M0000501.1

Problem:

L (L) Power up sequence 1 not activated: No PS-ON1+ signal on J105-3.

n (n) Power up sequence 2 not activated: No PS-ON2+ signal on J105-4.

Solution:

- Change the RSI board.

8.7.4.26 - Case 26

M0000500.1

Problem:

o (o) Power up sequence 1 not effective: 0V on J105-7: relay PS1 not activated or faulty contact.

P (P) Power up sequence 2 not effective: 0V on J105-8: relay PS2 not activated or faulty contact.

Solution:

PS1/PS2 relays are not operating or there is a bad connection in the wiring between RSI and PS1/PS2.

The error codes above can be obtained from MCP Control Panel.

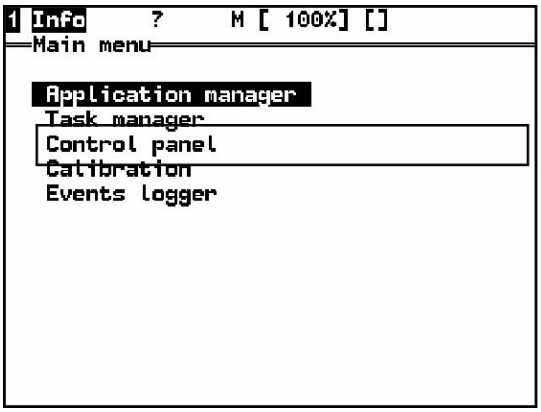


Figure 8.67

- Screen display:
  - Controller status
    - I/O
      - SystemIO
        - Inputs

          ...

          4 HPE2 = On

          5 HPE1 = On (Step 2)

          ...

          17 BUS_ON1 = On

          18 BUS_ON2 = On (Step 3)
        - Outputs

          0 HPE1_EN = On

          1 HPE2_EN = On (Step 1)

          ...

Step 2:
Enable power command applied to PS1 and PS2 relays
(RSI error codes o, P)

Step 3:
Feedback signals from PS1 and PS2 relays
(RSI error codes L, n)

Step 1:
Command signals to enable power on arm
(RSI error codes i, J)

HPEx_EN	HPEx	BUS_ONx	Action
Arm is Off			
OFF	OFF	OFF	Normal status when arm is Off
ON	OFF	-	Change the RSI board
OFF	**ON**	-	Change the RSI board
OFF	OFF	**ON**	Bad connection in the wiring between RSI and PS1/PS2 or PS1/PS2 defective
Arm enable power sequence requested			
OFF	OFF	OFF	Change the RSI board
ON	OFF	OFF	Change the RSI board
ON	**ON**	OFF	PS1/PS2 relays are not operating or there is a bad connection in the wiring between RSI and PS1/PS2
Arm is On			
ON	**ON**	**ON**	Correct status when arm is powered on

8.7.4.27 - Case 27

M0000499.1

Problem:

r (r) RPS power supply not activated: Contact J105, 1-2 open.

Solution:

In that case, the most common error displayed on MCP popup message is:

Internal error: DRIVE-BusUnderVoltage

Refer to RPS chapter (8.6) to verify F1-F3 fuses, 3-phase circuit breaker and RPS.

8.7.4.28 - Case 28

M0000498.1

Problem:

t (t) Braking command not activated by the computer.

Solution:

The enable arm power sequence has stopped because brake release command is not activated: Change the RSI board.

8.7.4.29 - Case 29

M0000497.1

Problem:

u (u) Power supply to the brakes not activated: Contact J104, 5-10 open.

Solution:

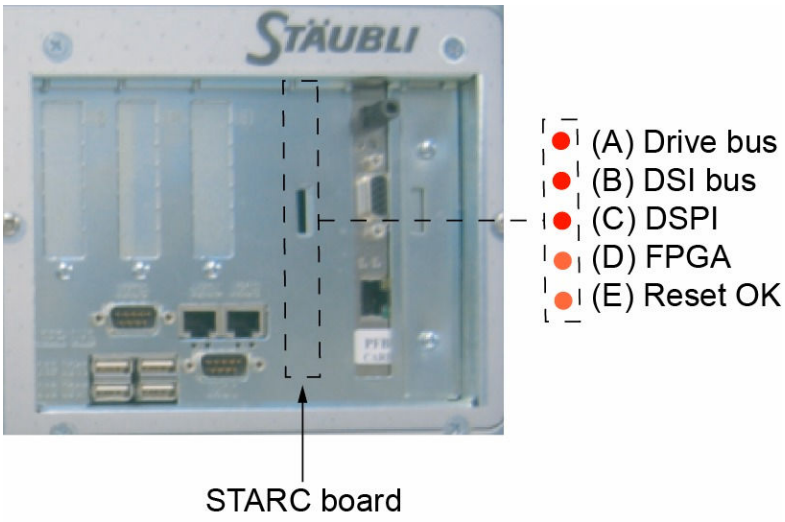
Refer to ARPS chapter (8.5) to differentiate between ARPS issue or RSI issue.

8.8 - STARC BOARD

8.8.1 - DESCRIPTION

M0000353.1

STARC board does motion control, communicating with encoders for axes positions and amplifiers for motors commands.



I0000243

Figure 8.68

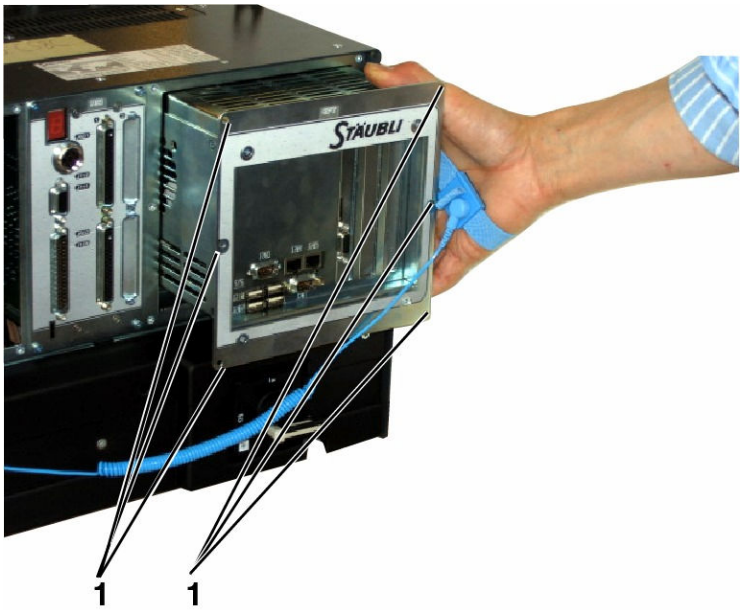
Leds A and B are blinking slowly (~ 1 s period) during boot sequence of controller.

They are blinking fast (~ 0.5 s period) when ready to operate.

8.8.2 - ACCESS

M0000354.1

- Turn off main switch S1 to position 0.
- Set aside the computer after removing the 6 fixing screws **(1)**.
- Remove the 6 screws **(2)** and open the computer card cage.

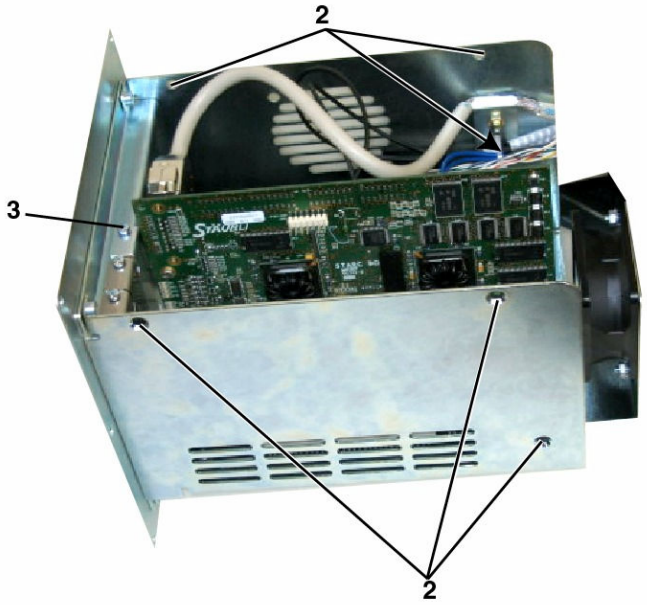


I0000291

Figure 8.69

DANGER
The heat-sink of the CPU board may be hot, especially in the event of a ventilation failure.

- The STARC board can be removed after taking out the fastening screw **(3)**.
- The PCI boards are locked on back edge with an adjustable plate. This plate can be re-ajusted if necessary when board is replaced.



I0000376

Figure 8.70

8.8.3 - TROUBLE SHOOTING

8.8.3.1 - Case 1

M0000496.1

Problem:

No lights at all on STARC.

- ○ (A)
- ○ (B)
- ○ (C)
- ○ (D)
- ○ (E)

Solution:

- Check first 24V1 on ARPS (see chapter 8.5).
- Check STARC internal power supplies leds.
  - If 5 V or 3.3 V led is off, change the CPU.
  - If one of the other leds is off, change the STARC.

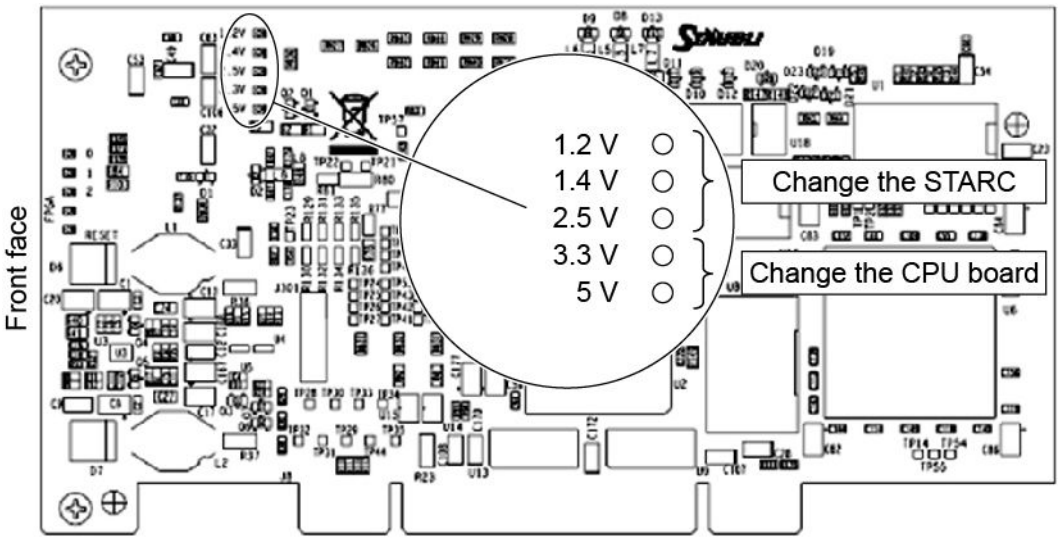


Figure 8.71

8.8.3.2 - Case 2

M0000495.1

Problem:

Led E is on, Led D is off: Initialization of the board is faulty (FPGA programming issue).

- ○ (A)
- ○ (B)
- ○ (C)
- ○ (D)
- ● (E)

Solution:

- Reboot controller.
- If problem remains, change STARC board.

8.8.3.3 - Case 3

M0000494.1

Problem:

Led E and D are on, Led C is off: STARC processor was not able to start.

- ○ (A)
- ○ (B)
- ○ (C)
- ● (D)
- ● (E)

Solution:

- Reboot controller.
- If problem remains, change STARC or CPU board.

Changing a CPU board requires an intervention level 2 or 3 to set up parameters of the new CPU (level 2 if CPU is preconfigured, level 3 if not).

8.8.3.4 - Case 4

M0000490.1

Problem:

Leds E, D and C are on, Led B is blinking slowly (~ 1s period) whatever the status is of led A (blinking slow or fast): There is a communication problem between STARC and the DSI in the arm base.

In that case, it is not possible to enable power on the arm.

○ (A)

● (B) Slow

● (C)

● (D)

● (E)

Solution:

When led B is blinking slowly, all encoders (sensor) and all PTC sensors are not operating. The information displayed on MCP control panel is the following:

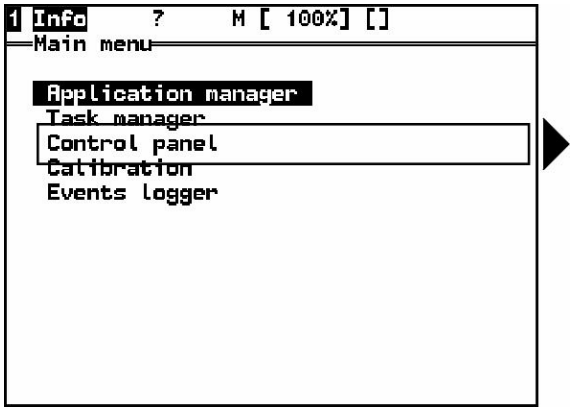


Figure 8.72

- Screen display:
  - Controller status

    ...
  - Sensor status
    - Sensor 1: Failed
    - Sensor 2: Failed

      ...
    - Sensor 6: Failed
  - PTC sensor
    - PTC sensor 1 = ?
    - PTC sensor 2 = ?

      ...
    - PTC sensor 6 = ?

It is a generic problem which is not likely due to a specific encoder or a specific PTC sensor.

- Either the communication through optical fibber between STARC and DSI is not operating:
  - Change interconnection cable from CS8C to arm.
- Or the DSI board in the arm base is not operating:
  - Change the DSI board.

Changing a DSI board requires an intervention level 2.

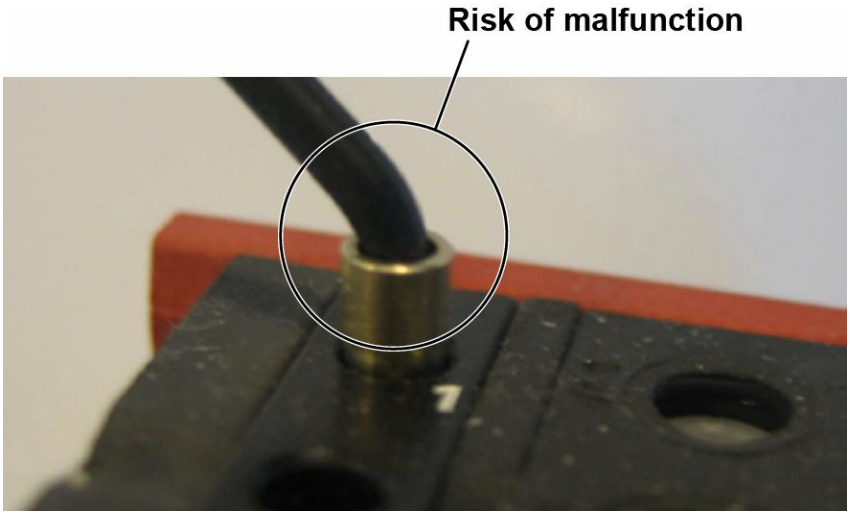
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Optical fiber

- Check optical fiber between STARC board and arm to ensure that there is no visible damage and no dust on optical fibber connectors, especially for the interconnection cable.

DANGER
Do not stand with your eyes directly opposite the optical fiber when it is lit, in order to avoid damage to the eyes.

- Check that there is red light coming out from STARC J306 connector. If there is not light or if it seems very dim:
  - Change STARC board.
- Check that there is not an angle on optical fiber inside CS8C or inside arm.



I0000190

Figure 8.73

- Check that there is red light coming back on STARC optical fiber connector J307.
- Change defective part if necessary.

Changing arm harness requires an intervention level 3.

DSI

If optical fiber looks good:

- Check that the 13 V led on ARPS is On (see chapter 8.5).

If ARPS 13 V light is on:

- Open the robot base and check the D1 light on DSI.

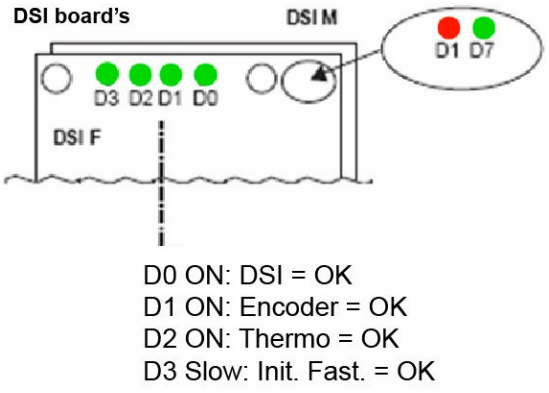


Figure 8.74

If D1 is off:

- Check the 13 V wiring between ARPS and DSI.

If D1 is on:

- Change the DSI board.

Changing a DSI board requires an intervention level 2.

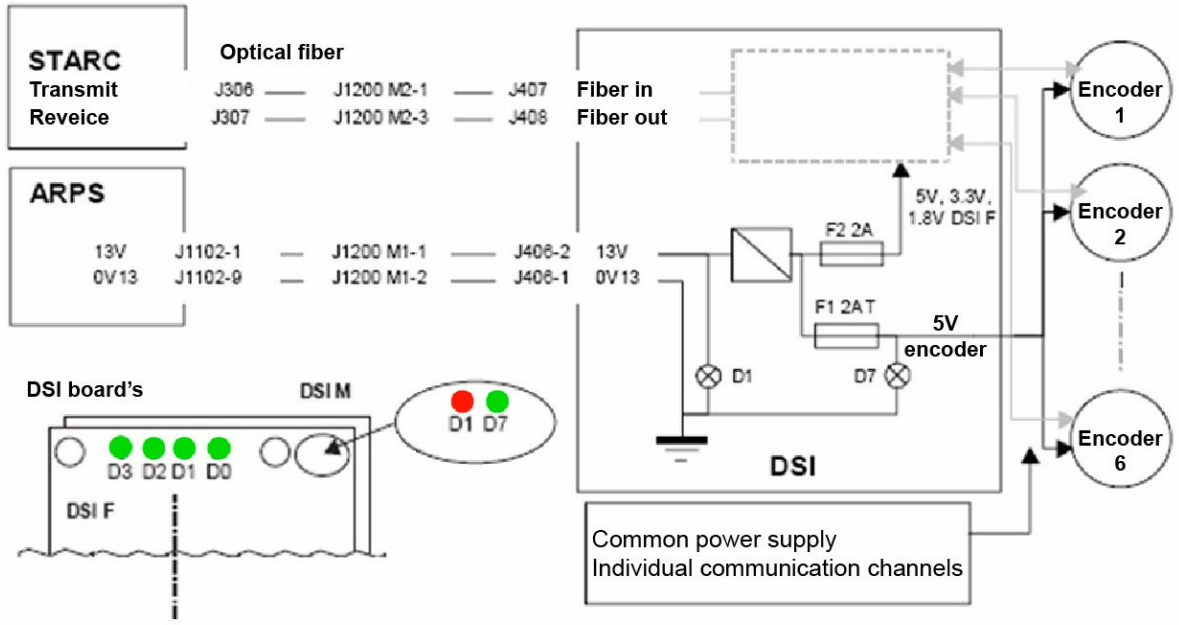
DSI advanced information

Common lines for all encoders:

- 13 V power supply.
- DSI.
- Optical fiber.
- 5 V from DSI to encoders.

Individual lines:

- DSI to encoder communication.



D0 ON: DSI = OK
D1 ON: Encoder = OK
D2 ON: Thermo = OK
D3 Slow: Init. Fast. = OK

I0000312

Figure 8.75

8.8.3.5 - Case 5

M0000493.1

Problem:

Led A is blinking slowly (~ 1 s period) after the boot sequence: There is a communication problem with the amplifiers.

- (A) Slow
- (B) Fast
- (C)
- (E)
- (D)

If both leds A and B are blinking slowly, check first the led B issue (previous chapter).

Solution:

When only led A is blinking slowly the STARC to drives communication is not operating, the information displayed on MCP control panel is the following:

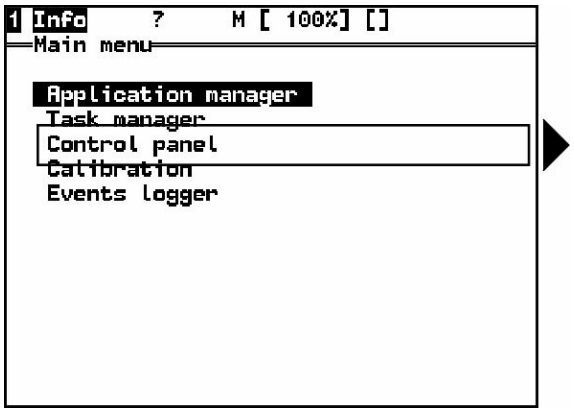


Figure 8.76

- Screen display:
  - Controller status

    ...
    - Drives information
      - Drive 1: Error
      - Drive 2: Error

        ...
      - Drive 6: Error
  - Sensor status
    - Sensor 1 = Working
    - Sensor 2 = Working

      ...
    - Sensor 6 = Working

Step 1

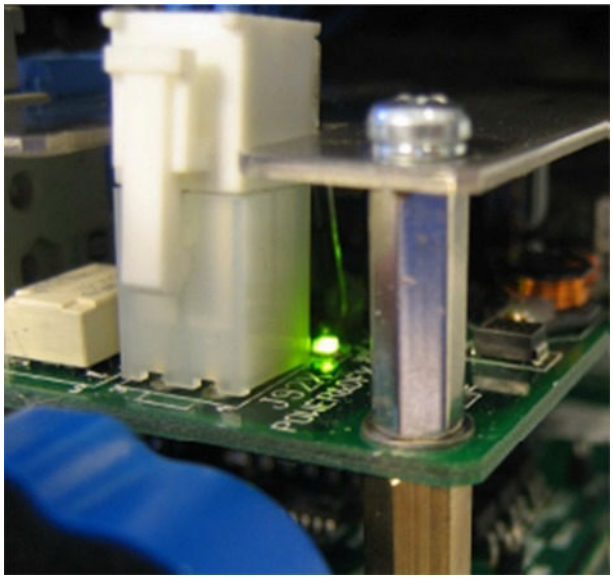
- Check flat cable between STARC board and the drives.

If cable has some damage:

- Change the cable.

Step 2

- Check that each drive is properly powered with 24 V: Led = On.
  - If one drive has a led off, check 24 V power supplier cables.
  - If power supplies are OK, change drive(s).



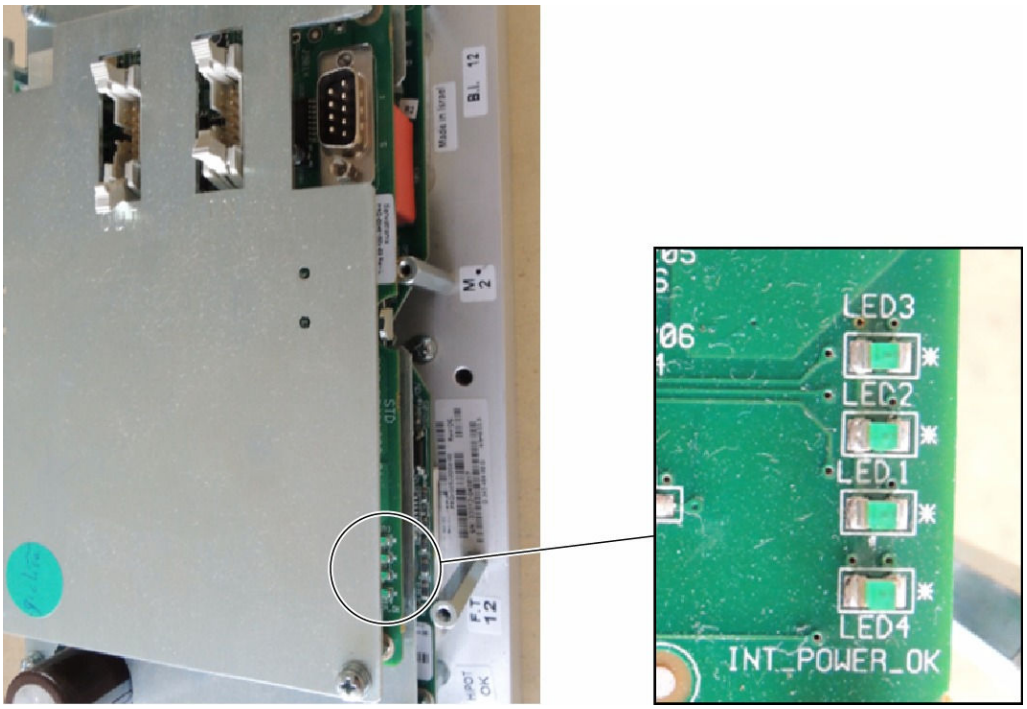
I0000374

Figure 8.77

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Step 3

- Check lights on drive side to get more information on the fault.



I0000344

Figure 8.78

LED 1 - Axis 0 status: Steady ON = Fault, slow blinking = disable, fast blinking = enable.

LED 2 - Axis 1 status: Steady ON = Fault, slow blinking = disable, fast blinking = enable.

LED 3 - Communication status: Steady ON = No communication (checksum error, ring broken), slow blinking (0.5 Hz) = initialization phase, fast blinking (2 Hz) = synchronous.

LED 4 - Indicate internal Power OK.

Step 4

- Change the STARC board.

8.9 - PREVENTIVE MAINTENANCE

M0000372.1

DANGER
Disconnect all the electrical and pneumatic power supplies before carrying out any work on the controller or the arm. Wait for at least 1 mn minutes before starting to work.

8.9.1 - VENTILATION

M0000373.1

The air filter on the ventilation system must be cleaned and/or replaced as and when necessary, depending on its level of fouling.

8.9.2 - RECOMMENDED SPARE PARTS

M0000374.1

- 10.3 x 38 fuse, rating depending on the voltage and the type of arm:

	Three-phase 400–480 V	Three-phase 200–230 V	Single phase 230 V	Single phase 115 V
TX40	4Am	6Am	10Am	16Am
TX60 - TS	4Am	8Am	10Am	16Am
TX90	6Am	12Am	–	–
RX160, TX250, TP80	8Am	16Am	–	–

- 250 mA and 500 mA microfuse.
- STARC board.
- Amplifier (drive): one of each type.

TX40	TX60	TX90	RX160/TP80/ TX250	TS
4/9 + 4/9	4/9 + 8/22	4/9 + 8/22	8/22 + 15/45	4/9 + 8/22

- BIO board.
- RSI board.
- CPU board.
- MCP.
- Air filter.

9 - APPENDIX

9.1 - PROTECTION OF THE POWER LINE FOR THE CS8C CONTROLLER

9.1.1 - CONTROLLER CHARACTERISTICS

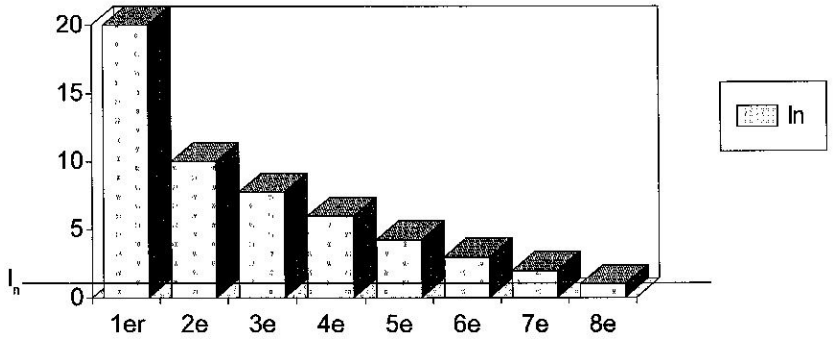
M0000264.1

The CS8C controller is protected at its input from risks of short-circuits by Am type fuses.

The load on the primary circuit depends on the type of arm installed, the voltage rating of the power supply and the type of network (single phase or three-phase) (load = current in the controller power supply circuit when the arm is operating).

Robot	Power installed
TX40	1.5 kVA
TX60	1.7 kVA
TX90	2 kVA
RX160/TP80/TX250	3 kVA
TS	1.7 kVA

When the controller is powered on, a surge current is generated. The current reaches the level of continuous duty after a 8 periods, i.e. ≈ 160 ms for a 50 Hz power frequency. The first peak of the surge current is a 20 x In.



I0000425

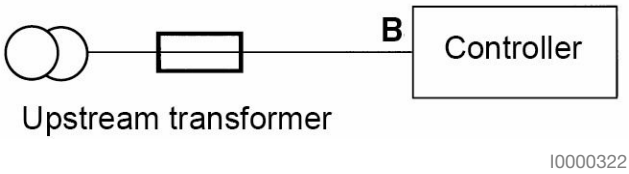
Figure 9.1

9.1.2 - PROTECTION UPSTREAM FROM THE CONTROLLER

M0000472.1

The calculation method below is provided for information and is applicable with NF C standard, it has to be adapted to relevant local standard when necessary.

The protections upstream from the controller are used to protect the secondary circuits of the upstream transformer from overcurrents and short-circuits.



I0000322

Figure 9.2

You need to install one of the following:

- The gl fuses
- A magneto thermic breaker, type U. Calculate minimum short-circuit current at the furthest point in the installation (in B) and choose the protection level so as to get a disconnection time < 5 s for this current.
- A magneto thermic breaker, type D

Choice of the protection: For example the french NF C 15-100 standard provides the calculation method for the protection against surges and short-circuits.

Summary:

1) Protection against surges:

The degree of protection must be less or equal to secondary surge current of the upstream transformer. This current depends on the electrical installation of the customer.

2) Protection against short-circuits:

Calculate minimum short-circuit current at the furthest point in the installation (in B) and choose the protection level so as to get a disconnection time < 5 s for this current.

$$IR\ mini = \frac{Us}{\frac{Us^2}{P} \times \frac{Ucc\%}{100} + \frac{2\rho l}{S}}$$

I0004259

Us = secondary voltage of the upstream transformer (1)

P = power of the upstream transformer (1)

Ucc% = short-circuit voltage of the upstream transformer in % (1)

I = length of the line (in metres).

S = section of the line (in mm^2). The wire sections must be chosen according to the current demand, the temperature rise, the potential difference on the line,...

ρ= 0.027 Ω mm^2/m for copper

(1) These characteristics are printed on the identification plate of the upstream transformer

Size of gl fuse:	In<=IR min. / 4
Breaker, type U:	In<=IR min. / 8
Breaker, type D:	In<=IR min. / 3.5

3) Use the functioning curves of both the breakers and the fuses to check that the power disconnection happens in less than 5 s for the selected size:

Example:

Us	= 400 V
P	= 10 kVA
Ucc%	= 4%
l	= 20 m
S	= 6 mm^2
ρ	= 0.027 Ω mm^2/m

$$IR\ mini = \frac{400}{\left(\frac{400^2}{10^4} \times \frac{4}{100}\right) + \frac{2 \times 0.027 \times 20}{6}} = 488\ A$$

For a breaker, type U: In<=488 / 8 = 61 A.

the size immediately lower is 52 A.

At the short-circuit current, the disconnection happens in less than 5 s.

9.2 - SYSTEM EVENTS CODES

M0000473.1

the character # in this list below denotes a number computed dynamically. They are generaly related to a joint, a motor, an encoder a channel or any devices number identification.

0x0300	768	DRIVE-NoFault:Drive axis # could not power motor because No Fault.
0x0301	769	DRIVE-DspNotReady:Drive axis # could not power motor because drive software dsp is not ready. (Please update your software).
0x0302	770	DRIVE-PowerBoardNotDefined:Drive axis # could not power motor because drive software does not recognize drive jumper configuration. (Please check jumper or update your software).
0x0303	771	DRIVE-BusOverVoltage:Drive axis # could not power motor because drive bus is Over Voltage. (Either your mouvement generates too many energy, either the dissipation resistance is out of order).
0x0304	772	DRIVE-IpmFault:Drive axis # could not power motor because drive IGBT can not generate requested current without damage. (Either the IGBT temperature is too high, the IGBT can not deliver such a current, or IGBT is already destroyed).
0x0305	773	DRIVE-OverTemperature:Drive axis # could not power motor because drive skin temperature excessive. (Please check fans and ventilation).
0x0306	774	DRIVE-Watchdog:Drive axis # could not power motor because drive software watchdog occured. (Please update your software).
0x0307	775	DRIVE-PllNotLocked:Drive axis # could not power motor because drive software could not synchronize its internal clock (10kHz or 20kHz) with the starc bus Frame Synchro. (Check starc bus and software update).
0x030a	778	DRIVE-SsiNotSynchro:Drive axis # could not power motor because the sensor bus (from direct link) is not synchronous. (Check starc sensor bus).
0x030d	781	DRIVE-PositionError:Drive axis # could not power motor because the dsi provides no position to the drive. (Check why dsi is in error).

0x030e	782	DRIVE-PositionIncoherence:Drive axis # could not power motor because the position provided by dsi is not in the same quarter than previous position. (Please check drive parameter configuration or sensor failure).
0x030f	783	DRIVE-NoCompensation:Drive axis # could not power motor because the controller did not send 'config' command after changing parameters. (Check why controller was interrupted during its init).
0x0311	785	DRIVE-CurrentSensorOffset:Drive axis # could not power motor because autocalibration detects that drive current sensor if out of order. (Please change the drive).
0x0312	786	DRIVE-CurrentSensorNoise:Drive axis # could not power motor because autocalibration detects that drive noise sensor if out of order. (Please change the drive or check EMC noise).
0x0313	787	DRIVE-OverSpeed:Drive axis # could not power motor because the difference between two positions provided by dsi is larger than overspeed threshold defined by controller. (Please check drive parameter configuration).
0x0314	788	DRIVE-BusUnderVoltage:Drive axis # could not power motor because the drive is not able to provide enough power to the motor. (Check RPS and power cables).
0x0315	789	DRIVE-DriveFoldback:Drive axis # could not power motor because the drive has to limit the current to avoid dammage by excessive heat. (Either controller wants excessive acceleration, or motor has poor performance, or arm is pushing on an obstacle).
0x0316	790	DRIVE-MotorFoldback:Drive axis # could not power motor because the temperature model signals that motor is too hot.
0x0317	791	DRIVE-HighPowerStillOn:Drive axis # could not power motor because the drive is in hardware fault. (Please change the drive).
0x0318	792	DRIVE-FaultLogFull:Drive axis # could not power motor because the drive logger is full. (Firsts events should be enough for diagnostic).
0x0a01	2561	ENCODER-Alarm:The encoder axis # provides no position because encoder is in internal error 'Signal amplitude' or 'Position value'. (Please check encoder).
0x0b01	2817	HARNESS-NotConnected:The harness could not allow communication between dsi and encoder axis # because encoder is not connected to dsi board. (Please check harness and dsi fuse).
0x0b02	2818	HARNESS-HwError:The harness could not allow communication between dsi and encoder axis # because dsi received no enDat answer from encoder. (Please check harness and encoder).
0x0b03	2819	HARNESS-CrcError:The harness could not allow communication between dsi and encoder axis # because dsi detected an invalid checksum from sensor data. (Please check harness).
0x0b04	2820	HARNESS-Timeout:The harness could not allow communication between dsi and encoder axis # because dsi received no answer from encoder before time out. (Please check harness and encoder).
0x0b05	2821	SENSOR-CrcError:The harness could not allow communication between dsi and encoder axis # because dsi detected an invalid checksum from sensor data. (Please check harness).

0x0c01	3073	DSI-Resident:The dsi board could not provide position for encoder axis # because dsp firmware is not present on flash (or CS8 does not request dsi to switch from resident to firmware).
0x0c02	3074	DSI-EndatDisable:The dsi board could not provide position for encoder axis # because (or more) other encoder is in error and dsi board decided not to enable this axis for safety reason. (Please correct all other encoder fault first).
0x0c03	3075	DSI-OverSpeedError:The dsi board could not provide position for encoder axis # because the encoder speed reached its limit.
0x0c04	3076	DSI-NoToggle:The dsi board could not provide position for encoder axis # because on board dsp-fpga communication failed. (Please contact the Stäubli Robotics Customer Support).
0x0c05	3077	SENSOR-OverSpeedError:The dsi board could not provide position for encoder axis # because the encoder speed reached its limit.
0x0d01	3329	DRIVE-Fault:The drive axis # just goes in 'Fault' state and could not power motor. Reasons will follow in next asynchronous message...
0x0d02	3330	DRIVE-Warning:The drive axis # just goes in 'Warning' state and if you continue such sollicitation a Drive Fault will occur.
0x0e01	3585	STARC-SlaveNotSynchronous: (SlaveSynchroMask=0x63)The Starc bus could not communicate with sensor because the bus is not synchronous [ASYNCHRO evt]. (Please check Starc-Device link and power on device).
0x0e02	3586	STARC-BusMasterOverrun:The Starc bus could not communicate with sensor because an overrun occured. (Please update your software).
0x0e03	3587	STARC-BusMasterChecksumError:The Starc bus could not communicate with sensor because a checksum error occured. (Please check Starc-Device link and power on device).
0x0e04	3588	STARC-BusMasterRingBroken:The Starc bus could not communicate with sensor because the bus is RingBroken. (Please check Starc-Device link and power on device).
0x0e07	3591	DRIVE-SensorNotReady:The Starc bus could not communicate with sensor because the bus for axis # is not synchronous [SYNCHRO bit]. (Please check Starc-Device link and power on device).
0x0f01	3841	STARC-SlaveNotSynchronous: (SlaveSynchroMask=0x63)The Starc bus could not communicate with drive because the bus is not synchronous [ASYNCHRO evt]. (Please check Starc-Device link and power on device).
0x0f02	3842	STARC-BusMasterOverrun:The Starc bus could not communicate with drive because an overrun occured. (Please update your software).
0x0f03	3843	STARC-BusMasterChecksumError:The Starc bus could not communicate with drive because a checksum error occured. (Please check Starc-Device link and power on device).
0x0f04	3844	STARC-BusMasterRingBroken:The Starc bus could not communicate with drive because the bus is RingBroken. (Please check Starc-Device link and power on device).
0x0f07	3847	DRIVE-NotReady:The Starc bus could not communicate with drive because the bus for axis # is not synchronous [SYNCHRO bit]. (Please check Starc-Device link and power on device).

0x1001	4097	STARC-CannotEnableNotSynchronous:The Starc board stopped processing because starc bus should be synchronous to enable the drive axis #. (Please check first why bus is not synchronous).
0x1002	4098	STARC-PositionSensorIllegalSequence:The Starc board stopped processing because the position axis # provided by dsi is not in the same quarter than previous position. (Please check sensor).
0x0000	0	Events
0x1003	4099	SERVO-ReadFeedbackError:The Starc board stopped processing because drive axis # goes in disable state since sensor is in error. (Check sensor status for details).
0x1004	4100	SERVO-ParametersNotDefined:The Starc board stopped processing because drive axis # can not go in enabling state since controller did not send servo parameters. (Check if controller send param and tuning.
0x1005	4101	SERVO-DriveDisabled:The Starc board stopped processing because drive axis # switched to disable state.
0x1006	4102	SERVO-TimeoutEnabling:The Starc board stopped processing because drive axis # did not succeed to go in enable state after controller request.
0x1007	4103	SERVO-TimeoutDisabling:The Starc board stopped processing because drive axis # did not succeed to go in disable state after controller request.
0x1008	4104	SERVO-EnveloppeError:The Starc board stopped processing because difference between command and feedback axis # position is larger than threshold sent by controller.
0x1009	4105	SERVO-DriveNotDisabled:The Starc board stopped processing because controller requested to switch drive axis # to enable state while it is not disable yet.
0x100a	4106	SERVO-Overspeed:The Starc board stopped processing because speed axis # is larger than threshold sent by controller.
0x100b	4107	STARC-Dsp1NotToggle:The Starc board could not communicate with device because the Dsp1 did not toggle. (Please reboot system or change starc board).
0x100c	4108	STARC LoggerErrorReadingNumberOfEvents:The starc event logger fails to get info from axis # because could not read nbr of data from device.
0x100d	4109	STARC LoggerErrorReadingEvent:The starc event logger fails to get info from axis # because device did not send all expected data.
0x100e	4110	STARC LoggerAnswerErrorOnAsynchronousCommand:The starc event logger fails to get info from axis # because device is in error.
0x100f	4111	STARC LoggerAsynchronousCommandAnswerLenghtError:The starc event logger fails to get info from axis # because could not read data from device.
0x1010	4112	STARC LoggerErrorSendingAsynchronousCommand:The starc event logger fails to get info from axis # because could not communicate with device.
0x1011	4113	STARC-CurrentLoopNotOptimal:The Starc board could not communicate with device because the direct link between sensor and drive bus is not synchronous. (Please reboot system).
0x1012	4114	STARC-SensorAndDriveBusNotSynchronized:The Starc board could not communicate with device because the drive and sensor buses do not have same axis bit at same time. (Please reboot system).

0x1013	4115	STARC-LoggerFull:The Starc event logger could not send messages to controller because The Starc event logger is full
0x1014	4116	STARC-Cs8NotToggle:The STARC PCI communication failed because starc received two consecutive frames from CPU with same toggle bit.
0x1032	4146	STARC-Safety:
0x1201	4609	INIT-DsiNoOffset:The controller initialization was interrupted and so dsi did not receive from controller the OFFPOS command for axis # . (Please correct previous error before).
0x1202	4610	INIT-DsiNotEnabled:The controller initialization was interrupted and so dsi did not receive from controller the DEVSEL command for axis #. (Please correct previous error before).
0x1301	4865	Invalid arm calibration data: arm configuration file arm.cf is missing or invalid.
0x1302	4866	Calibration battery low voltage.
0x1303	4867	Not able to read the motor position. Calibration lost on axis #
0x1304	4868	System event DAPS-IllegalSequence on axis #
0x1305	4869	System event DAPS-Overflow on axis #
0x1306	4870	DSI board and arm.cfx zero offsets do not match, axis #
0x1307	4871	DSI board and arm.cfx phase offsets do not match for axis #
0x1308	4872	Zero offset has never been written or is invalid, axis #
0x1309	4873	Phase offset has never been written or is invalid, axis #
0x130a	4874	Internal error STARC-CalibrationCommError
0x130b	4875	Internal error STARC-CalibrationCommandFailed
0x130c	4876	System event DRIVE-InvalidConfiguration axis #
0x130d	4877	Position was not saved at controller shutdown. Calibration lost axis #
0x130e	4878	Motor position has changed since last controller shutdown. Calibration lost axis #
0x130f	4879	Encoder analog input error. Calibration lost axis #
0x1310	4880	Drive is not initialized. Calibration lost axis #
0x1311	4881	Over speed error. Calibration lost axis #
0x1312	4882	Encoder serial communication error. Calibration lost axis #
0x1313	4883	System event DRIVE-UnexpectedCalibrationError axis #
0x1314	4884	Calibration not initialized
0x1402	5122	Internal error BRAKE BrakeOff
0x1502	5378	Internal error Position out of range axis #axis
0x1503	5379	Envelop error axis #joint
0x1504	5380	ROBOT-OverForce Over force: axis #, 999.
0x1505	5381	ROBOT-OverVelocityMax axis2, 999.
0x1506	5382	Internal error ROBOT-DrivesNotDisabled
0x1507	5383	System event ROBOT-Init
0x1508	5384	System event Calibration disabled on axis #

0x1509	5385	Internal error ROBOT-DrivesNotInitialized
0x150a	5386	Cannot calibrate robot: arm power is on.
0x150b	5387	Please, calibrate axis # before axis #
0x1601	5633	Arm power settled.
0x1602	5634	End of arm power shutdown phase.
0x1603	5635	Settling arm power...
0x1604	5636	Arm power shutdown...
0x1605	5637	Cannot settle arm power: working mode change.
0x1606	5638	Cannot settle arm power: system event ROBOT-SafetyInit.
0x1607	5639	Cannot settle arm power: system event SAFETY-NoSignal.
0x1608	5640	Cannot settle arm power: system event POWER-Error
0x1609	5641	Cannot settle arm power: system event POWER-PWR OK-Error
0x160a	5642	Cannot settle arm power: simultaneous arm power shutdown request.
0x160b	5643	Internal error ROBOT-Power-
0x160c	5644	Cannot settle arm power: system event SAFETY-Unknown.
0x160d	5645	Cannot settle arm power: joint limit electrical switch activated.
0x160e	5646	Cannot settle arm power: drive error SAFETY-DriveFault.
0x160f	5647	Cannot settle arm power: system event SAFETY-ShortrOpen.
0x1610	5648	Cannot settle arm power: system event SAFETY-Watchdog.
0x1611	5649	Cannot settle arm power: system event SAFETY-ThermoOpened.
0x1612	5650	Cannot settle arm power: internal power error (fuse F3 on USBI safety board ?)
0x1613	5651	Cannot settle arm power: internal power error (fuse F2 on RSI board ?)
0x1614	5652	Cannot settle arm power: teach pendant emergency stop activated.
0x1615	5653	Cannot settle arm power: remote emergency stop activated
0x1616	5654	Cannot settle arm power: cell emergency stop chain open (User ES1-2)
0x1617	5655	Cannot settle arm power: deadman switch released or emergency stop chain for manual mode open (User EN).
0x1618	5656	Cannot settle arm power: emergency stop chain DOOR open (door open).
0x1619	5657	Cannot settle arm power: cell emergency stop chain open (User ES3-4)
0x161a	5658	Cannot settle arm power: brake select switch turned from normal (0) position.
0x161b	5659	Cannot settle arm power: brake release button pressed.
0x161c	5660	Cannot settle arm power: system event ROBOT-SafetyInit.
0x161d	5661	Cannot settle arm power: system event SAFETY-UnstableSignal.
0x161e	5662	Cannot settle arm power: system event SAFETY-NoSignal.
0x161f	5663	Cannot settle arm power: system event SAFETY-Transient.
0x1620	5664	Cannot settle arm power: phasing signal is enabled
0x1621	5665	Cannot settle arm power: system event SAFETY-EstopSoft

0x1622	5666	Cannot settle arm power: system event ROBOT-Safety.
0x1626	5670	Cannot settle arm power:
0x1627	5671	Arm power shutdown:
0x1701	5889	Detected: system event SAFETY-Unknown.
0x1702	5890	Detected: joint limit electrical switch activated.
0x1703	5891	Detected: drive error SAFETY-DriveFault.
0x1704	5892	Detected: system event SAFETY-ShortrOpen.
0x1705	5893	Detected: system event SAFETY-Watchdog.
0x1706	5894	Detected: system event SAFETY-ThermoOpened.
0x1707	5895	Detected: internal power error (fuse F3 on USBI safety board ?)
0x1708	5896	Detected: internal power error (fuse F2 on RSI board ?)
0x1709	5897	Detected: teach pendant emergency stop activated.
0x170a	5898	Detected: remote emergency stop activated
0x170b	5899	Detected: cell emergency stop chain open (User ES1-2)
0x170c	5900	Detected: deadman switch released or emergency stop chain for manual mode open (User EN).
0x170d	5901	Detected: emergency stop chain DOOR open (door open).
0x170e	5902	Detected: cell emergency stop chain open (User ES3-4)
0x170f	5903	Detected: brake select switch turned from normal (0) position.
0x1710	5904	Detected: brake release button pressed.
0x1711	5905	Detected: system event ROBOT-SafetyInit.
0x1712	5906	Detected: system event SAFETY-UnstableSignal.
0x1713	5907	Detected: system event SAFETY-NoSignal.
0x1714	5908	Detected: system event SAFETY-Transient.
0x1715	5909	Detected: phasing signal is enabled
0x1716	5910	Detected: system event SAFETY-EstopSoft
0x1717	5911	Detected: system event ROBOT-Safety.
0x171a	5914	Detected end of: system event SAFETY-Unknown.
0x171b	5915	Detected end of: joint limit electrical switch activated.
0x171c	5916	Detected end of: drive error SAFETY-DriveFault.
0x171d	5917	Detected end of: system event SAFETY-ShortrOpen.
0x171e	5918	Detected end of: system event SAFETY-Watchdog.
0x171f	5919	Detected end of: system event SAFETY-ThermoOpened.
0x1720	5920	Detected end of: internal power error (fuse F3 on USBI safety board ?)
0x1721	5921	Detected end of: internal power error (fuse F2 on RSI board ?)
0x1722	5922	Detected end of: teach pendant emergency stop activated.
0x1723	5923	Detected end of: remote emergency stop activated
0x1724	5924	Detected end of: cell emergency stop chain open (User ES1-2)

0x1725	5925	Detected end of: deadman switch released or emergency stop chain for manual mode open (User EN).
0x1726	5926	Detected end of: emergency stop chain DOOR open (door open).
0x1727	5927	Detected end of: cell emergency stop chain open (User ES3-4)
0x1728	5928	Detected end of: brake select switch turned from normal (0) position.
0x1729	5929	Detected end of: brake release button pressed.
0x172a	5930	Detected end of: system event ROBOT-SafetyInit.
0x172b	5931	Detected end of: system event SAFETY-UnstableSignal.
0x172c	5932	Detected end of: system event SAFETY-NoSignal.
0x172d	5933	Detected end of: system event SAFETY-Transient.
0x172e	5934	Detected end of: phasing signal is enabled
0x172f	5935	Detected end of: system event SAFETY-EstopSoft
0x1730	5936	Detected end of: system event ROBOT-Safety.
0x1733	5939	Arm power shutdown: system event SAFETY-Unknown.
0x1734	5940	Arm power shutdown: joint limit electrical switch activated.
0x1735	5941	Arm power shutdown: drive error SAFETY-DriveFault.
0x1736	5942	Arm power shutdown: system event SAFETY-ShortrOpen.
0x1737	5943	Arm power shutdown: system event SAFETY-Watchdog.
0x1738	5944	Arm power shutdown: system event SAFETY-ThermoOpened.
0x1739	5945	Arm power shutdown: internal power error (fuse F3 on USBI safety board ?)
0x173a	5946	Arm power shutdown: internal power error (fuse F2 on RSI board ?)
0x173b	5947	Arm power shutdown: teach pendant emergency stop activated.
0x173c	5948	Arm power shutdown: remote emergency stop activated
0x173d	5949	Arm power shutdown: cell emergency stop chain open (User ES1-2)
0x173e	5950	Arm power shutdown: deadman switch released or emergency stop chain for manual mode open (User EN).
0x173f	5951	Arm power shutdown: emergency stop chain DOOR open (door open).
0x1740	5952	Arm power shutdown: cell emergency stop chain open (User ES3-4)
0x1741	5953	Arm power shutdown: brake select switch turned from normal (0) position.
0x1742	5954	Arm power shutdown: brake release button pressed.
0x1743	5955	Arm power shutdown: system event ROBOT-SafetyInit.
0x1744	5956	Arm power shutdown: system event SAFETY-UnstableSignal.
0x1745	5957	Arm power shutdown: system event SAFETY-NoSignal.
0x1746	5958	Arm power shutdown: system event SAFETY-Transient.
0x1747	5959	Arm power shutdown: phasing signal is enabled
0x1748	5960	Arm power shutdown: system event SAFETY-EstopSoft
0x1749	5961	Arm power shutdown: system event ROBOT-Safety.
0x1750	5968	Internal error SAFETY-

0x1755	5973	Emergency stop chain miss wired for signal #
0x175a	5978	Emergency signal # has been repaired.
0x175f	5983	Emergency signal # has been acknowledged.
0x1764	5988	-- Not published --
0x1769	5993	Emergency signal # has been acknowledged with integrator privileges.
0x1801	6145	Working mode:automatic.
0x1802	6146	Working mode:manual.
0x1803	6147	Working mode:test.
0x1804	6148	Working mode:remote.
0x1805	6149	Working mode:invalid.
0x1806	6150	Internal error ROBOT-WorkingMode-MANUoff
0x1807	6151	Working mode button has no effect
0x1808	6152	Test mode is not allowed
0x1809	6153	Your profile does not allow you to change the working mode.
0x180a	6154	Use of WMS is not allowed
0x1902	6402	Internal error DRIVE-CannotEnable Channel number
0x1903	6403	Internal error DRIVE-CannotDisable
0x1904	6404	Position sensor error on axis #
0x1905	6405	Internal error DRIVE-RingDead
0x190b	6411	Internal error DRIVE-AxisShutdown
0x190c	6412	Internal error DRIVE-AxisFault
0x190d	6413	Internal error DRIVE-AxisWarning
0x190e	6414	Internal error DRIVE-AxisNotInitialized
0x190f	6415	Internal error DRIVE-NotInitialized
0x1910	6416	Internal error DRIVE-Error
0x1911	6417	Internal error DRIVE-NotReadyDrive
0x1912	6418	Internal error DRIVE-ShutdownDrive
0x1913	6419	Internal error DRIVE-WriteTimeout-drive #-IDN-0
0x1914	6420	Internal error DRIVE-ReadTimeout-drive #-IDN-0
0x1915	6421	Internal error DRIVE-WriteError-drive #-IDN-0
0x1916	6422	Internal error DRIVE-ReadError-drive #-IDN-0
0x1917	6423	Internal error DRIVE-Process-drive #-IDN-0
0x1918	6424	Internal error DRIVE-Report-drive #-CD1-0x3e7 msg
0x1919	6425	Internal error DRIVE-NoConfigurationVersion-drive #-IDN-0
0x191a	6426	Internal error DRIVE-NoDriveConfiguration
0x191b	6427	DRIVE-InvalidFirmwareVersion-
0x191c	6428	DRIVE-FirmwareVersion-
0x191d	6429	DRIVE-SoftwareVersion-

0x1a01	6657	The following application could not be opened at boot:
0x1a02	6658	The following application could not be run at boot:
0x1a03	6659	System event INTERFACE-Stopped ()
0x1a04	6660	System event INTERFACE-Stopped (Application). 1 events lost.
0x1a05	6661	Error in system interface code file line:
0x1c01	7169	Recorder parameters error : recorder not found
0x1c02	7170	Recorder parameters error : invalid duration
0x1c03	7171	Recorder configuration error : invalid frequency
0x1c04	7172	Recorder configuration error : recorder variable Variable not known
0x1e01	7681	Arm power shutdown: internal power error (CS8: 24V 2, CS8C: ALIM OK in ARPS module). Reboot is needed to recover.
0x1e02	7682	Arm power shutdown: controller power supply shutdown detected (SECTEUR OK signal). Reboot is needed to recover.
0x1f01	7937	system event CPU-TemperatureLimit. Temperature limit is reached, behaviour of CPU board is uncertain.
0x1f02	7938	system event CPU-TemperatureLow. Unexpected temperature on CPU board.
0x1f03	7939	system event CPU-NoTemperature. Temperature cannot be read on CPU board.
0x1f04	7940	system event RSI-TemperatureLimit. Temperature limit is reached, behaviour of RSI board is uncertain.
0x1f05	7941	-- Not published --
0x1f06	7942	system event RSI-NoTemperature. Temperature cannot be read on RSI board.
0x2001	8193	Simulated MCP connected (0.0.0.0)
0x2101	8449	The communication protocol for this USB device is not supported. The CS8 driver supports the RBC, UFI and SCSI USB 1.0 protocols.
0x2201	8705	New profile : User
0x2301	8961	Your profile does not allow you to use the Menu key.
0x2302	8962	Your profile does not allow you to acknowledge safety fault.
0x2303	8963	Your profile does not allow you to change language.
0x2401	9217	cs8 message
0x2402	9218	cs8 message
0x2403	9219	cs8 message
0x2404	9220	Cs8 Diagnostic
0x2405	9221	cs8 message
0x2501	9473	Your profile does not allow you to acknowledge safety fault.
0x2601	9729	New automatic IP : 0.0.0.0
0x2801	10241	Internal error DAPS-Error-1-9999
0x2802	10242	Internal error DAPS-Communication-Unknown

0x2901	10497	System time successfully synchronized with 0.0.0.0 server
0x2902	10498	System time synchronization has failed with 0.0.0.0 server
0x2a01	10753	STARC-UnknownMessage :
0x2a02	10754	Internal error STARC-Logger:0
0x2b01	11009	-- Not published --
0x2b02	11010	-- Not published --
0x2c01	11265	STX-CannotConnectSerialLine
0x2d01	11521	Internal error FIELDBUS-InitSoftware-30
0x2d02	11522	FIELDBUS-BuildItem-
0x2d0a	11530	Internal error FIELDBUS-BuildPort
0x2d0b	11531	Internal error FIELDBUS-Write Board=# Equipment=# Channel=# Status=#
0x2d0c	11532	Internal error FIELDBUS-Read Board=# Equipment=# Channel=# Status=#
0x2d0d	11533	Internal error FIELDBUS-EquipmentStatus Board=# Equipment=# Channel=# Status=#
0x2d0e	11534	Internal error Equipment error on fieldbus Board=# Equipment=# Channel=# Status=#
0x2d0f	11535	Internal error FIELDBUS-RefreshIn Board=# Equipment=# Channel=# Status=#
0x2d10	11536	Internal error FIELDBUS-RefreshOut Board=# Equipment=# Channel=# Status=#
0x2d11	11537	Internal error FIELDBUS-InitBoard Board=# Equipment=# Channel=# Status=#
0x2d12	11538	Internal error FIELDBUS-ExitBoard Board=# Equipment=# Channel=# Status=#
0x2d13	11539	Internal error FIELDBUS-BoardId Board=# Equipment=# Channel=# Status=#
0x2d14	11540	Internal error FIELDBUS-ConfigBoard-30
0x2d15	11541	Internal error FIELDBUS-NetworkStatus Board=# Equipment=# Channel=# Status=#
0x2d16	11542	Internal error Network error on fieldbus Board=# Equipment=# Channel=# Status=#
0x2d17	11543	FIELDBUS-Version-
0x2d18	11544	FIELDBUS-StartServer
0x2d19	11545	Internal error FIELDBUS-ExitAllBoard Board=# Equipment=# Channel=# Status=#
0x2e01	11777	Internal error MODBUS-InitLib Channel=# Status=#
0x2e02	11778	MODBUS-BuildItem-
0x2e0a	11786	Internal error MODBUS-Write Channel=# Status=#
0x2e0b	11787	Internal error MODBUS-Read Channel=# Status=#
0x2e0c	11788	Internal error MODBUS-StartServer Channel=# Status=#
0x2e0d	11789	Internal error MODBUS-StopServer Channel=# Status=#

0x2f01	12033	USB-DriverNotFound
0x3001	12289	STARC-FpgaVersion:
0x3002	12290	STARC-FlashCounter:
0x3003	12291	STARC-SerialNumber:
0x3101	12545	Internal error RSI-I2cNoAcknoledge-State:0
0x3103	12547	Internal error RSI-DspNotToggle-State:0
0x3104	12548	Internal error RSI-FpgaNotToggle-State:0
0x3201	12801	Internal error STARC-UpdateFirmwareError:0
0x3202	12802	STARC-UpdateFirmwareInfo:0
0x3203	12803	Internal error STARC-UpdateFirmwareError:
0x3204	12804	New Starc software
0x3205	12805	Starc software update, Please reboot the controller
0x3206	12806	Internal error STARC-UpdateFirmwareAsyncCommError:
0x3301	13057	Internal error USB-Write-0
0x3302	13058	Internal error USB-Read-0
0x3303	13059	Internal error USB-Timeout
0x3304	13060	Internal error USB-Start
0x3305	13061	Internal error USB-Access
0x3401	13313	Internal error MCP-Read
0x3402	13314	Internal error MCP-BufferFull
0x3403	13315	Internal error MCP-UnknownEvent: char 0
0x3404	13316	Internal error MCP-Checksum: char 0
0x3405	13317	Internal error MCP-UnknownCommmand
0x3406	13318	Internal error MCP-NoAcknowledge
0x3407	13319	Internal error MCP-NoAnswer: char 0
0x3408	13320	Internal error MCP-NotConnected
0x3409	13321	Internal error MCP-ResetTimeout
0x340a	13322	Mcp communication established
0x340b	13323	-- Not published --
0x3601	13825	Internal error DAPS-Protocol-1-Unknown
0x3602	13826	DAPS board version
0x3701	14081	Internal error STX-NoHardware
0x3702	14082	Internal error STX-Version
0x3703	14083	Internal error STX-MessageQueue
0x3704	14084	Internal error STX-Semaphore
0x3705	14085	Internal error STX-Mutex
0x3706	14086	Internal error STX-Overrun
0x3707	14087	Internal error STX-Synchro

0x3708	14088	Internal error STX-IsrOverlap
0x3709	14089	Internal error STX-Shutdown
0x370a	14090	Internal error STX-Device
0x370b	14091	Internal error STX-AtTelegram
0x370c	14092	Internal error STX-InvalidData
0x370d	14093	Internal error STX-NoCyclicIdn
0x370e	14094	Internal error STX-Adress
0x370f	14095	Internal error STX-InvalidLength
0x3710	14096	Internal error STX-Unexpected
0x3901	14593	-- Not published --
0x3902	14594	-- Not published --
0x3903	14595	-- Not published --
0x3b01	15105	Internal error MOTION-NoRelativeMovej
0x3b02	15106	Internal error MOTION-NoRelativeMovel
0x3b03	15107	Internal error MOTION-InvalidConfig
0x3b04	15108	Invalid blending in mdesc: reason
0x3b05	15109	Cannot move: destination is out of software axis limits
0x3b06	15110	Cannot complete move: destination is out of reach.(out of software joint limits)
0x3b07	15111	Cannot move: cannot cross singularity in this case.
0x3b08	15112	Compliant Move too long
0x3b09	15113	Invalid vel value in mdesc
0x3b0a	15114	Internal error MOTION-TranslationVelTooLarge
0x3b0b	15115	Internal error MOTION-RotationVelTooLarge
0x3b0c	15116	Internal error MOTION-CutOffFreqTooSmall
0x3b0d	15117	Internal error Angle between circle start and intermediate points exceeds 180 deg.
0x3b0e	15118	Internal error Angle between circle intermediate and end points exceeds 180 deg.
0x3b0f	15119	Internal error Circle start, intermediate and end points are too close.
0x3b10	15120	Internal error Orientation change between start and intermediate, or intermediate and end points is exactly 180°: direction of move is undetermined.
0x3b11	15121	Internal error MOTION-CartCurveProblem
0x3b12	15122	MOTION-AlterOff
0x3b13	15123	MOTION-CartBlending-RotTooBig
0x3b14	15124	MOTION-Tracking-BadSequence Incorrect tracking sequence. TrackOn needed
0x3b15	15125	MOTION-Tracking-Stopped Tracking stopped: The motion was stopped but the frame has moved. Cannot track it.
0x3b16	15126	Tracking solutions failed

0x3b17	15127	Motion internal tracking error
0x3b18	15128	Target velocity is too high
0x3b19	15129	Robot not at trajectory start position
0x3b1a	15130	Part mounting error:error
0x3c01	15361	Internal error MOTION-NoRelativeMovej
0x3c02	15362	Internal error MOTION-NoRelativeMovel
0x3c03	15363	Internal error MOTION-InvalidConfig
0x3c04	15364	Invalid blending in mdesc: reason
0x3c05	15365	Cannot move: destination is out of software axis limits
0x3c06	15366	Cannot complete move: destination is out of reach.(out of software joint limits)
0x3c07	15367	Cannot move: cannot cross singularity in this case.
0x3c08	15368	Compliant Move too long
0x3c09	15369	Invalid vel value in mdesc
0x3c0a	15370	Internal error MOTION-TranslationVelTooLarge
0x3c0b	15371	Internal error MOTION-RotationVelTooLarge
0x3c0c	15372	Internal error MOTION-CutOffFreqTooSmall
0x3c0d	15373	Internal error Angle between circle start and intermediate points exceeds 180 deg.
0x3c0e	15374	Internal error Angle between circle intermediate and end points exceeds 180 deg.
0x3c0f	15375	Internal error Circle start, intermediate and end points are too close.
0x3c10	15376	Internal error Orientation change between start and intermediate, or intermediate and end points is exactly 180 deg: direction of move is undetermined.
0x3c11	15377	Internal error MOTION-CartCurveProblem
0x3c12	15378	MOTION-AlterOff
0x3c13	15379	MOTION-CartBlending-RotTooBig
0x3c14	15380	MOTION-Tracking-BadSequence Incorrect tracking sequence. TrackOn needed
0x3c15	15381	MOTION-Tracking-Stopped Tracking stopped: The motion was stopped but the frame has moved. Cannot track it.
0x3c16	15382	Tracking solutions failed
0x3c17	15383	Motion internal tracking error
0x3c18	15384	Target velocity is too high
0x3c19	15385	Robot not at trajectory start position
0x3c1a	15386	Part mounting error:error
0x3d01	15617	Your profile does not allow you to change monitor speed
0x3d02	15618	Your profile does not allow you to use move/hold in remote mode.
0x3e01	15873	Deadman error: check that the deadman switch is pressed.

0x3e02	15874	Teach pendant parking error: check that the pendant is correctly parked in the cradle.
0x3e04	15876	RemoteMcp timeout
0x3f01	16129	-- Not published --
0x3f02	16130	Cannot apply arm power: after an emergency stop, the teach pendant must be parked again.
0x3f03	16131	Your profile does not allow you to disable power in remote mode.
0x3f04	16132	Cannot apply arm power: arm power is still in disabling phase.
0x3f05	16133	Attention: the arm could not be stopped in time during power disabling phase.
0x3f06	16134	ROBOT-PtcWarning:
0x3f07	16135	ROBOT-PtcFault:
0x3f08	16136	Cannot apply arm power: press and hold the deadman switch.
0x3f09	16137	Internal error ROBOT-Software
0x3f0a	16138	The robot is not calibrated: only joint jog move is possible.
0x3f0c	16140	-- Not published --
0x3f0e	16142	Fast test mode enabled. Use monitor speed keys to increase maximum speed.
0x3f0f	16143	Remote fast speed request refused: cabinet working mode is not test.
0x3f10	16144	End of connection move.
0x3f11	16145	Internal error ROBOT-Motion-Unexpected
0x3f12	16146	Cannot apply arm power: release & press deadman to enable arm power request.
0x3f13	16147	Internal error ROBOT-NoManuControl
0x3f14	16148	Internal error ROBOT-NoAutoControl
0x3f15	16149	Jog mode cannot be selected: first enable arm power.
0x3f16	16150	Turn off arm power or switch to manual mode. Jogging is not possible in this mode
0x3f17	16151	Robot-NoStartStopKeys
0x3f18	16152	Your profile does not allow you to stop the running application.
0x3f19	16153	Jog move impossible: first select a jog mode.
0x3f1a	16154	Invalid move axis or direction
0x3f1b	16155	Manual motion mode:Disabled
0x3f1c	16156	Output write impossible: first select manual working mode.
0x3f1e	16158	ROBOT-JogMoveRefused
0x3f1f	16159	No target defined
0x3f20	16160	No target defined
0x3f21	16161	No target defined
0x3f22	16162	Cannot apply arm power: park the pendant on the cradle.
0x4101	16641	Recorder cannot open data

0x4102	16642	Recorder cannot close data
0x4103	16643	No recorder data available.
0x4104	16644	Invalid record file name.
0x4105	16645	Cannot open record file.
0x4301	17153	A record is already pending.
0x4303	17155	No record data to store.
0x4304	17156	Record stopped: record memory is full
0x4305	17157	End of record.
0x4306	17158	Record stopped: a variable could not be read
0x4307	17159	Recorder configuration error : invalid frequency
0x4308	17160	Recorder frequency was rounded
0x4309	17161	Recorder internal error: timeout
0x430a	17162	Recorder cannot open data
0x430b	17163	Recorder cannot close data
0x4401	17409	Internal error ROBOT-Overrun-
0x4402	17410	Internal error ROBOT-Watchdog-
0x4501	17665	Recorder configuration error : invalid frequency
0x4601	17921	no sercos ring
0x4602	17922	argument not supported
0x4603	17923	idn not found
0x4604	17924	wrong number of parameters
0x4605	17925	cannot open file
0x4606	17926	memory full
0x4607	17927	invalid type
0x4608	17928	unhandled exeption
0x4701	18177	loop param not defined
0x4703	18179	loop record not defined
0x4704	18180	loop recorder not defined
0x4705	18181	loop incorrect name
0x4706	18182	loop param range
0x4707	18183	loop cannot restore param
0x4708	18184	loop cannot memorize param
0x4709	18185	loop invalid index
0x4801	18433	Tun: timeout not ready
0x4802	18434	Tun: cannot move
0x4803	18435	Tun: timeout movement tool long
0x4901	18689	Tun: param not memorized
0x4902	18690	GPVFR out of range

0x4903	18691	GPAFR out of range
0x4904	18692	GPAFR2 out of range
0x4905	18693	Cannot read sercos DIPEAK
0x4906	18694	Cannot read sercos MICONT
0x4907	18695	IFRIC out of range
0x4908	18696	cannot read sercos GPVFR
0x4909	18697	cannot read sercos GPAFR
0x490a	18698	cannot read sercos GPAFR2
0x490b	18699	Cannot read sercos IFRIC
0x490c	18700	Incorrect drive number
0x4a01	18945	Internal error INTERFACE-EventLost : some system event messages will be lost.
0x4a02	18946	End of internal error INTERFACE-EventLost.
0x4b00	19200	Unival: ND
0x4b01	19201	Unival: ND
0x4b02	19202	Unival: ND
0x4b03	19203	Unival: ND
0x4b04	19204	Unival: ND
0x4b05	19205	Unival: ND
0x4b06	19206	Unival: ND
0x4b07	19207	Unival: ND
0x4b08	19208	Unival: ND
0x4b09	19209	Unival: ND
0x4b0a	19210	Unival: ND
0x4b0b	19211	Unival: ND
0x4b0c	19212	No board found for <Powerlink>
0x4b0d	19213	Unival: ND
0x4b0e	19214	Unival: ND
0x4b0f	19215	Unival: ND
0x4b10	19216	Unival: ND
0x4b11	19217	Unival: ND
0x4b12	19218	Unival: ND
0x4b13	19219	Unival: ND
0x4b14	19220	Unival: ND
0x4b15	19221	Unival: ND
0x4b16	19222	Unival: ND
0x4b17	19223	Unival: ND
0x4b18	19224	Unival: ND

0x4b33	19251	Unival: ND
0x4b34	19252	Unival: ND
0x4b35	19253	Unival: ND
0x4c01	19457	While building CoDeSys IO Driver : Cannot create directory Data
0x4c02	19458	While building CoDeSys IO Driver : Cannot create directory Data
0x4c03	19459	While building CoDeSys IO Driver : Cannot load module Data
0x4c04	19460	While building CoDeSys IO Driver : Cannot start module Data
0x4c05	19461	While building CoDeSys IO Driver : Cannot end module Data
0x4c06	19462	While building CoDeSys IO Driver : Found in XML file an invalid IEC TYPE (neither BOOL, BYTE, WORD, DWORD, SINT, USINT, INT, UINT, DINT, UDINT, nor REAL) for Data
0x4c07	19463	While building CoDeSys IO Driver : Found in XML file an invalid DIRECTION (neither 'I' nor 'Q') for Data
0x4c08	19464	While building CoDeSys IO Driver : Found in XML file an invalid ADDRESS (not a numeric value) for Data
0x4c0a	19466	While building CoDeSys IO Driver : Found an IO description in XML file but not in CODESYS runtime for Data
0x4c0b	19467	While building CoDeSys IO Driver : Found in XML file an invalid OFFSET (not match with IEC TYPE) for Data
0x4c0c	19468	While building CoDeSys IO Driver : Found an IO description in CODESYS runtime but not in XML file for Data
0x4c0d	19469	While building CoDeSys IO Driver : Found a TYPE that does not match between XML file and CODESYS runtime for Data
0x4c0e	19470	While building CoDeSys IO Driver : Found a DIRECTION that does not match between XML file and CODESYS runtime for Data
0x4c0f	19471	While building CoDeSys IO Driver : Found an OFFSET that does not match between XML file and CODESYS runtime for Data
0x4c10	19472	While building CoDeSys IO Driver : Found IO with name already used Data
0x4c11	19473	While building CoDeSys IO Driver : CoDeSys feature is only available on real target Data
0x4c13	19475	While building CoDeSys IO Driver : Downloaded PLC automaton has a different IO maping than the XML config file used at boot time. Data
0x4c14	19476	While building CoDeSys IO Driver : Refuse to enable PLC because IO maping is not valid. Data
0x4d01	19713	Build error because found incoherency while building Platform : Starc Firmware version is not valid : Data
0x4e01	19969	Project Data saved.
0x4f01	20225	Starting system backup...
0x4f02	20226	Restoring contents from archive...
0x4f03	20227	Updating system...
0x4f0a	20234	System backup successfully completed.
0x4f0b	20235	Archive contents successfully restored.

0x4f0c	20236	Please reboot controller to complete system restoration.
0x4f14	20244	Backup operation failed (Data).
0x4f15	20245	Restore operation failed (Data).
0x4f16	20246	System update failed (Data).
0x4f17	20247	Archive is invalid (Data).
0x4f18	20248	Path is invalid (Data).
0x4f19	20249	Unable to backup specified items (Data).
0x4f1a	20250	Unable to restore specified items (Data).
0x4f1b	20251	Error while writing to filesystem (Data).
0x5a01	23041	Deadman enabled.
0x5a02	23042	Deadman disabled.
0x5a03	23043	Deadman validated
0x5a04	23044	Deadman lock to validated
0x5304	21252	Autoload of VAL3 applications cancelled (unrecoverable system error at boot time)
0x5302	21250	Memory full
0x5307	21255	Cannot step: task not stopped.
0x5303	21251	Autoload error:
0x5305	21253	Autostart error:
0x5301	21249	Project %s deleted.
0x5306	21254	An application is already running. Stop it to start another one.
0x5101	20737	No application running.
0x5102	20738	Application is already stopping
0x5402	21506	application started
0x5403	21507	application stopped
0x5401	21505	Application has ended, pending motion was cancelled
0x5404	21508	VAL3 runtime error.
0x5001	20481	Adjustment on all axes
0x5002	20482	Adjustment on axis #
0x5006	20486	To calibrate robot for the first time, place the robot close to the marks and press Rec.
0x5005	20485	An error occured during boot. Robot is not operational. See logger for more details.
0x5003	20483	Position recovery on all axes
0x5004	20484	Position recovery on axis #